September 4, 2025

Dresner Advisory Services, LLC

2025 Edition

AI, Data Science, and Machine Learning Market Study

Wisdom of Crowds® Series

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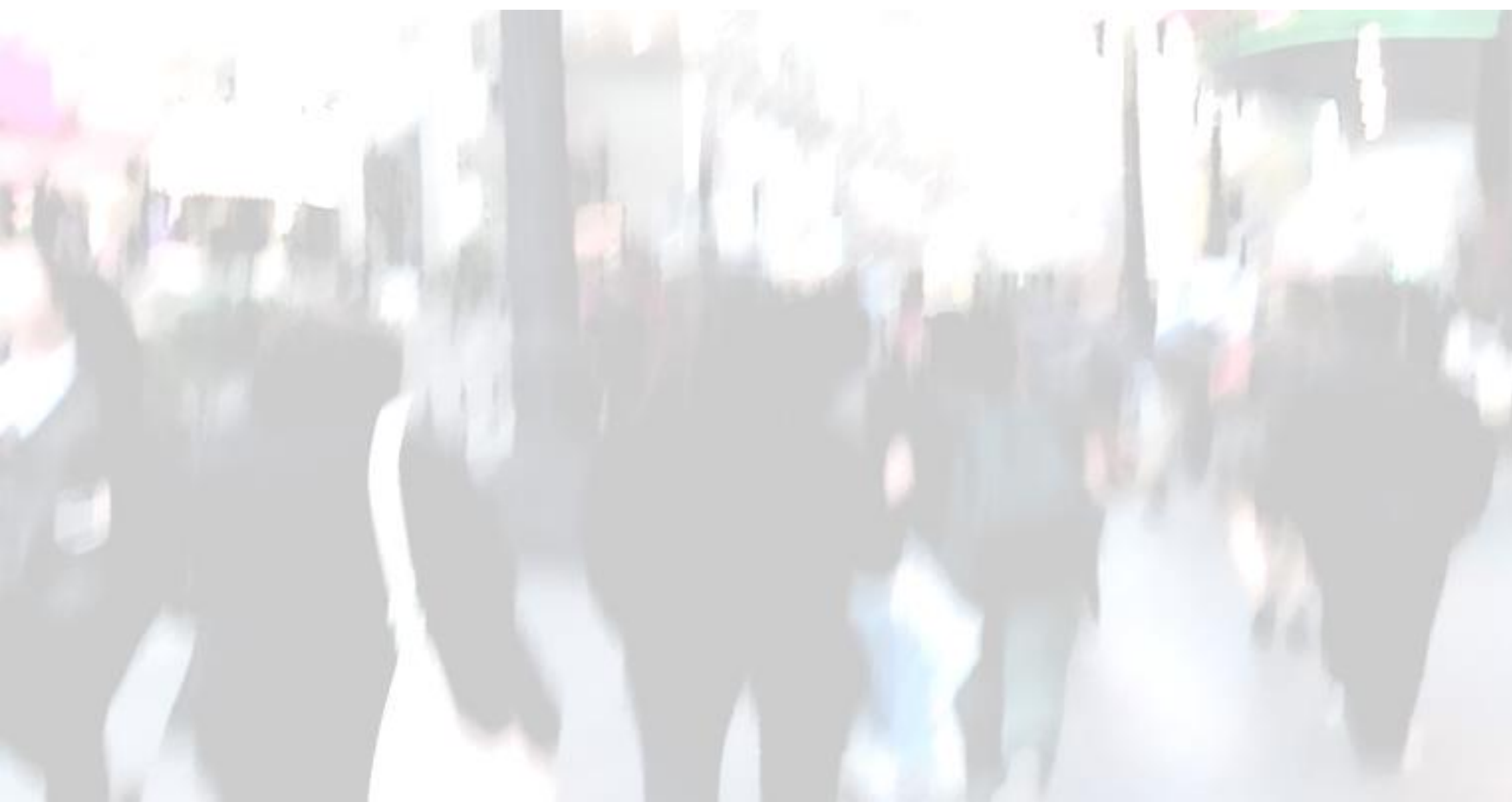


Definitions

Business Intelligence Defined

Business intelligence (BI) is “knowledge gained through the access and analysis of business information.

Business intelligence tools and technologies include query and reporting, online analytical processing (OLAP), data mining and advanced analytics, end-user tools for ad hoc query and analysis, and dashboards for performance monitoring.”

Howard Dresner*, The Performance Management Revolution: Business Results Through Insight and Action* (John Wiley & Sons, 2007).

AI, Data Science, and Machine Learning Defined

“AI, data science, and machine learning” includes statistics, modeling, machine learning, neural networks, and data mining to analyze facts to make predictions about future or otherwise unknown events.

Generative AI Defined

Generative artificial intelligence (also called generative AI) is artificial intelligence capable of generating text, images, or other media, using generative models. Generative AI models learn the patterns and structure of their input training data and then generate new data that has similar characteristics

Agentic AI Defined

Agentic AI solutions deliver measurable business outcomes by wrapping together three foundational components: a clearly defined business process, a large language model (LLM), and access to relevant, cloud-hosted data. These components work together to enable intelligent task execution-automating actions, making decisions, and adapting in real time with minimal human intervention. The business process at the core of each solution can take the form of a structured, rule-based workflow or a more adaptive business script, depending on the nature of the task. Workflows are ideal for repeatable, high-precision operations, while scripts offer flexibility for more variable or context-sensitive activities.

Introduction

As we start our 19th year as an independent, objective, and data-driven research organization, our commitment to providing insights into data, analytics, and performance management remains steadfast. This year's report marks the 12th edition of our annual research into AI, data science, and machine learning (AI/DS/ML), reflecting the growing significance of these technologies as organizations seek to enhance operations, improve forecasting, and drive innovation.

Artificial intelligence (AI), data science (DS), and machine learning (ML) are shifting from experimental to strategic enablers. Most organizations report AI is playing either a direct or supporting role in their business, with investment driven by the need to solve inefficiencies, experiment, and prepare for disruption. At the same time, AI governance remains inconsistent, with approaches ranging from centralized oversight to ad hoc or absent controls, reflecting an uneven path to maturity.

Generative and agentic AI are fueling a new wave of interest. While only 15% of organizations report generative AI in production and 7% report the same for agentic AI, experimentation is widespread—more than one-third for generative AI and more than one-quarter for agentic AI.

Despite modest current deployment, momentum is building. Larger organizations are sustaining long-term AI use, correlating with experimentation in generative and agentic approaches. Features such as outlier detection and model explainability remain essential, underscoring the need for transparency and trust. Overall, organizations are cautious but optimistic, advancing at different speeds toward more mature and pervasive use of AI, DS, and ML to drive business value.

As always, our research aims to provide valuable, actionable insights that empower organizations to make informed decisions in their AI, DS, and ML initiatives.

We hope you enjoy this report!

Best,

Howard

Howard Dresner
Chief Research Officer
Dresner Advisory Services

Contents

Benefits of the Study

The Dresner Advisory Services AI, Data Science, and Machine Learning Market Study provides a wealth of information and analysis, offering value to both consumers and producers of business intelligence technology and services.

Consumer Guide

As an objective source of industry research, the Dresner Advisory Services AI, Data Science, and Machine Learning Market Study helps consumers to understand how their peers leverage and invest in business intelligence and related technologies.

Using relevant criteria to evaluate vendors and products, users glean key insights into software supplier performance, enabling:

- Comparisons of current vendor performance to industry norms
- Identification and selection of new vendors

Supplier Tool

Vendor Licensees use the Dresner Advisory Services AI, Data Science, and Machine Learning Market Study in several important ways. For example:

External Awareness

- Build awareness for the business intelligence market and supplier brand, citing Dresner Advisory Services AI, Data Science, and Machine Learning Market Study trends and vendor performance
- Create lead and demand generation for supplier offerings through association with the Dresner Advisory Services AI, Data Science, and Machine Learning Market Study findings, webinars, etc.

Internal Planning

- Refine internal product plans and align with market priorities and realities as identified in the Dresner Advisory Services AI, Data Science, and Machine Learning Market Study
- Better understand customer priorities, concerns, and issues
- Identify competitive pressures and opportunities

About Howard Dresner and Dresner Advisory Services

The Dresner Advisory Services AI, Data Science, and Machine Learning Market Study was conceived, designed, and executed by Dresner Advisory Services, LLC—an independent advisory firm—and Howard Dresner, its founder and chief research officer.

Howard Dresner is one of the foremost thought leaders in business intelligence and performance management, having coined the term “business intelligence” in 1989. He has published two books on the subject, *The Performance Management Revolution – Business Results through Insight and Action* (John Wiley & Sons, Nov. 2007) and *Profiles in Performance – Business Intelligence Journeys and the Roadmap for Change* (John Wiley & Sons, Nov. 2009). He lectures at forums around the world and is often cited by the business and trade press.

Prior to Dresner Advisory Services, Howard served as chief strategy officer at Hyperion Solutions and was a research fellow at Gartner, where he led its business intelligence research practice for 13 years.

Howard has conducted and directed numerous in-depth primary research studies over the past three decades and is an expert in analyzing these markets.

Through the Wisdom of Crowds® Business Intelligence market research reports, we engage with a global community to redefine how research is created and shared. Other research reports include:

- Wisdom of Crowds® Flagship BI Market Study
- Active Data Architecture®
- Agentic AI
- Analytical Data Infrastructure
- Analytical Platforms
- Data Engineering
- Embedded Business Intelligence
- Generative AI
- Guided Analytics®
- Master Data Management
- ModelOps
- Semantic Layer

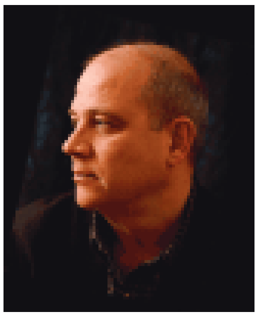
You can find more information about Dresner Advisory Services at www.dresneradvisory.com.

About Jim Ericson

Jim Ericson is vice president and distinguished analyst with Dresner Advisory Services.

Jim has served as a consultant and journalist who studies end-user management practices and industry trending in the data and information management fields.

From 2004 to 2013 he was the editorial director at *Information Management* magazine (formerly *DM Review*), where he created architectures for user and industry coverage for hundreds of contributors across the breadth of the data and information management industry.



As lead writer, he interviewed and profiled more than 100 CIOs, CTOs, and program directors in a program called "25 Top Information Managers." His related feature articles earned ASBPE national bronze and multiple Mid-Atlantic region gold and silver awards for Technical Article and for Case History feature writing.

A panelist, interviewer, blogger, community liaison, conference co-chair, and speaker in the data-management community, he also sponsored and co-hosted a weekly podcast in continuous production for more than five years.

Jim's earlier background as senior morning news producer at NBC/Mutual Radio Networks and as the first managing editor of MSNBC's Washington, D.C. online news bureau cemented his understanding of fact-finding, topical reporting, and serving broad audiences.

The Dresner Team

About Elizabeth Espinoza

Elizabeth is director of analytics at Dresner Advisory and is responsible for the data preparation, analysis, and creation of charts for Dresner Advisory reports.

About Sherry Fairchok

Sherry is senior editor at Dresner Advisory, ensuring the quality and consistency of all research publications.

About Danielle Guinebertiere

Danielle is vice president of client services at Dresner Advisory. She supports the ongoing research process through her work with executives at companies included in Dresner market reports.

About Michelle Whitson-Lorenzi

Michelle is director of research operations and is responsible for managing software company survey activity and our internal market research data.

Survey Method and Data Collection

As with all our Wisdom of Crowds® market studies, we constructed a survey instrument to collect data and used social media and crowdsourcing techniques to recruit participants.

Data Quality

We carefully scrutinized and verified all respondent entries to ensure that only qualified participants were included in the study.

Executive Summary

Executive Summary

- A large majority of survey respondents say artificial intelligence (AI) is taking a direct strategic or supporting role, sometimes driving the business; or are at minimum in exploratory stages of development (figs. 5-9).
- The greatest AI investment drivers are specific business challenges or inefficiencies, followed by experimentation and by anticipation of disruption (figs. 10-12).
- The top AI objectives are operational excellence, data-driven decision making, and enhanced customer service (figs. 13-15).
- AI policy management falls to a mix of controls and often to an alarming lack thereof, from centralized to ad hoc (figs. 16-18).
- A majority describe AI maturity as “emerging;” small numbers of small and very large organizations are most advanced (figs. 19-23).
- Users are “cautiously optimistic” to “excited about prospects” for generative and agentic AI; more than one-third are “experimenting today” with generative AI, more than one-quarter with agentic AI; fewer are in production (figs. 24-25). The most common response to AI budgeting questions is “don’t know,” or “no budget allocated” (fig. 26).
- In-production use of generative AI is 15%, a slight year-over-year increase; use is front-office sales and back-office development/deployment; use scales with organization size; customer and unstructured are most common data sources (figs. 27-33).
- Just 7% are using agentic AI in production, though experimentation and future plans are expanding; sales & marketing, IT, and operations are top users. Use increases with organization size, and customer and finance data are most referenced (figs. 34-41).
- The roles of statistician/data scientist and BI expert are the most likely constant or often users of AI, data science, and machine learning. User activity is at historically average or slightly higher levels (figs. 42-48).
- AI, data science, and machine learning ranks 24th, and cognitive BI/artificial intelligence-based BI ranks 35th among 65 topics under study. Its historic importance remains in a range greater than important and increases with organization size, and its importance correlates to generative/agentic AI use (figs. 48-57).
- Current deployment of use cases for AI, data science, and machine learning is fairly low but broad, with predictions of strong future uptake. The most important include demand forecasting, customer segmentation and predictive maintenance (figs. 58-67).

- Current deployment and adoption of AI, data science, and machine learning is at a steady rate, showing historic gains above the level of "important." In-production deployments are highest in R&D and in larger organizations; deployment correlates strongly to strategic importance (figs. 68-77).
- Longevity of AI, data science, and machine learning use is rising, most sustained in the largest organizations; this correlates to generative and agentic AI adoption (figs. 78-82).
- The most important features for AI, data science, and machine learning include outlier detection and model explainability. Feature interest is near or below historic levels. Industry feature support is very strong (figs. 83-88).
- Top usability features include Python support, low code/no code, and advanced analytics (figs. 89-93). Industry support for usability is strong (fig. 111).
- The most important scalability features are in-database analytics and in-memory analytics (figs.94-97). Industry support is strong (fig. 102).
- The most important neural networks include recursive and artificial neural networks; overall importance is highest year over year (figs. 98-99). Industry support is incomplete (fig. 110).
- Top industry data sources include Snowflake, Amazon S3, and RedShift (fig. 114).
- AI, data science, and machine learning vendor ratings are shown in fig. 115.

Study Demographics

Study Demographics

Our study includes a cross-section of respondents across geographies, functions, organization sizes, and vertical industries. We believe that, unlike other industry research, this supports a more representative sample that is a better indicator of true market dynamics. We constructed cross-tab analyses using these demographics to identify and illustrate important industry trends.

Geographies

North America (which includes the United States, Canada, and Puerto Rico) represents the largest group in our 2025 user study, with 60% of all respondents. EMEA is the second-largest group (about 18%), followed by Asia Pacific (18%) and Latin America (5%; fig.1).

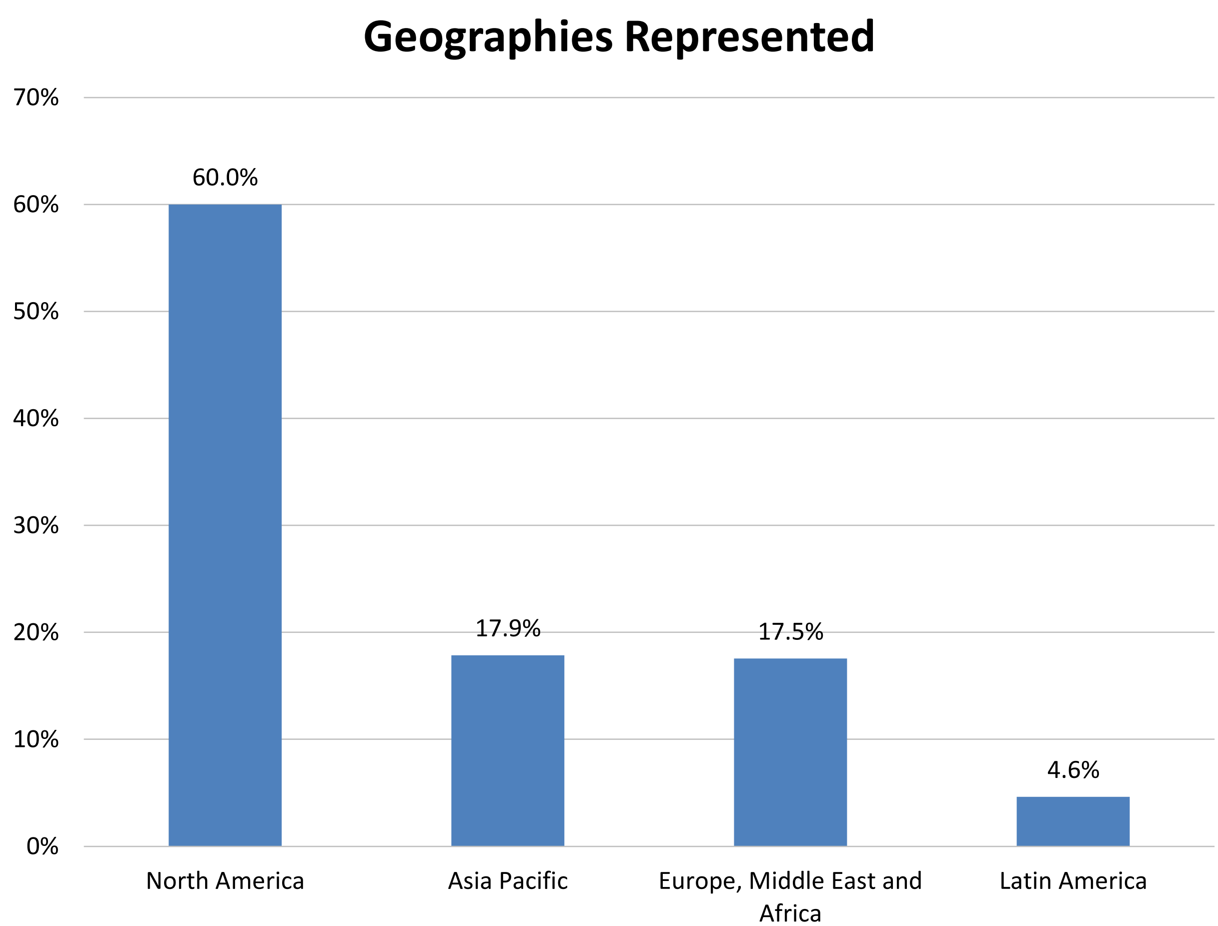


Figure 1 – Geographies represented

Functions

Our 2025 user sample includes a cross section of functional roles in respondent organizations. Information technology (35%) is the most represented, followed by finance (16%), executive management (15%), and the BI/analytics competency center (14%; fig. 2).

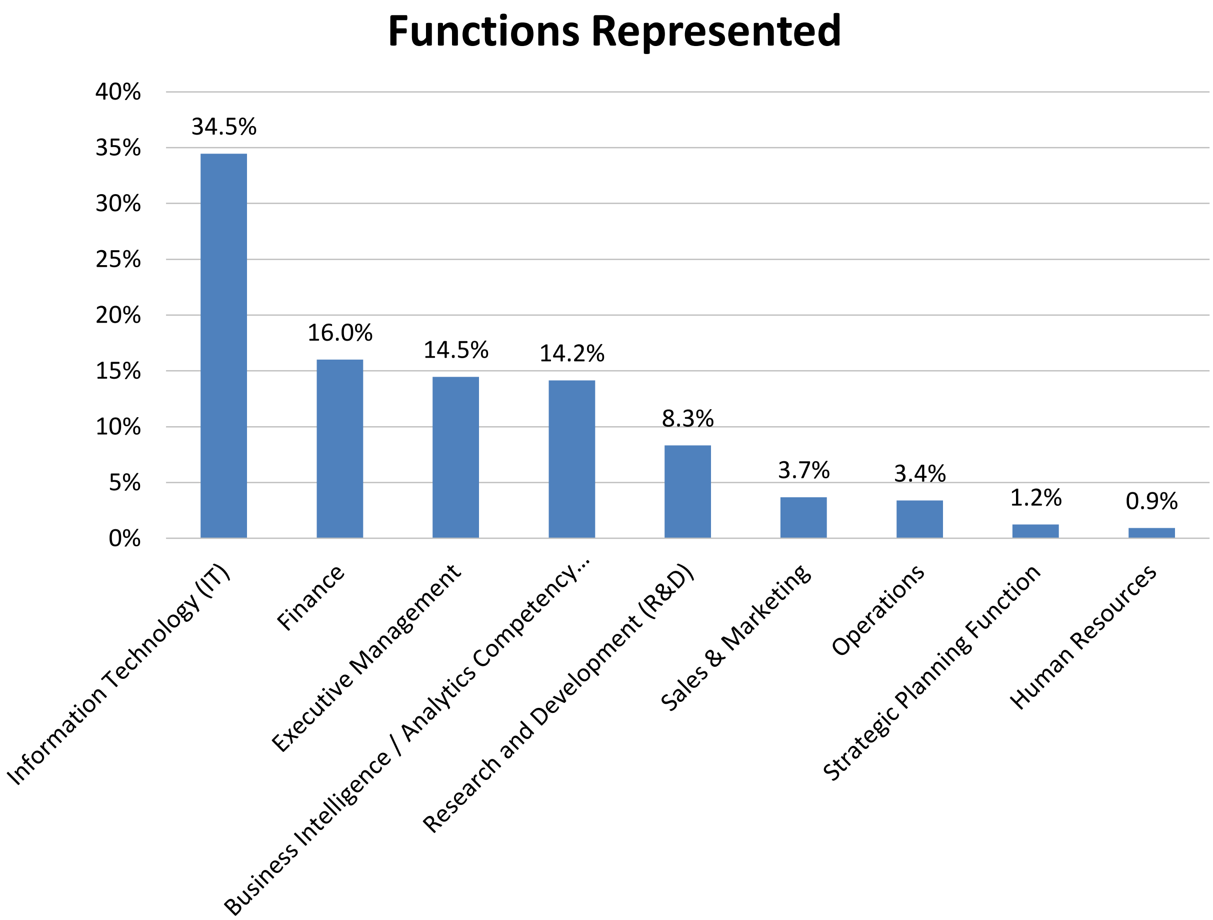


Figure 2 – Functions represented

Vertical Industries

Technology (24%), business services (16%), and manufacturing (16%) lead industry participation in our 2025 user sample (fig. 3). Financial services, healthcare, education, and consumer services are the next-most represented.

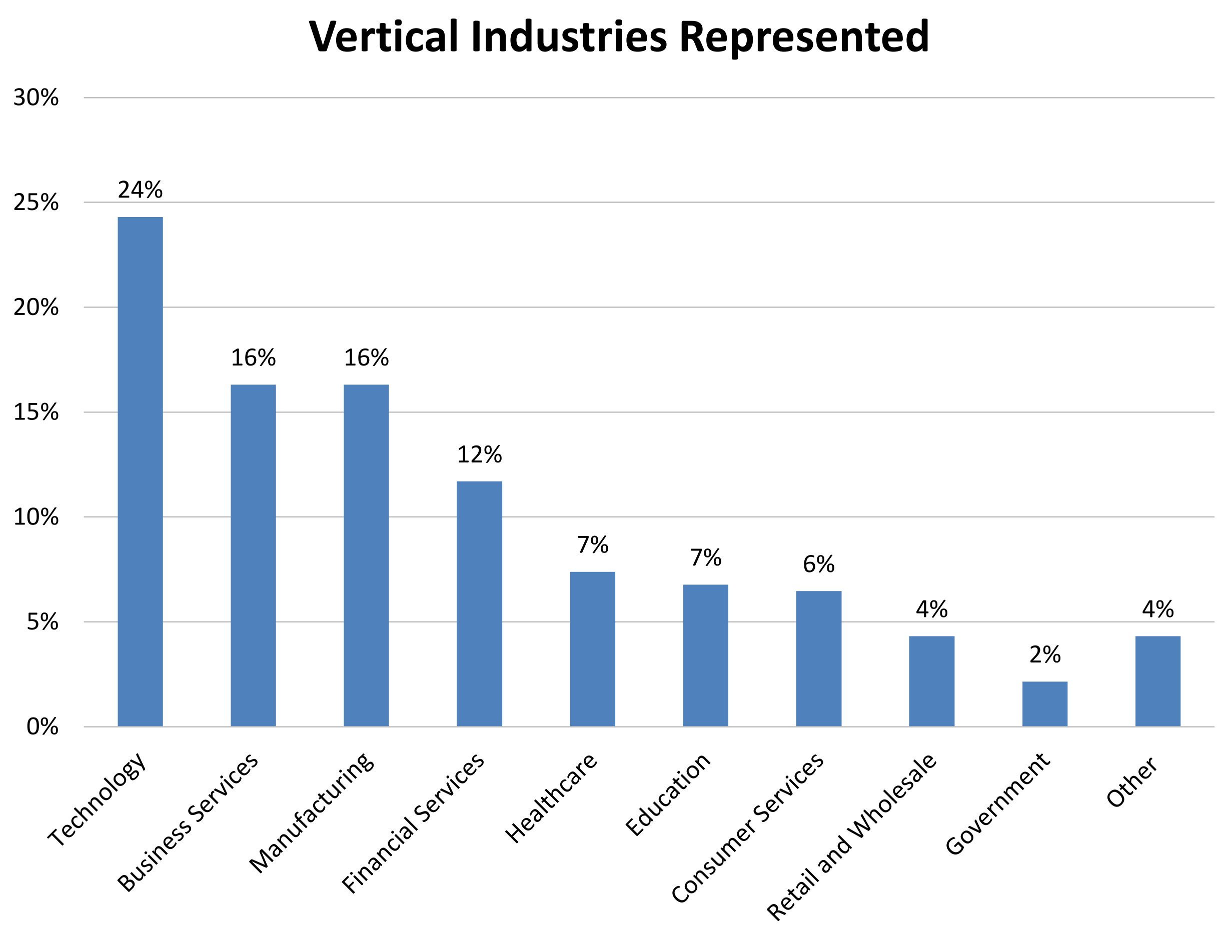


Figure 3 – Vertical industries represented

Organization Sizes

Our 2025 user sample includes a mix of small, midsize, and large organizations (fig. 4). Small organizations (1-100 employees) and midsize organizations (101-1,000 employees) account for 24% and 28% of the sample, respectively. Large organizations (more than 1,000 employees) account for the remaining 48%. The chart below further breaks out these organizations by global headcount.

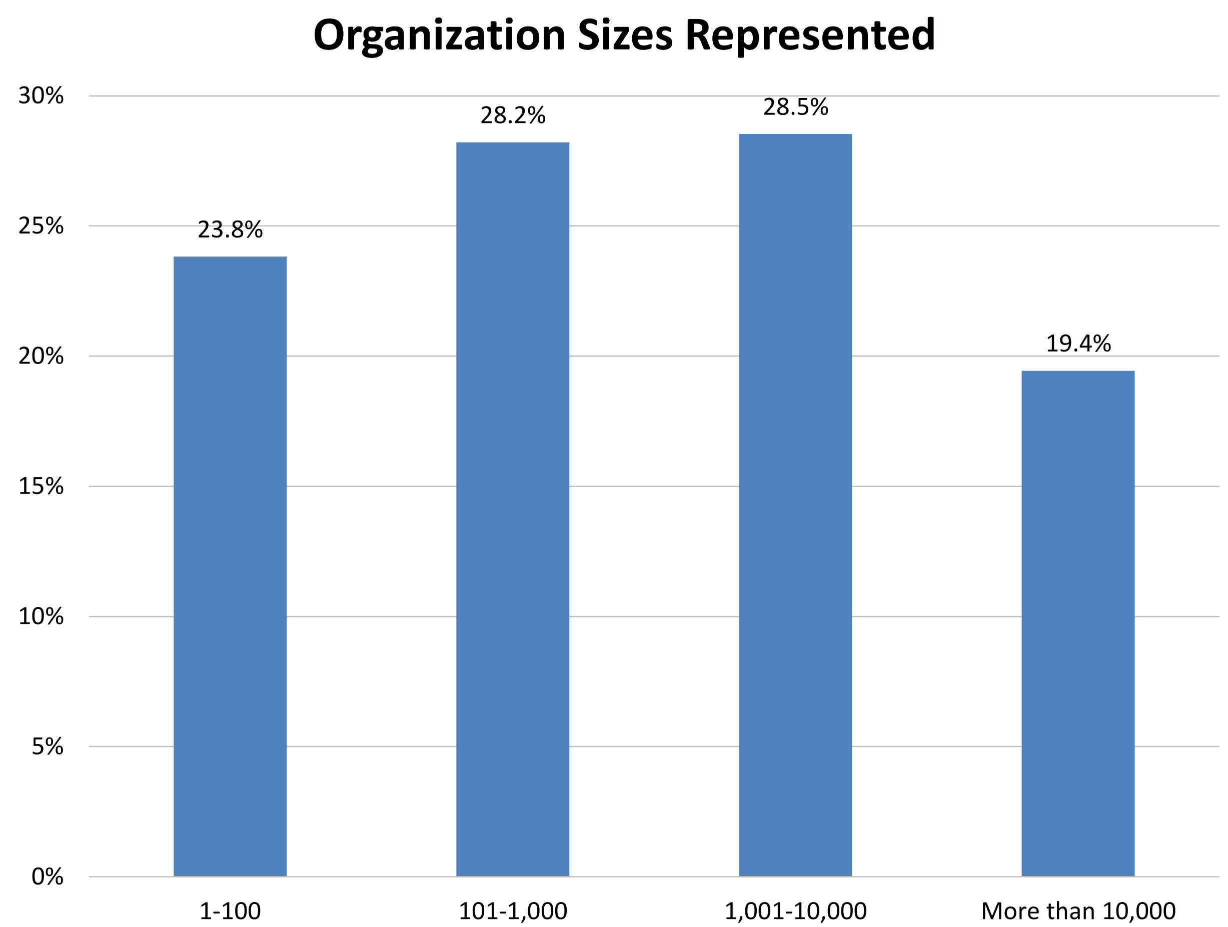


Figure 4 – Organization sizes represented

Analysis and Trends

Analysis: AI, Data Science, and Machine Learning

Current Strategic Role of AI

We asked respondents, “How integrated is AI to your organization’s overall business?” A large majority of user respondents (83%) say AI is either taking a direct or supporting role or, at minimum, is in exploratory stages of development (fig. 5). Eleven percent indicate a direct role for AI driving or shaping the business, while 24% describe an indirectly supporting role, likely in personal productivity or routine task automation. The greatest percentage (48%) indicate exploratory use, while 17% are largely or completely uninvolved with AI. These findings are further colored by functional, cultural, and other demographics examined in figs. 6-9.

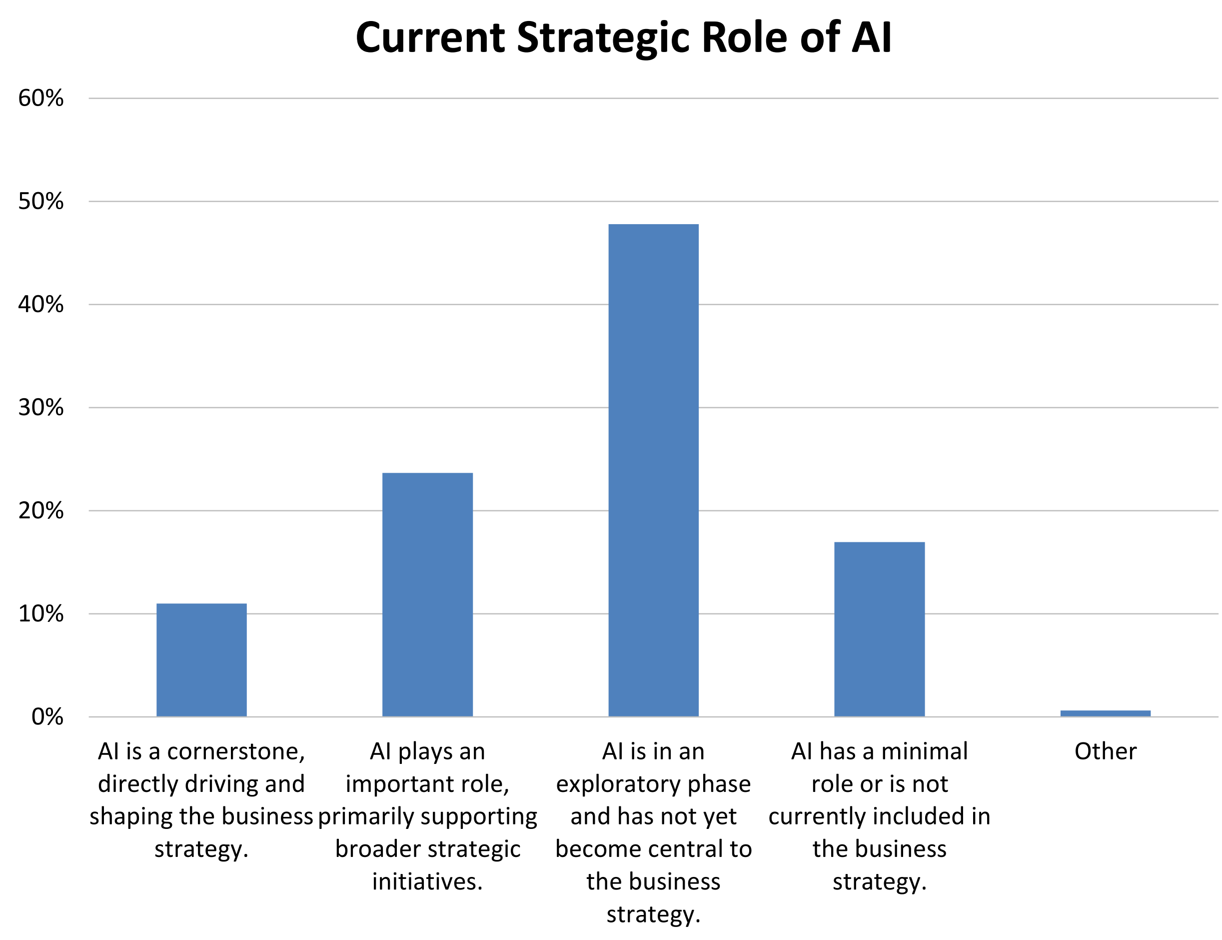


Figure 5 - Current strategic role of AI

Current Strategic Role of AI by Function

Viewed by function, AI finds 2025 mixed use among both front- and back-office roles (fig. 6). The greatest number of "cornerstone" users (31%) is found in sales & marketing. This represents almost three times the number of all enterprise users who describe AI as having a "direct role" driving the business (fig. 5). Respondents in R&D are second-most likely to describe AI as taking a cornerstone or "important" role and serve to support custom development as well as productivity or other uses. Respondents in IT are the third-strongest advocate of strategic AI, served in part by code writing and other productivity automation. Lower estimations from BICC respondents likely indicate we still see early-stage provision and deployment of AI. This extends to operations, executive management, and last of all to finance, where cornerstone use is lowest, below 10% of user responses.

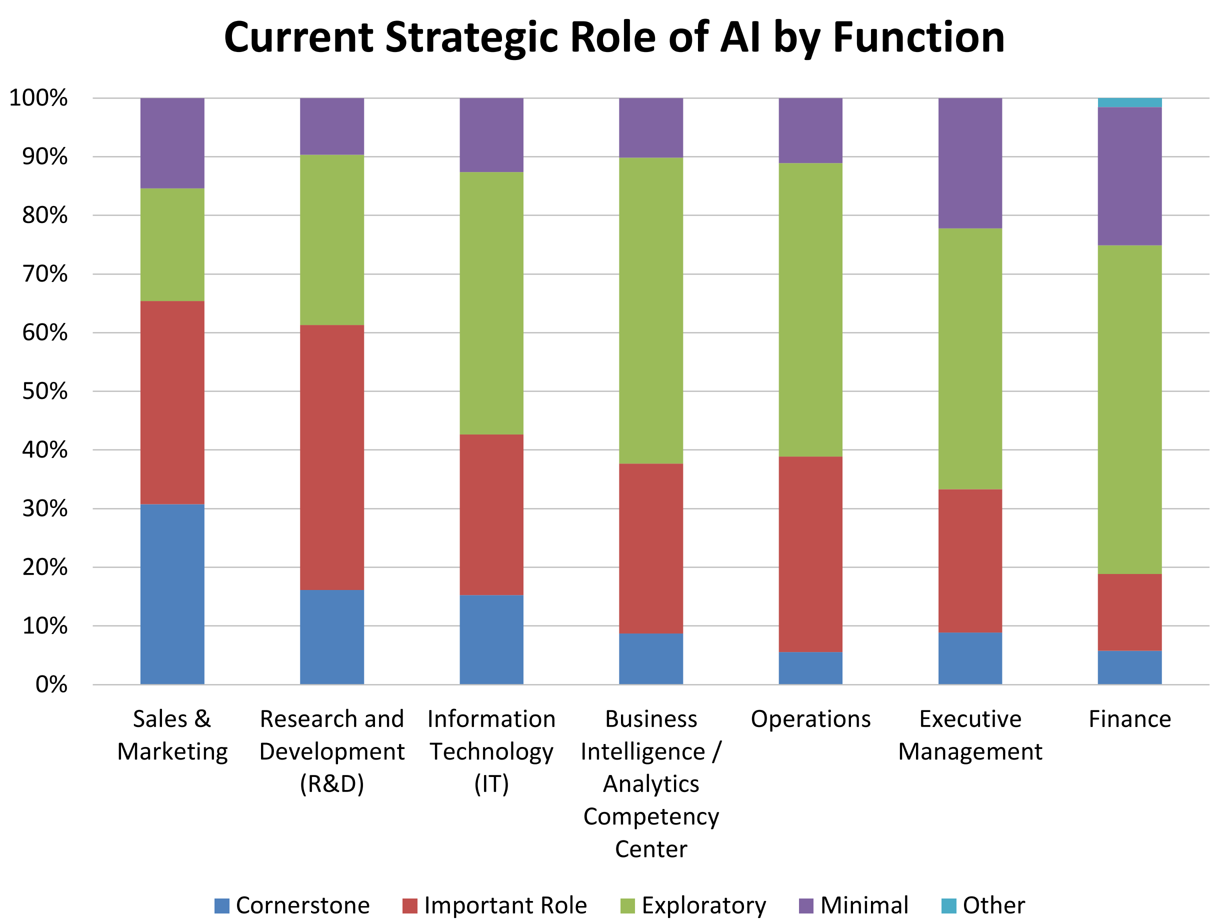


Figure 6 - Current strategic role of AI by function

Current Strategic Role of AI by Organization Size

In 2025, the largest (more than 10,000 employees) and smallest (1-100 employees) organizations are most likely to assign a more critical strategic role to AI (fig.7). Among respondent organizations with 100 or more employees, we also observe that the strategic role of AI scales incrementally with headcount. The greatest number of combined "cornerstone" and "important role" users are found in very large (47%) and small (36 %) organizations. In contrast, the number of organizations that assign only minimal strategic importance to AI decreases with organization headcount, which implies a perceived economy of scale with AI use.

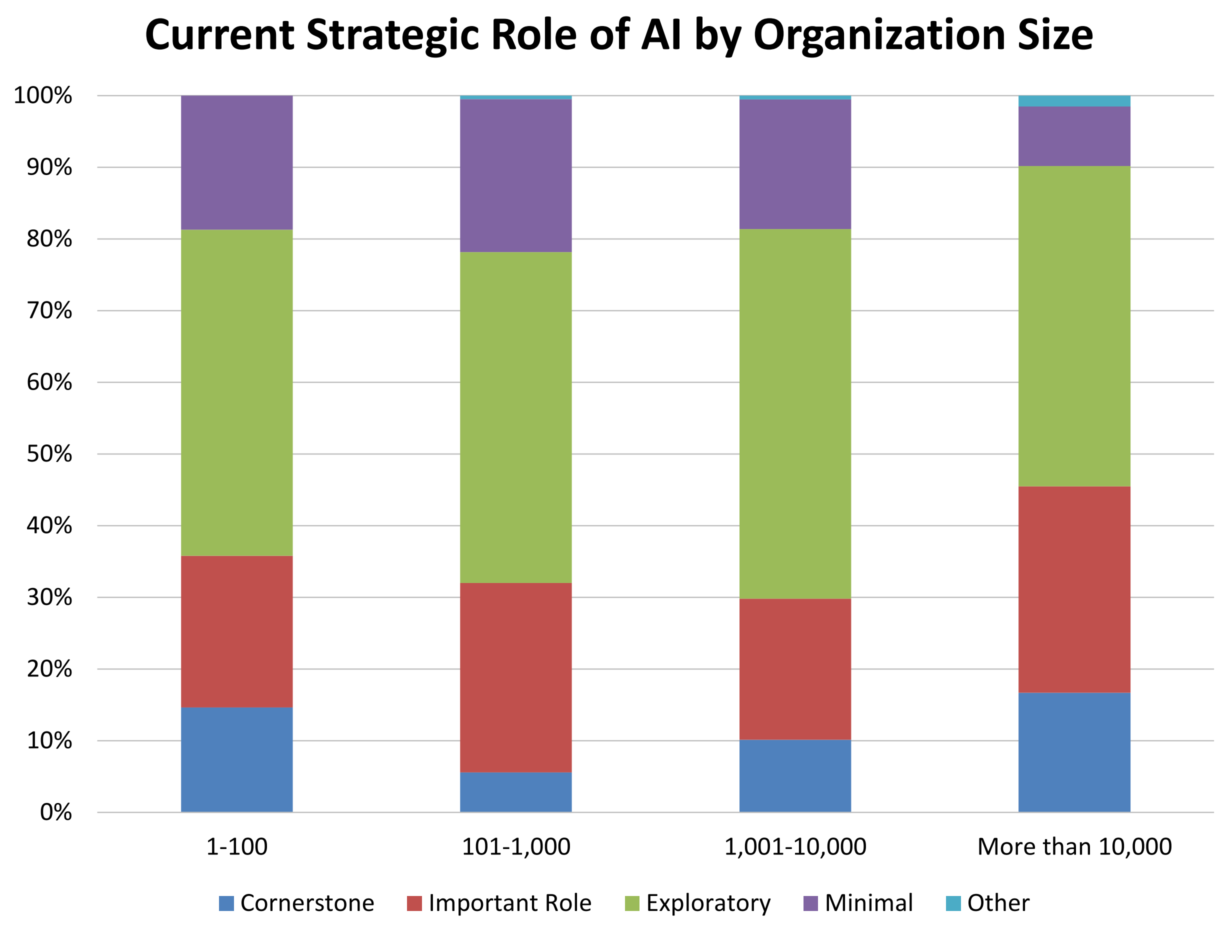


Figure 7 - Current strategic role of AI by organization size

Current Strategic Role of AI by Difficulty Finding Analytic Content

In 2025, strategic importance assigned to AI scales inversely with an organization's expressed difficulty finding analytic content (fig. 8). Depending on the organization, this finding might appear either logical or counterintuitive. Given the early stage of AI deployment, we might expect organizations that find it difficult to access analytic content are more likely to embrace AI to augment traditional search and improve future visibility to analytic content. Conversely, we might assume early-stage AI adopters have already improved their ability to access analytic content. In either event, the correlation is visible. Organizations that find it extremely easy to find analytic content are far more likely to embrace AI in a "cornerstone" or "important" role (50%), compared with only 29% of organizations that find it "difficult" to access analytic content.

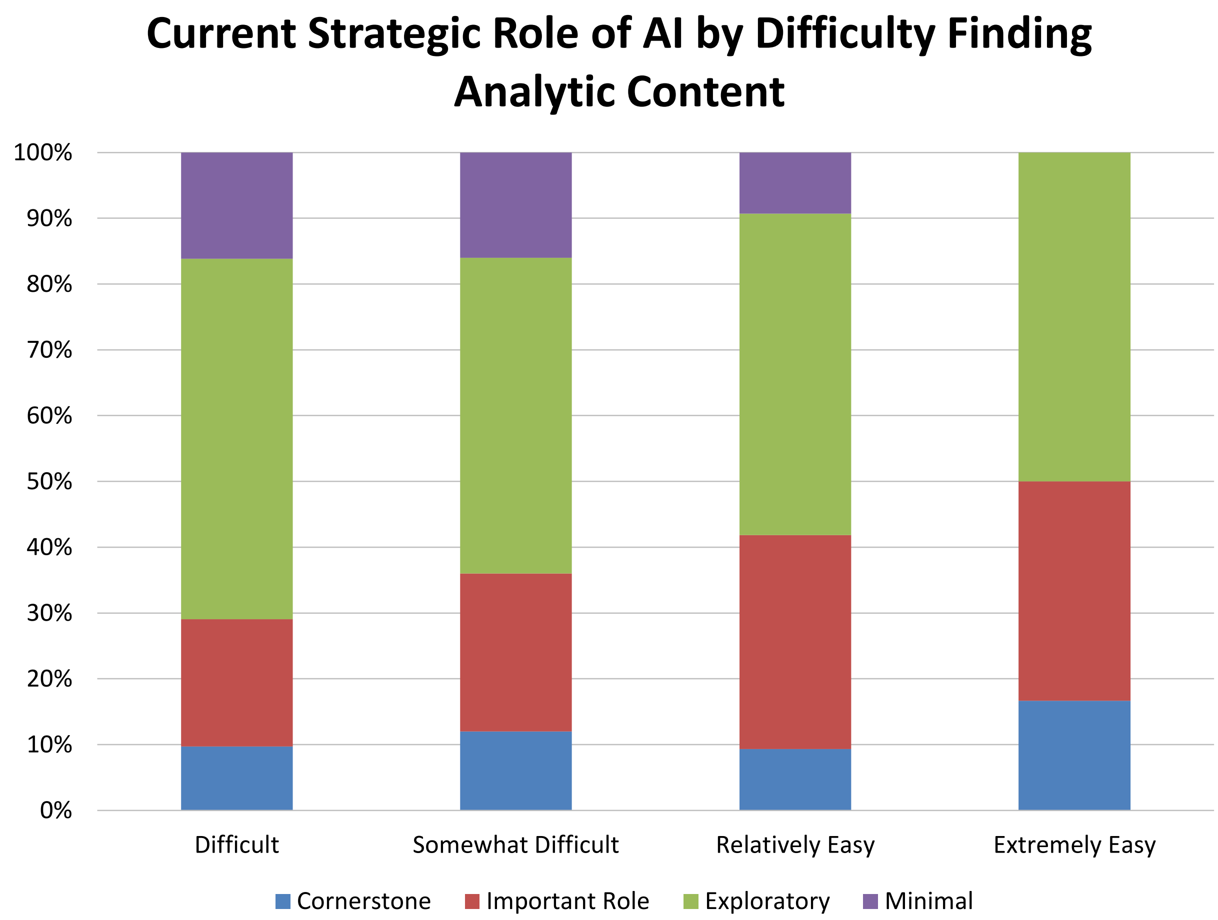


Figure 8 - Current strategic role of AI by difficulty finding analytic content

Current Strategic Role of AI by Data Leadership

In 2025, organizations with identified data leaders (chief data officer, chief analytical officer, or another designated role) are more likely to describe their current strategic state of AI as "cornerstone" or "important" (fig. 9). This implies that organizations which embrace the value of treating data as an asset through dedicated practice and program oversight more highly value AI as a part of their proficiency. Organizations that assign "cornerstone" value to AI are 16% likely to have a data leader in place, compared with 15% of those with future plans for data leadership and just 5% of those with no data leader. The findings extrapolate inversely to those that assign a minimal or exploratory role to AI. Organizations with no data leadership are by far most likely (27%) to have "minimal" future plans for AI.

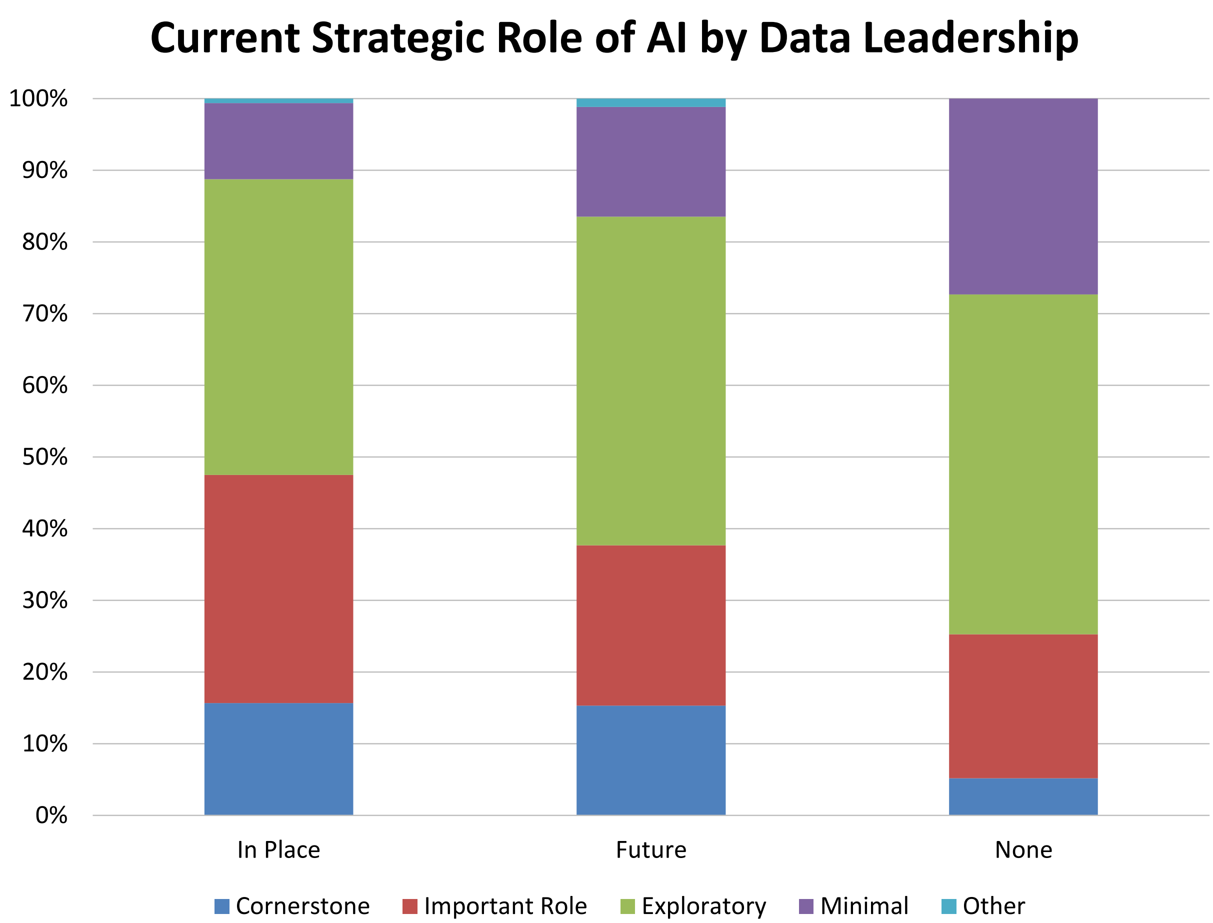


Figure 9 - Current strategic role of AI by data leadership

Primary Driver for AI Investments

We asked respondents, “What is the primary driver for your AI investments in 2025?” The findings suggest that most enterprise users are acting selectively with both targeted investments and acknowledgement of a learning curve for future AI use cases (fig. 10). The greatest number, 42%, are targeting specific business challenges or inefficiencies with their AI investments. The next-greatest cohort (35%) says the primary driver is experimentation for future use, and another 13% are anticipating future industry disruption. About 7% say they are simply attempting to maintain parity with competitors. We note that this finding does not include sentiment from the 17% of the overall sample who say AI has a minimal role or is not included in the business strategy (fig. 10).

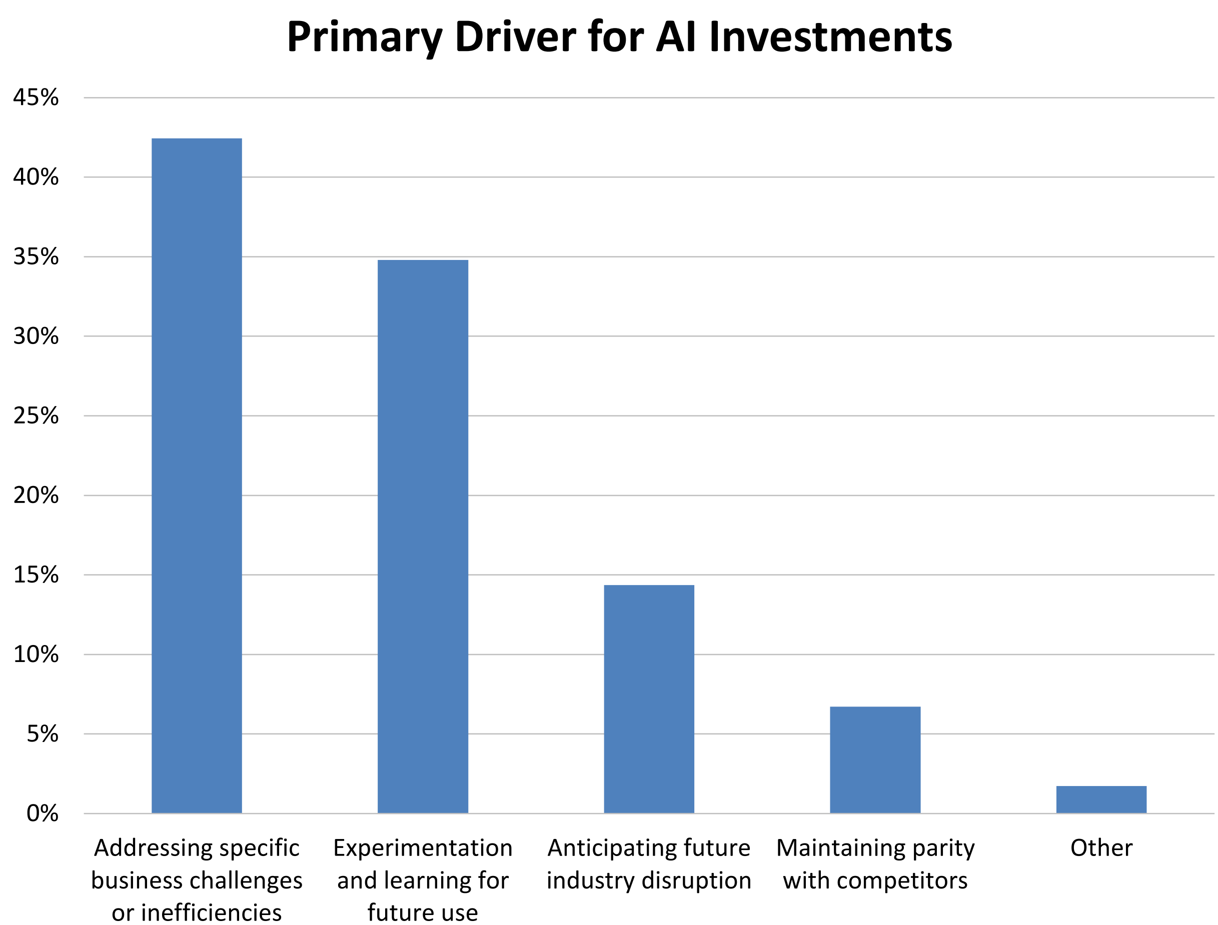


Figure 10 - Primary driver for AI investments

Primary Driver for AI Investments by Industry

When we rank-sort survey responses by industries most actively "addressing specific business challenges or inefficiencies," we observe that financial services and manufacturing respondents are most likely to be applying AI today (51% and 47% respectively; fig. 11). Close behind, respondents in technology (43%), consumer services (41%), business services (40%), and healthcare (39%) are actively using AI to address specific business problems. When we combine active use with experimentation and anticipation of future industry disruption, we see very strong industry buy-in, with fewer than 20% in any industry only maintaining parity or not otherwise seeking to leverage AI productivity and other benefits.

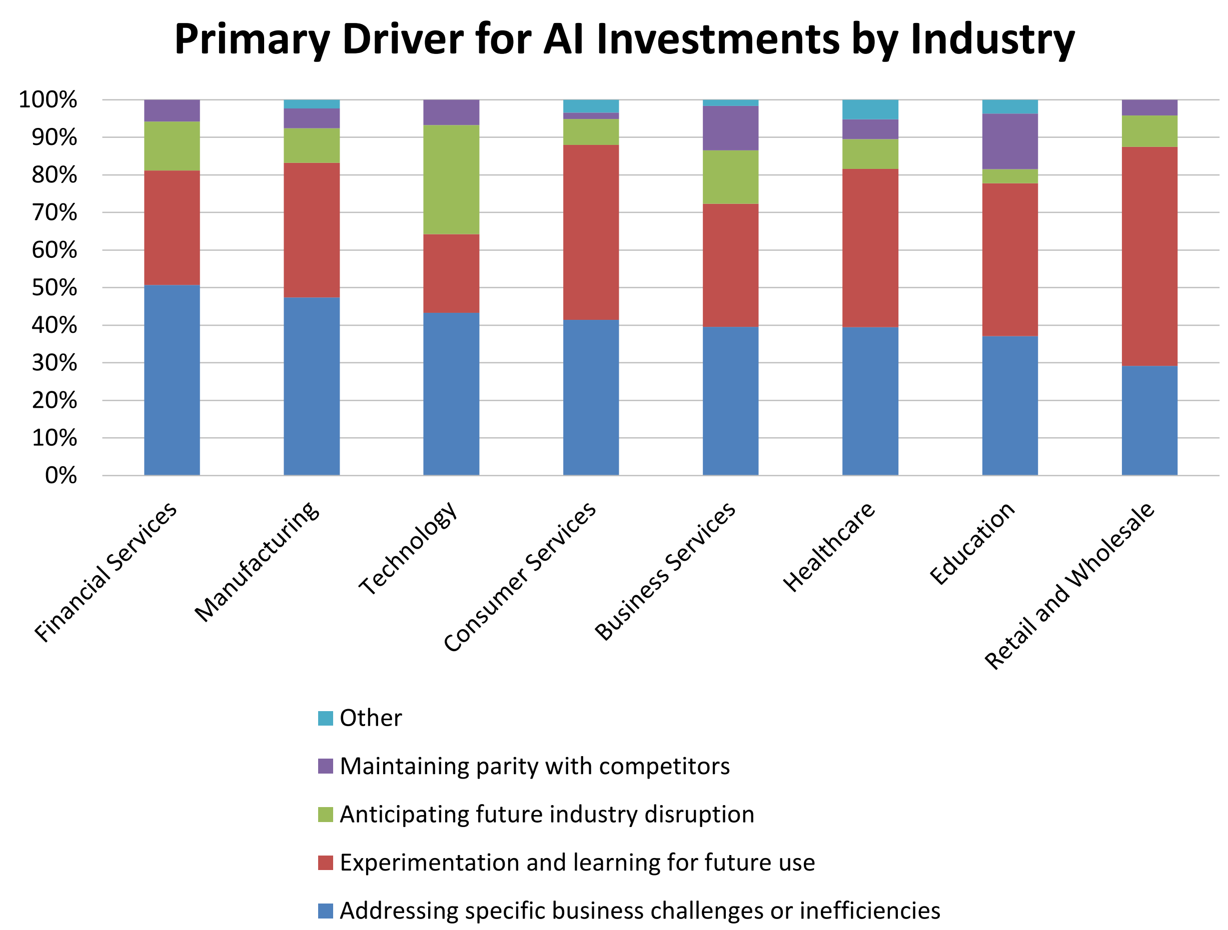


Figure 11 - Primary driver for AI investments by industry

Primary Driver for AI Investments by Organization Size

Active use and experimentation with AI investments is strong in organizations of all sizes and increases with global headcount (fig. 12). In 2025, very large (more than 10,000 employees) are 51% likely to be using AI to address specific business problems and inefficiencies, compared to 40% in large (1,001-10,000), 41% in midsize (101-1,000), and 37% of small (1-100) organizations. More than 90% of all organizations of any size report either current use, experimentation, or anticipation of future industry disruption as investment drivers for AI.

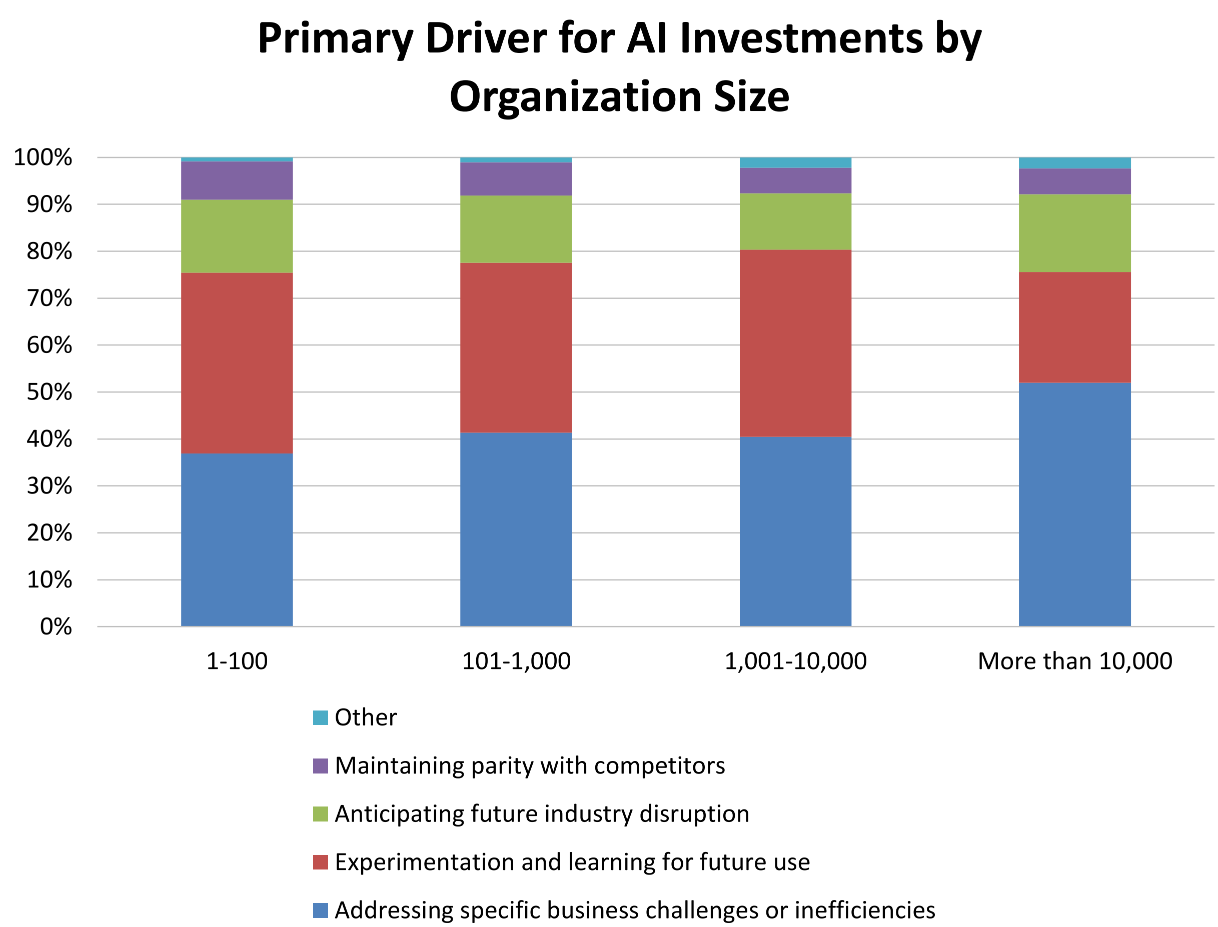


Figure 12 - Primary driver for AI investments by organization size

AI Objectives

We asked respondents, “What strategic business outcomes is your organization aiming to achieve with AI in 2025?” This year, more than 50% of respondents report active use or experimentation with the top three objectives, “driving operational excellence,” “enabling data-driven decision making,” and “enhancing customer experience and satisfaction in a mix of efficiency and new opportunity” (fig. 13). A strong second tier of using or experimenting with gaining competitive market advantage and accelerating revenue growth is embraced by more than 40% of all respondents. The lowest-ranked objective, “expanding into new products or markets,” is nonetheless seen as an area of active use, experimentation, or anticipation of future market disruption by more than half of all respondents. (Not shown in this chart is the correlation between these numbers and specific findings for generative and agentic BI we collected for separate studies.)

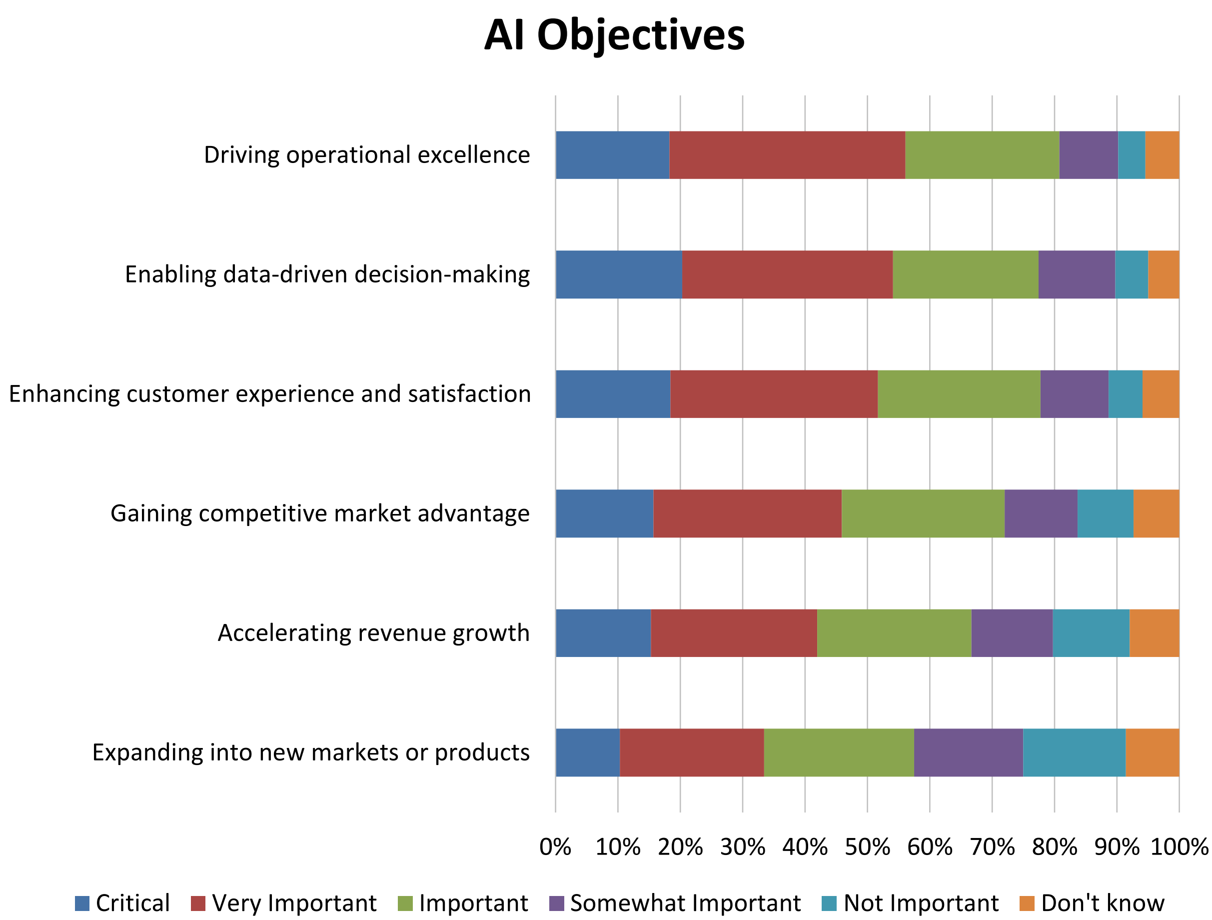


Figure 13 - AI objectives

AI Objectives by Organization Size

Viewed by organization size, all objectives for AI are most prioritized by very large (more than 10,000 employees), followed by small (1-100), large (1,001-10,000), and midsize (101-1,000) organizations (fig. 14). The top priority in very large organizations is enhancing customer service. Smaller peers' top objectives are driving operational excellence and enabling data-driven decision making, along with enhancing customer experience and satisfaction. The most outsized gaps in very large and small organizations' emphasis appear in lower-ranked priorities, including accelerating revenue growth, expanding into new products or markets, and gaining competitive market advantage.

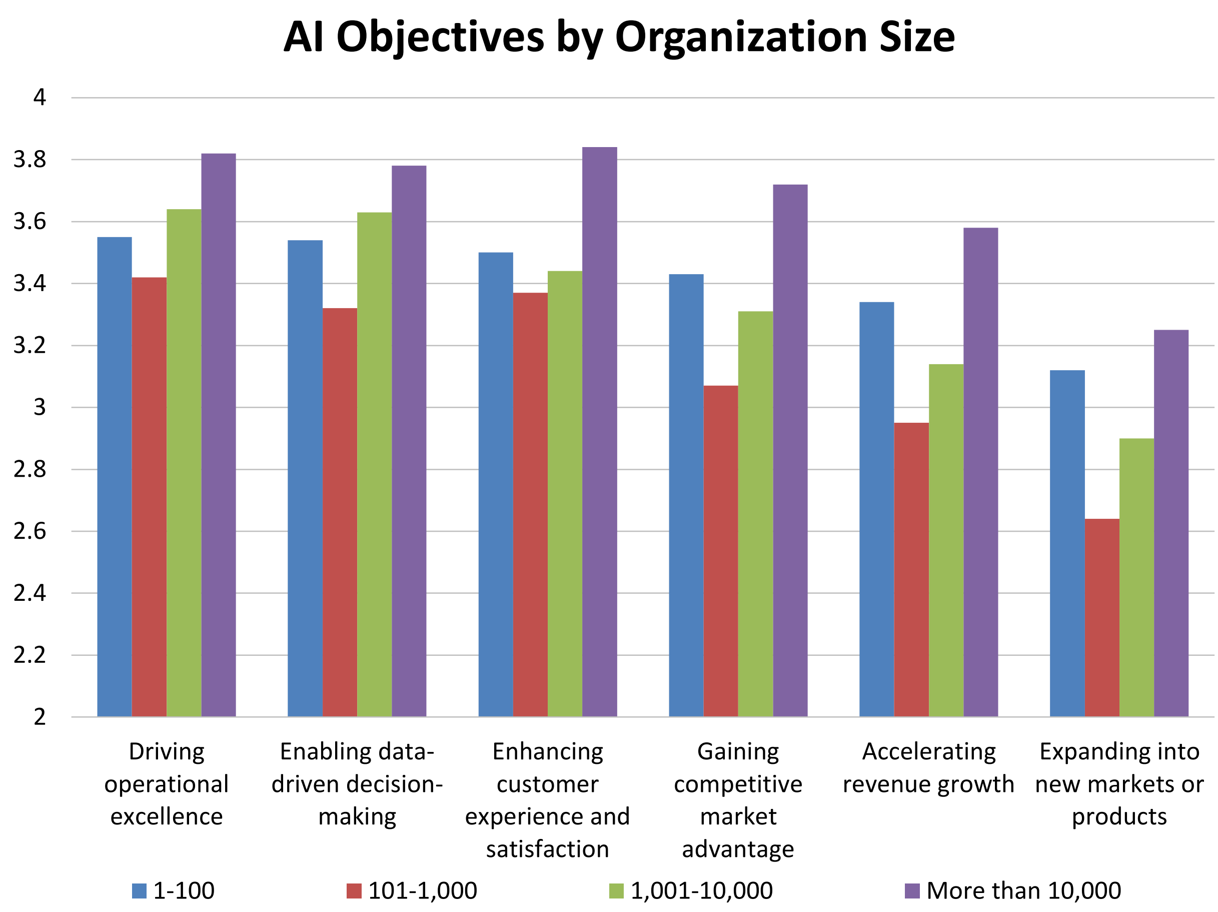


Figure 14 - AI objectives by organization size

AI Objectives by Company Age

Organizations of different ages are more and less likely to prioritize specific AI objectives in 2025 (fig. 15). This year, "youngish" organizations, led by those five to 10 years old, most emphasize all AI objectives. The youngest organizations of less than five years are next-most likely to endorse all AI objectives. These youngest organizations most prioritize enhancing customer service and satisfaction, along with gaining competitive market advantage. Older organizations of 11-16 years are most interested in the top overall priorities: driving operational excellence, enabling data-driven decision making, and enhancing customer experience and satisfaction.

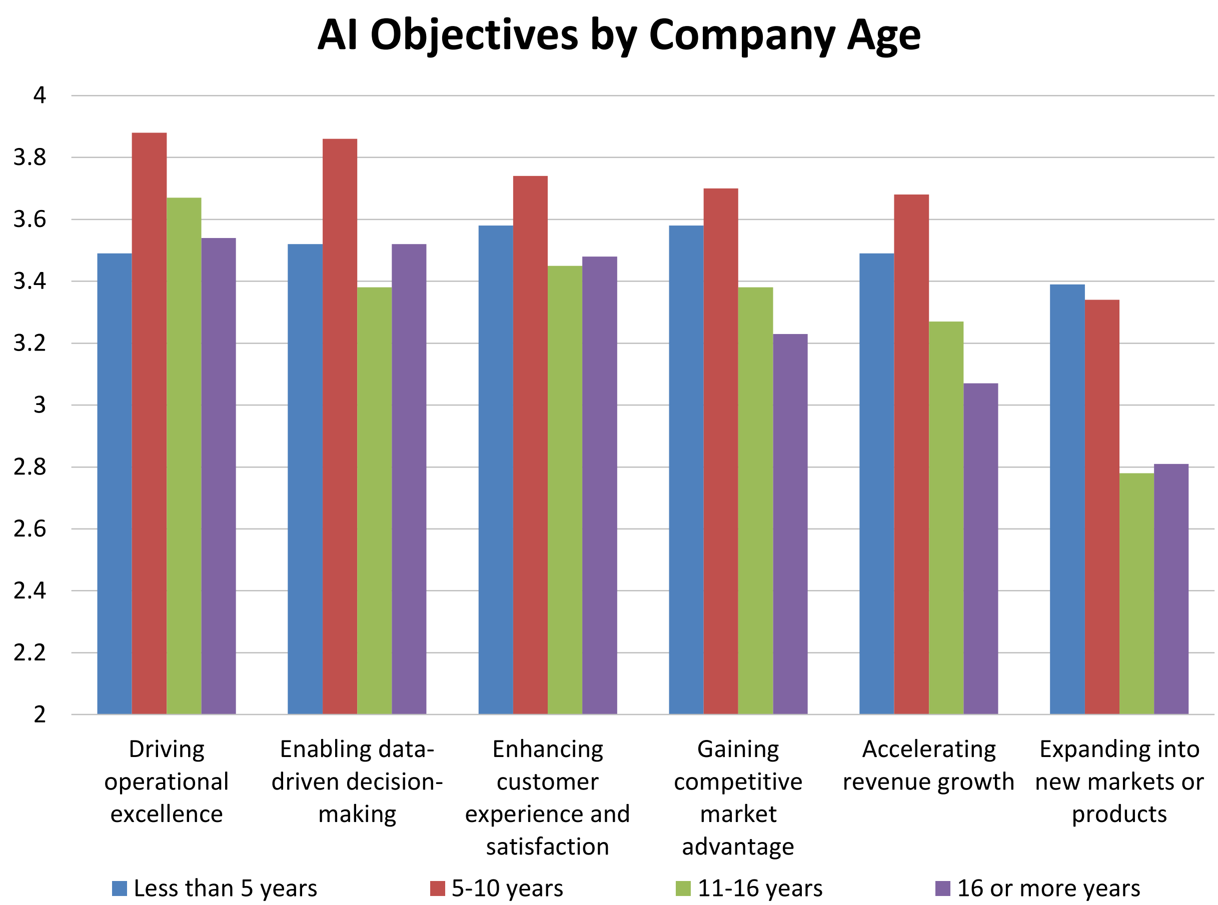


Figure 15 - AI objectives by company age

AI Policy Management

We asked respondents, “How is the coordination and control of AI policies managed within your organization?” and offered six responses plus “other.” Figure 16 shows responses sorted from most to least coordination and centralization. The top three (most coordinated), “centralized within existing GRC,” “centralized specifically for AI” and “coordinated across separate governance, legal, risk and compliance” account for about 60% of responses. But the replies reveal a broad range of responses indicative of governance immaturity in AI and, in many cases, holes in policy oversight expressed in terms that would alarm any risk manager. About 37% report conditions that include “uncoordinated,” “managed independently,” and “leading to potential overlaps or gaps” or “unmanaged or handled on an ad hoc basis.”

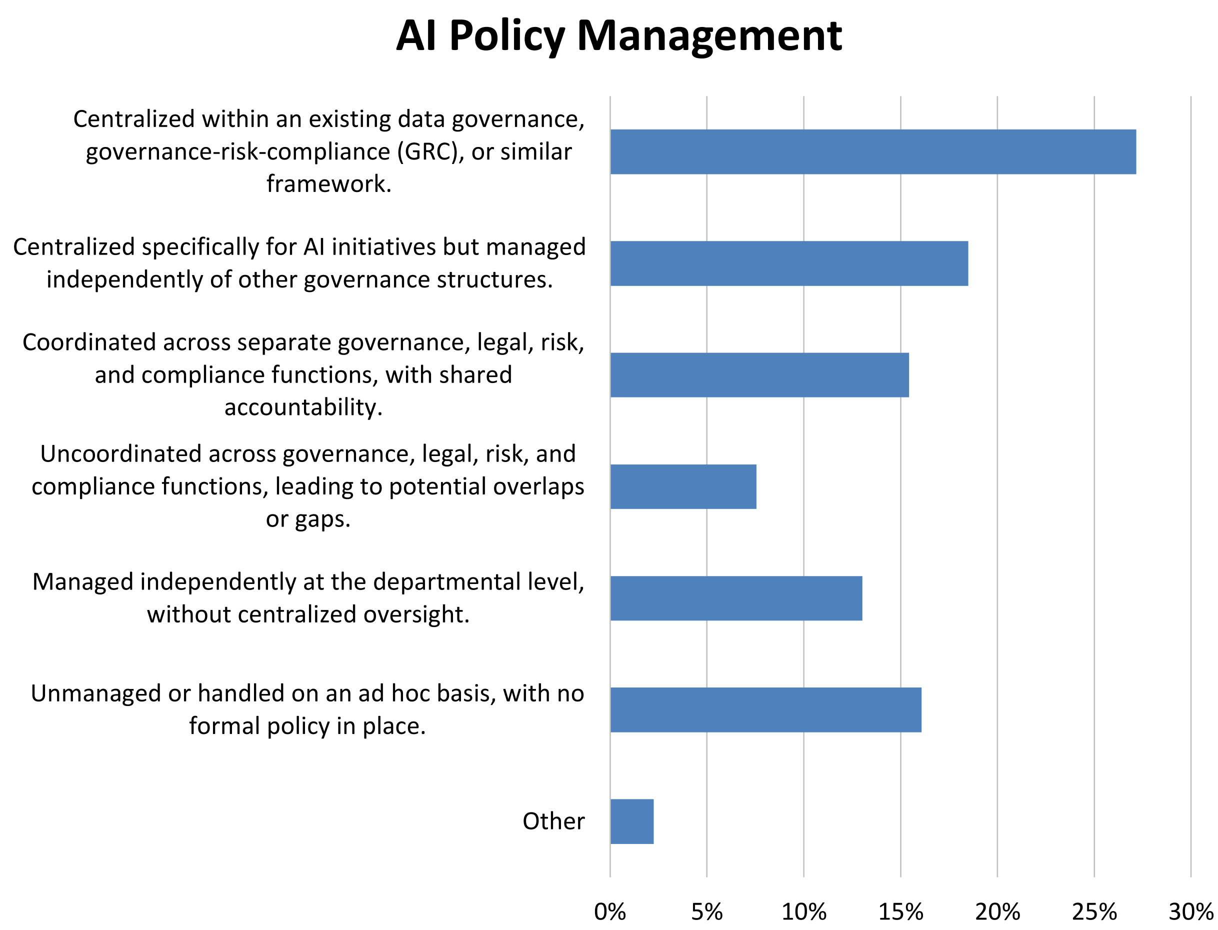


Figure 16 - AI policy management

AI Policy Management by Organization Size

Coordination and control of AI policies correlates to and increases with organization size in 2025 (fig. 17). Very large (more than 10,000 employees) organizations are 60% likely to report either “centralized specifically for AI” or “centralized within GRC,” compared to 48% of large, and 40% of midsize or small peers. This finding is intuitive and expected, given that controls and centralization are associated with scale and bespoke functions for governance, risk management, and compliance. We also sense that, compared with more formal rollouts, the ubiquity of AI tools and free resources makes them easy and tempting extensions to existing enterprise technologies, creating an outsized opportunity for experimentation and adjacent use.

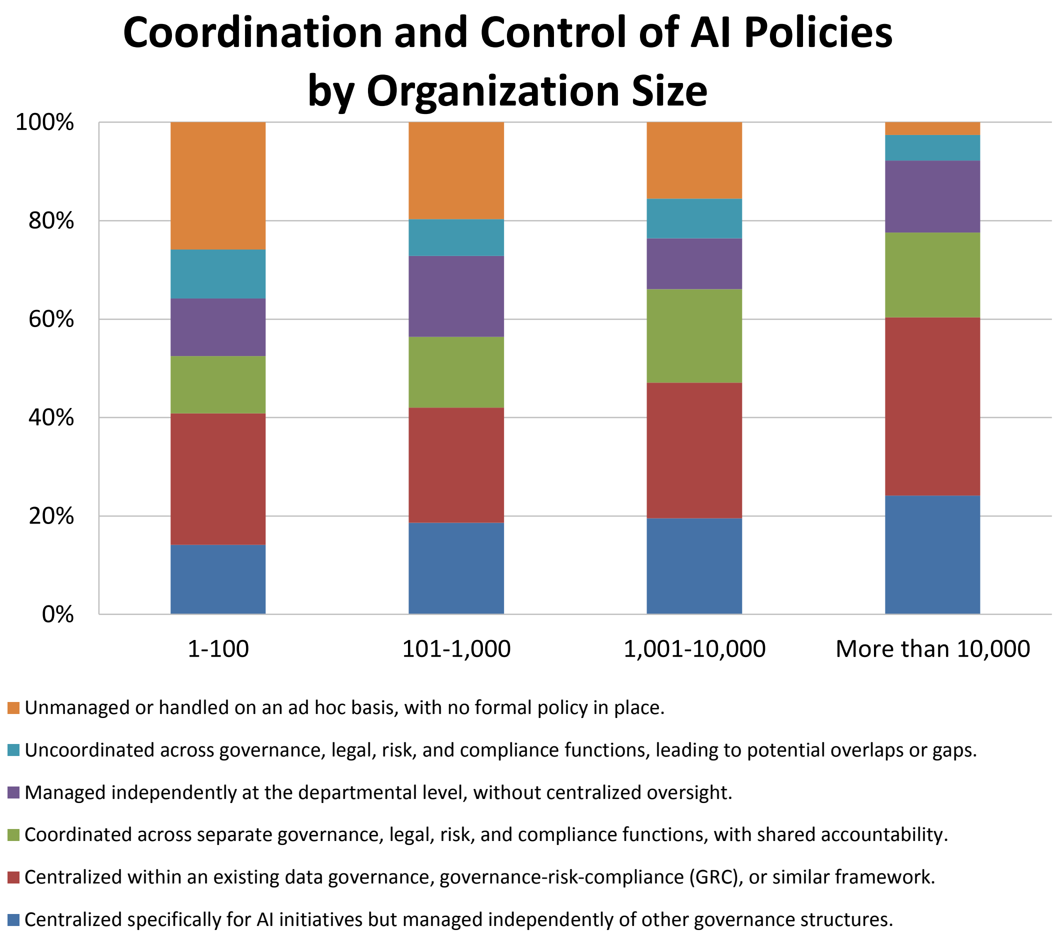


Figure 17 - Coordination and control of AI policies by organization size

AI Policy Management by Data Leadership

In 2025, organizations with identified data leaders (chief data officer, chief analytical officer, or another designated role) are more likely to report higher levels of coordination and control of AI policies (fig. 18). Organizations with data leadership "in place" are 57% likely to report either "centralized specifically for AI" or "centralized within GRC," compared to 40% of those with "future plans" and 34% of those without data leadership or future plans. This is another intuitive and expected finding, since the presence of data leadership is a likely indicator of best practices in multiple areas including policy, data governance, data literacy, education, and systematic controls. Organizations with no plans for data leadership are 26% likely to report uncoordinated or independently managed, departmental, or ad hoc oversight.

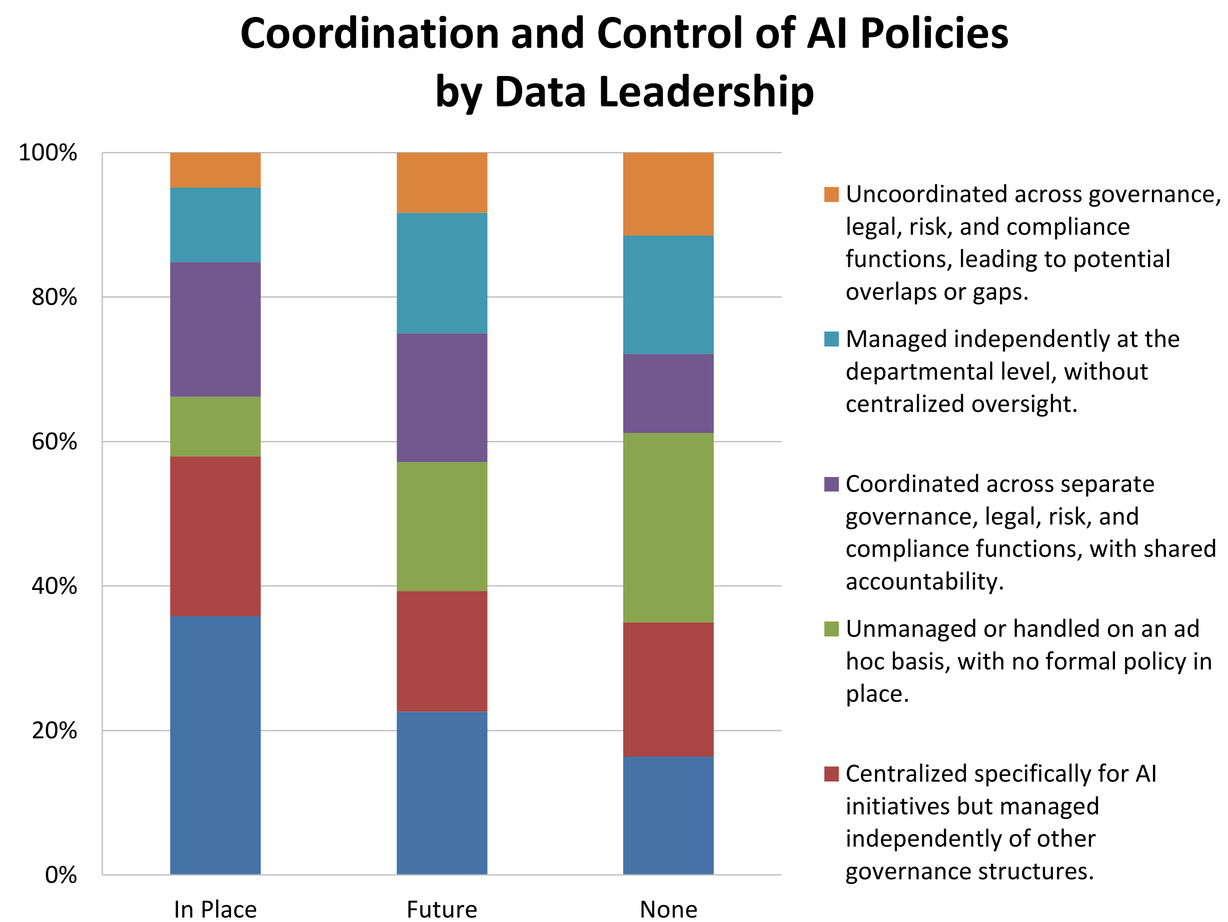


Figure 18 - Coordination and control of AI policies by data leadership

AI Maturity

We asked respondents, “How would you describe your organization's AI maturity in 2025?” This year, among five possible choices, just more than half (51%) of respondents describe AI maturity as “emerging,” and primarily in pilot or experimental phases (fig. 19). About 16% say AI is “nascent” and not yet actively pursued or implemented. Notably, 25% say AI maturity is “intermediate” and currently used in impactful areas. Combined with another 7% that claim “advanced” use of AI in core business processes, nearly one-third of the sample claims intermediate to advanced use of AI, which we describe as strong early uptake for new technology and services. We also expect that organizations with more extensive data management maturity will be better equipped with practices to expedite AI maturity.

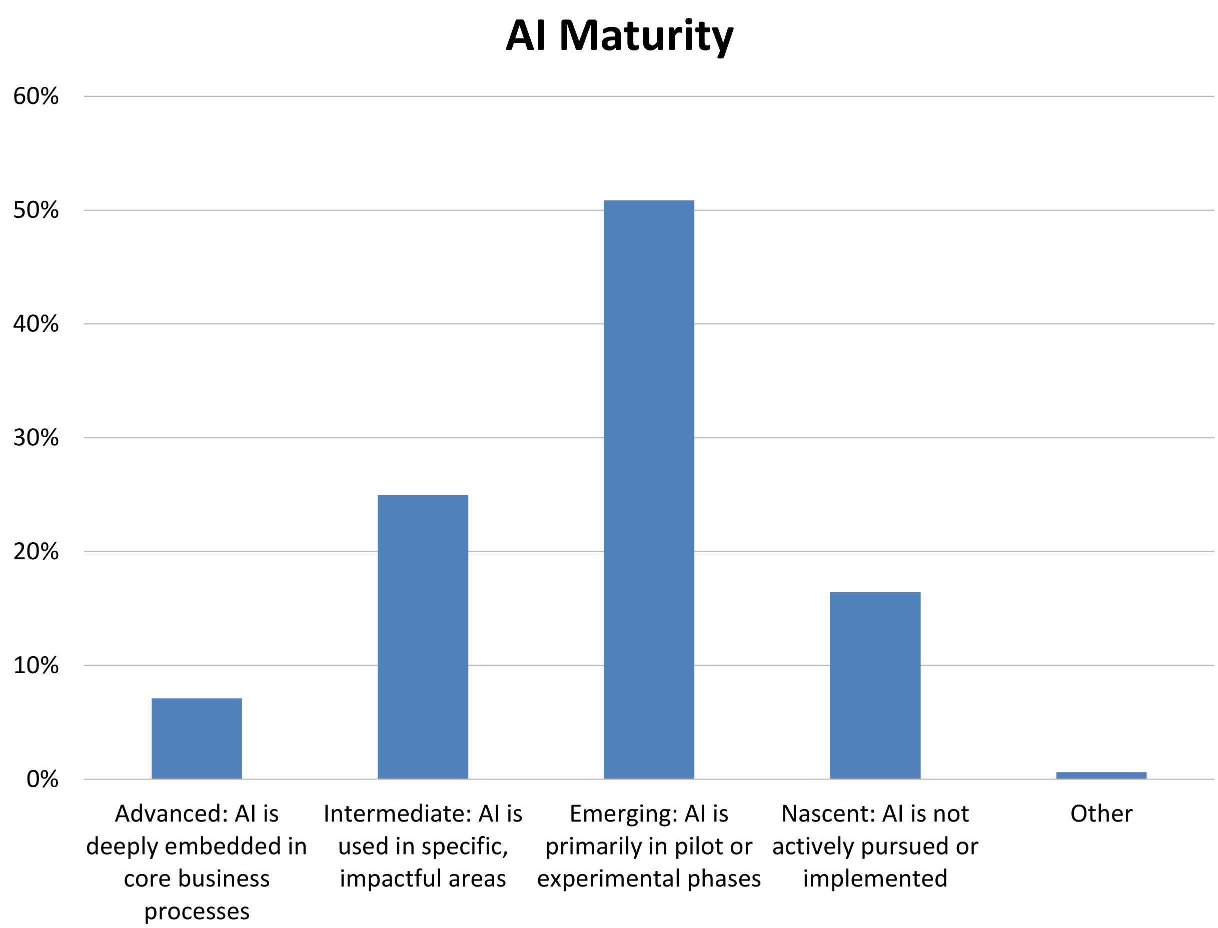


Figure 19 - AI maturity

AI Maturity by Geography

In 2025, weighted-mean maturity of AI is highest in Asia Pacific, followed by North America, EMEA, and Latin America (fig. 20). This year, combined advanced and intermediate use in Asia Pacific stands at 48%, compared to 29% in North America, 28% in Latin America, and 27% in EMEA. In another measure, estimations of "nascent" AI maturity are lowest in Asia Pacific (8%) and almost four times higher in Latin America (30%). That said, between 70%-92% of organizations in any region describe AI maturity as advanced, intermediate, or emerging.

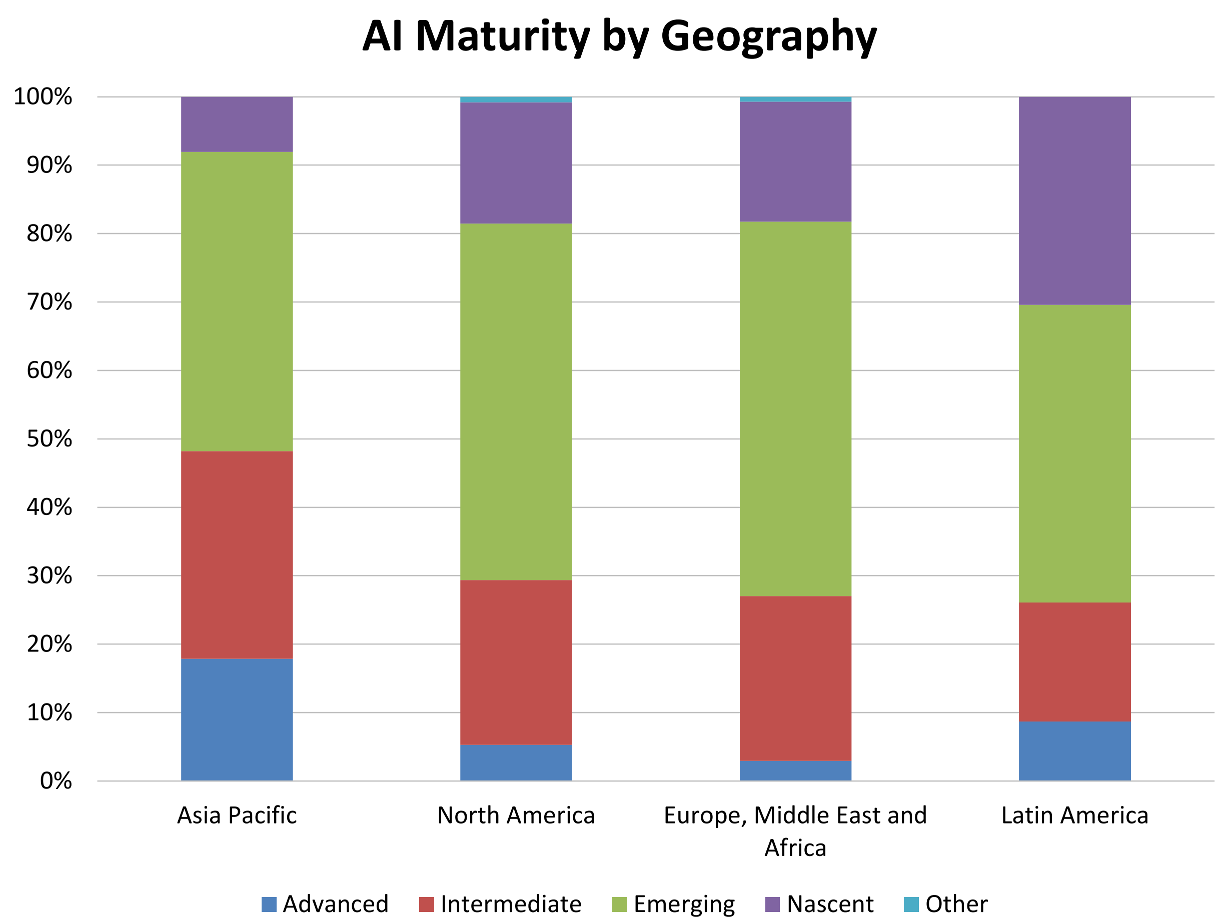


Figure 20 - AI maturity by geography

AI Maturity by Organization Size

In 2025, the largest (more than 10,000 employees) and smallest (1-100 employees) organizations are most likely to describe AI maturity as “advanced,” albeit in small numbers of 13% and 9%, respectively (fig. 21). Sentiment around higher AI maturity is undeniably strongest in very large organizations, where 96% of respondents describe it as “advanced,” “intermediate,” or “emerging.” Among respondent organizations with 100 or more employees, we also observe that the strategic role of AI scales incrementally with headcount. Between 19%-21% of respondents in small, midsize, and large organizations describe AI maturity as “nascent.”

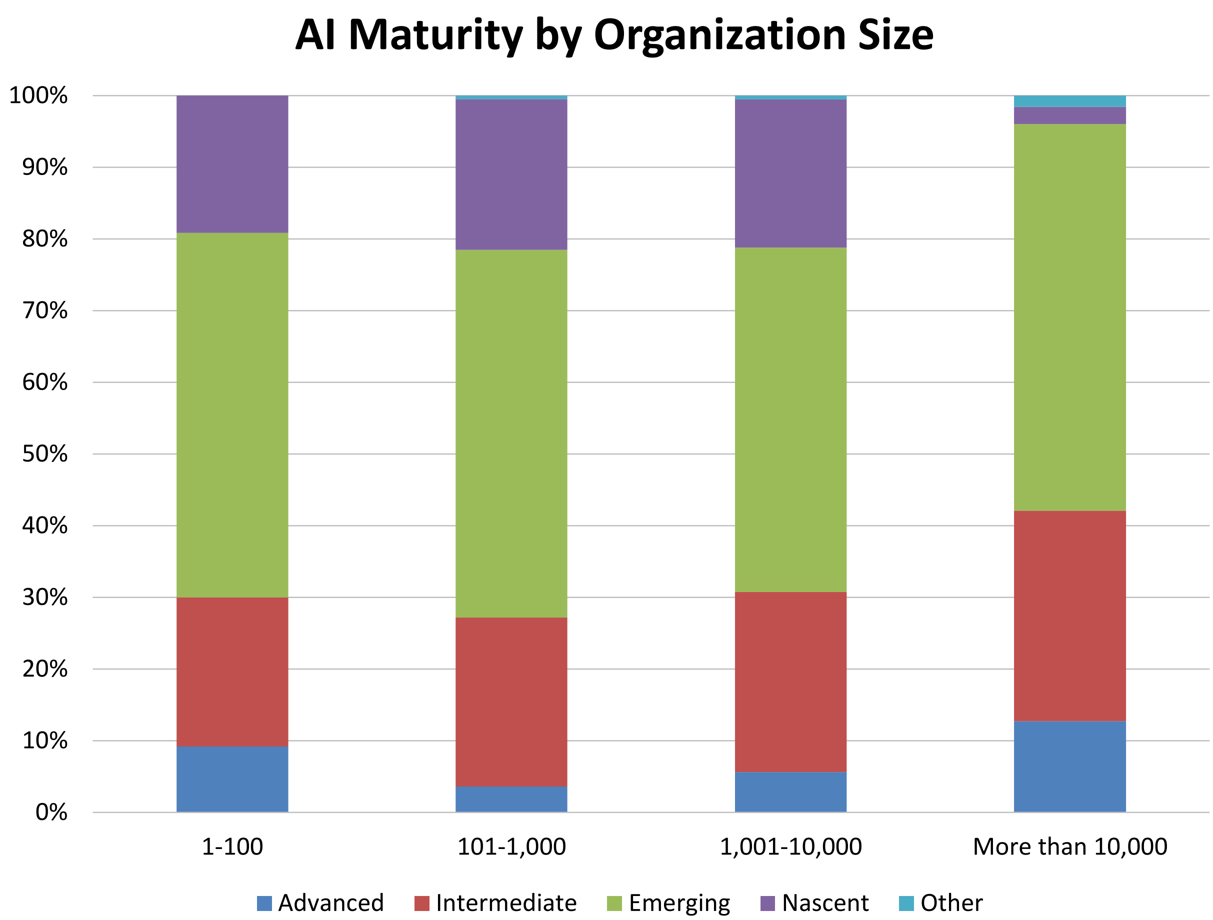


Figure 21 - AI maturity by organization size

AI Maturity by Company Age

Estimations of AI maturity vary by company age, and “younger” companies are more likely to describe the technology or their initiatives as more mature (fig. 22). In 2025, organizations of less than five years and five to 10 years are 42% and 45% likely to describe AI maturity as “advanced” or “intermediate,” compared to 38% at organizations of 11-16 years and 26% at organizations of 16 or more years. This perception might be partly due to a startup mentality at younger organizations, and perhaps to the difficulties of addressing more complex entrenched legacy processes and infrastructure in older companies. We also expect cultural and regulatory issues to weigh on individual companies’ and industries’ estimation of their own AI maturity.

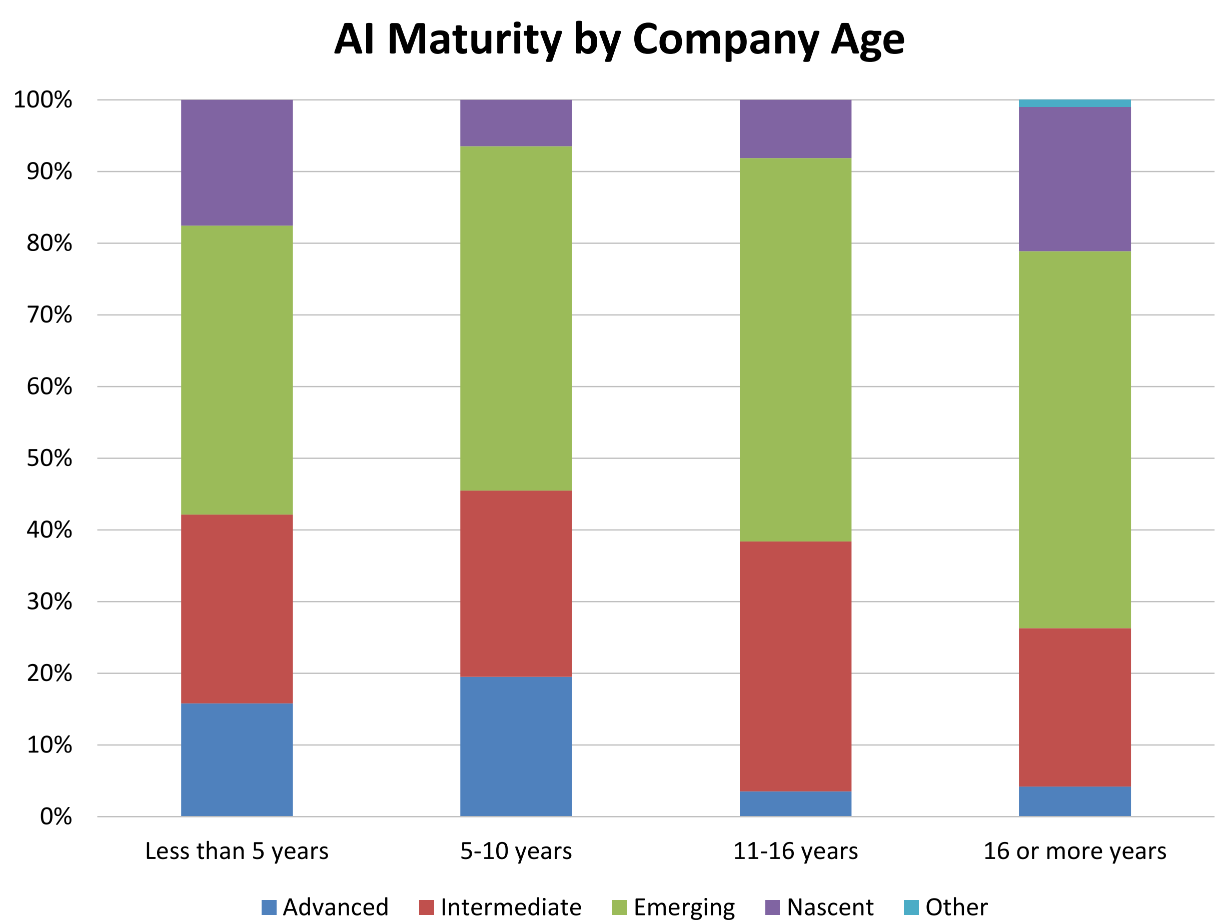


Figure 22 - AI maturity by company age

AI Maturity by Data Leadership

In 2025, organizations with identified data leaders (chief data officer, chief analytical officer, or another designated role) are more likely to describe their current strategic state of AI as "cornerstone" or "important" (fig. 23). This implies that organizations which embrace the value of treating data as an asset through dedicated practice and program oversight are more likely to have or describe AI practices that are more mature. This year, fully 40% of organizations with data leadership in place say AI is of advanced or intermediate maturity, compared to 28% of organizations with future plans for data leadership and only 28% of organizations with no data leadership plans. The findings extrapolate to estimations of only "nascent" AI maturity, which is nearly twice as high at organizations with no data leadership than at other cohorts.

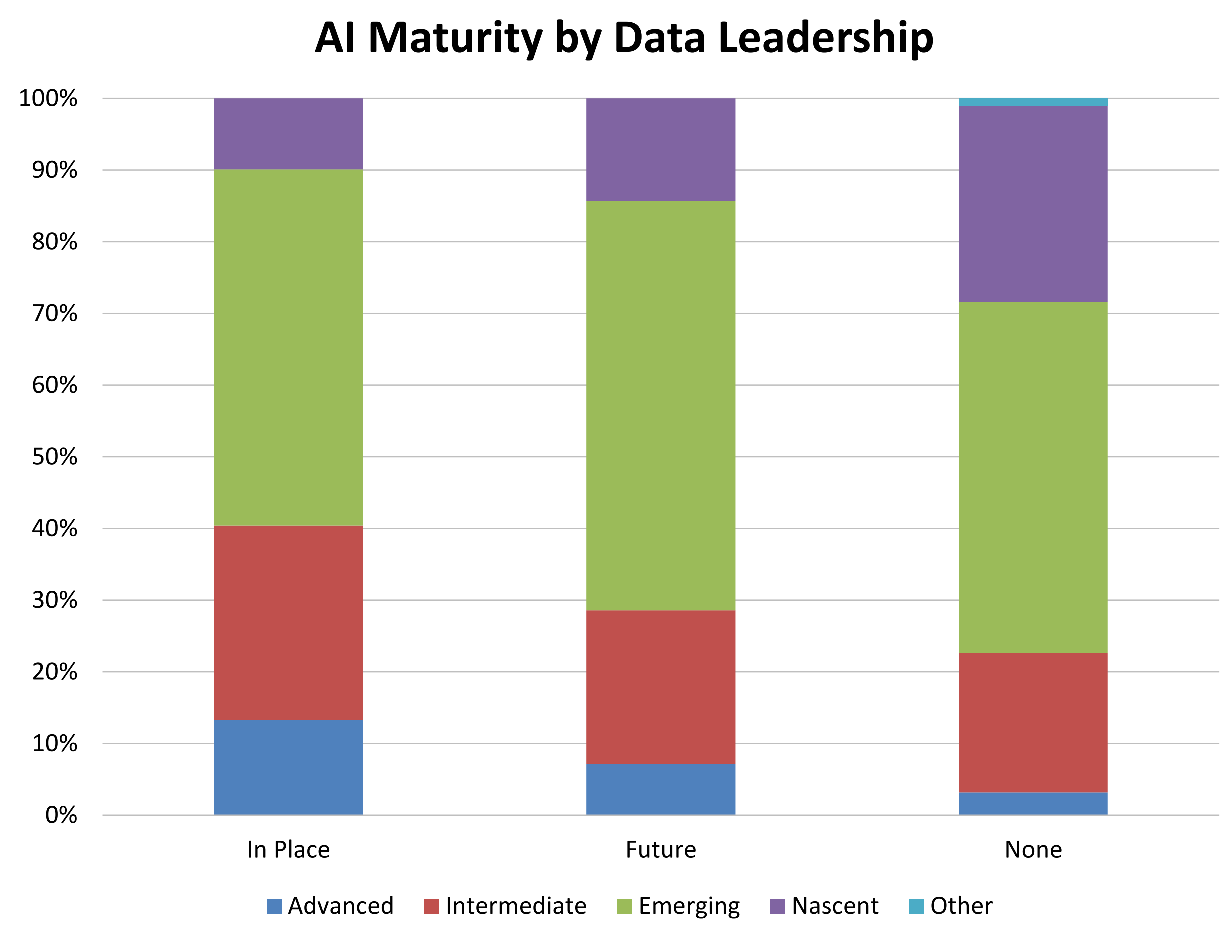


Figure 23 - AI maturity by data leadership

Outlook on Generative and Agentic AI

We asked respondents, "What is your perspective on agentic and generative AI?" In 2025, among five offered responses, the most popular range from "cautiously optimistic" to "excited about the possibilities" (fig. 24). When we combine the cautiously optimistic and excited responses with a smaller group that is actively adopting, a surprising 77% report a positive outlook on generative AI and another 63% report the same on agentic AI. Indifferent, negative, or unsure responses comprise the attitude of the remaining 23% toward generative AI and 37% toward agentic AI. We can mentally connect users' positive attitudes and early strong confidence in AI to the massive capital investment occurring on the supply side of generative and agentic AI, but we agree that it is too soon to determine many outcomes.

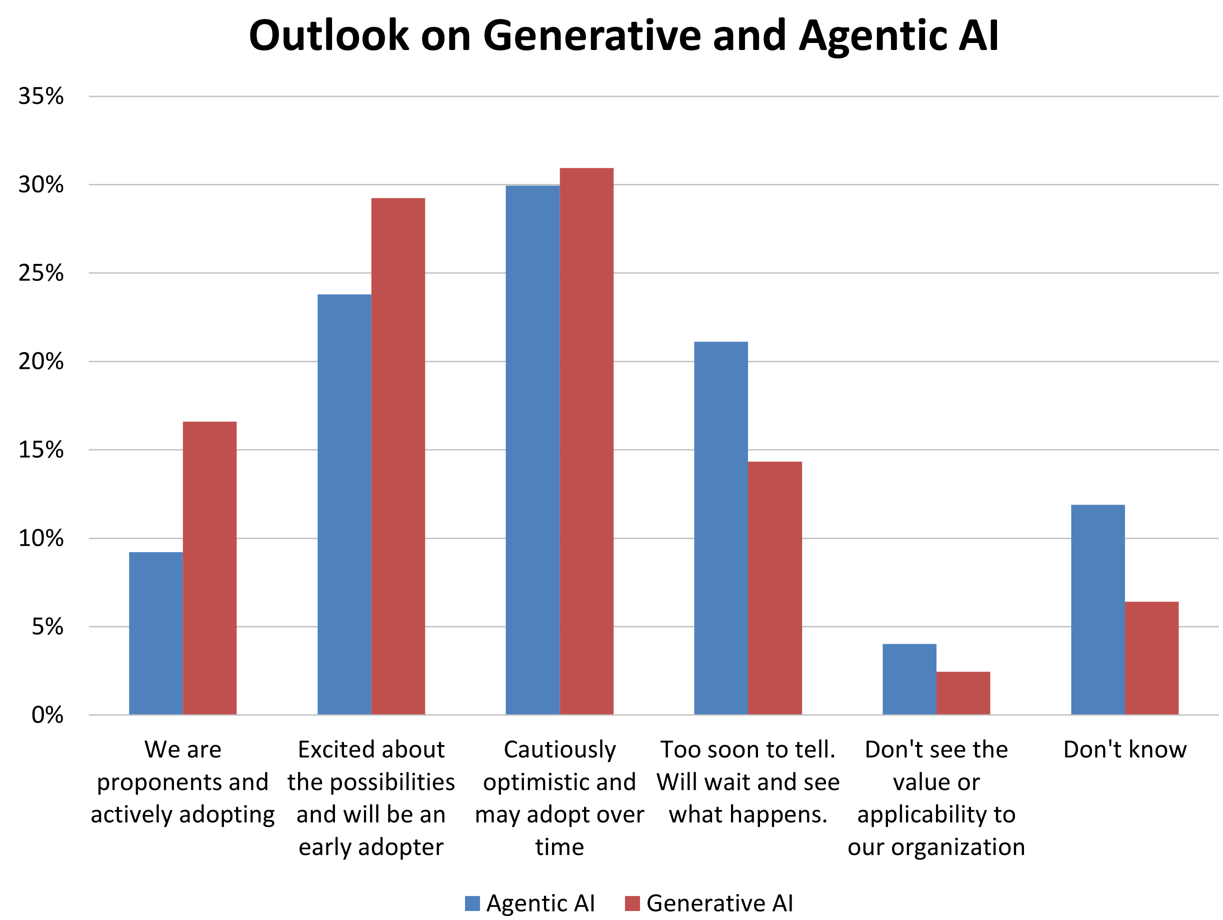


Figure 24 - Outlook on generative and agentic AI

Adoption of Generative and Agentic AI

We asked respondents, "What are your plans surrounding generative and agentic AI, if any?" In 2025, the strongest among five offered responses is "experimenting today," where about one-third (34%) cite generative AI and 28% cite agentic AI (fig. 25). Generative AI is in production today for 15%, while agentic AI is in production among 5% of respondents. Among the remaining non-adopters, 25% have 12- or 24-month plans for generative AI and 31% have plans for agentic AI. The latter finding suggests agentic AI adoption is more likely to occur within either currently experimenting or in-production users. Even so, more than one-quarter (26%) of respondents we asked about adoption of generative AI and 39% we asked about agentic AI have no plans or don't know their outlook.

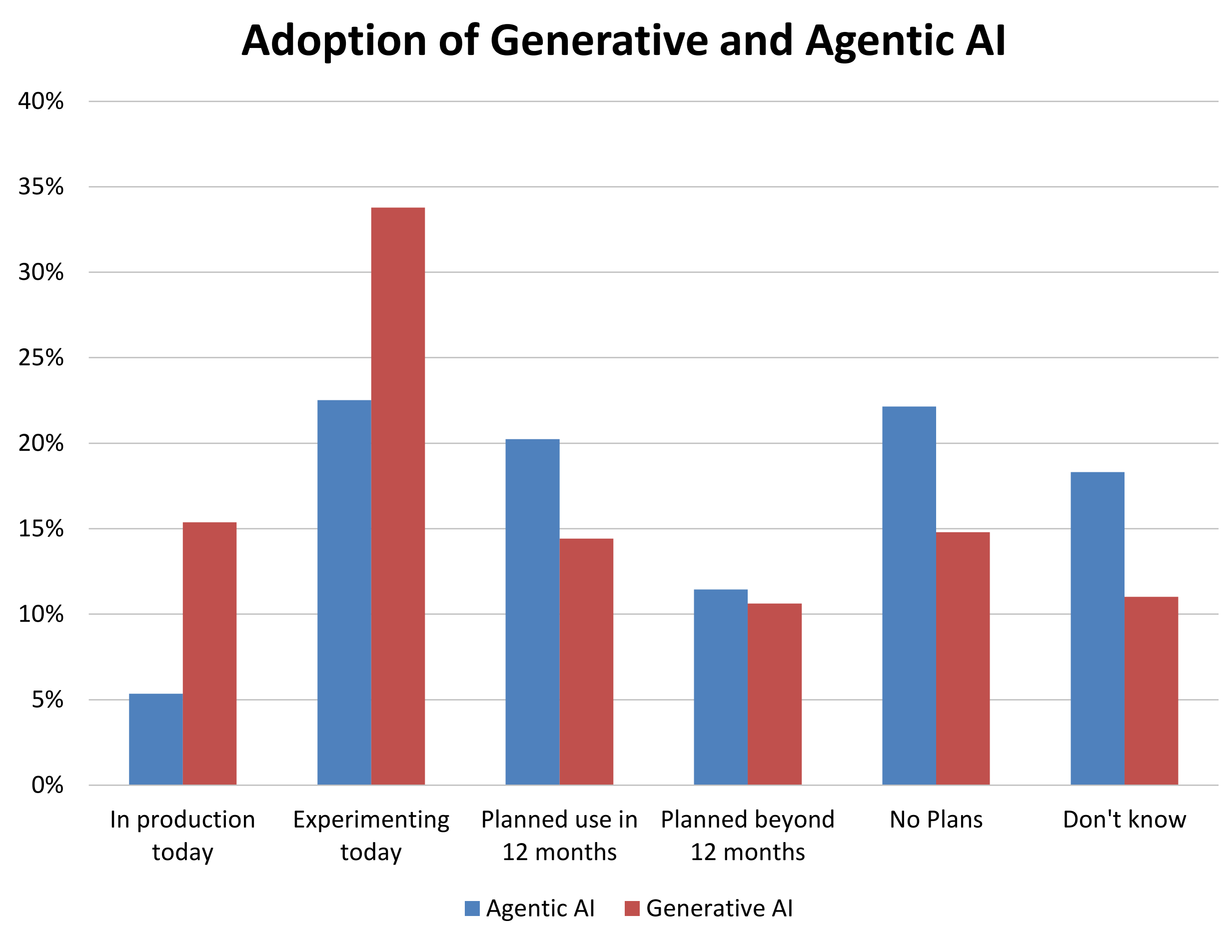


Figure 25 - Adoption of generative and agentic AI

AI Budget Plans

We asked respondents, "How much budget has been established to support generative or agentic AI projects?" Tellingly, the most popular answer for both generative and agentic AI is "don't know" (fig. 26). This might suggest compartmentalization, a lack of program transparency, soft rollouts, or another cause, and is drawn from a large and diverse audience of enterprise users. The second-most common response, "no budget has been allocated or reallocated," applies especially to agentic AI. The least-selected responses, that a "substantial" or "significant percentage of our technology budget has been allocated or reallocated," might indicate an overall lack of budget visibility among users, the use of discretionary funds, or something else. Understanding budgeting is important when users allocate or consume technology initiatives, so this discrepancy might be considered an early-stage shortfall in education and visibility.

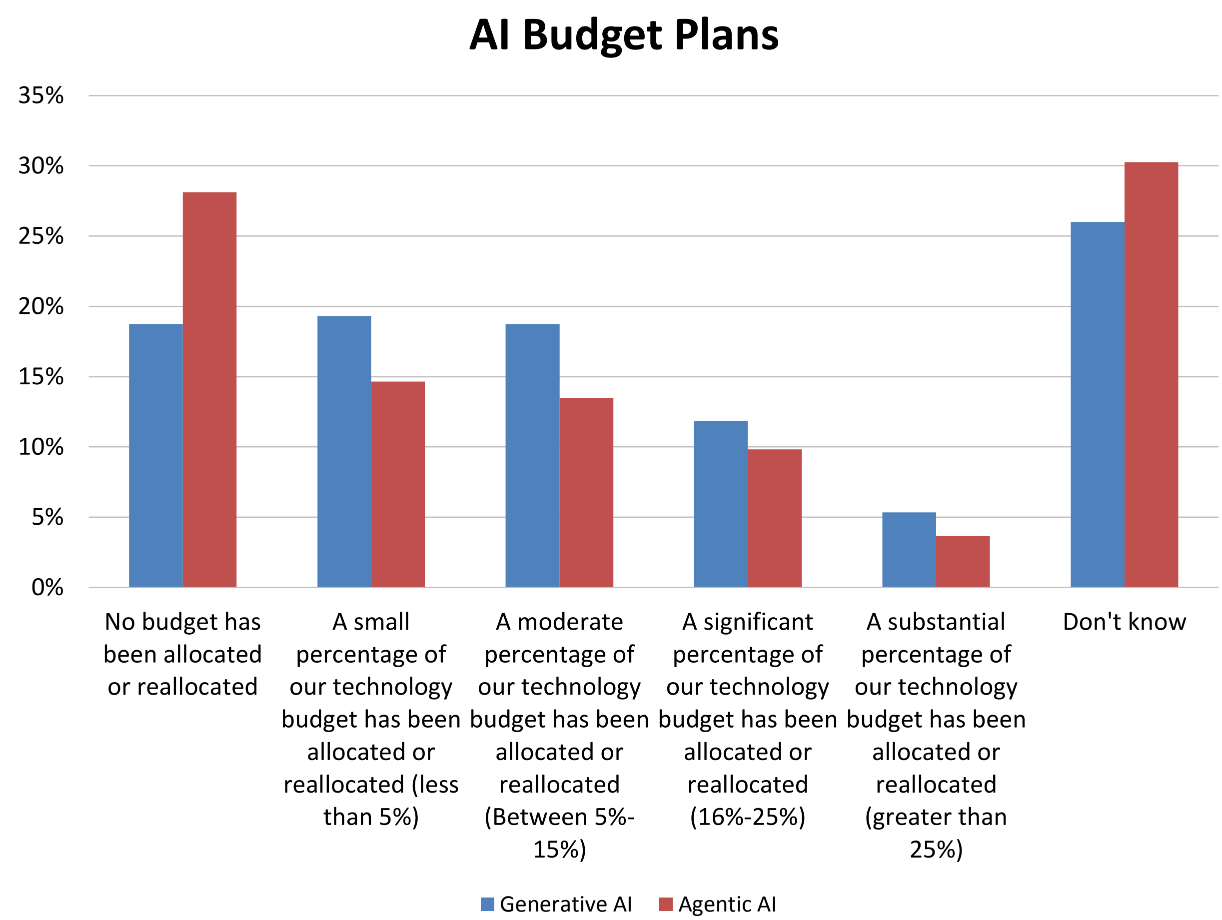


Figure 26 - AI budget plans

Generative AI Adoption

Generative AI uses machine-learning algorithms (e.g., neural networks) to create large language models (LLMs), enabling the generation of new and original data, images, text, or programming code that best approximates the training data used. In a rapidly expanding class of products and services, prominent examples such as ChatGPT are available as a stand-alone offering or extension with and plug-ins for enterprise applications—including many data and analytics products.

Year over year, 2025 current in-production use of generative AI has expanded to about 15%, while experimental use and 12-month plans are slightly lower than in 2024 (fig. 27). Additionally, while more plans have been pushed out beyond 12 months, a correspondingly lower number report no future plans for generative AI.

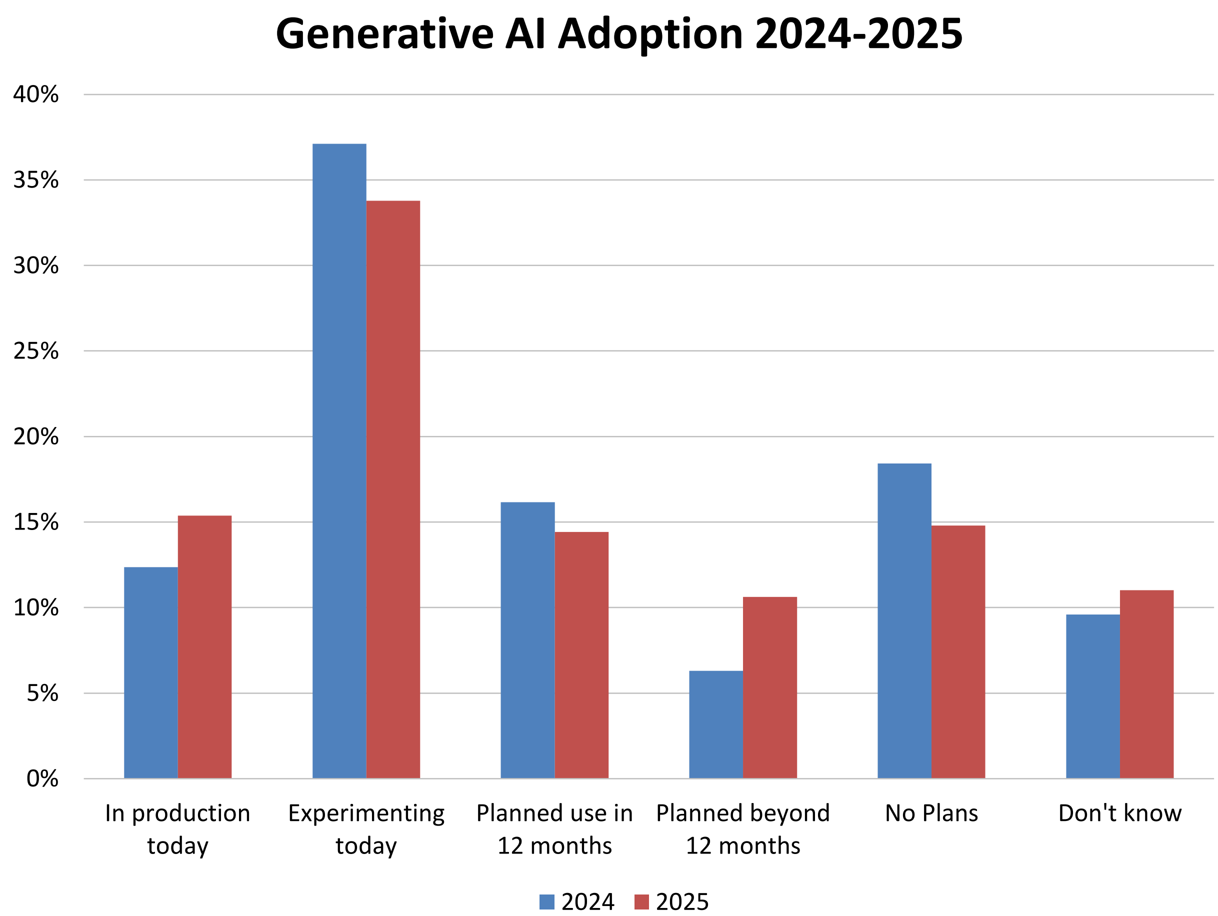


Figure 27 – Generative AI adoption 2024-2025

Generative AI Adoption by Geography

Adoption plans for generative AI vary by geography in 2025, with combined in-production and experimental use highest in Latin America (70%), followed by Asia Pacific (68%), EMEA (60%), and finally North America (49%; fig. 28). This year, still-nascent in-production adoption is also highest in Latin America (29%), followed by Asia Pacific (19%), Asia Pacific (9%), North America (18%), and EMEA (12%). “Don’t know” responses tellingly declined to just 7%-11%. Interestingly, “no plans” scores are highest in North America (20%) and EMEA (16%).

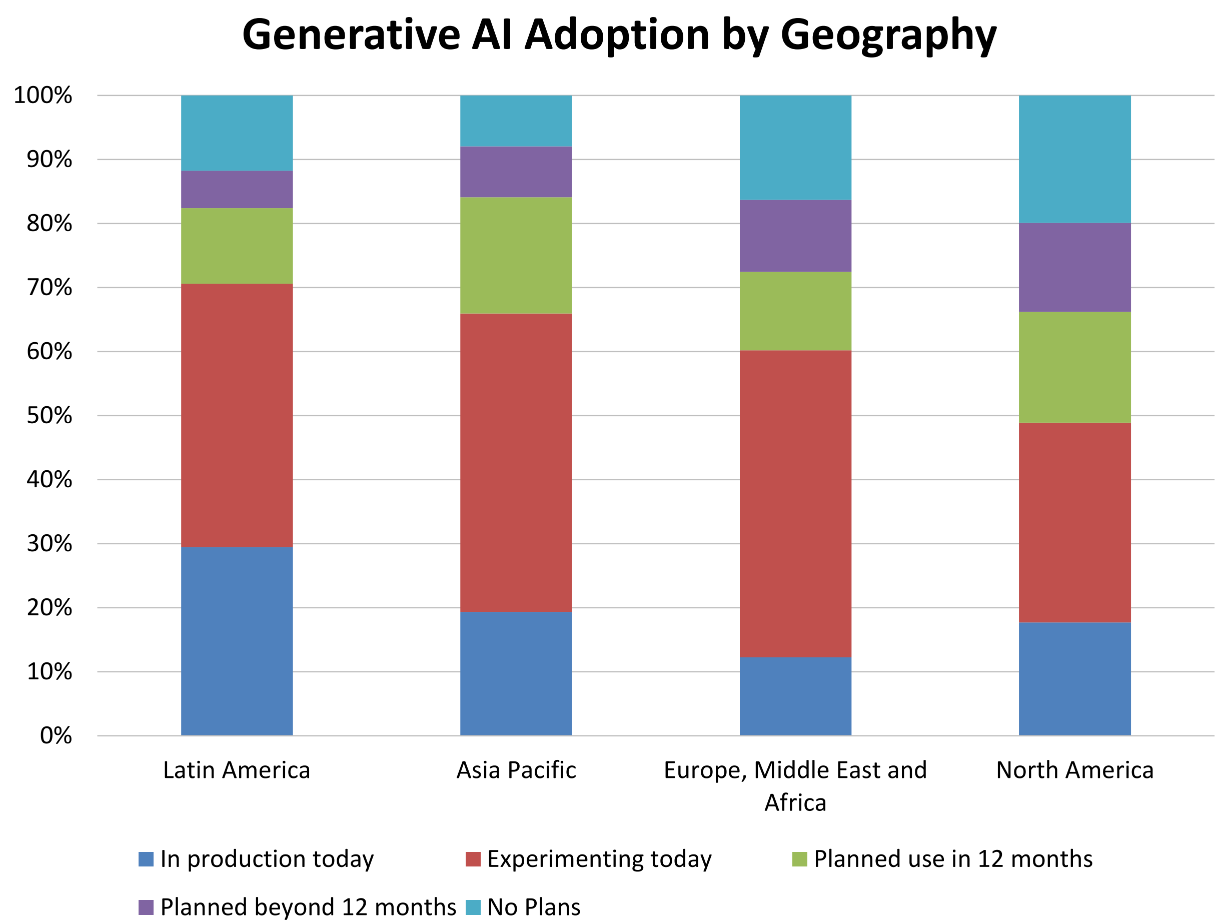


Figure 28 – Generative AI adoption by geography

Generative AI Adoption by Function

A view of adoption plans by function reveals mostly back-office development and deployment support in R&D, the BICC, and IT (fig. 29). Between 74%-81% of respondents in all three functions report in-production use, experimentation, or planned use in 12 months. Sales & marketing respondents are next-most interested: More than 70% report in-production, experimental, or 12-month plans. This year, trailing even finance respondents, users in operations (53%) are least likely to have generative AI in production, in experimental use, or in 12-month plans.

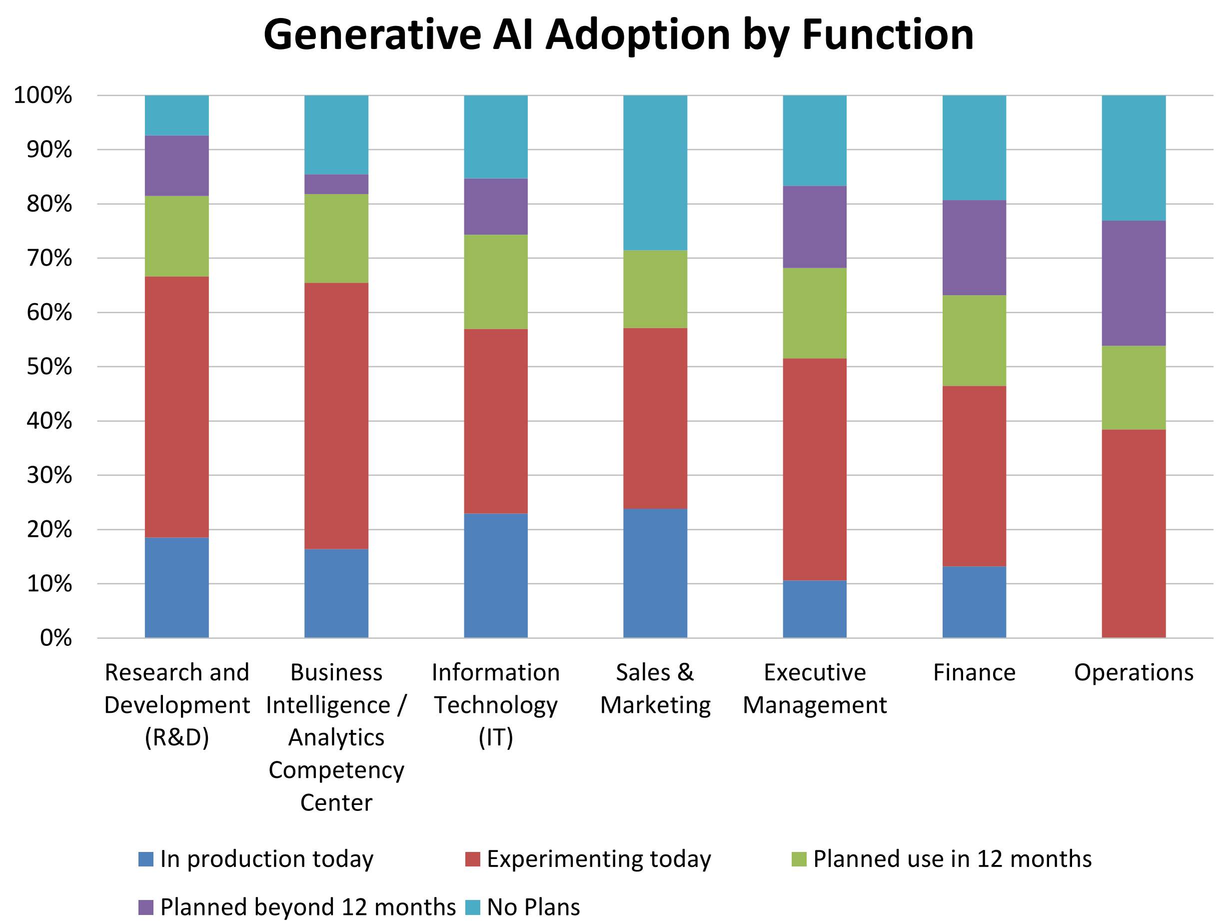


Figure 29 – Generative AI adoption by function

Generative AI Adoption by Industry

Though in-production use is 15% or lower outside of technology (31%) and manufacturing (19%), 2025 experimental use and 12-month plans raise collective plans for adoption of generative AI to 60% and up to 80% across all industries sampled (except for retail & wholesale; see fig. 30). This year, technology, financial services, and manufacturing are the most likely users, followed by a second tier of education, business services, and healthcare. Consumer services and retail & wholesale are the slowest adopters of generative AI to date.

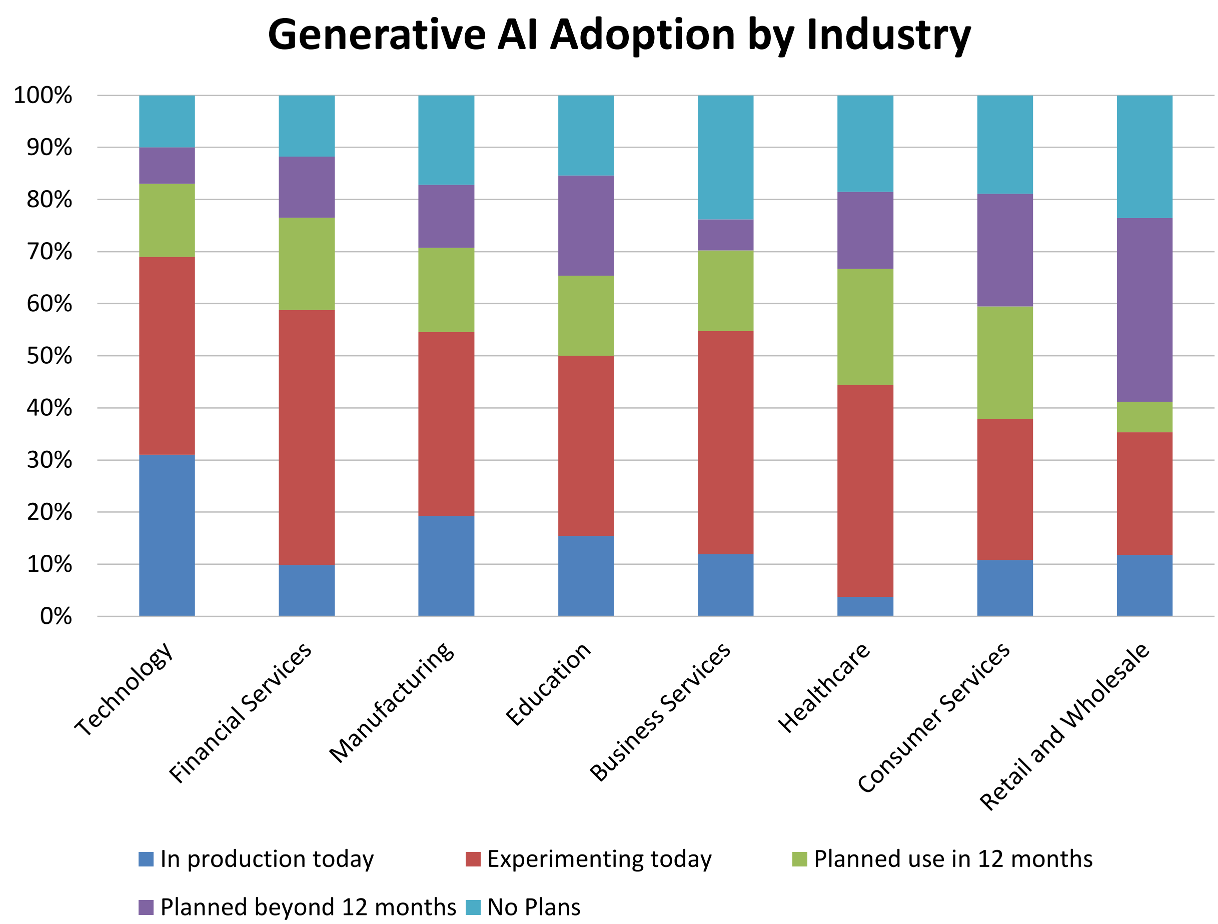


Figure 30 – Generative AI adoption by industry

Generative AI Adoption by Organization Size

Viewed by organization size, adoption plans for generative AI correlate positively to and increase with global headcount in 2025 (fig. 31). The largest organizations (more than 10,000 employees) report the highest combined in-production and experimental use (64%), followed by large (1,001-10,000 employees) at 58%, midsize (101-1,000 employees) also at 48%, and small (1-100 employees) organizations at 51%. Very large organizations also report the lowest likelihood of having no plans (6%), compared to 16% of large, 20% of midsize, and 23% of small organizations.

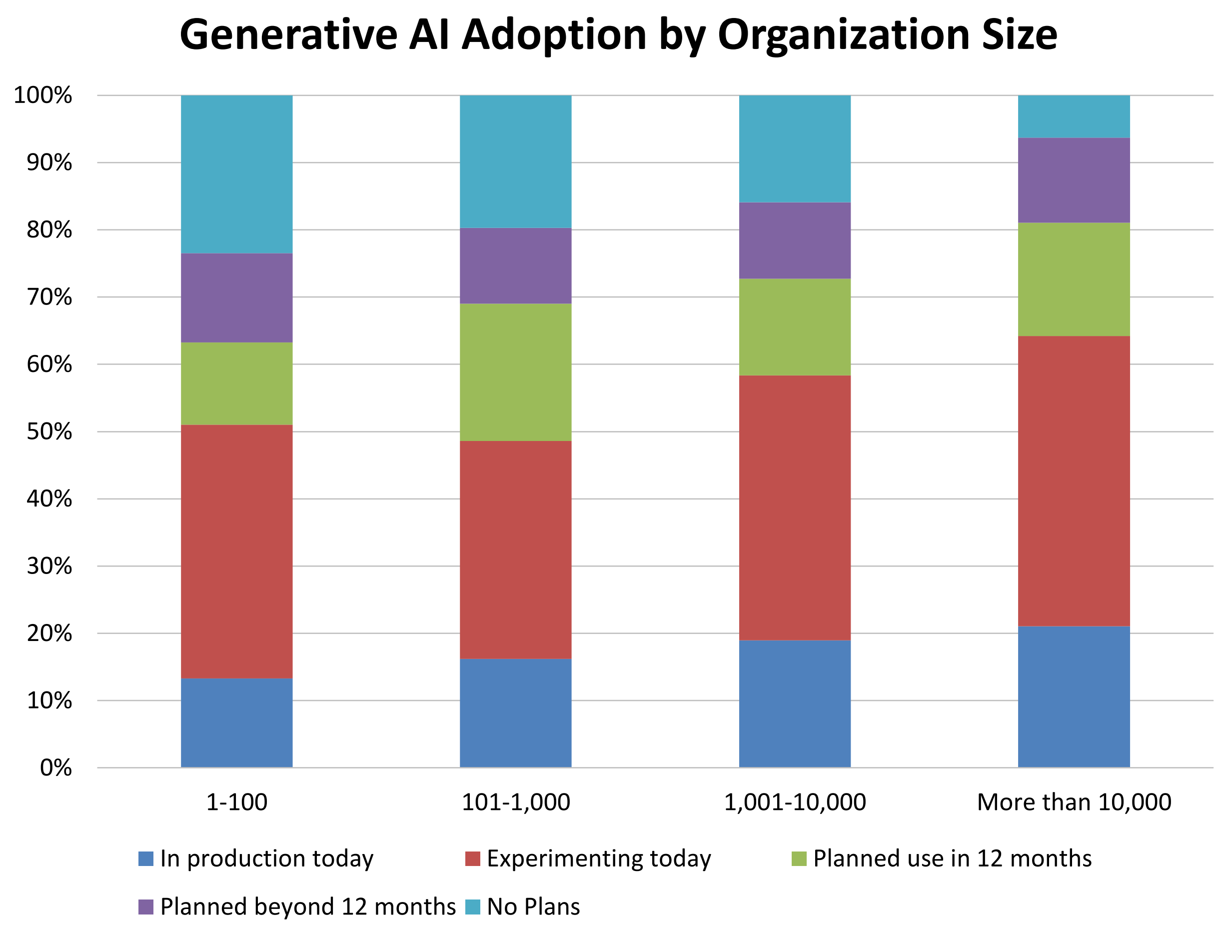


Figure 31 – Generative A adoption by organization size

Generative AI Adoption by Difficulty Finding Analytic Content

Organizations that find it difficult to find analytic content in 2025 are least likely to have generative AI in production today (13%) but are the most likely (52%) to be experimenting with generative AI today (fig. 32). In comparison, 18%-20% of organizations that say it is only somewhat difficult or relatively easy to find analytic content are likely to have generative AI in production and 33%-35% are likely to be experimenting with generative AI.

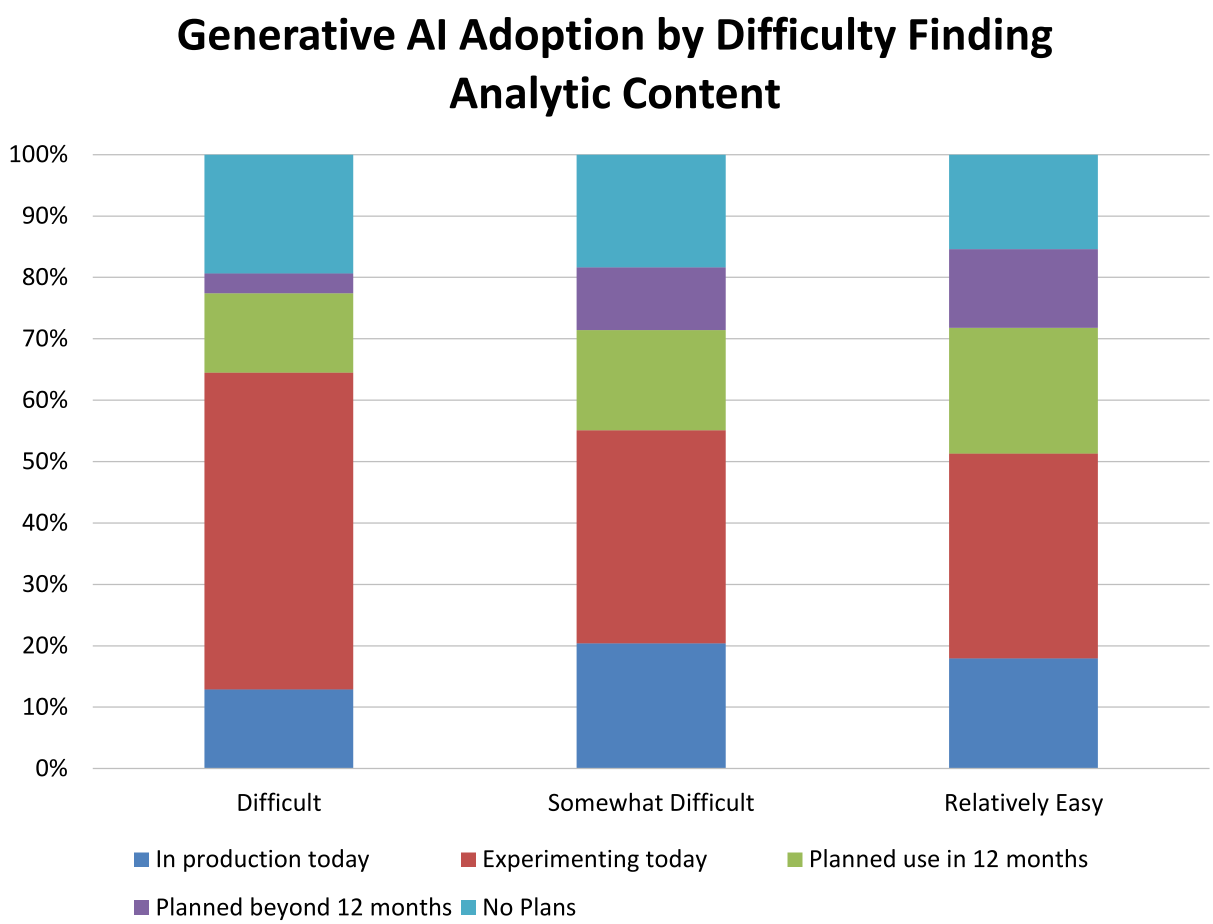


Figure 32 – Generative AI adoption by difficulty finding analytical content

Generative AI Data Sources

We asked respondents, “Which data sources are important to leverage with generative AI?” (fig. 33). The answers fell first and most often to customer relationship management (CRM) records, and documents and unstructured data (though many or most data sources pointed at large language models are likely to consist of structured and/or unstructured materials). This year, documents and unstructured data are considered critical or important by 58% of respondents, indicating generative AI’s attraction for use with non-tabular data. High combined critical and very important attitudes are also seen in competitive data (53%) and finance and accounting data (53%), and the remaining percentages are considerable and relevant for every data source sampled.

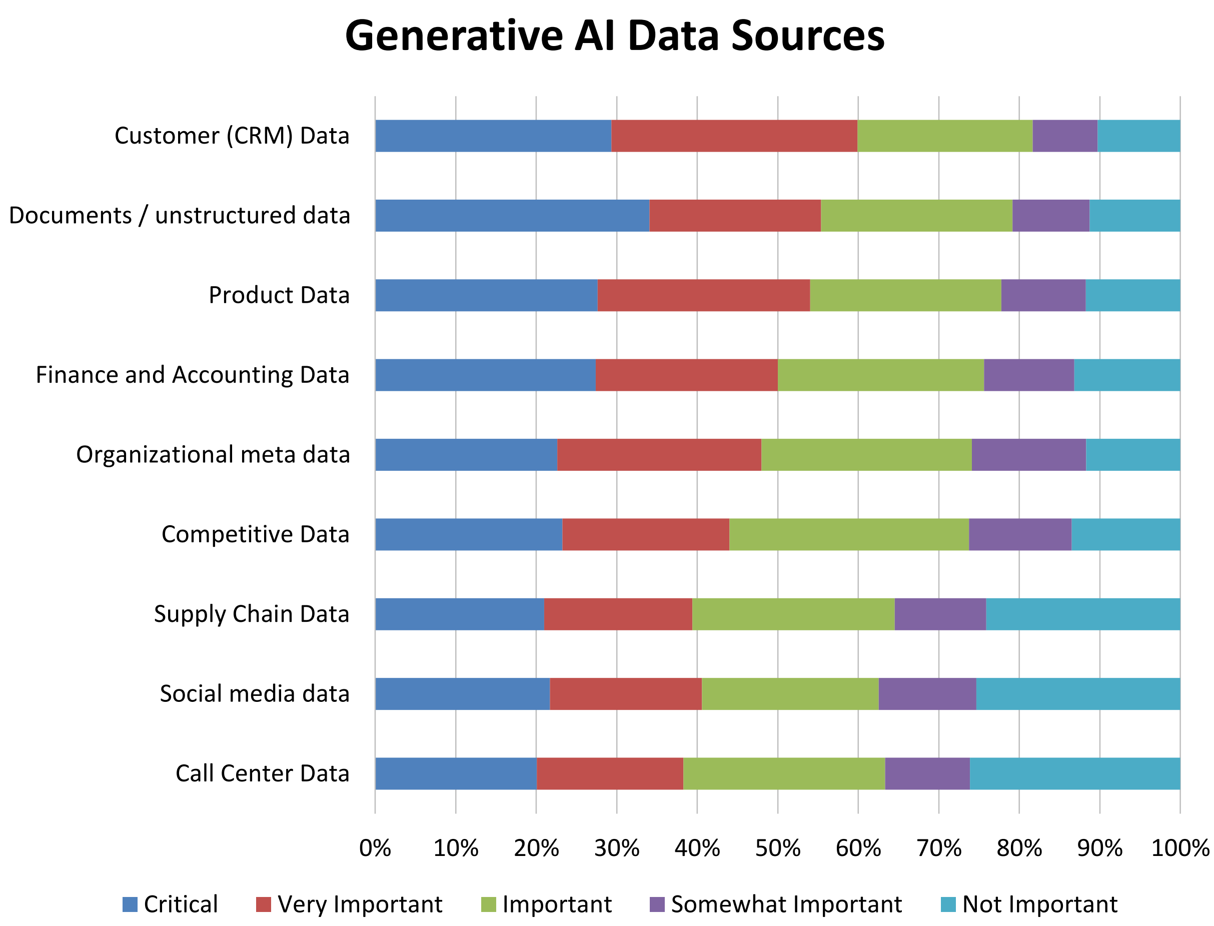


Figure 33 - Generative AI data sources

Agentic AI Adoption

Despite growing interest, only 7% of respondent organizations have successfully transitioned from experimentation to production deployment of agentic AI (fig. 34). This figure excludes embedded use cases, illustrating the distinctive adoption path of using vendor-provided functionality in an application or solution context. Although many organizations faced barriers in moving from experimentation to deployment with earlier genAI efforts, the clarity of use cases and stronger value propositions for agentic AI suggest that history may not repeat itself. Encouragingly, 28% of respondents have advanced from excitement to active experimentation, signaling that a meaningful portion of organizations are dedicating resources to validate and scale this innovation. Together, these two groups represent a clear early majority.

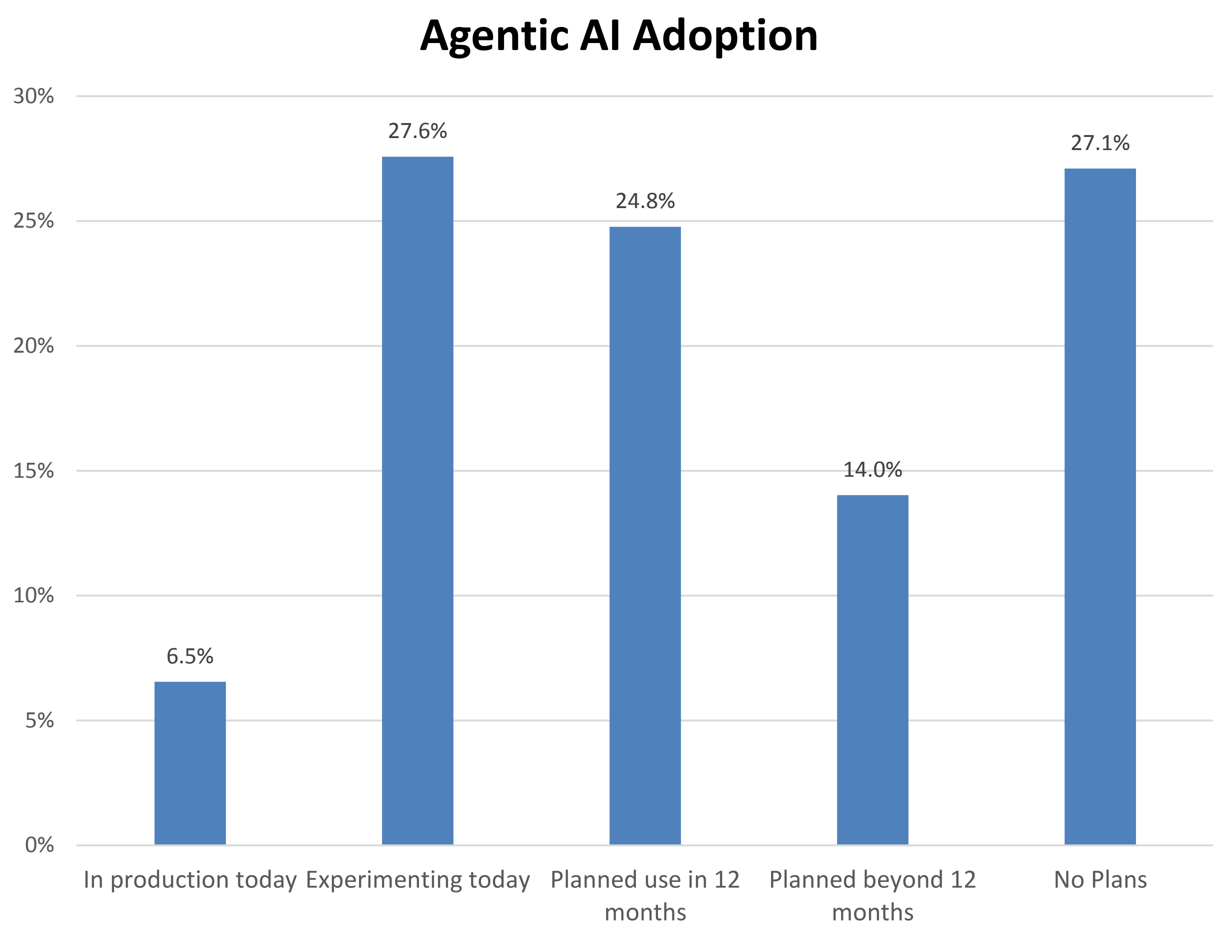


Figure 34-- Agentic AI adoption

Agentic AI Adoption by Geography

Adoption of agentic AI varies by region (fig. 35). Organizations in North America report the greatest production use of agentic AI (9%). Latin America and Asia Pacific follow, while EMEA significantly lags (with essentially no in-production use among our respondents). These figures mirror broader generative AI adoption trends, where Latin America leads in production deployment, followed by Asia Pacific and North America, with EMEA again at the bottom at just 12%. The consistently lower adoption rates in EMEA reflect general regulatory caution and a more conservative approach to emerging technologies as EU AI laws restrict use. Slower investment cycles may also play a role. Regional maturity, innovation appetite, and risk tolerance continue to shape global adoption patterns for agentic AI.

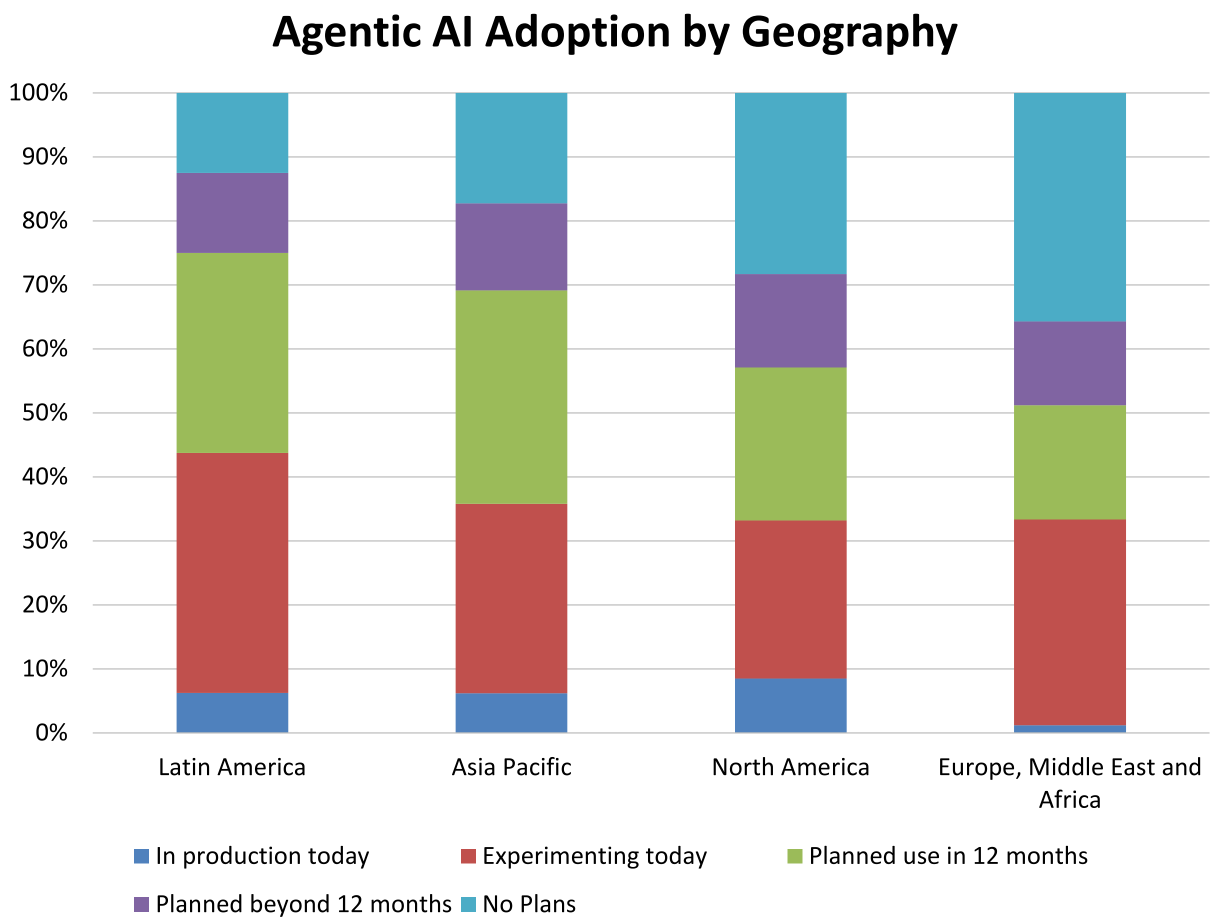


Figure 35 - Agentic AI adoption by geography

Agentic AI Adoption by Function

By business function, sales and marketing (unsurprisingly) lead the way in adoption of agentic AI, with 14% of organizations having moved to production—reflecting high demand for personalization, customer engagement, and campaign optimization (fig. 36). IT and operations follow (both at 9%), aligning with the broader enterprise outlook that positions agentic AI as a tool for both front-office innovation and back-office efficiency. The data indicates that agentic AI adoption occurs where measurable value and faster returns are most evident, particularly in functions that benefit from automation, decision support, and data-driven insights.

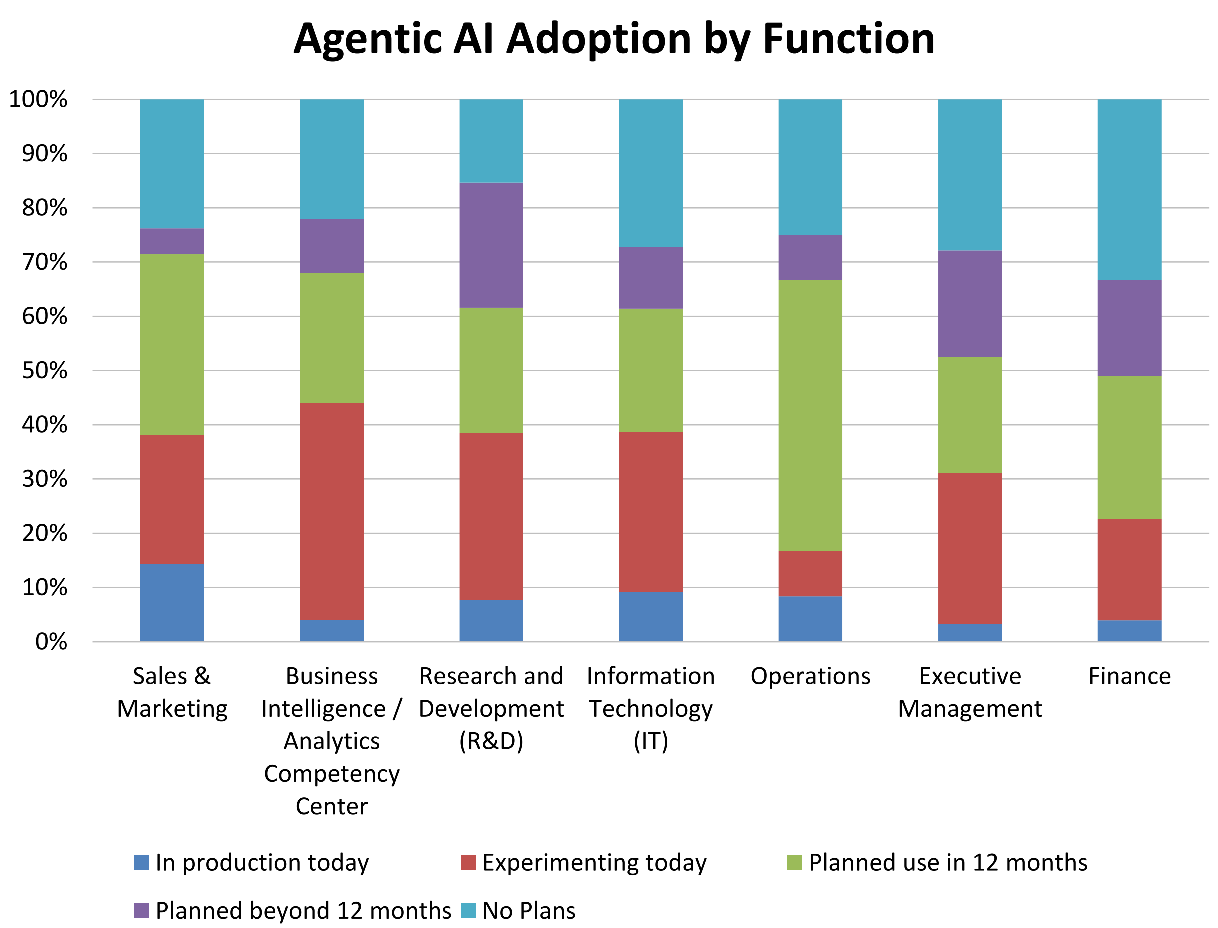


Figure 36 - Agentic AI adoption by function

Agentic AI Adoption by Industry

By industry, the highest levels of production deployment of agentic AI occur in technology, where 16% of organizations have moved beyond experimentation (fig. 37). Technology leadership is expected, given the sector's agility, technical expertise, and culture of rapid innovation. Manufacturing and business services next-most-often report production use of agentic AI (5% each), respectively leveraging it in their industries to drive automation, quality control, and supply chain optimization, and enhance client delivery and operational efficiency. These trends highlight that early adoption is strongest in industries with both technical readiness and clear operational use cases to capitalize on agentic AI's capabilities.

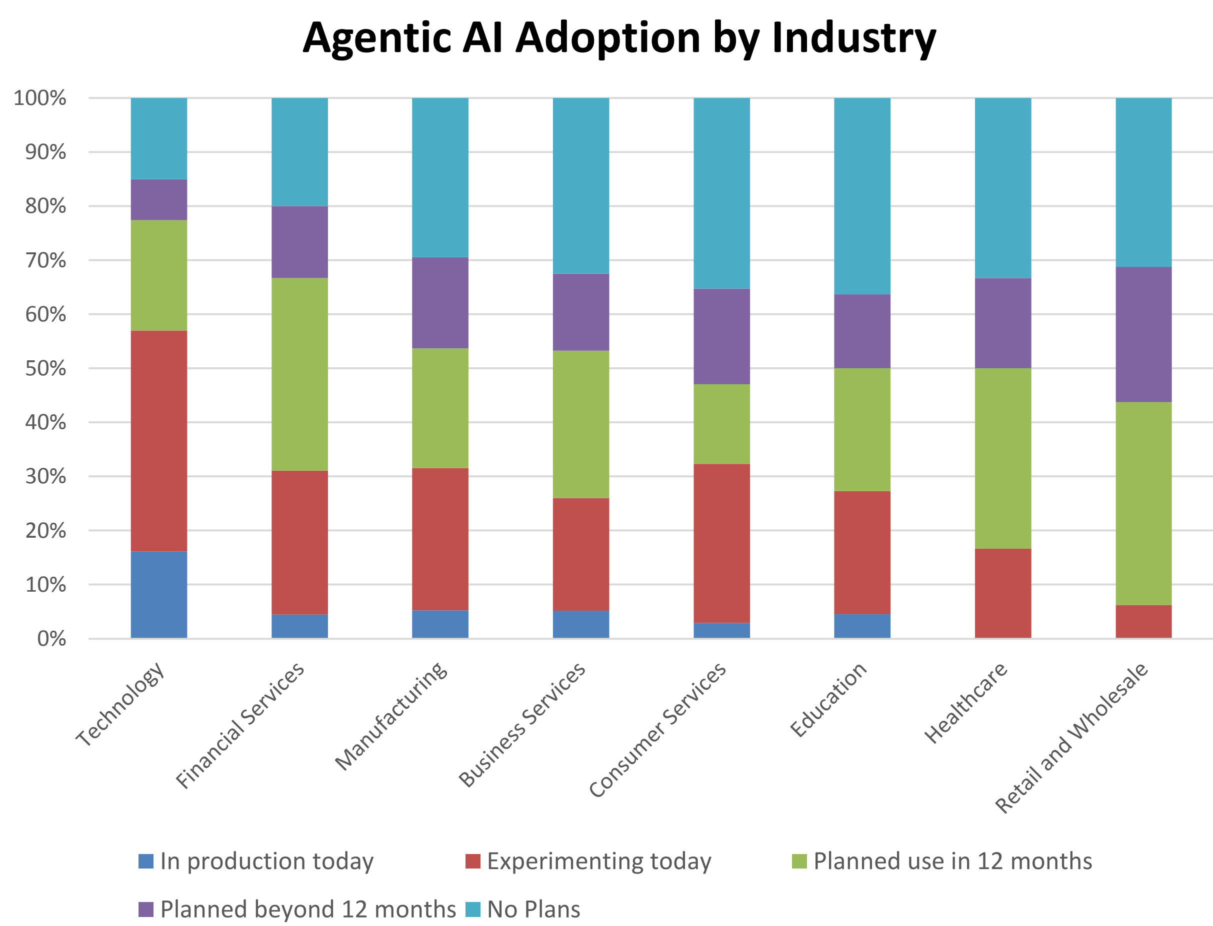


Figure 37 - Agentic AI adoption by industry

Agentic AI Adoption by Organization Size

Viewed by organization size, adoption plans for agentic AI correlate positively to and increase with global headcount in 2025 (fig. 38). The largest organizations (more than 10,000 employees) report the highest combined in-production and experimental use (45%), followed by large (1,001-10,000 employees) at 32%, midsize (101-1,000 employees) at 30%, and small (1-100 employees) organizations at 32%. Very large organizations also report the lowest likelihood of having no plans (15%), compared to 30% of large, 28% of midsize, and 32% of small organizations.

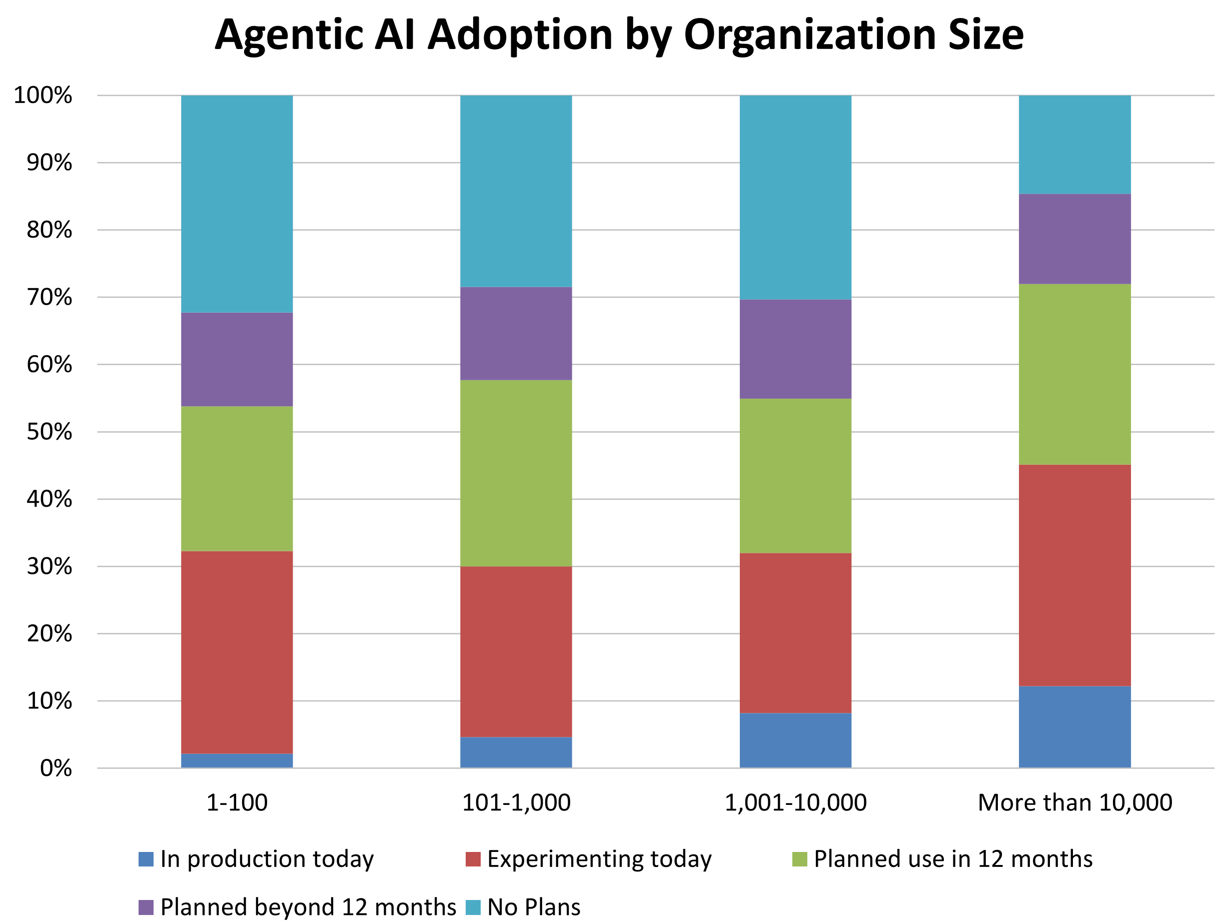


Figure 38 - Agentic AI adoption by organization size

Agentic AI Adoption by Difficulty Finding Analytic Content

Organizations that find it difficult to find analytic content in 2025 are least likely to have agentic AI in production today, though in-production use is also extremely weak in organizations that find it somewhat difficult (4%) and relatively easy (11%) to find analytic content (fig. 39). Possible implications are that organizations experiencing difficulty are likely candidates for agentic AI, or that those who find content more easily have already benefited from it. But with early adoption so low, it appears more likely that organizations with any level of difficulty are behaving opportunistically by actively experimenting with agentic AI or have 12-month plans.

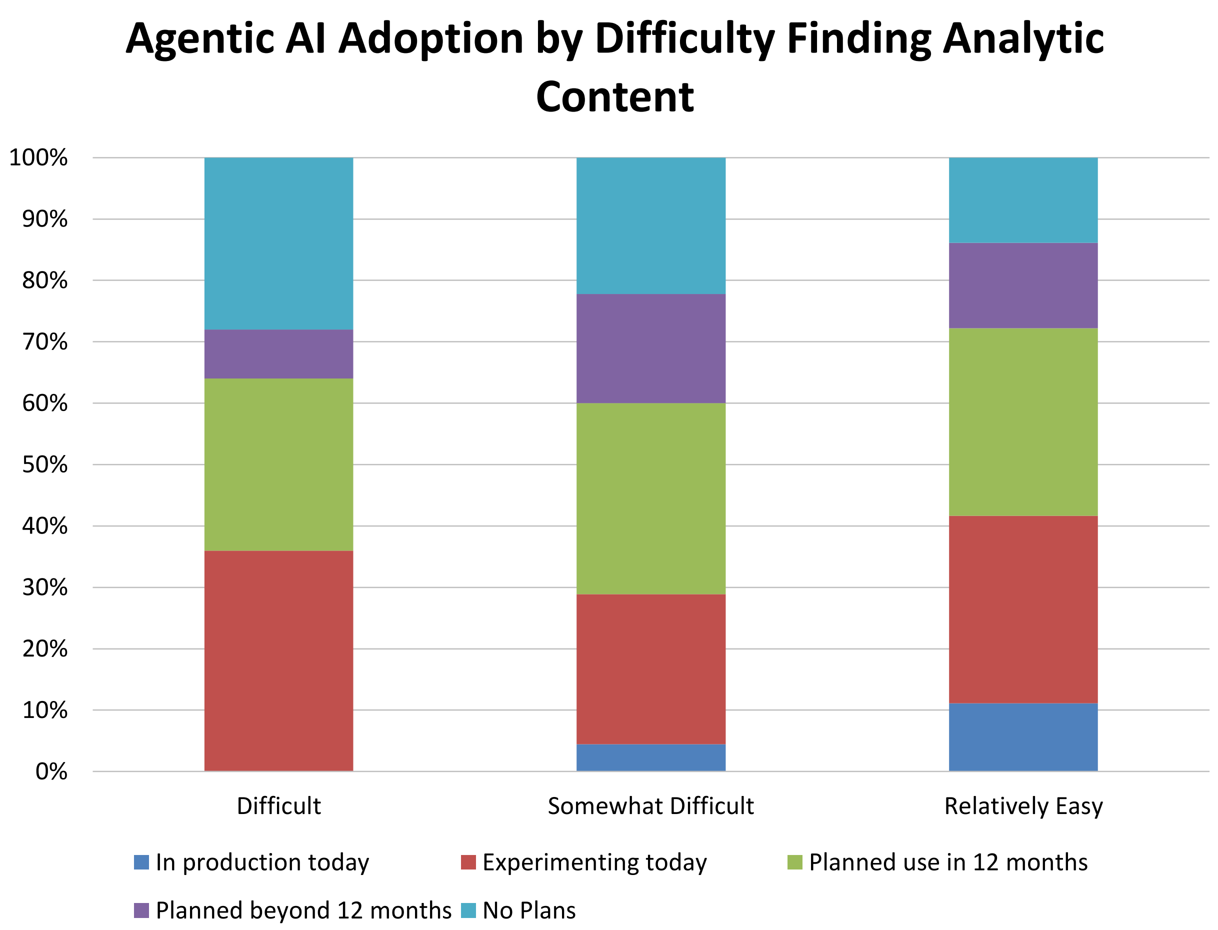


Figure 39 - Agentic AI adoption by difficulty finding analytic content

Agentic AI Adoption by Success with BI

AI adoption, particularly for agentic AI, closely correlates with an organization's success with BI. Ten percent of organizations that consider their BI initiatives completely successful (the highest level of achievement) are likely to be in production with agentic AI, while another 5% of in-production users consider their BI efforts somewhat successful (fig. 40). Together, these two groups make up the entirety of current adopters, indicating that a strong data foundation and analytical maturity are prerequisites for effectively operationalizing advanced AI capabilities. This alignment underscores the importance of a well-established BI strategy as a stepping stone for AI-driven transformation.

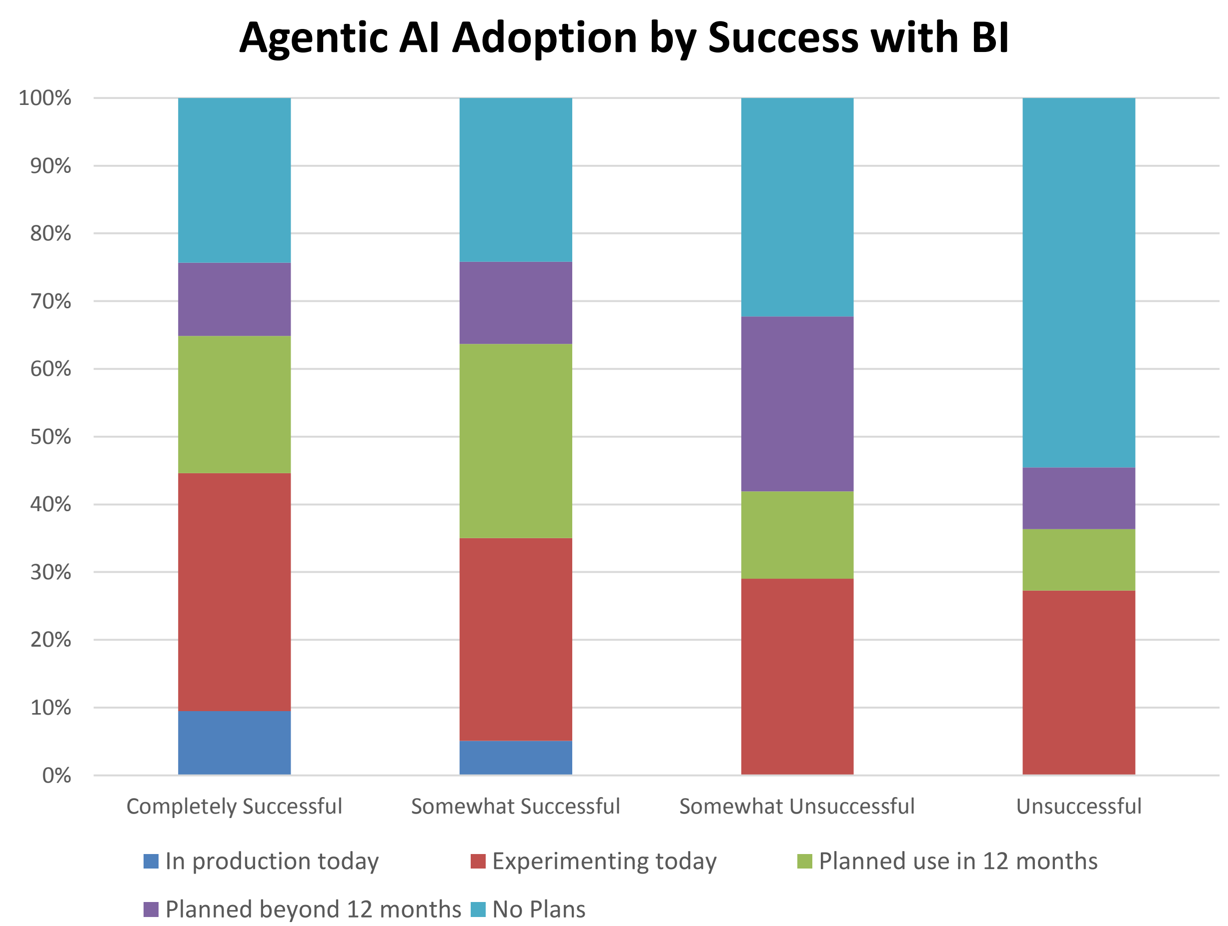


Figure 40 - Agentic AI adoption by success with BI

Agentic AI Data Sources

We asked respondents, “Which data sources are important to leverage with agentic AI?” (fig. 45). As was the case with generative AI (fig. 41), responses most often cited customer relationship management (CRM) records. The second-most-popular source for agentic AI, however, is finance and accounting data, allowing association and analysis of both tabular and unstructured formats contained in accounting records, statements, and filings. Product data and documents/unstructured data are the next-most cited sources; along with organizational metadata, the top five picks are critical or very important to 40% or far more of respondents, and all nine sampled data sources are at least important to close to 60% and up to 80% of respondents.

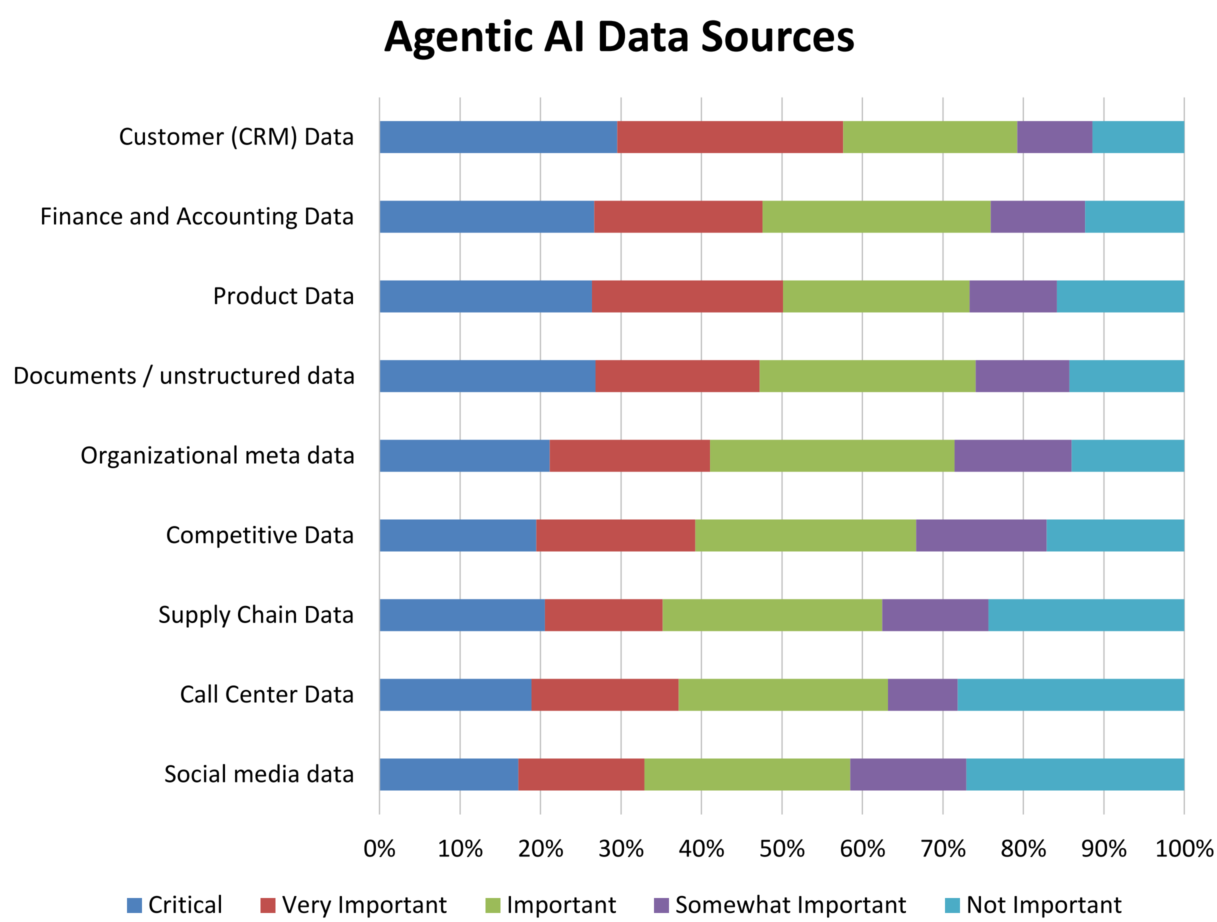


Figure 41 - Agentic AI data sources

Users of AI, Data Science, and Machine Learning

Fig. 42 describes the functional users of AI, data science, and machine learning. We include citizen data scientist, a role that might overlap with other users but generally describes users that can generate models. In 2025, the role of statistician/data scientist is the most likely “constant” or “often” user of AI, data science, and machine learning (63%), followed by BI expert (59%), business analyst (55%), IT staff (48%), and citizen data scientist (45%). Constant use thereafter drops to 33% for marketing analysts and financial analysts, 28% for C-level executives, and 23% for third-party consultants. Overall, the mix of user responses represents a broad mix of deployment and back- and front-office support.

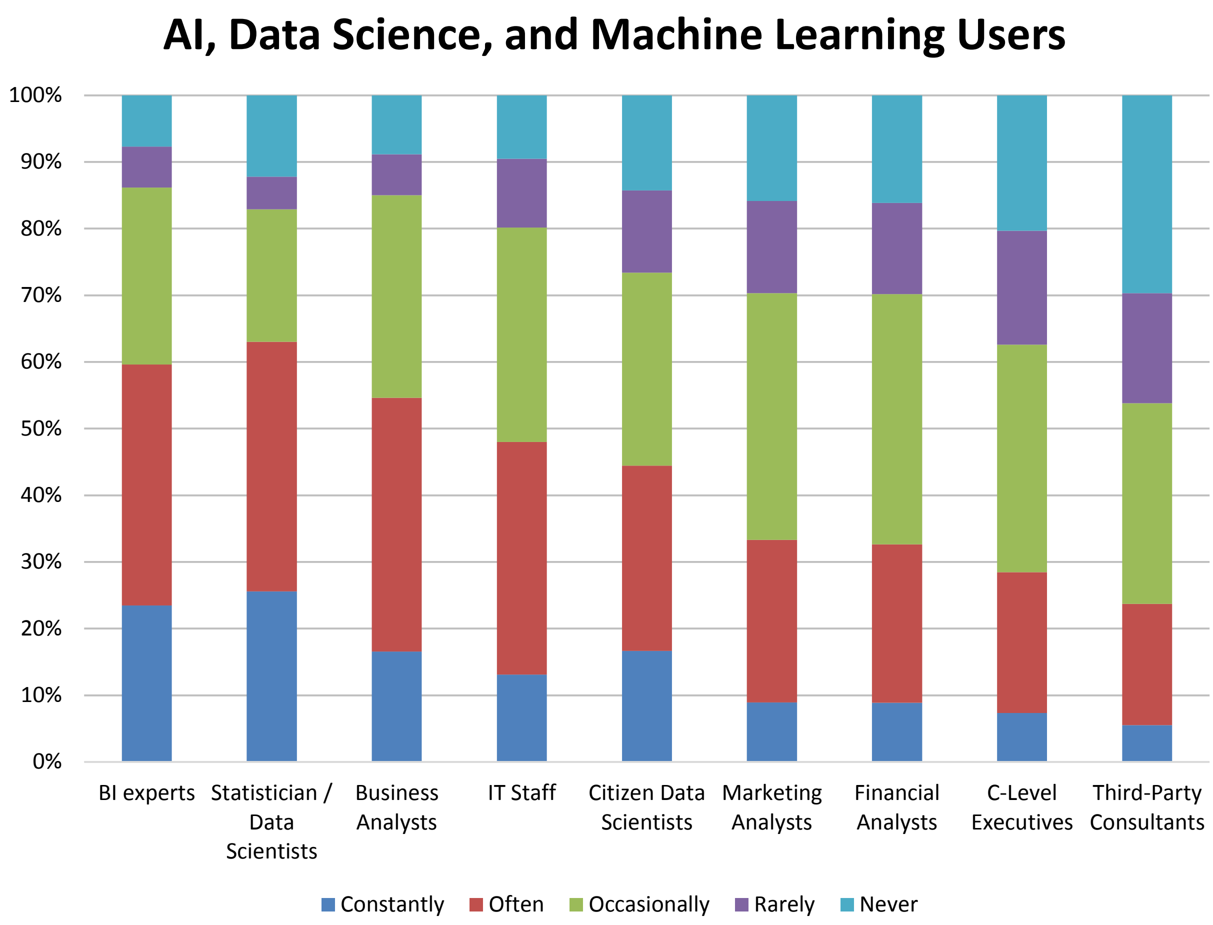


Figure 42 – AI, data science, and machine learning users

Users of AI, Data Science, and Machine Learning Users 2018–2025

Viewed across eight years of data collection, most user roles for AI, data science, and machine learning report average to above-average levels of use, with some reaching historic highs as user bases grow and spread (fig. 43). For example, IT staff use is at a relative all-time high, as are the four lowest-ranked roles: marketing analysts, financial analysts, C-level executives, and third-party consultants. Perhaps most interesting is a relative two-year decline in citations of statistician/data science users after a 2022-2023 heyday. This finding might reflect a rethinking of job titles or absorption into other named roles. Relative levels of BI experts and business analysts are also flat or somewhat lower in 2025 and closer to average historic levels.

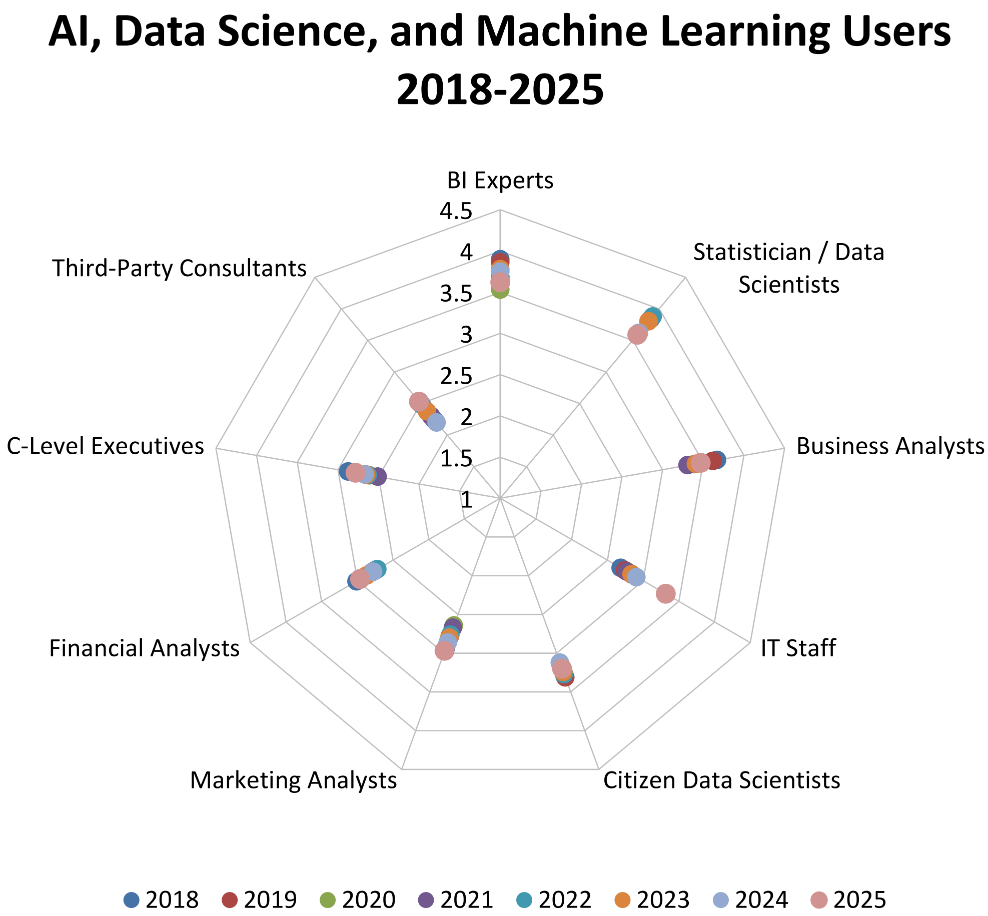


Figure 43 – AI, data science, and machine learning users 2018-2025

Users of AI, Data Science, and Machine Learning by Geography

Viewed by geography, all roles receive the highest 2025 importance scores from respondents in Asia Pacific (fig. 44). Scores for the top five choices (BI expert, statistician/data scientist, business analyst, IT staff, citizen data scientist) all correspond to greater than “important” across all regions. Excluding Asia Pacific, respondents narrowly post the highest importance for BI experts, business analysts, and IT staff, while EMEA gives a slightly higher score to statisticians/data scientists. Excluding Asia Pacific, marketing analysts, financial analysts, C-level executives, and third-party consultants all rank slightly below the level of important.

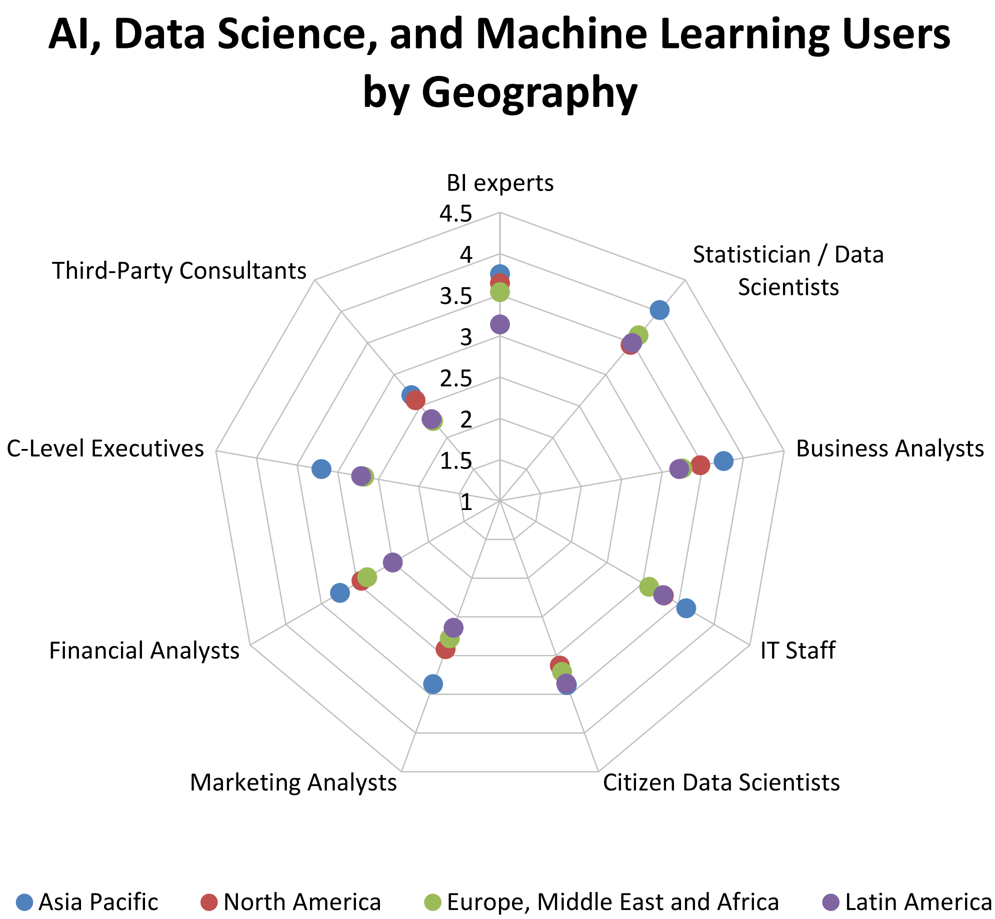


Figure 44 – AI, data science, and machine learning users by geography

Users of AI, Data Science, and Machine Learning by Organization Size

The likelihood of various and multiple user roles for AI, data science, and machine learning most often correlates directly with increasing organization size in 2025 (fig. 45). Roles that are relatively more important in very large (more than 10,000 employees) and large (1,001-10,000) organizations include statistician/data scientist, IT staff, marketing analyst, and third-party consultants. Though the overall correlation favors scale, small (1-100) organizations often report higher scores than midsize (101-1,000) peers. The top five user roles are at least important to all organizations regardless of size.

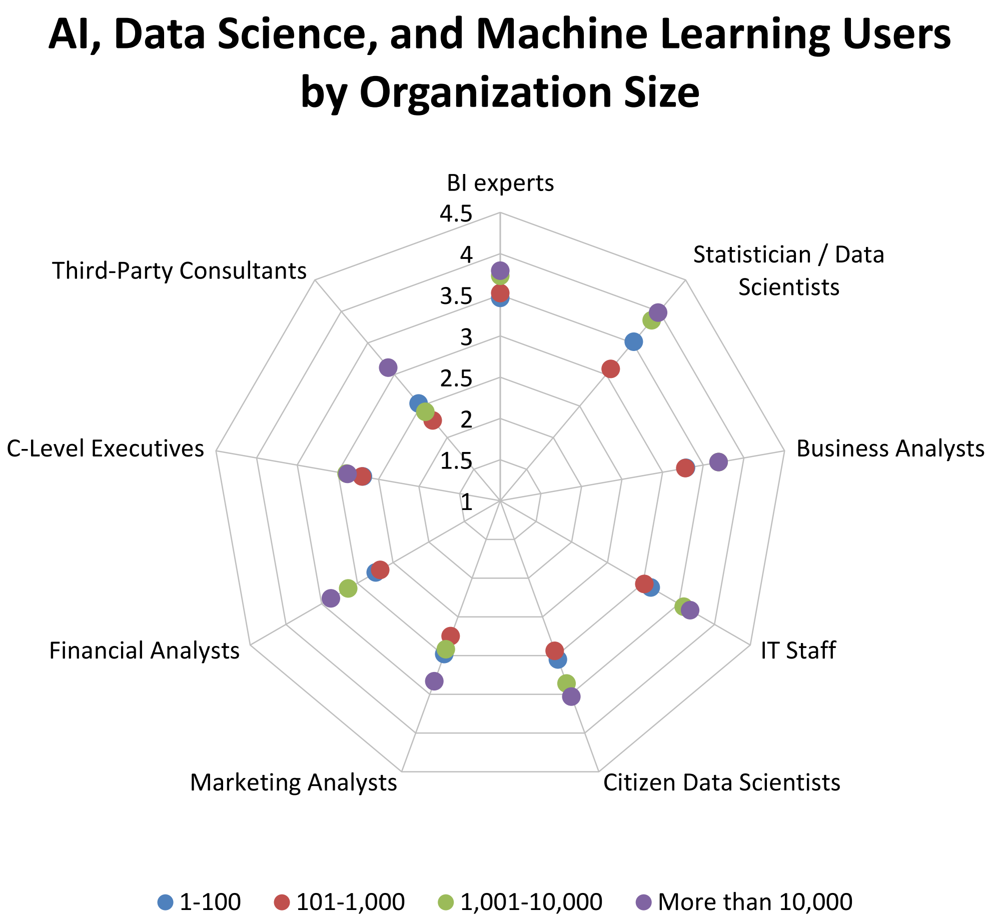


Figure 45 – AI, data science, and machine learning users by organization size

Users of AI, Data Science, and Machine Learning by Industry

Taken in sum, multiple industries assign an uneven mix of high to medium and occasionally lower usage by roles for AI, data science, and machine learning in 2025 (fig. 46). This year, the highest overall mean scores are reported by respondents in consumer services, manufacturing, healthcare, financial services, technology, and business services. The lowest scores come from retail & wholesale, government, and education. That said, unique standout scores are seen, such as the high importance of BI experts in government (4.0). Only the top two roles, BI expert and statisticians/data scientists, are at least “important” to respondents in all industries.

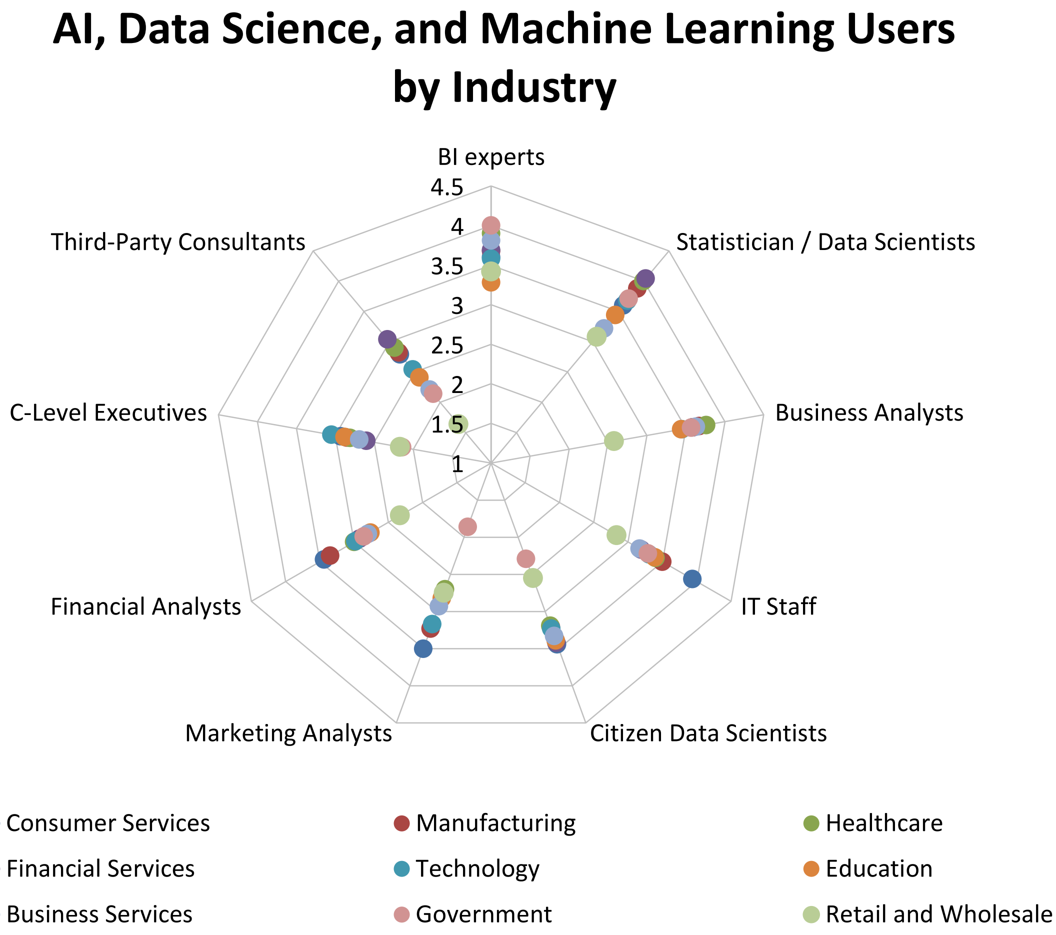


Figure 46 – AI, data science, and machine learning users by industry

Users of AI, Data Science, and Machine Learning by Company Age

The frequency of data science and machine learning use by role varies unevenly by company age in 2025 but offers some interesting specific observations (fig. 47). This year, for example, older organizations of 11-16 years are most or near equally most likely to report users among BI experts, statisticians/data scientists, business analysts, marketing and financial analysts, and C-level executives. But the same older organizations are among the least likely to report users among IT staff, who are reported as much more frequent users within the youngest organizations of less than five years. Among many other observations, the oldest companies of more than 16 years are least likely to report citizen data science users, and they report decreasing numbers of business analyst users.

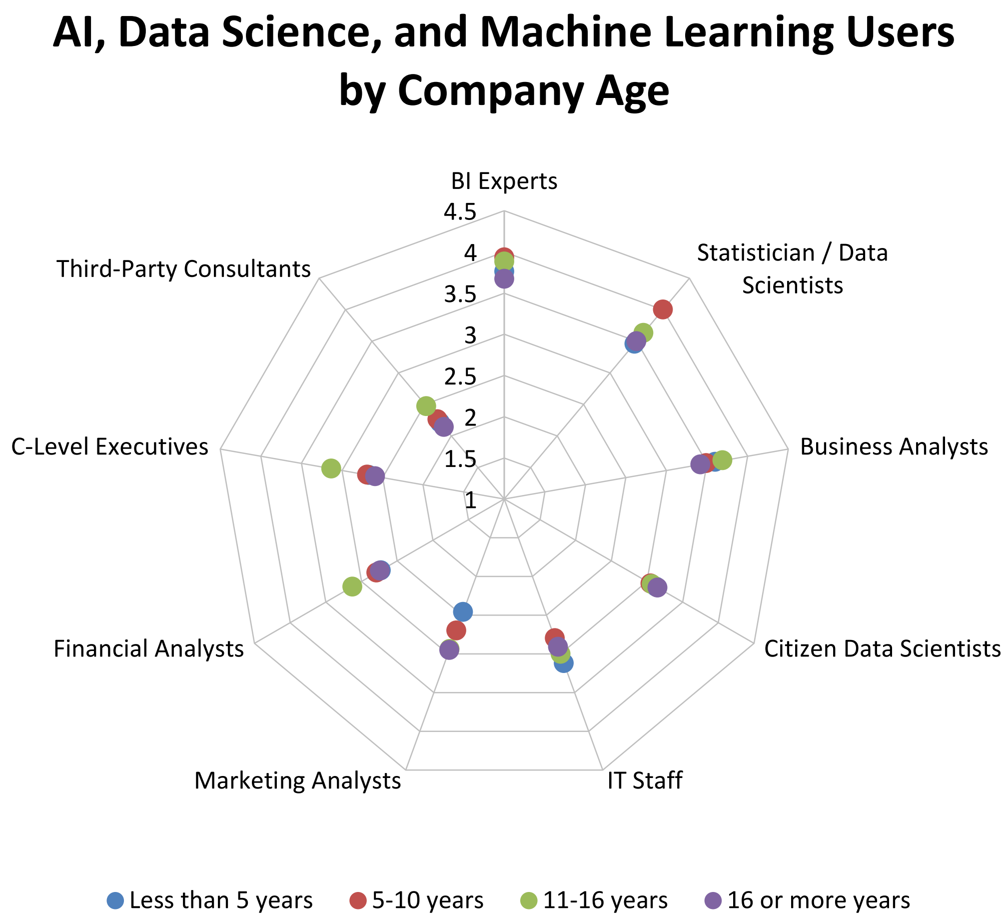


Figure 47 – AI, data science, and machine learning users by company age

Users of AI, Data Science, and Machine Learning by Success with BI

The use of AI, data science, and machine learning correlates to success with BI in 2025 (fig. 48). Organizations that are completely successful either lead or rank near equally highest in all eight categories sampled. Conversely, organizations reporting the lowest score of “unsuccessful” trail in all sampled roles. The top four roles, statistician/data scientists, BI experts, business analysts and IT staff, are at least “important” to all organizations that are not “unsuccessful.” Areas where completely successful organizations stand out among peers include BI experts: business analysts; citizen data scientists; marketing and financial analysts; and C-level executives.

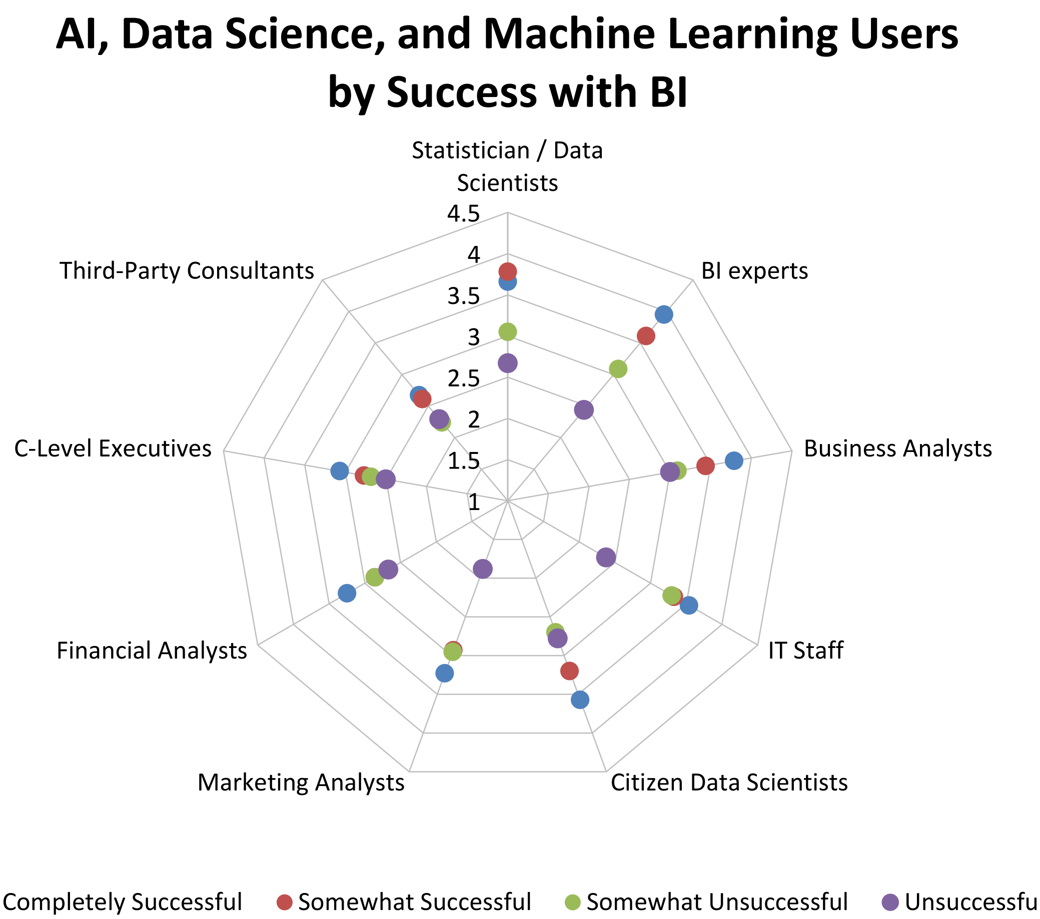


Figure 48 – AI, data science and machine learning users by success with BI

Importance of AI, Data Science, and Machine Learning

AI, data science, and machine learning ranks 24nd, and cognitive BI/artificial intelligence-based BI ranks 35nd among 65 topics under our study, as defined in our 2025 user survey. Generative AI ranks 37th and agentic AI ranks 41st (fig. 49).

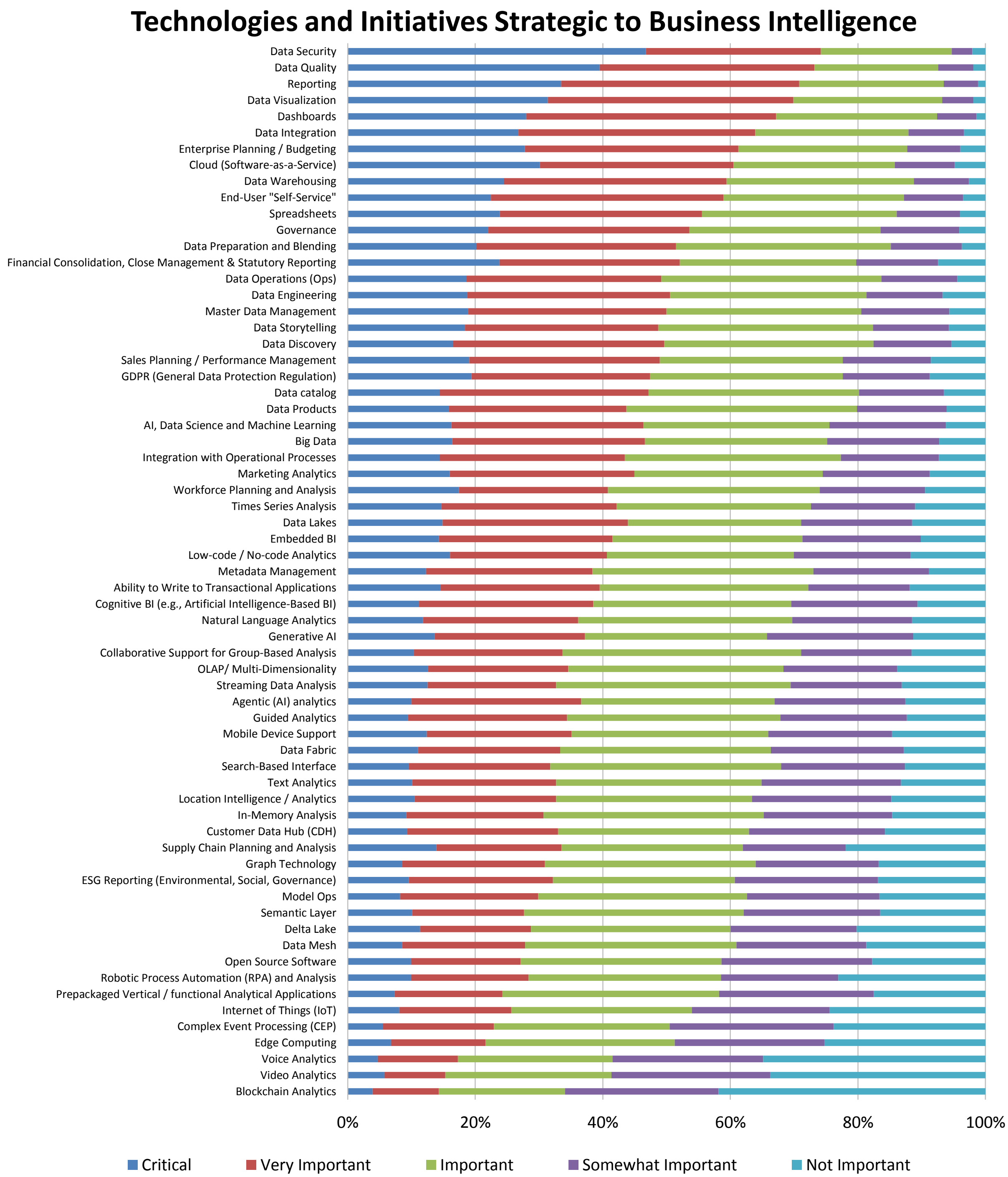


Figure 49 – Technologies and initiatives strategic to business intelligence

Importance of AI, Data Science, and Machine Learning 2014-2025

In our 2025 study, the weighted-mean perceived importance of AI, data science, and machine learning stands at 3.3, below last year's 3.6, and below the historic high of 3.7 in 2022 (fig. 50). This year's score is the lowest reported since 2017, which may indicate better understanding or some burnout from overexposure. Scores of "critical" and "very important" are also lower than seen in 2022-2024. Nonetheless, a narrow range of 3.3-3.7 held from 2018 to 2025, and the current score is well within one standard deviation for the history of the study. This sustained measure of interest, which has never been close to the level of very important, holds considerable room for future growth amid market interest and capex spending. (Also see industry importance, fig. 107.)

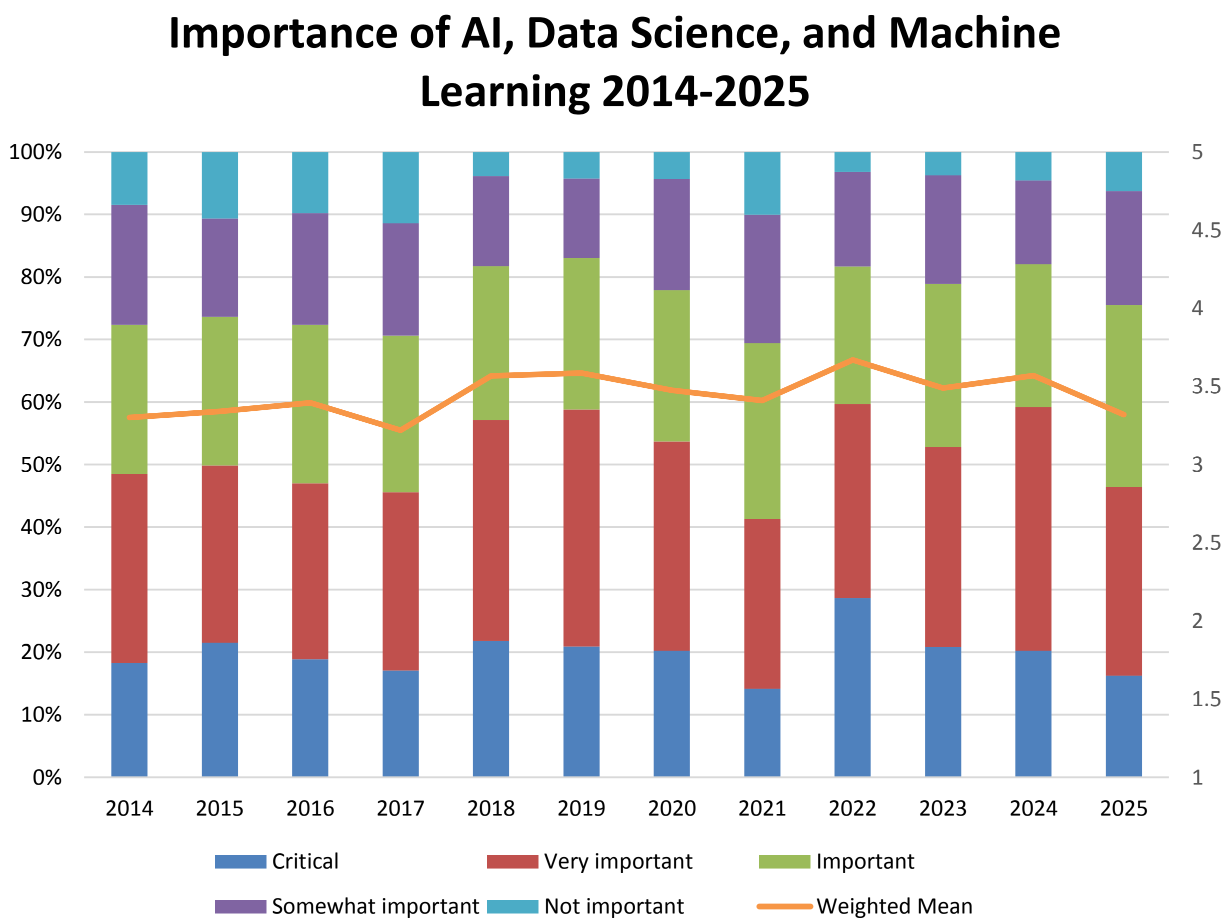


Figure 50 – Importance of AI, data science, and machine learning 2014-2025

Importance of AI, Data Science, and Machine Learning by Geography

Interest in AI, data science, and machine learning varies by geography in 2025, with weighted-mean interest highest in Asia Pacific (3.9), followed by Latin America (3.5), North America (3.4), and EMEA (3.4; fig. 51). Thus, all regions consider AI, data science, and machine learning to be close to midway or well above the level between important and very important. Levels of combined critical and very important criticality are also highest in Asia Pacific (68%), compared to North America (49%), Latin America (47%), and EMEA (44%).

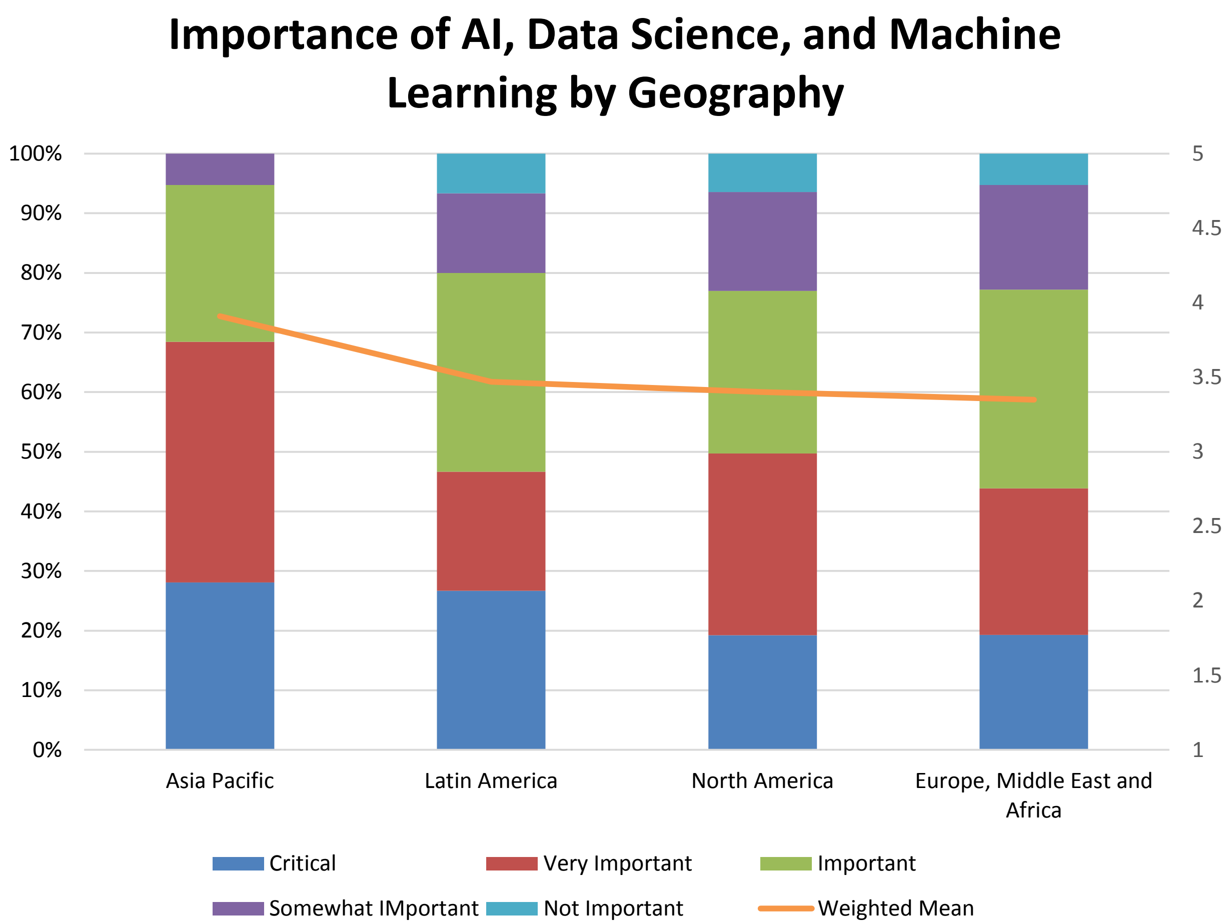


Figure 51 – Importance of AI, data science, and machine learning by geography

Importance of AI, Data Science, and Machine Learning by Function

Interest in AI, data science, and machine learning is well above the level of "important" for all but one function in 2025 (fig. 52). This year, in a mix of both use and active development, weighted-mean sentiment is highest in front-office sales & marketing (3.7) and back-office R&D (3.7). These are followed closely by respondents in operations (3.6) and the BICC (3.5). This finding is more varied than is usually found in new technology rollouts that are led early by development and deployment and followed later by in-production adoption. In another sense, it is a different representation of the coordination and lack thereof in AI policy management (fig. 15). Only in finance is AI, data science, and machine learning only at the 3.0 middle level of important.

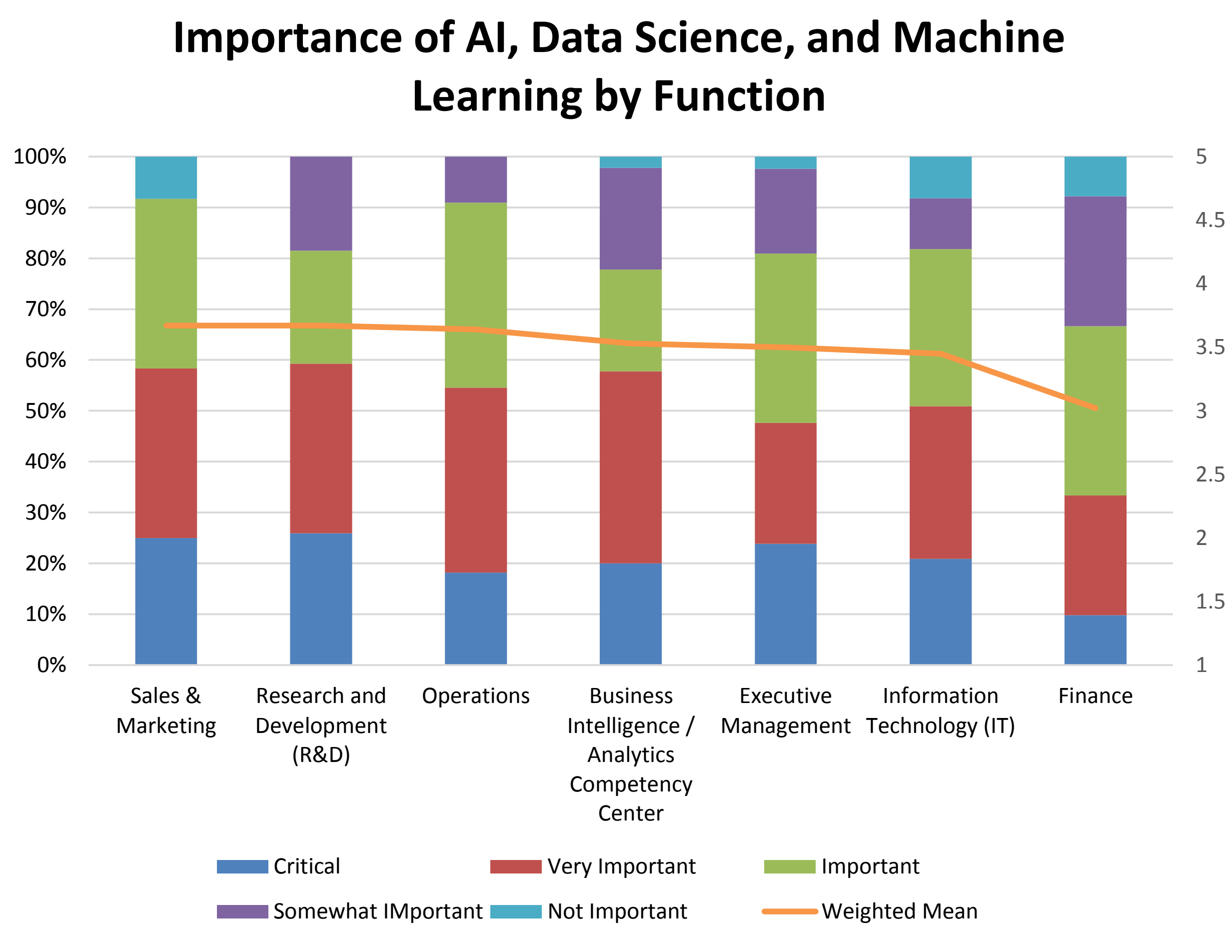


Figure 52 – Importance of AI, data science, and machine learning by function

Importance of AI, Data Science, and Machine Learning by Industry

Among vertical industries sampled in 2025, all give AI, data science, and machine learning higher to far higher than important scores ranging from 3.1-3.7 (fig. 53). This year, respondents in technology give the high score of 3.7, followed by healthcare (3.6) and financial services (3.6). Close behind are manufacturing (3.5), retail & wholesale (3.4), education (3.4), and consumer services (3.4), combined scores of "somewhat important" and "not important" are highest in education and business services (29%-32%).

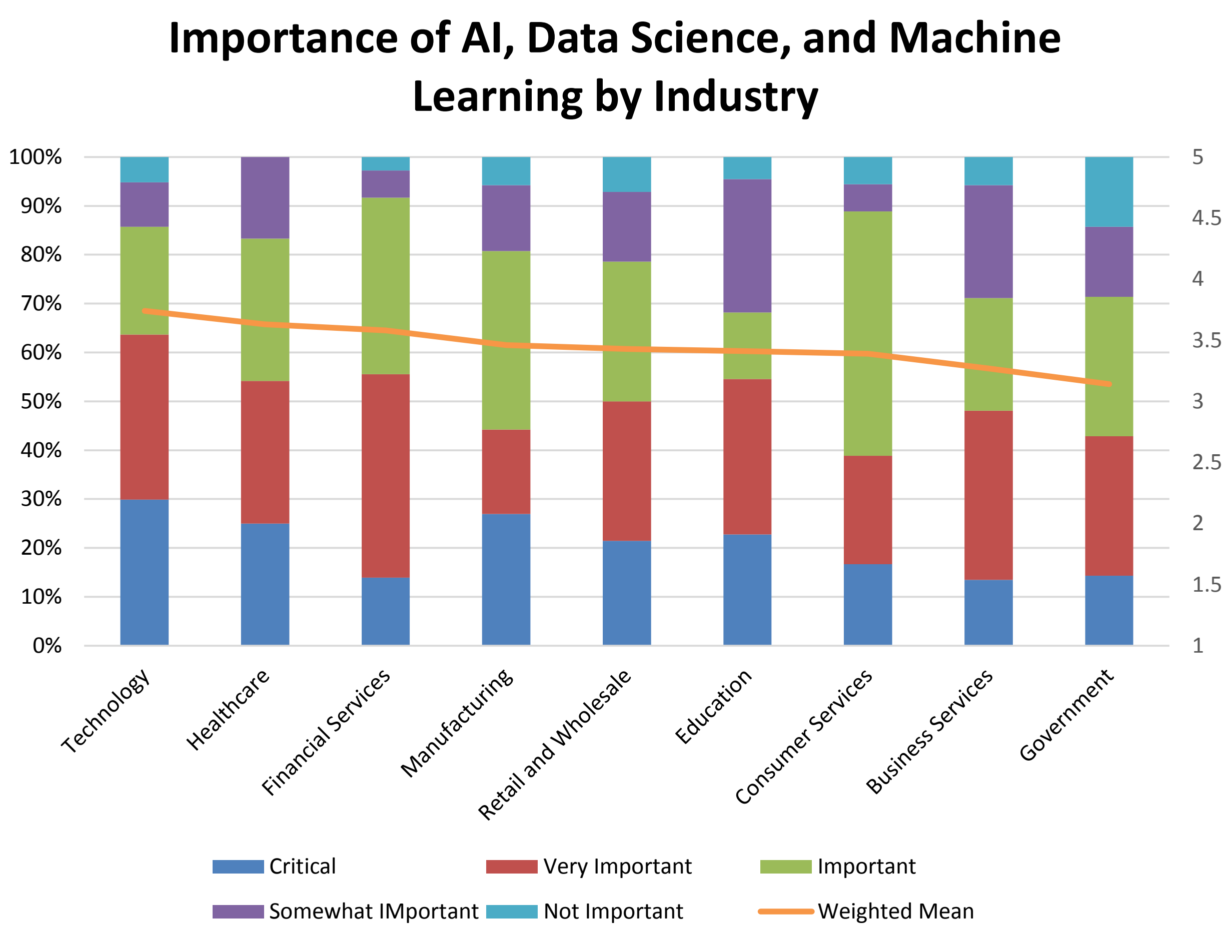


Figure 53 – Importance of AI, data science, and machine learning by industry

Importance of AI, Data Science, and Machine Learning by Organization Size

In 2025, the importance of AI, data science, and machine learning is highest by weighted mean in very large (more than 10,000 employees) and small (1-100 employees) organizations and grows among organizations with 100 or more employees (fig. 54). Fifty-seven percent of very large organization respondents say AI, data science, and machine learning is either critical or very important, compared to 55% of large, 40% of midsize, and 54% of small organizations.

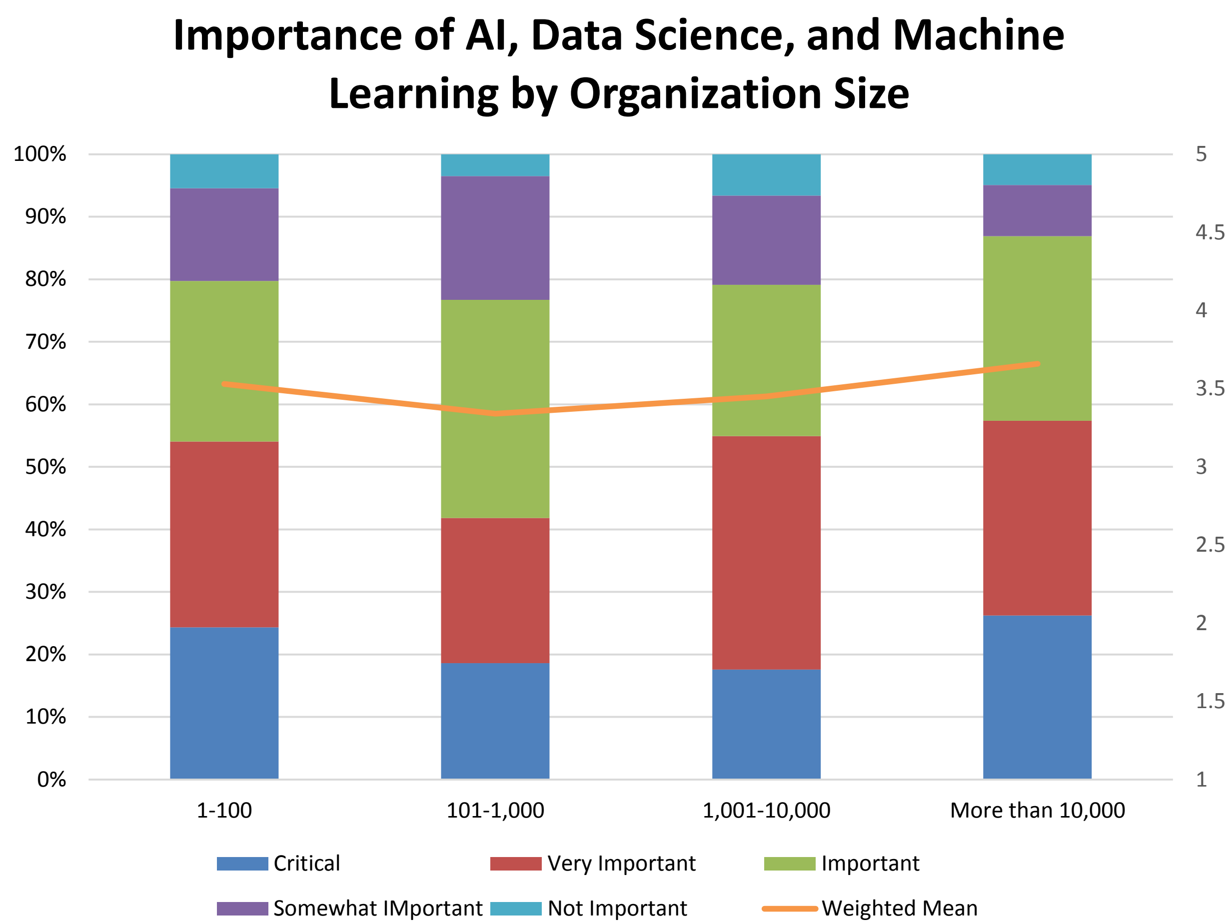


Figure 54 – Importance of AI, data science, and machine learning by organization size

Importance of AI, Data Science, and Machine Learning by Current Strategic Role of AI

Organizations that have fully embraced and incorporated AI into their business and technology strategies (considering it a cornerstone) most often perceive AI, data science, and machine learning as critical or very important (84%, fig. 55). This is notably higher than those considering AI to have an important role (72%) and significantly higher than respondents who are exploring (45%) or minimally integrating AI with their business and technology strategies (6%). Although the data and its correlation isn't surprising, the magnitude of lower prioritization by those taking a minimalist approach to AI is. With less than half of these organizations considering it at least "important" to incorporate AI into their business and technology strategies, we believe that many will struggle in the long term as competitors more aggressively incorporating AI into their strategies gain advantages that could be hard or impossible to overcome.

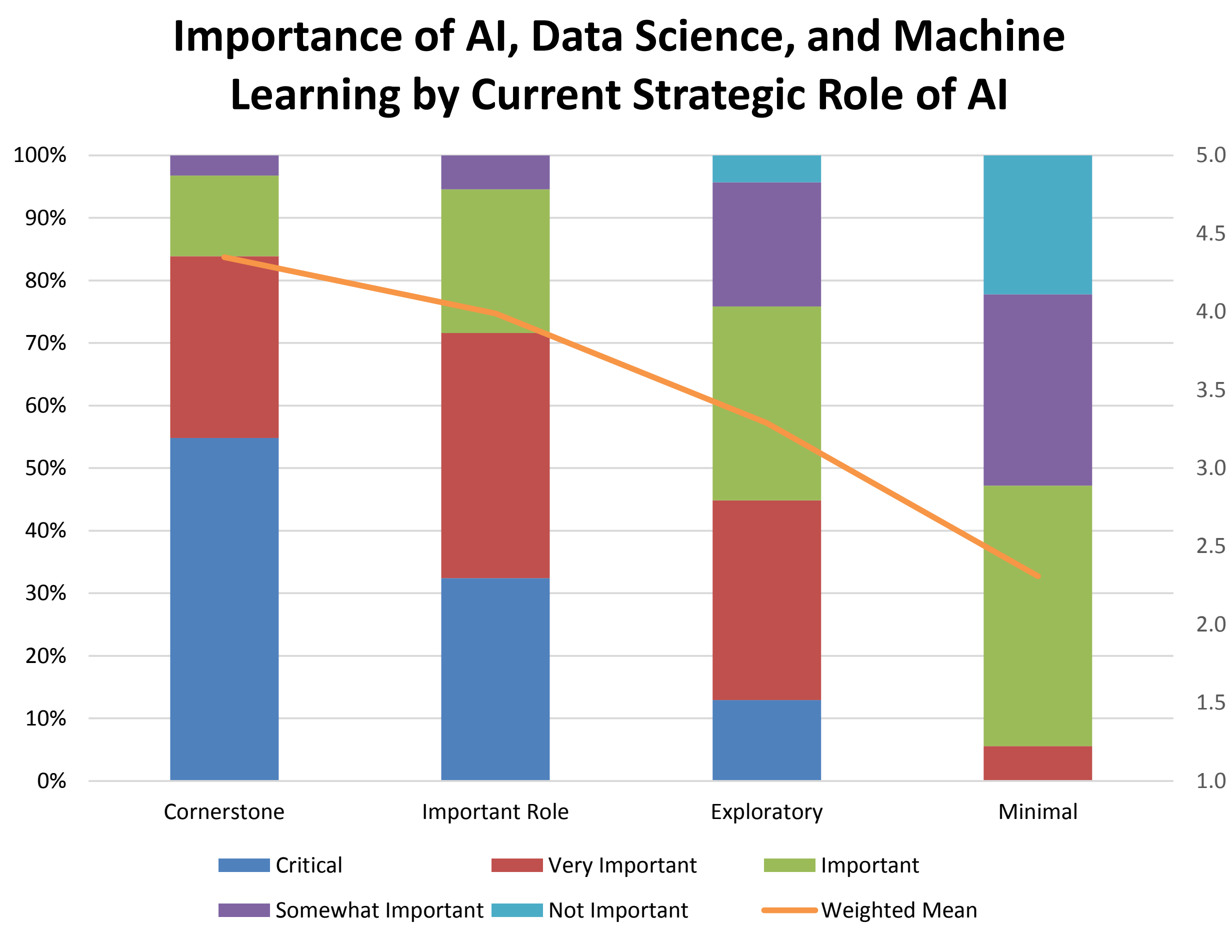


Figure 55 - Importance of AI, data science and machine learning by current strategic role of AI

Importance of AI, Data Science, and Machine Learning by Generative AI Adoption

Similar to the previous chart—and as indicated by the slope of the weighted-mean scores—the data shows a strong linear correlation between the frequency of the two highest perceptions of AI, data science, and machine learning, and time to deployment for generative AI (fig. 56). Organizations already deploying generative AI in production use cases most often see AI, data science, and machine learning as critical or very important (71%), while those with no plans to use generative AI least often hold the same view (19%). Although the data and its correlation are unsurprising, the high perceptions expressed by those experimenting with generative AI (66%) and those planning to use generative AI within a year (53%) indicate more of a question of when and to what extent—and not if—AI, data science, and/or machine learning will play more critical roles in these organizations.

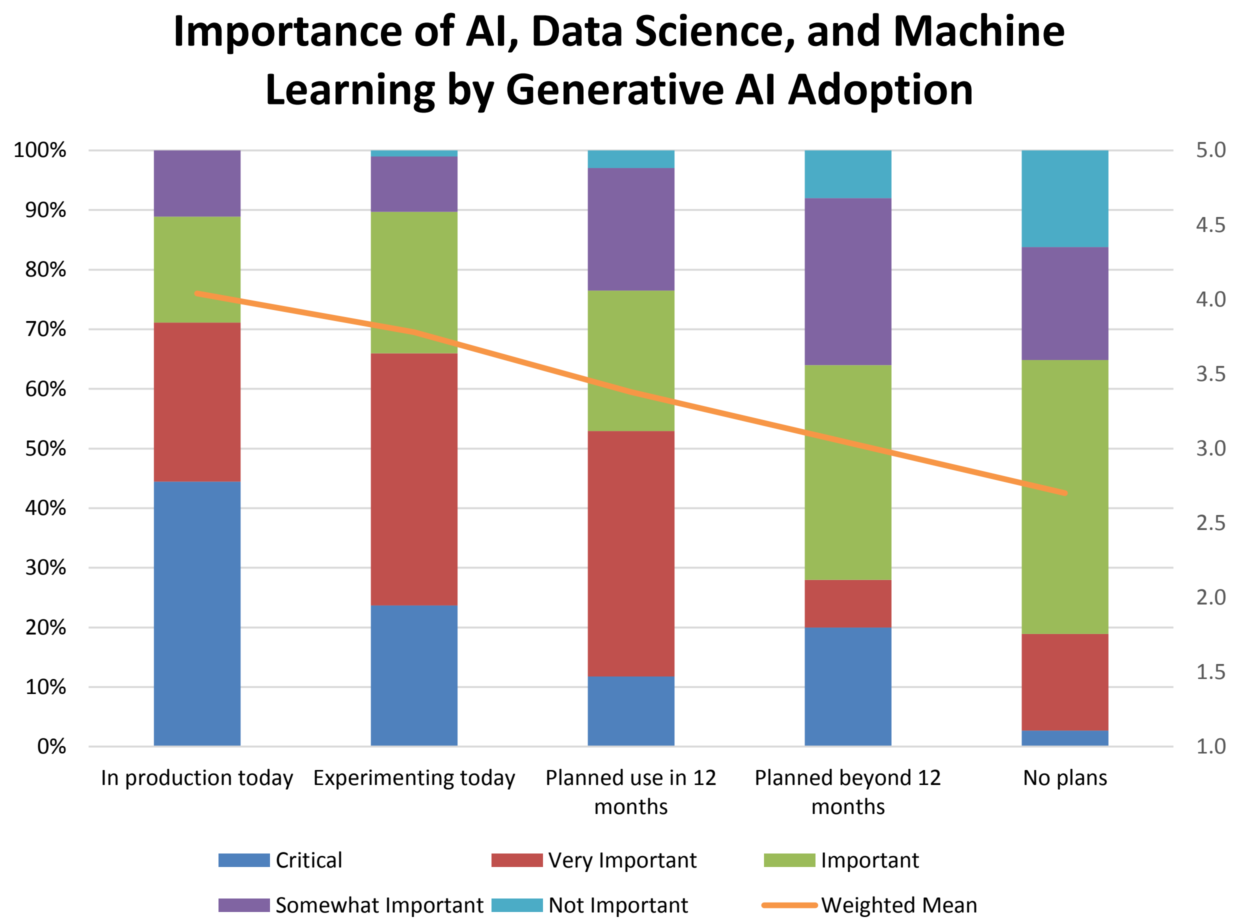


Figure 56 - Importance of AI, data science and machine learning by generative AI adoption

Importance of AI, Data Science, and Machine Learning by Agentic AI Adoption

The importance of AI, data science and machine learning correlates positively to the degree to which agentic AI has been adopted within organizations (fig. 57). Those already deploying agentic AI for in-production use cases most often see AI, data science, and machine learning as critical or very important (77%), while those with no plans to use generative AI least often hold the same view (30%). Although the data and its correlation are unsurprising, the same high combined estimations expressed by those experimenting with agentic AI (68%) and those planning to use agentic AI within a year (65%) indicate more of a question of when and to what extent—and not if—AI, data science, and/or machine learning will play more critical roles in these organizations.

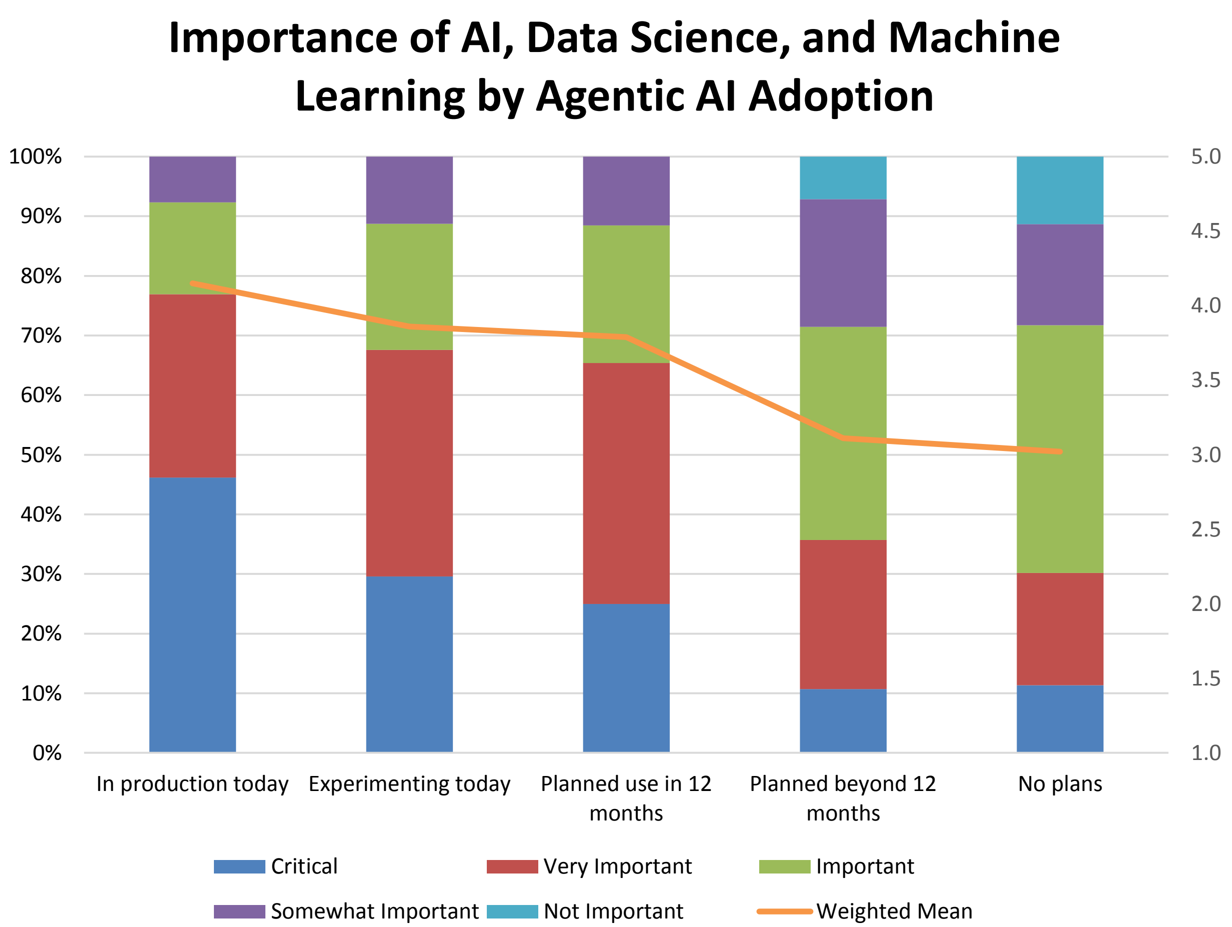


Figure 57 - Importance of AI, data science and machine learning by agentic AI adoption

Use Cases for AI, Data Science, and Machine Learning

On the much-debated execution end of AI, data science, and machine learning, we asked respondents to describe the relevance of 13 different use cases for the technologies (fig. 58). While current deployment of use cases can still be described as fairly low, we observed a dramatic year-over-year acceleration across all responses from 6%-19% in 2024 to 15%-28% in 2025. The top currently used use cases in 2025 include predictive maintenance (28%), demand forecasting (27%), customer segmentation (26%), and fraud detection (24%). Quality assurance and risk management are also currently in use by more than 20% of respondents. All use cases are targeted for at least 40% use within 12 months, and 12 of 13 use cases are targeted for at least 60% use within 24 months.

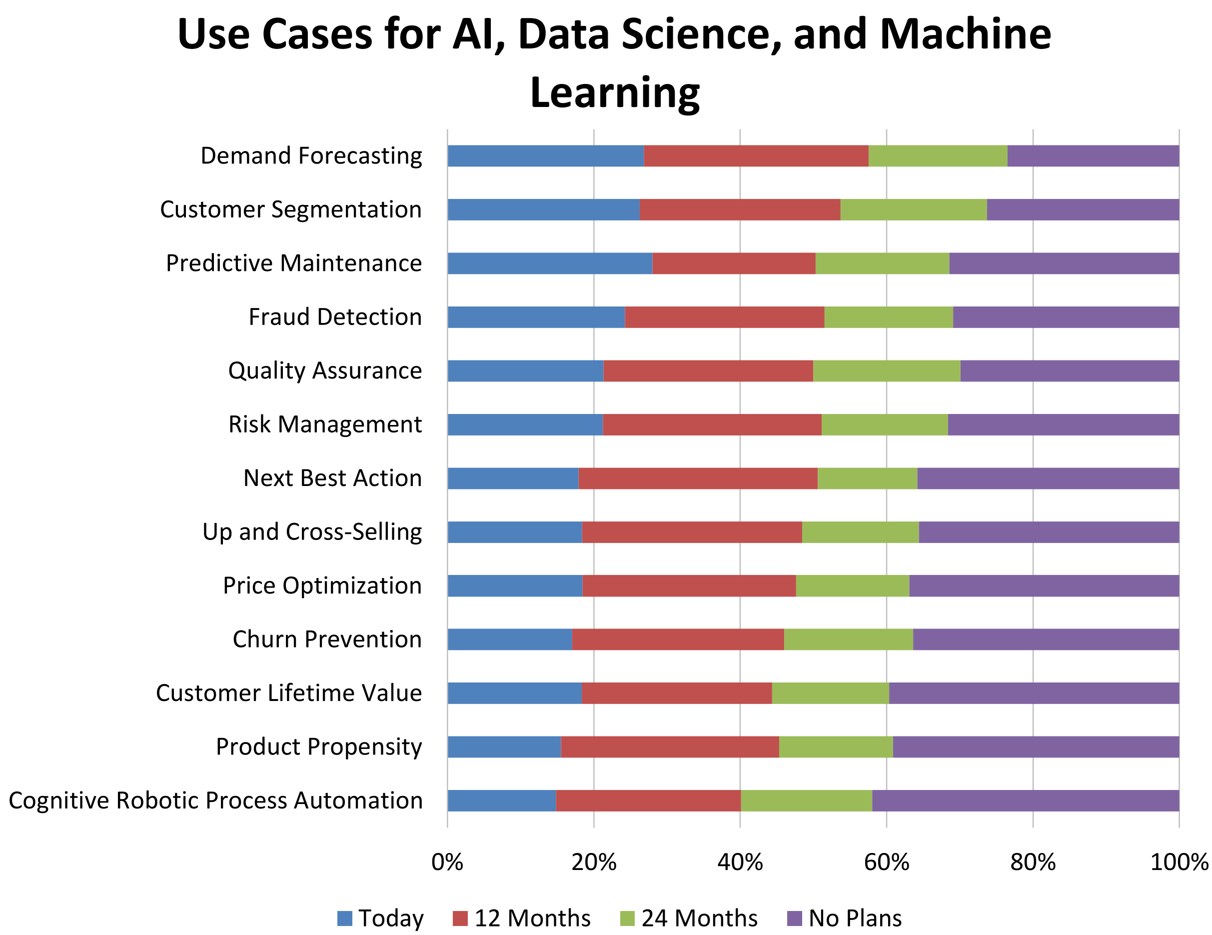


Figure 58 – Use cases for AI, data science, and machine learning

Use Cases for AI, Data Science, and Machine Learning by Geography

(Note: Figs. 59-67 all sample user response choices measured on a five-point scale by "currently use," "12 month," "24 month," "no plans," and "don't know.")

We observe regional preferences regarding the current and future use of AI, data science, and machine learning in 2025 (fig. 59). In all but two cases, Asia Pacific leads in use case importance, particularly in the areas of customer segmentation and predictive maintenance, while EMEA leads interest in the remaining two, fraud detection and risk management. Excluding Asia Pacific, EMEA respondents dominate interest in most categories, though North America respondents narrowly lead interest in demand forecasting and quality assurance. Latin America respondents most prioritize demand forecasting and customer segmentation. While this representation is interesting, we consider it a very early snapshot of regional use.

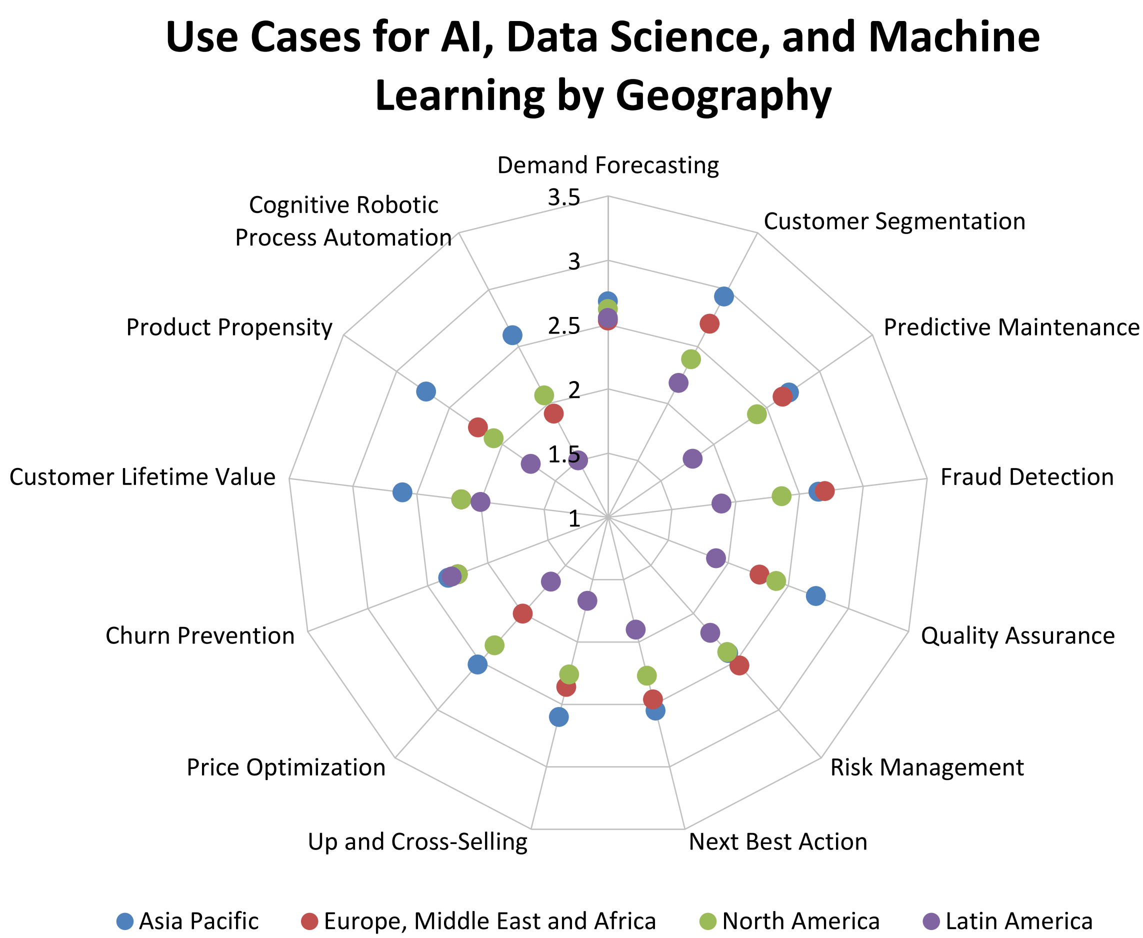


Figure 59 – Use cases for AI, data science, and machine learning by geography

Use Cases for AI, Data Science, and Machine Learning by Function

Current use and future uptake of use cases for AI, data science, and machine learning vary interestingly by function in 2025 and are plainly highest overall in sales & marketing, which leads interest in seven of 13 use cases (fig. 60). Standout areas of sales & marketing focus include customer segmentation, fraud detection, quality assurance, risk management up and cross-selling, and cognitive robotic process automation. This year, R&D posts the highest or equally highest scores for predictive maintenance and next best action. And operations posts the highest scores in four areas: price optimization, churn prevention, customer lifetime value, and product propensity. Interestingly, BICC and executives are among the lower-scoring respondents, which may indicate bundled or preconfigured use cases more in demand among other functions.

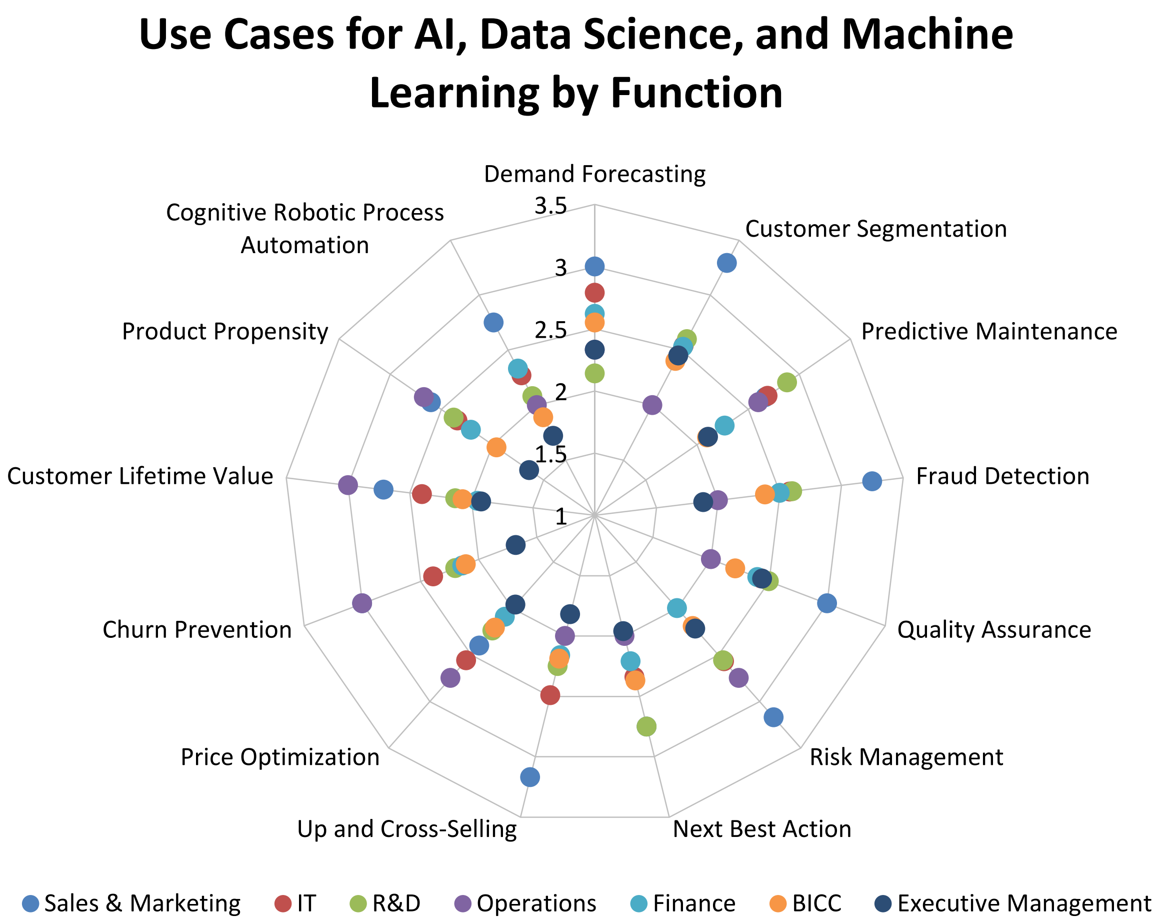


Figure 60 – Use cases for AI, data science, and machine learning by function

Use Cases for AI, Data Science, and Machine Learning by Industry

2025 current use and future uptake of use cases for AI, data science, and machine learning are highest overall in financial services and also includes notable interest among technology and healthcare respondents (fig. 61). This year, financial services organizations report the highest or near-highest current/future uptake in areas including demand forecasting, risk management, up- and cross-selling, price optimization, and churn prevention. Technology respondents report equally high or the highest scores for customer segmentation, predictive maintenance, next-best offer, churn prevention, customer lifetime value, and product propensity. And healthcare respondents report the highest scores for fraud detection, quality assurance, risk management, and customer lifetime values. All of these are among many other notable but early-stage observations of industry uptake of AI, data science, and machine learning.

Use Cases for AI, Data Science, and Machine Learning by Function

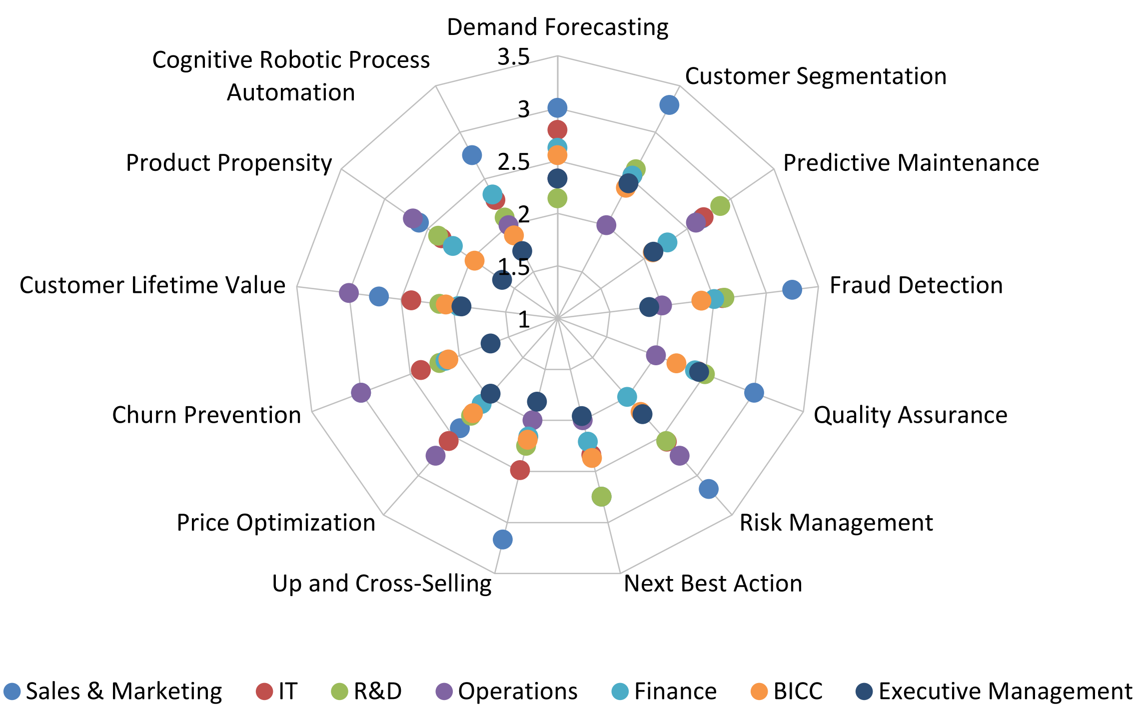


Figure 61 – Use cases for AI, data science, and machine learning by function

Use Cases for AI, Data Science, and Machine Learning by Organization Size

Interest in use cases for AI, data science, and machine learning in 2025 is dominantly a very large organization (more than 10,000 employees) phenomenon (fig. 62). This year, very large organizations distantly lead uptake in all use cases, with the very highest scores (more than 3.0) going to demand forecasting, predictive maintenance, risk management, up- and cross-selling, product propensity, and cognitive robotic process automation. Excluding very large organizations, large organizations almost always lead in interest, though some use cases sort unevenly according to size. Examples include customer segmentation, where midsize (101-1,000 employees) organizations lead all but very large peers, and quality assurance, where small (1-100 employees) organizations are second-most interested overall.

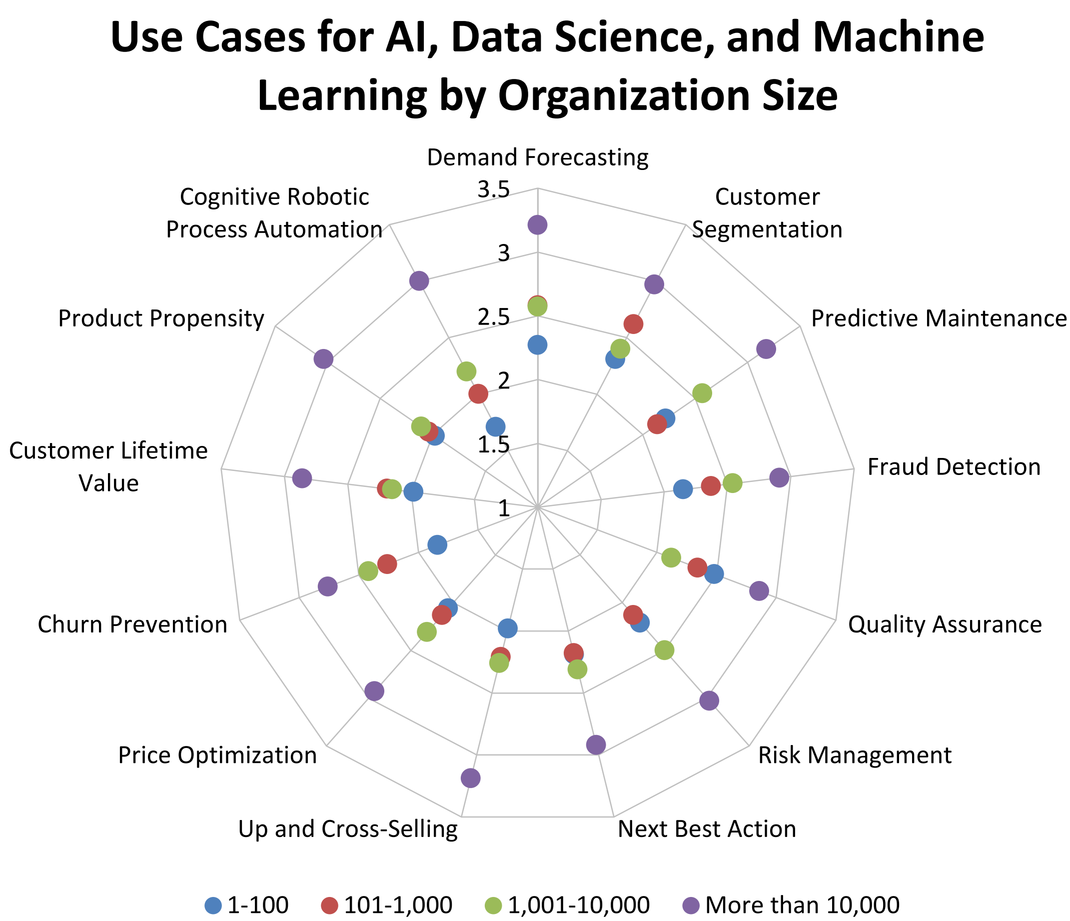


Figure 62 – Use cases for AI, data science, and machine learning by organization size

Use Cases for AI, Data Science, and Machine Learning by Company Age

Uptake of use cases for AI, data science, and machine learning varies unevenly between middle-aged and younger companies, and more rarely by the oldest (fig. 63). Most noticeably in 2025, "youngish" companies of five to 10 years post the highest overall use case scores for AI, data science, and machine learning, top scores for seven of 13 use cases, and standout high scores for customer segmentation, predictive maintenance, and quality assurance. The youngest companies of five years or less report the highest scores for next-best action, price optimization, and product propensity—use cases that might be optimized for B2B or direct to consumer. Interest thereafter reverts to back-office or high-level use cases. Companies of 11-16 years or more than 16 years post respectably high scores across the board, but top or near-top scores only in areas like demand forecasting, risk management, and separately, cognitive robotic process automation.

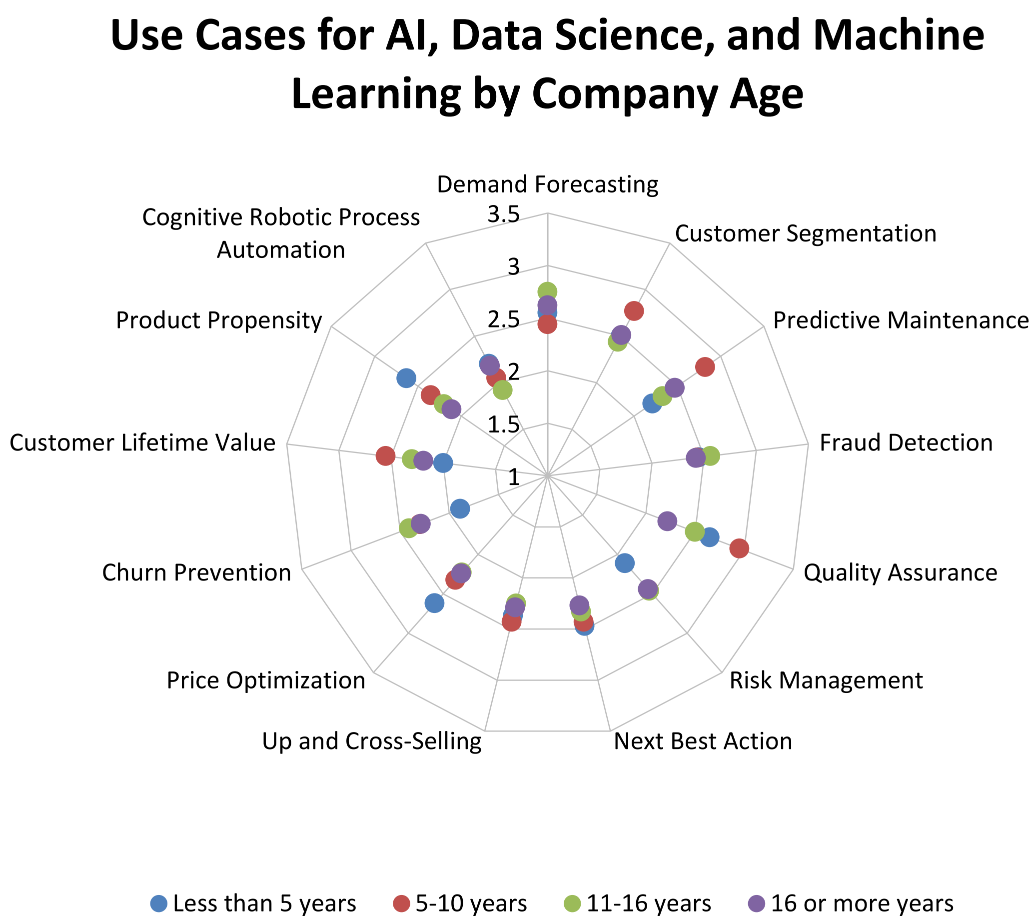


Figure 63 – Use cases for AI, data science, and machine learning by company age

Use Cases for AI, Data Science, and Machine Learning by Success with BI

Unlike some other measures, we observe a strong correlation between interest in use cases for AI, data science, and machine learning, and perceived success with business intelligence (fig. 64). Given the limited experience and a likely shortage of measured outcomes associated with the technology, we might conclude this finding largely reflects higher attention to and anticipation of use cases in companies that consider themselves successful BI organizations. Even so, we see a consistent and powerful distinction between completely successful, somewhat successful, and somewhat unsuccessful organizations.

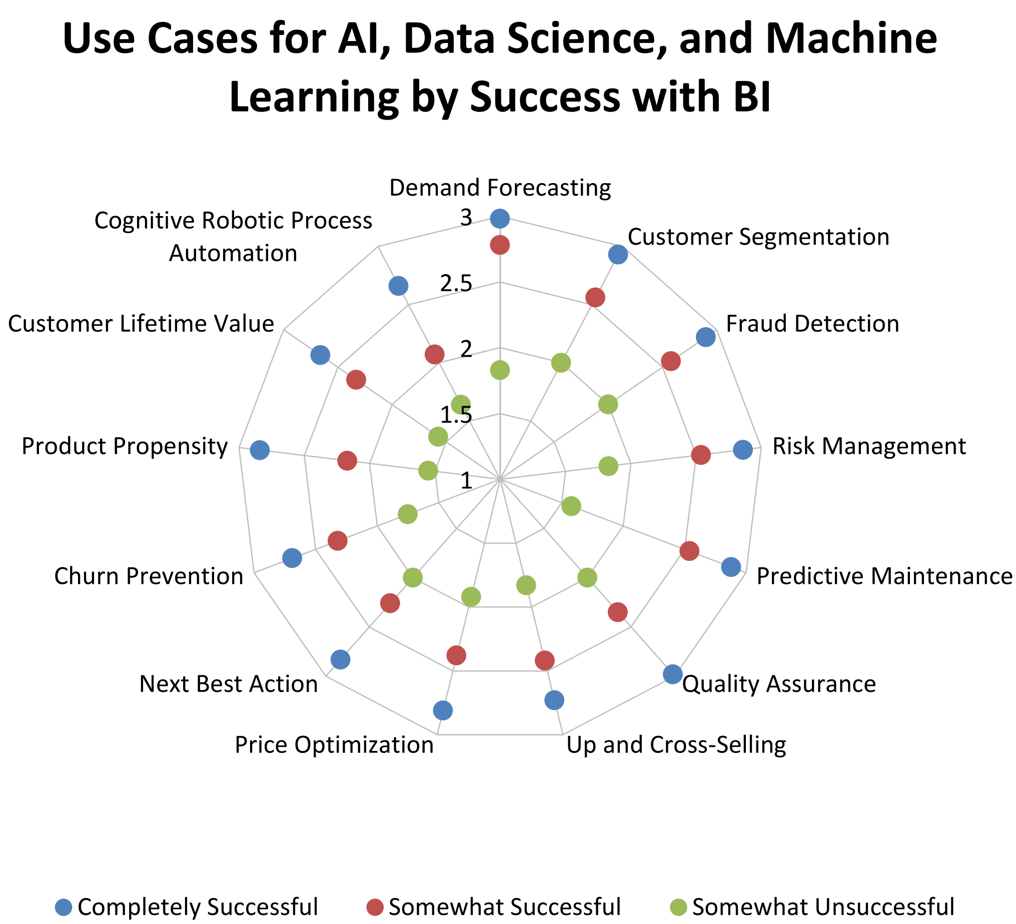


Figure 64 - Use cases for AI, data science, and machine learning by success with BI

Use Cases for AI, Data Science, and Machine Learning by Current Strategic Role of AI

Prioritization of the use cases for AI, data science, and machine learning correlates strongly with perceptions of the strategic role of AI (fig. 65). Organizations that view AI as a cornerstone most often emphasize all use cases; those that view AI as currently having a minimal strategic role least often stress all use cases; and those that ascribe important or exploratory roles to strategic use of AI fill the gap in between—with all data showing a clear linear relationship. Demand forecasting is the use case with the highest prioritization overall, receiving top emphasis from those who consider strategic use of AI important or exploratory, and showing the closest clustering of perceptions among the top three views of AI's strategic role.

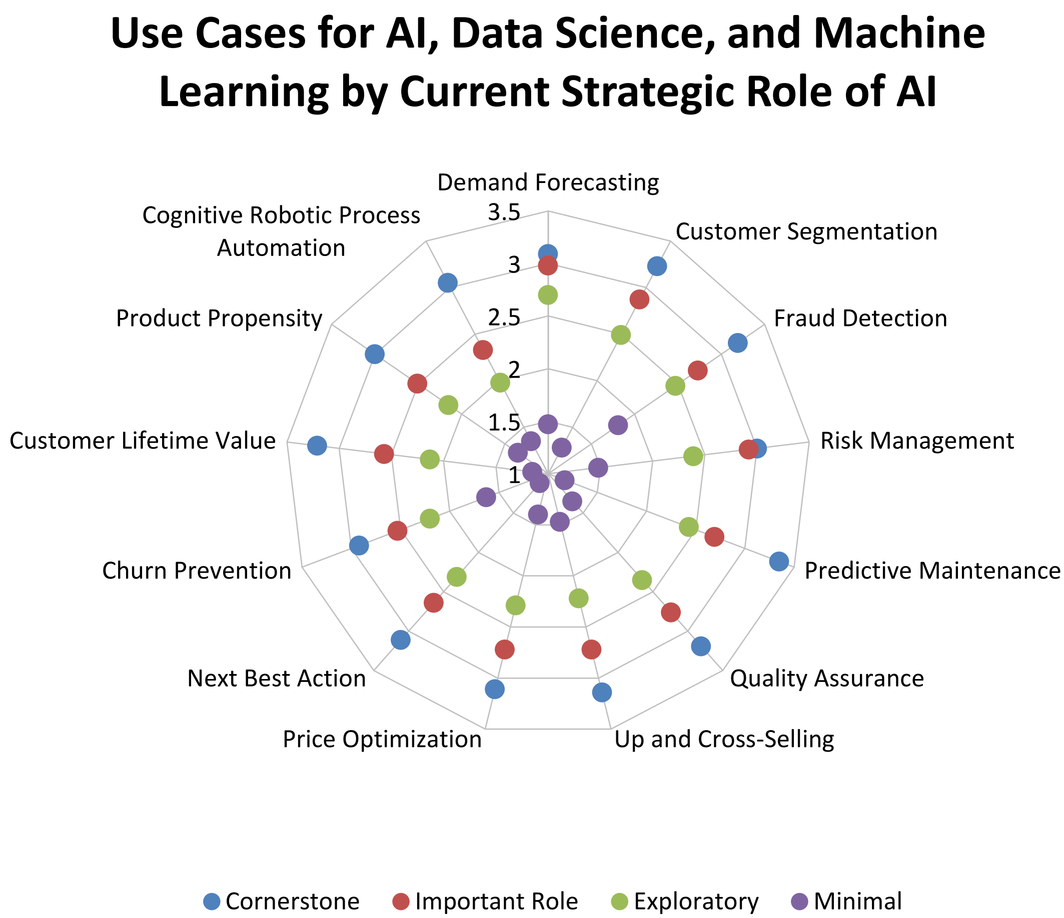


Figure 65 - Use cases for AI, data science, and machine learning by current strategic role of AI

Use Cases for AI, Data Science, and Machine Learning by Generative AI Adoption

Organizations that are more actively using or considering the use of generative AI correlate strongly to prioritization of any and all AI, data science, and machine learning use cases in 2025 (fig. 66). This year, organizations with generative AI in production report the highest priority for all use cases, led by demand forecasting, customer segmentation, and fraud detection, all of which rise above the level of "important." Organizations that are experimenting today are nearly almost always next-most likely to prioritize all use cases for AI, data science, and machine learning. Areas most likely to be prioritized in the next 12 months are the top two use cases, demand forecasting and customer segmentation.

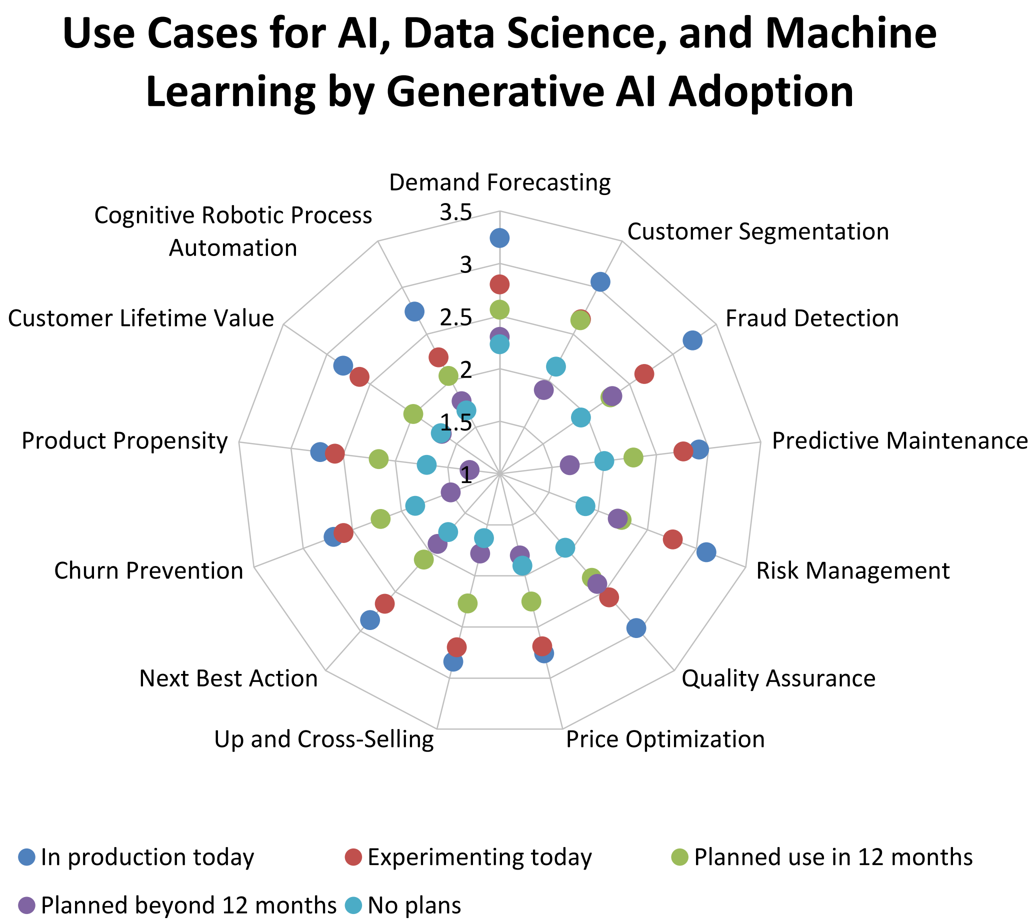


Figure 66 – Use cases for AI, data science, and machine learning by generative AI adoption

Use Cases for AI, Data Science, and Machine Learning by Agentic AI Adoption

Organizations actively using agentic AI most often prioritize all AI, data science, and machine learning use cases (fig. 67). Respondents experimenting with agentic AI—a phase that is often a precursor to nearer-term production deployment—prioritize all use cases at the second-highest rate, with a strong concentration of nearly identical and most frequent emphasis on demand forecasting, fraud detection, customer segmentation, risk management, quality assurance, and predictive maintenance (with each rating varying by less than 2%). Although all data often shows a high tendency to increase use-case prioritization as agentic AI deployment becomes more "real," responses by those with planned agentic AI use (as early as the next 12 months) and those with no usage plans show much more clustering of perceptions in general. Notably, those with no plans would most often prioritize the demand forecasting use case—at least 7% more prioritization than any other use case, and at almost the same level of perception as organizations planning to use agentic AI in the next year.

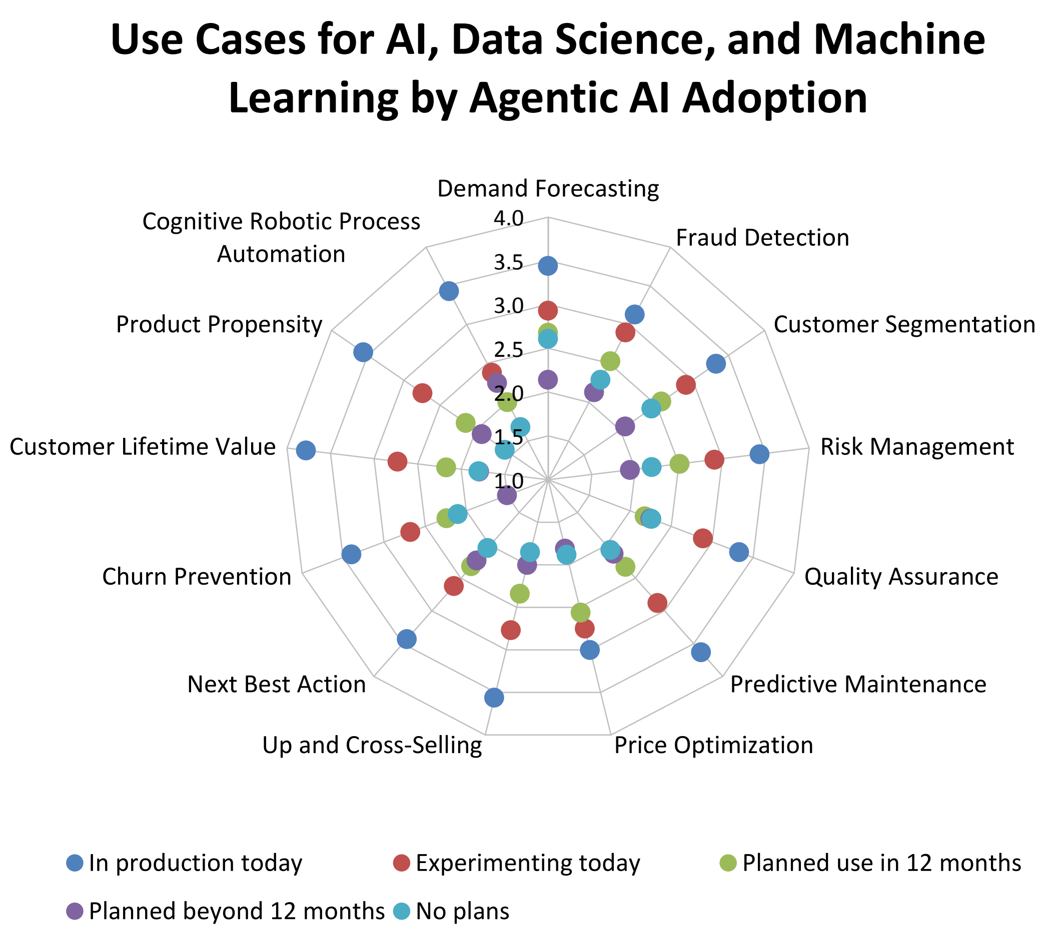


Figure 67 - Use cases for AI, data science and machine learning by agentic AI adoption

Deployment and Adoption Plans for AI, Data Science, and Machine Learning

Deployment of AI, Data Science, and Machine Learning 2016-2025

Across the last 10 years of our study, we have observed consideration of and actual deployment of AI, data science, and machine learning gain, flatten or decrease slightly, and begin to very mildly rebound in 2025 (fig. 68). (Note: Beginning in 2022, we divided “yes, we use today” into “in production” and “in very limited ways,” and combined the weighted mean.) From a 2017 weighted-mean low of 3.0 to a 2021 high of 3.6, 2025 sentiment holds at a rounded level of 3.3, the same as in 2023 and again in 2024. Despite this flattening, sentiment remains above the level of “important” for the last eight years. Those with no plans stand at 15% in 2025 compared to an all-time low of 9% in 2022, possibly indicating some hesitancy or mixed success amid market events. Even so, we expect ongoing momentum in the AI, data science, and machine learning space to bode well for wider adoption as more use cases and solutions become apparent (also see perceived importance, fig. 50).

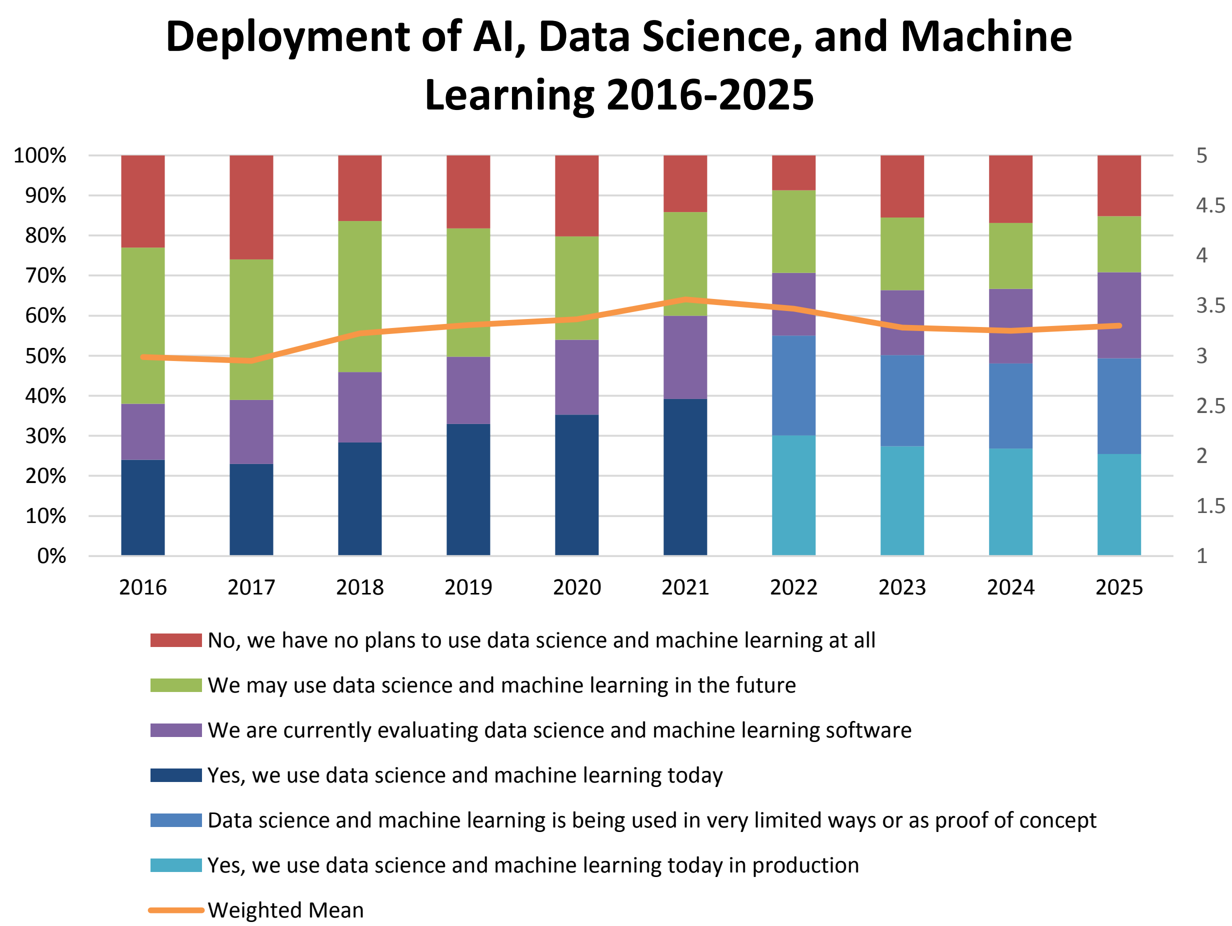


Figure 68 – Deployment of AI, data science, and machine learning 2016-2025

Deployment of AI, Data Science, and Machine Learning by Geography

Deployment rates of AI, data science, and machine learning vary by geography in 2025, with both current and weighted-mean use visibly highest in Asia Pacific, followed by North America, Latin America, and EMEA (fig. 69). The rate of current in-production use is highest in Asia Pacific (36%), followed by North America and EMEA (both at 24%) and finally, Latin America (13%). All regions are committed to future use, with combined current users and evaluators above 70% in Asia Pacific, North America, and EMEA.

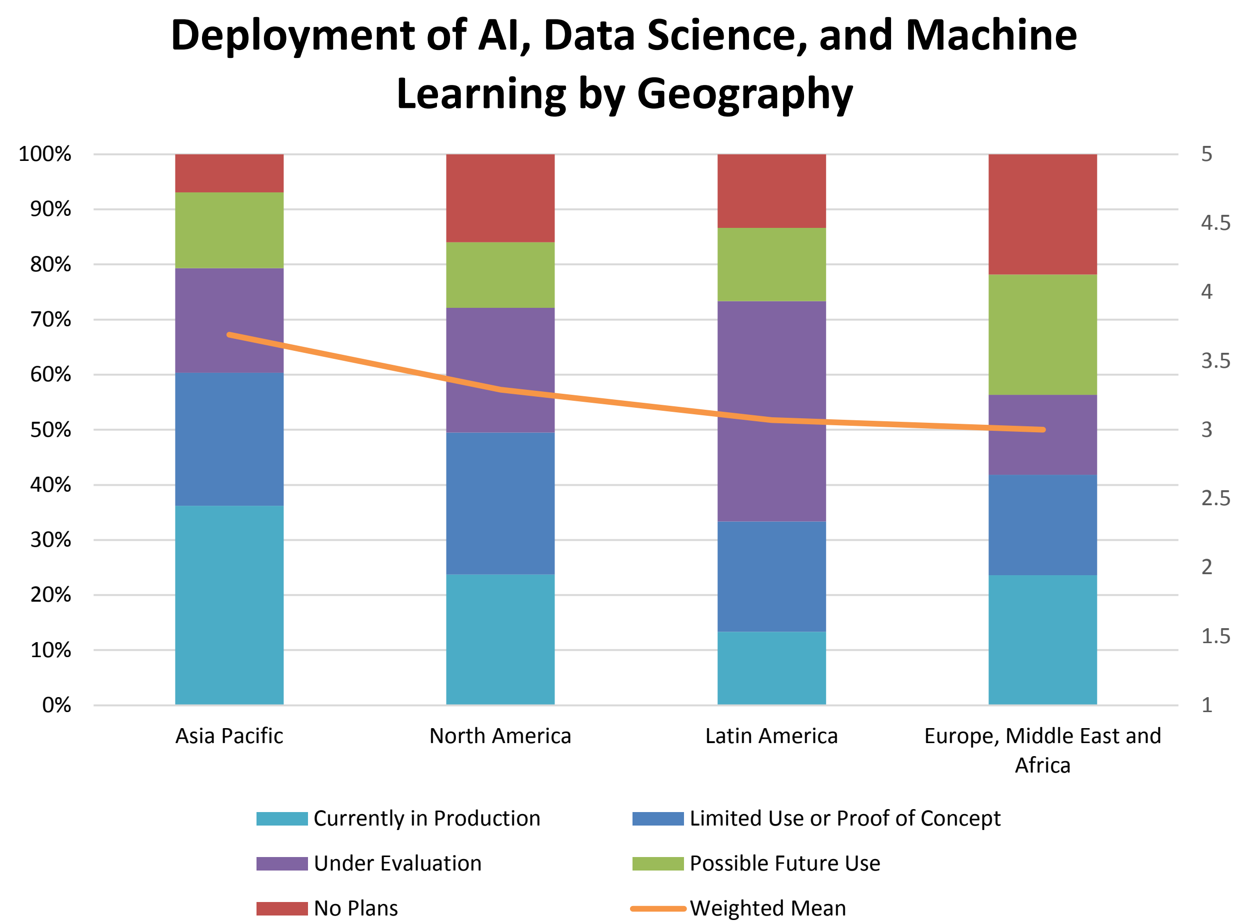


Figure 69 – Deployment of AI, data science, and machine learning by geography

Deployment of AI, Data Science, and Machine Learning by Function

Current deployment of AI, data science, and machine learning varies broadly according to function in a weighted-mean range of 2.9 (near important) in finance to 3.9 in R&D (near very important; see fig. 70). Current production and limited use in R&D stands at about 63%, followed by about 60% in operations. Current and limited use then falls to 50% in the BICC and IT, falling again to 42%-43% among executives and in sales & marketing. As is often the case, finance respondents are least likely to report deployment of AI, data science, and machine learning.

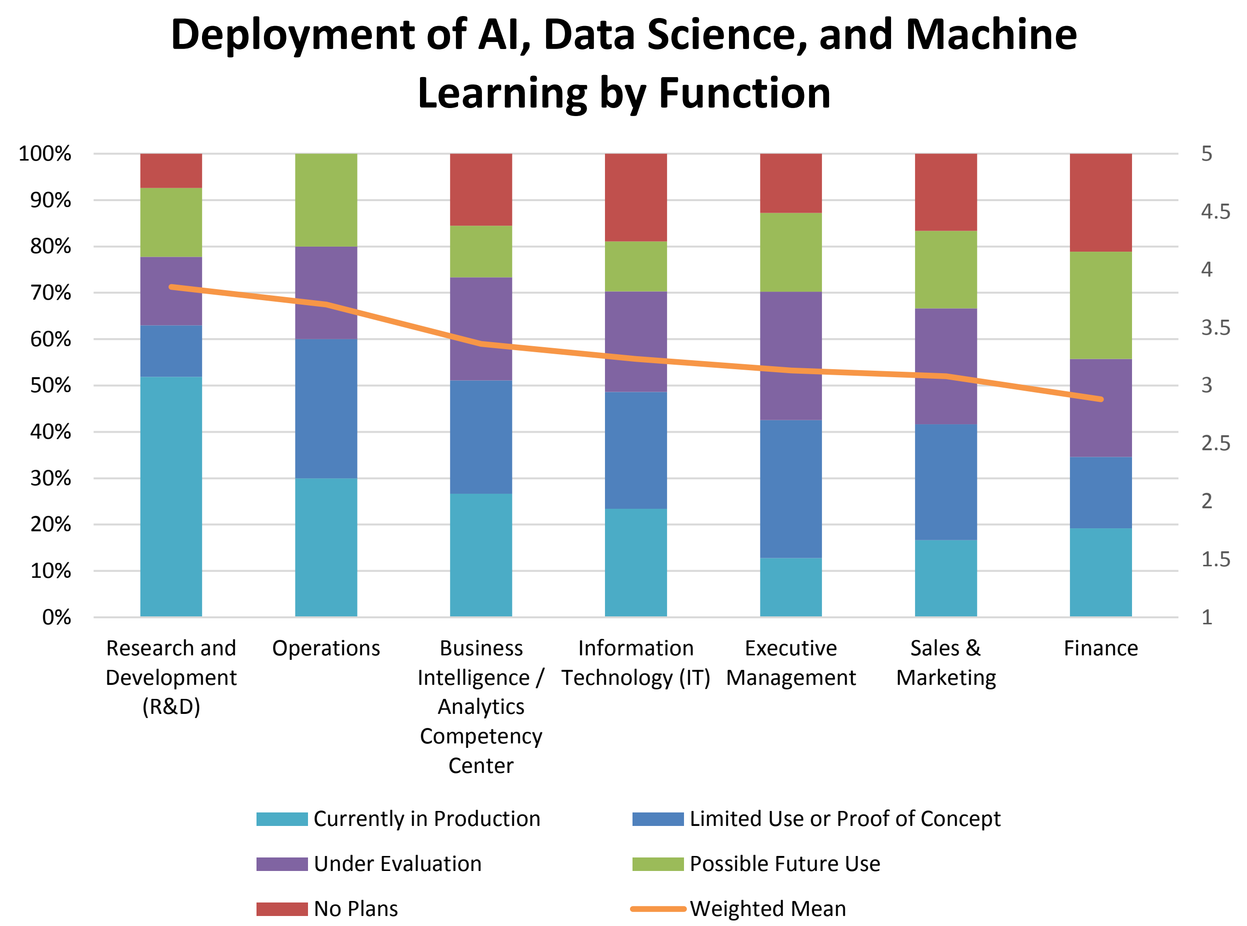


Figure 70 – Deployment of AI, data science, and machine learning by function

Deployment of AI, Data Science, and Machine Learning by Industry

Viewed by industry, respondents in technology report the highest weighted-mean deployment in 2025, while business services reports the lowest (2.9; fig. 71). The highest current in-production use is in technology (34%), followed by manufacturing (31%), consumer services (30%), and retail & wholesale (29%). Current in-production, limited, and evaluation use is above 60% in all industries sampled, and highest in healthcare, where current in-production use is relatively low (21%).

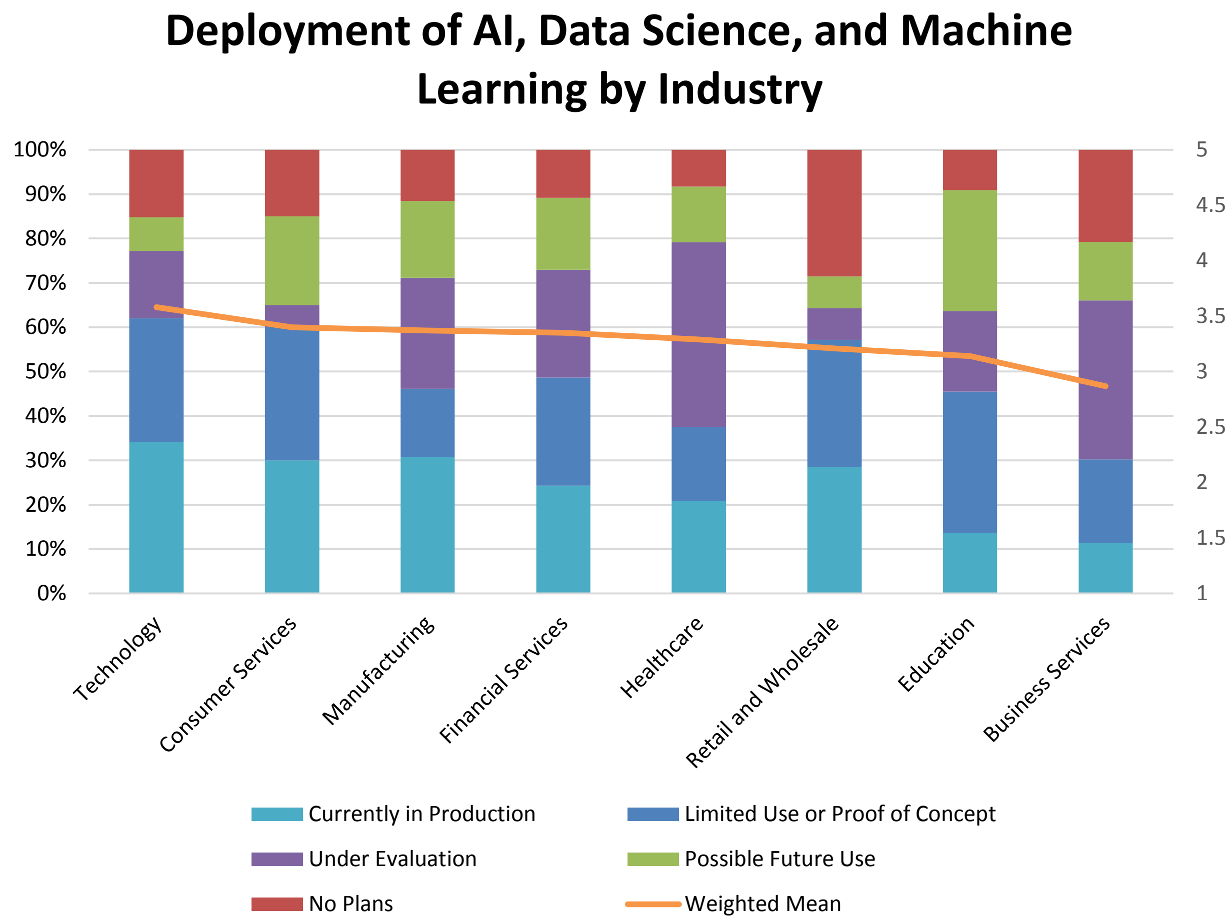


Figure 71 – Deployment of AI, data science, and machine learning by industry

Deployment of AI, Data Science, and Machine Learning by Organization Size

The number of deployments of AI, data science, and machine learning clearly increases with global headcount in 2025 and is by far highest in very large organizations (more than 10,000 employees; fig. 72). This year, 36% of small organizations (1-100 employees) report in-production deployments, compared to 17% at midsize (101-1,000 employees), 24% at large (1,001-10,000 employees), and 42% at very large organizations. Combined in-production, limited use, and evaluation responses boost very large organization participation to 85%. About 80%-81% of small and midsize organizations, 84% of large, and 85% of very large organizations say at minimum that they may use AI, data science, and machine learning in the future.

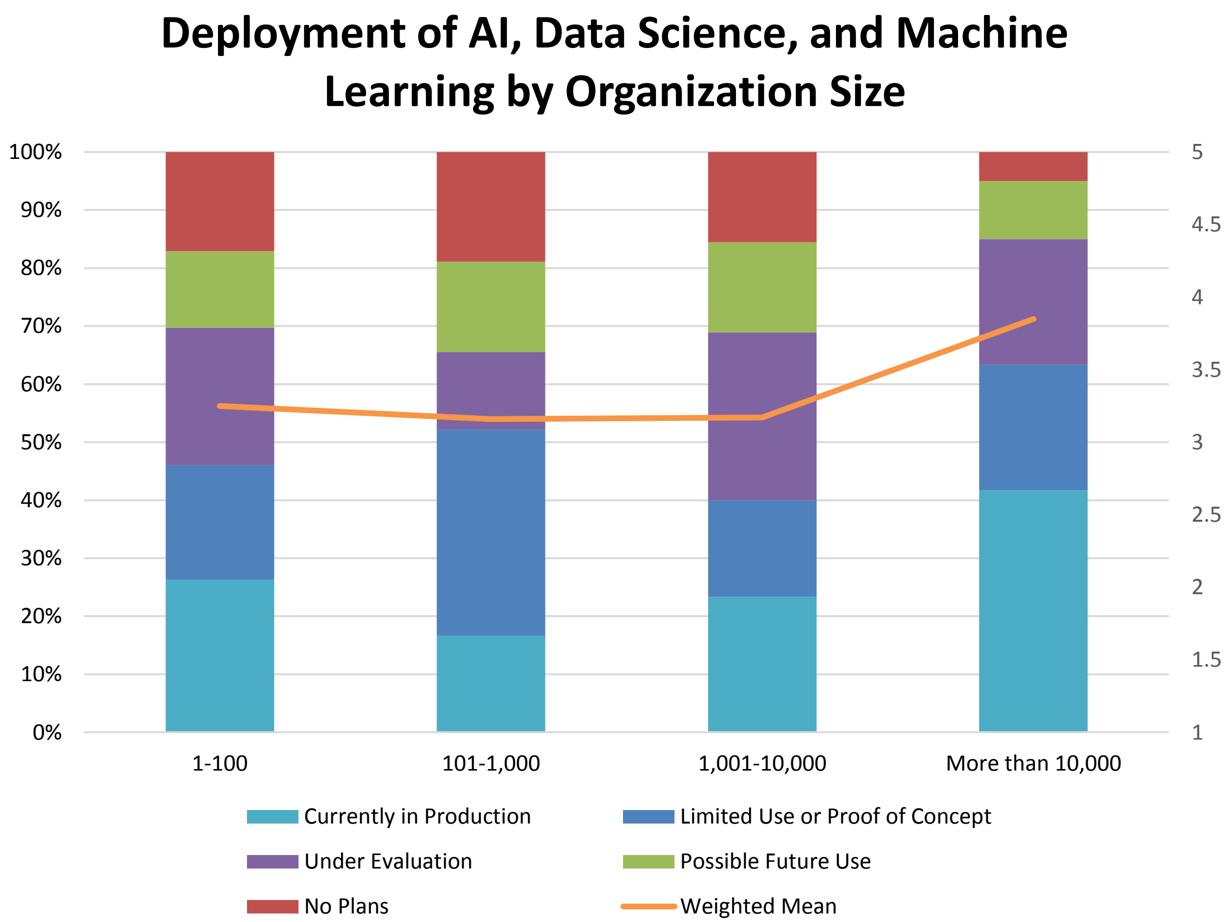


Figure 72 – Deployment of AI, data science, and machine learning by organization size

Deployment of AI, Data Science, and Machine Learning by Success with BI

Deployment of AI, data science, and machine learning correlates to success with BI in 2025 (fig. 73). Thirty-six percent of organizations that say they are completely successful with business intelligence report in-production use of AI, data science, and machine learning, compared to 24% of somewhat successful,10% of somewhat unsuccessful, and zero percent of unsuccessful organizations. Conversely, organizations that are somewhat unsuccessful and unsuccessful with BI are far more likely to say they have no plans (28%-30%).

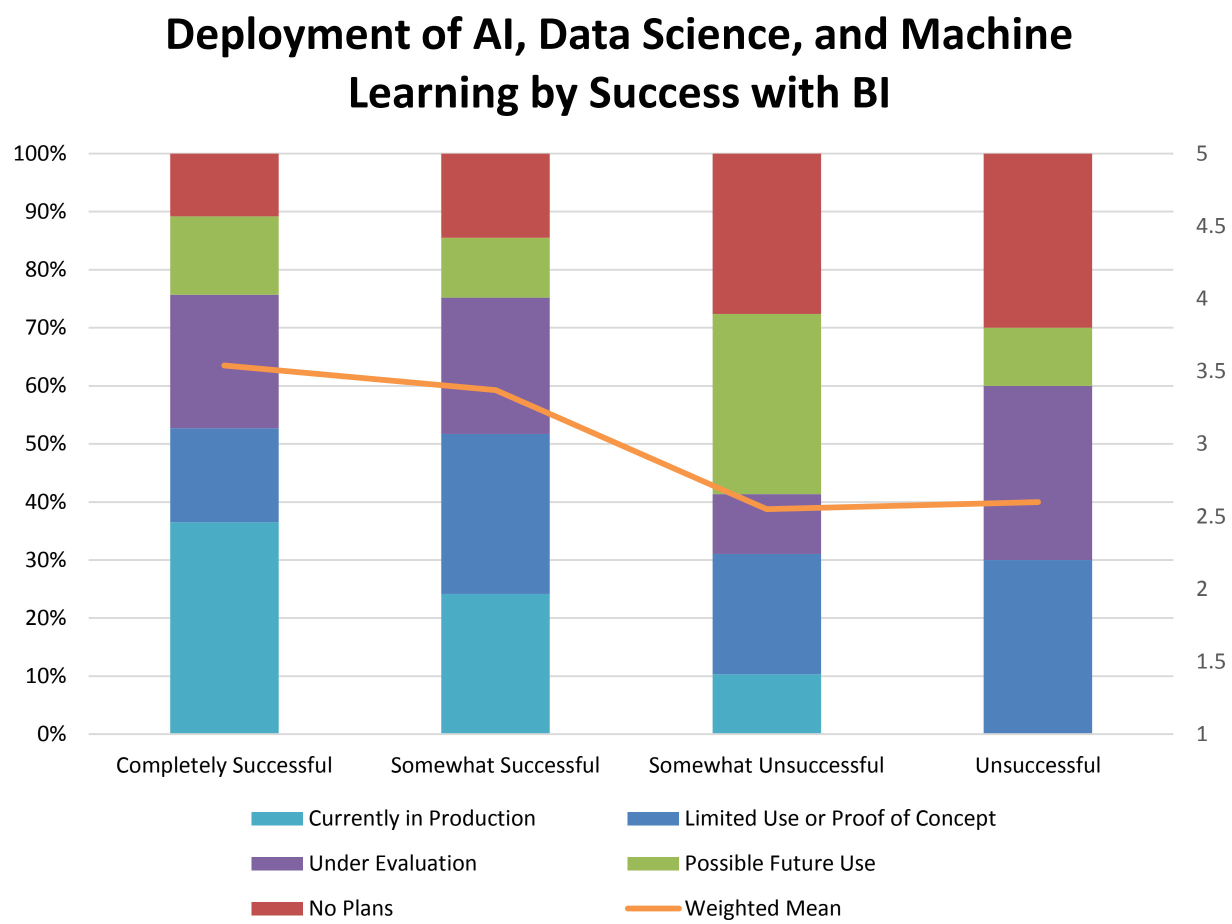


Figure 73 – Deployment of AI, data science, and machine learning by success with BI

Deployment of AI, Data Science, and Machine Learning by Difficulty Finding Analytic Content

The perceived level of adoption of AI, data science, and machine learning clearly correlates positively with the likelihood of difficulty finding analytic content in 2025 (fig. 74). Sixty-eight percent of organizations that say it is difficult to find analytic content have AI, data science, and machine learning either in production or in limited use, compared to 46% who say it is somewhat difficult, 44% who say it is relatively easy, and 13% that say it is extremely easy. The implication is that organizations that struggle to find analytic content are more likely to turn to AI, data science, and machine learning to address their difficulty. An interesting counterpoint is that 60% of organizations that find it extremely easy to find analytic content see no present or future use for AI, data science, and machine learning.

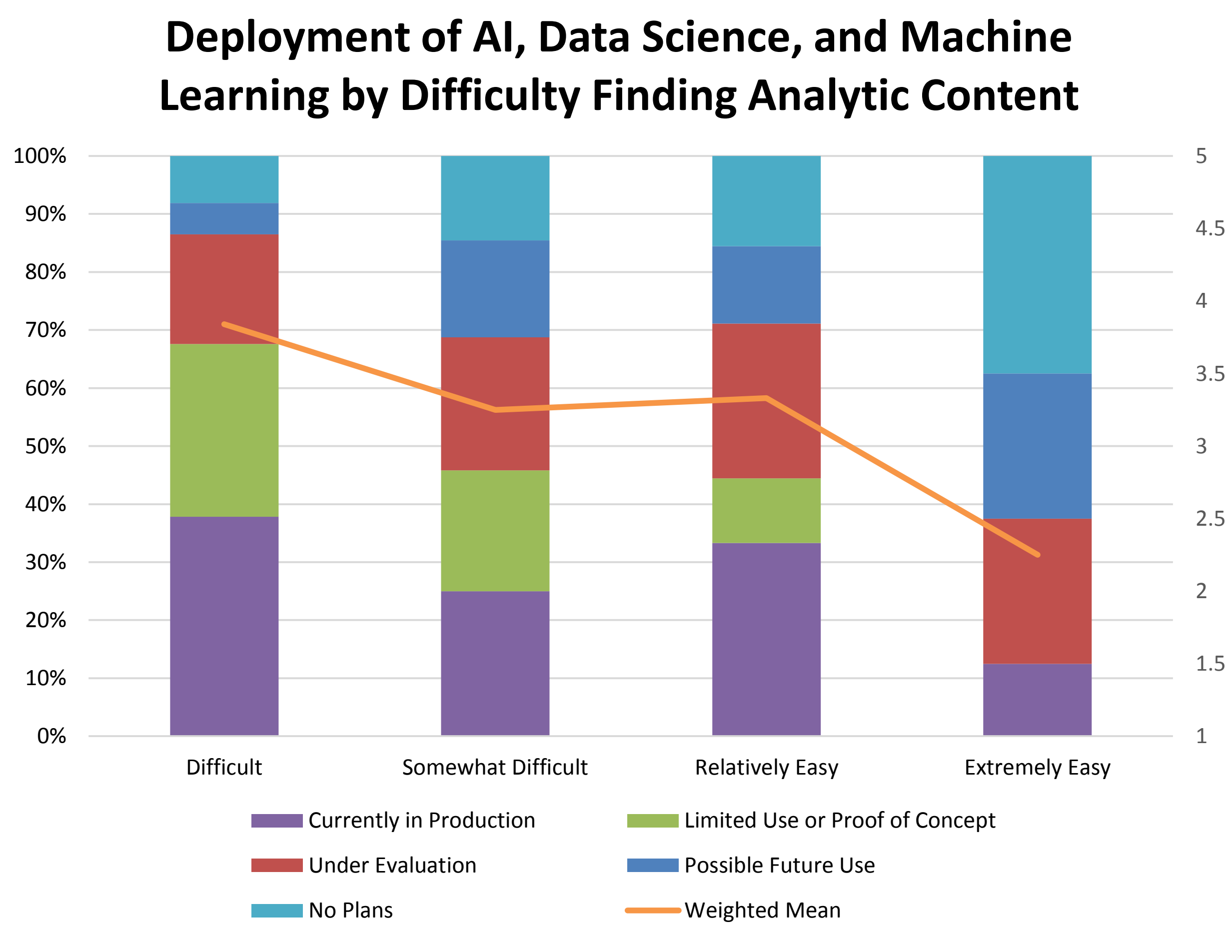


Figure 74 – Deployment of AI, data science, and machine learning by difficulty finding analytic content

Deployment of AI, Data Science, and Machine Learning by Current Strategic Role of AI

AI, data science, and machine learning deployment shows a strong correlation with strategic perceptions of AI (fig. 75). Organizations that consider AI's strategic role a cornerstone most frequently report production or limited use of AI, data science, and machine learning (90%). Frequency of response for these same two usage levels drops linearly as AI's strategic role decreases, from important (63%) to exploratory (41%) to minimal (8%). Across all levels of use except for having no plans, a similar linear relationship exists (albeit with less differences between levels of strategic perception).

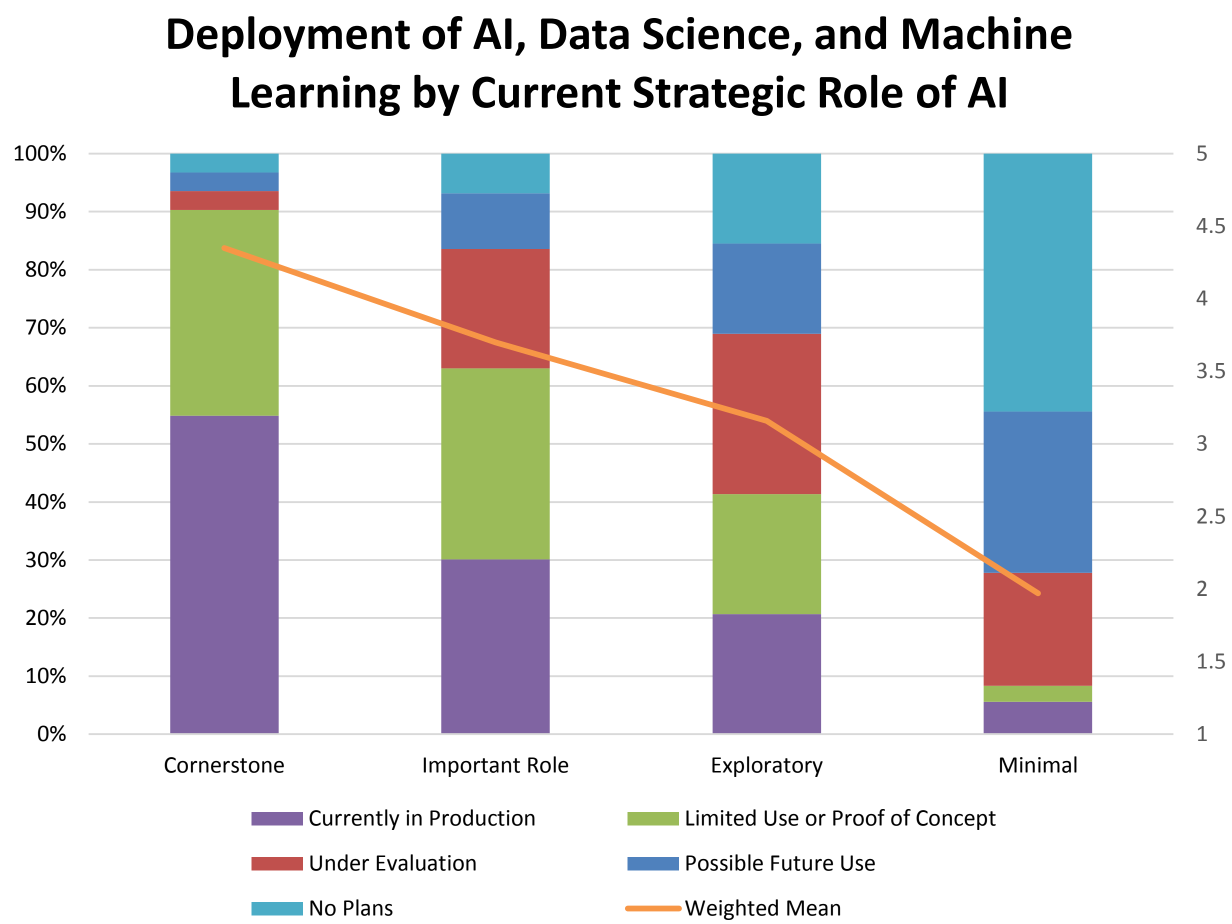


Figure 75 - Deployment of AI, data science and machine learning by current strategic role of AI

Deployment of AI, Data Science, and Machine Learning by Generative AI Adoption

AI, data science, and machine learning deployment also shows a strong correlation with generative AI adoption (fig. 76). Organizations with production use of or experimentation with generative AI most frequently report production or limited use of AI, data science, and machine learning (78% and 53%, respectively). A notable drop in response frequency occurs among those with planned use of generative AI (any time in the future) or no plans for its use. The data also shows little variation in these usage levels (a range of 32%-37%), indicating that these organizations believe that their AI, data science, and machine learning resources are best deployed to address functionality other than generative AI; that vendors may already be providing "good enough" generative AI capabilities in their solutions, so internal data science/machine learning teams do not need to focus on building generative AI capabilities; or both.

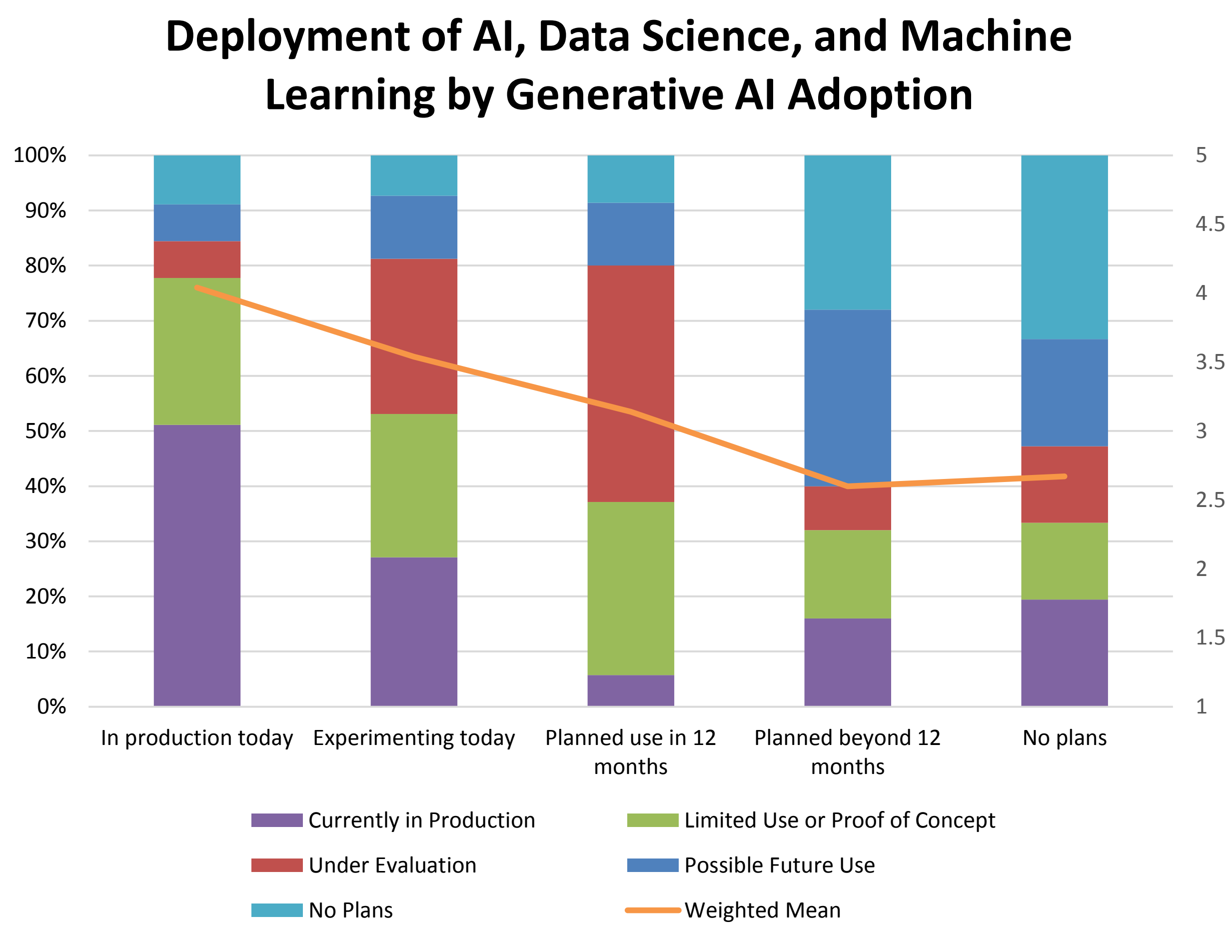


Figure 76 - Deployment of AI, data science, and machine learning by generative AI adoption

Deployment of AI, Data Science, and Machine Learning by Agentic AI Adoption

AI, data science, and machine learning deployment shows a similarly strong correlation with agentic AI adoption (fig. 77). Similar to the data on generative AI, organizations with production use of or experimentation with agentic AI most frequently report production or limited use of AI, data science, and machine learning (85% and 69%, respectively). However, those expecting to deploy agentic AI in the next year seem more committed to production or limited use of AI, data science, and machine learning (58%) than do their counterparts aiming to deploy generative AI (37%). Furthermore, respondents with no plans for production, or for limited or possible future use, or for evaluations more frequently report agentic AI activity than those in organizations that expect to use agentic AI at some point in the future. This difference could signify that many with future plans to use agentic AI at some indefinite time in the future also could be struggling to identify the right use cases against which to apply it, to determine the right approach for agentic AI for their organizations (build capabilities internally vs. deploy vendor-provided functionality in their solutions), or a combination of the two.

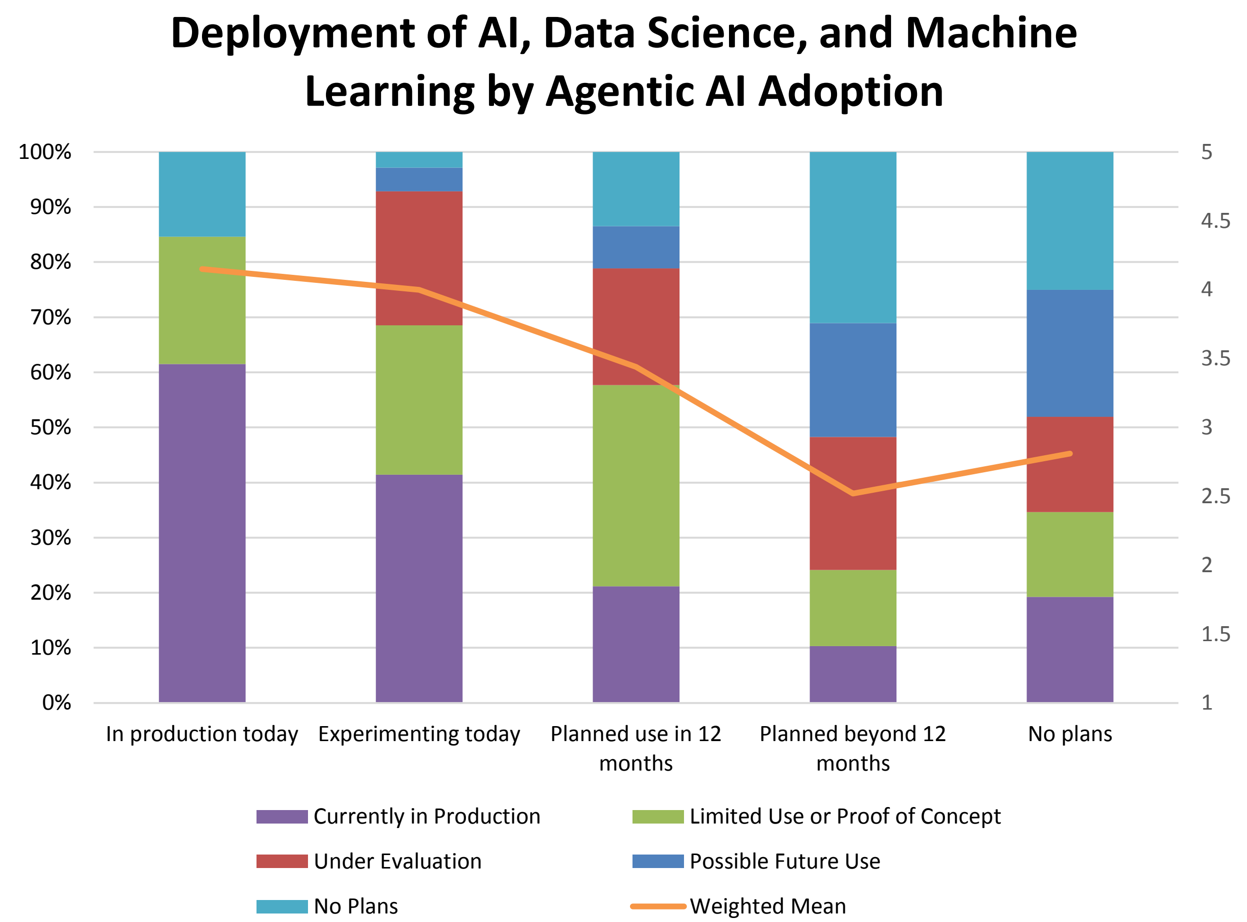


Figure 77 - Deployment of AI, data science, and machine learning by agentic AI adoption

Longevity of AI, Data Science, and Machine Learning

Longevity of AI, Data Science, and Machine Learning 2018-2025

We asked respondents, “How long has AI, data science, and machine learning been in use in your organization?” Viewed across eight years of survey data, longevity is sustained in a rising trend line that includes a varying mix of new adoption and ongoing use of AI, data science, and machine learning (fig. 78). Early use that led to an all-time high weighted mean in 2022 reversed somewhat in the following two years before resuming an upward trend line in 2025. Most notably, this year the percentage of the most mature AI, data science, and machine learning programs of more than five years is at a record level of 38%. Year over year, startups less than one year held steady at about 14%.

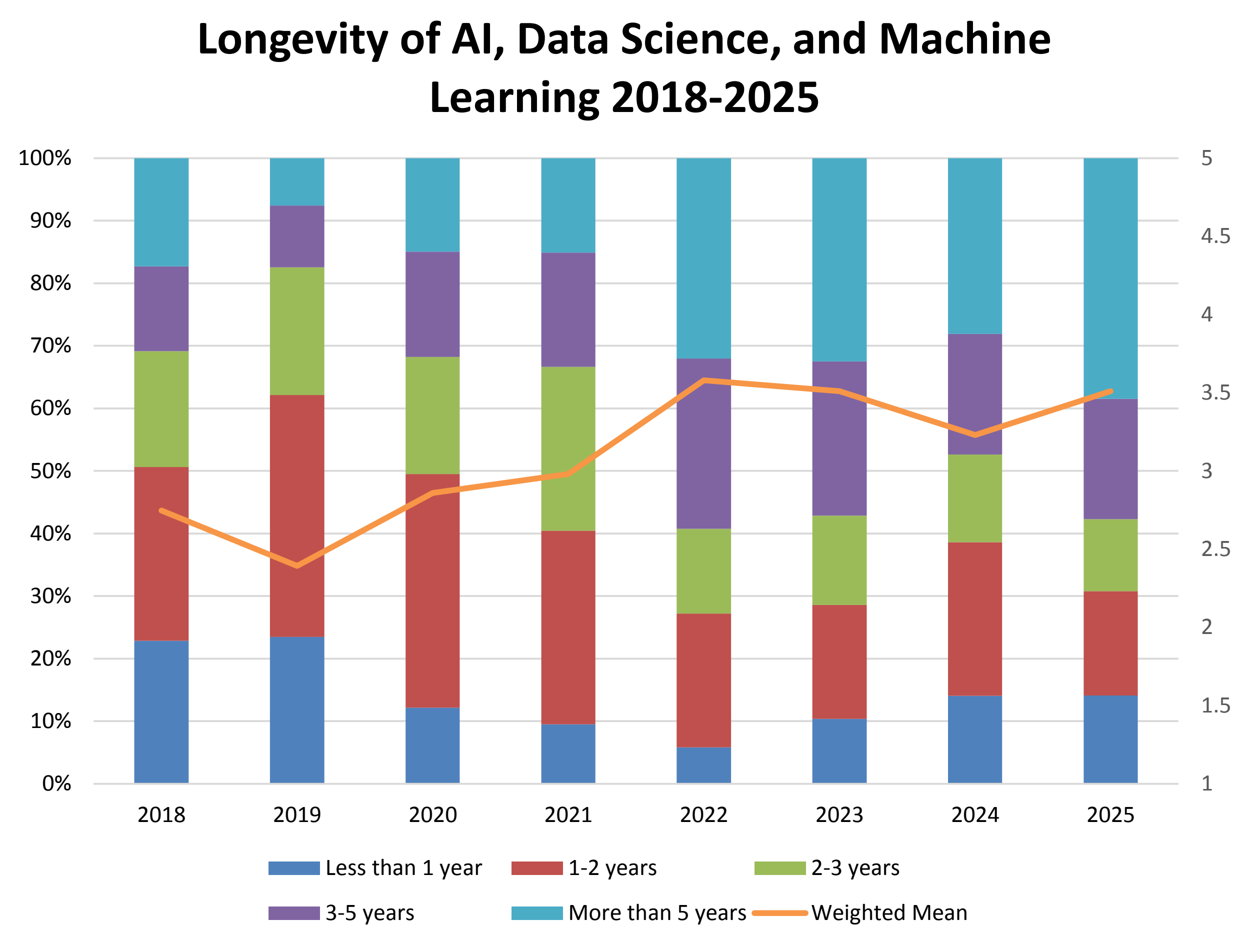


Figure 78 – Longevity of AI, data science, and machine learning 2018-2025

Longevity of AI, Data Science, and Machine Learning by Organization Size

The longevity of the use of AI, data science, and machine learning clearly increases with organization size in 2025 (fig. 79). This finding is unsurprising in the wake of very-large organization dominance of deployment and other measures. Nearly half (48%) of very-large organization involvement reports more than five years of experience, compared to 39% at large (1,001-10,000 employees), 38% at midsize (101-1,000 employees), and 30% at small (1-100 employees) peer organizations. We also observe the strongest catch-up rise in startup AI, data science, and machine learning activity of less than one year or one to two years in small organizations (45%), midsize (39%), and large (29%) peers.

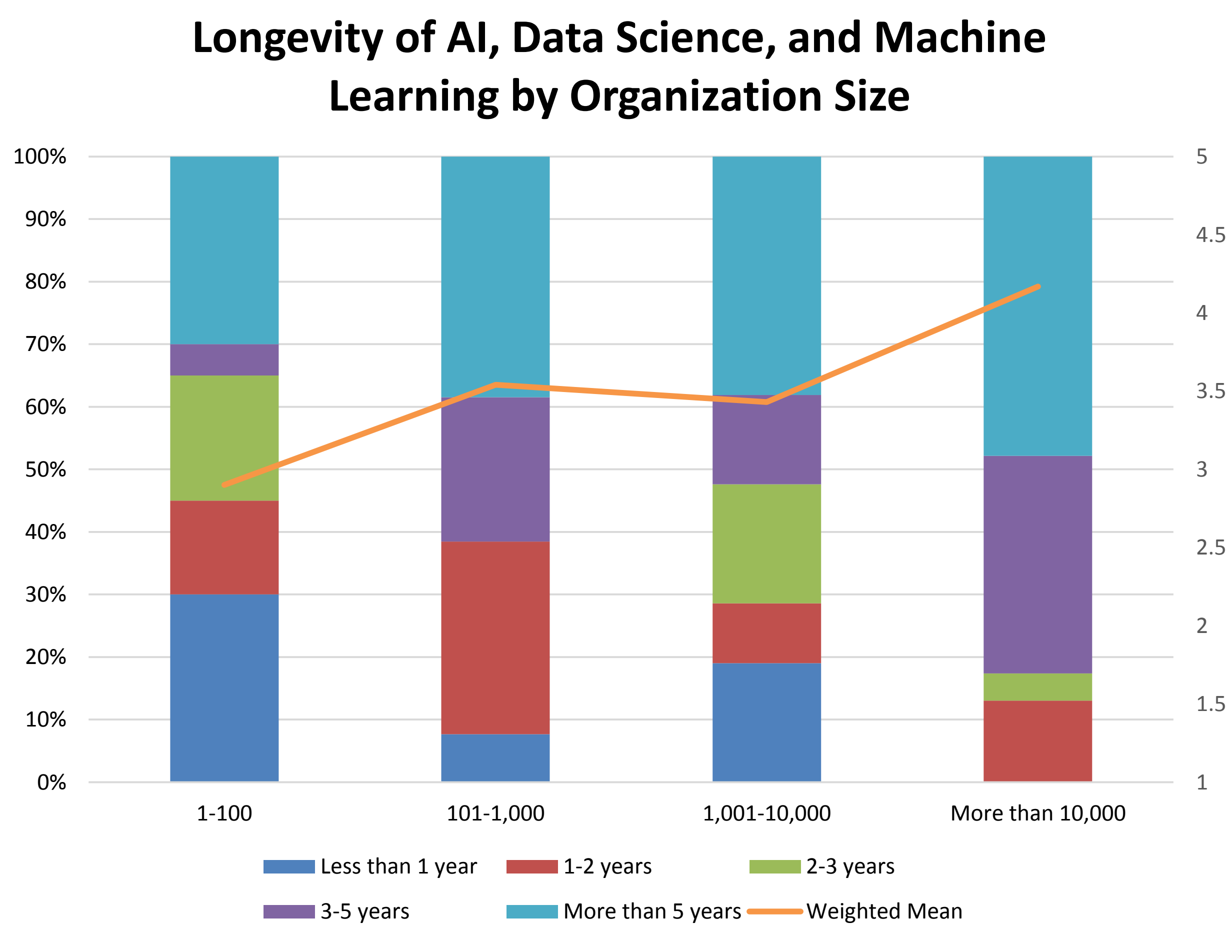


Figure 79 – Longevity of AI, data science, and machine learning by organization size

Longevity of AI, Data Science, and Machine Learning by Strategic Role of AI

Organizations with the longest-tenured AI, data science, and machine learning programs (in place for more than five years) most frequently consider AI as having an important strategic role (55%; fig. 80). When considering such programs tenured for more than three years, those that view AI as strategically important (70%) or worth exploring (61%) exceed the frequency of responses from those who see AI as a strategic cornerstone (59%). In addition, those with the most nascent programs (in place for less than a year) most often consider AI a strategic cornerstone (24%). Together, this data indicates that organizations with more experienced AI, data science, and machine learning programs are taking a slightly more conservative stance regarding the strategic role of AI in an organization, and that those in the least-experienced programs will tend to be most aggressive in applying AI strategically. Whether these organizations are "missing the bus" or "buying into the hype," respectively—or neither—remains to be seen.

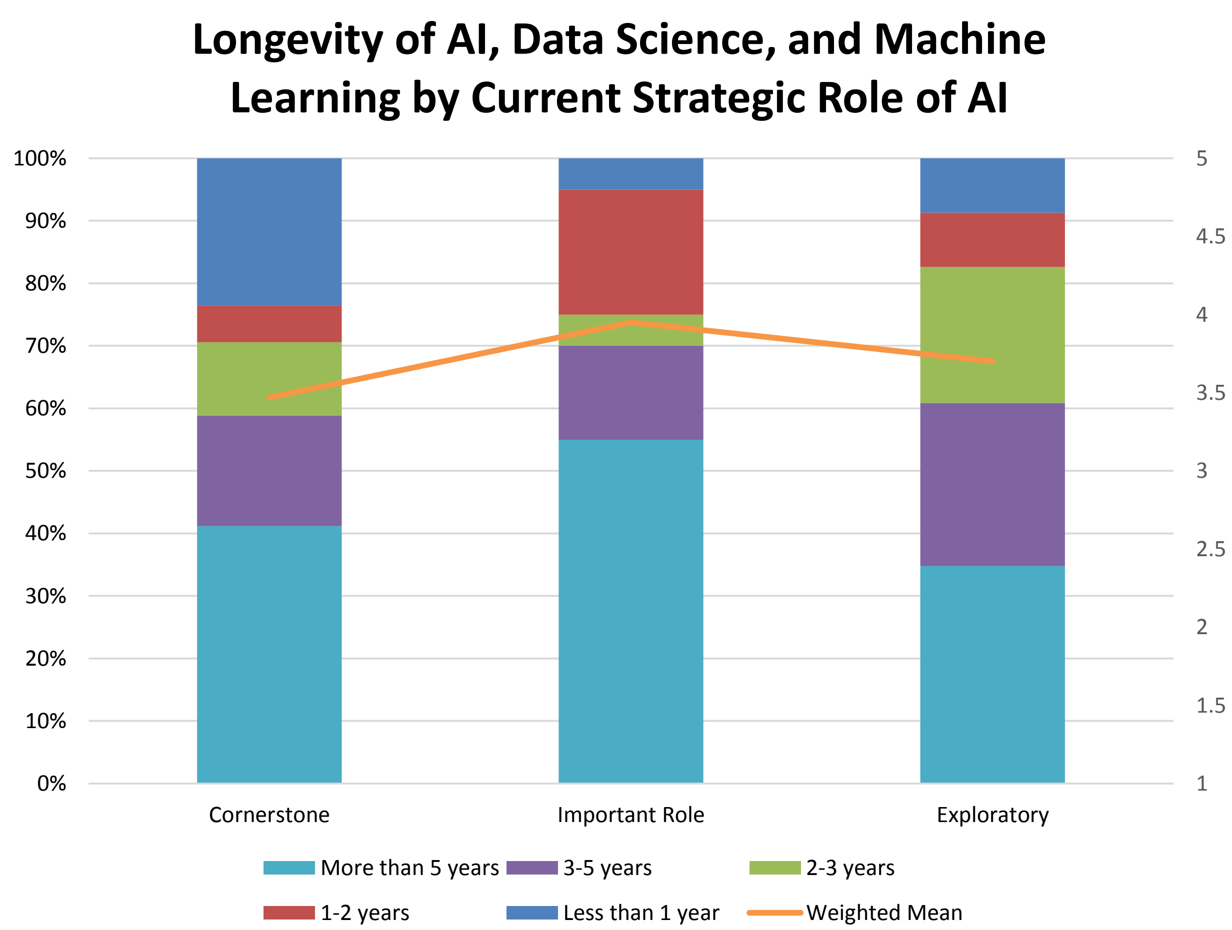


Figure 80 - Longevity of AI, data science, and machine learning by current strategic role of AI

Longevity of AI, Data Science, and Machine Learning by Generative AI Adoption

The longer an AI, data science, and machine learning program exists, the higher its propensity to deploy generative AI for production use (fig. 81). For generative AI, organizations with such programs in place for three or more years most often report its production use or experimentation with it at fairly similar levels (73% and 68%, respectively). The data then shows a significant drop among those with no plans to deploy generative AI (only 43%). In addition, the likelihood that an organization is using generative AI in production or experimenting with it tends to scale linearly with AI, data science, and machine learning program tenure—with the youngest programs doing so only 12% of the time; those in place one to two years doing so 26% of the time; programs tenured two to three years doing so 21% of the time; those in existence for three to five years doing so 43% of the time; and those in place for more than five years almost always doing so (98%).

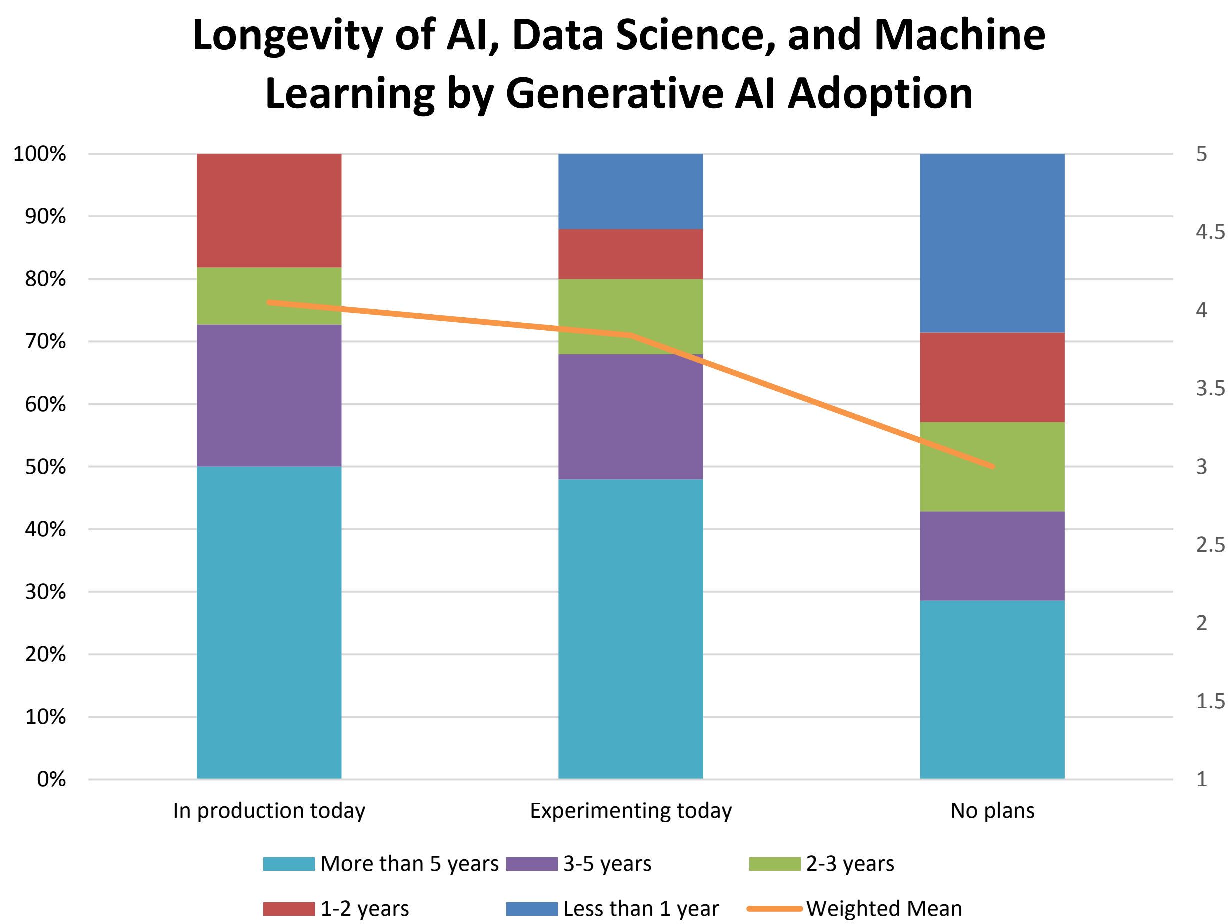


Figure 81 - Longevity of AI, data science, and machine learning by generative AI adoption

Longevity of AI, Data Science, and Machine Learning by Agentic AI Adoption

The two longest-tenured AI, data science, and machine learning programs (that is, those in existence for more than three years) show an almost equal propensity to report either production use of agentic AI (75%) or no plans to deploy it (70%; fig. 82). Among the oldest programs (in place for more than five years), the data shows an equal likelihood for production deployment and no plans (50% each). With relatively high and similar levels of experimentation and planned future use of agentic AI (64% each) by organizations with AI, data science, and machine learning programs in place for more than three years, we believe this data more likely indicates (at least for now) specific organizations' difficulty aligning agentic AI to specific use cases and business needs, rather than a "coin flip" that the longest-tenured programs will engage in a wholesale embrace or rejection of agentic AI.

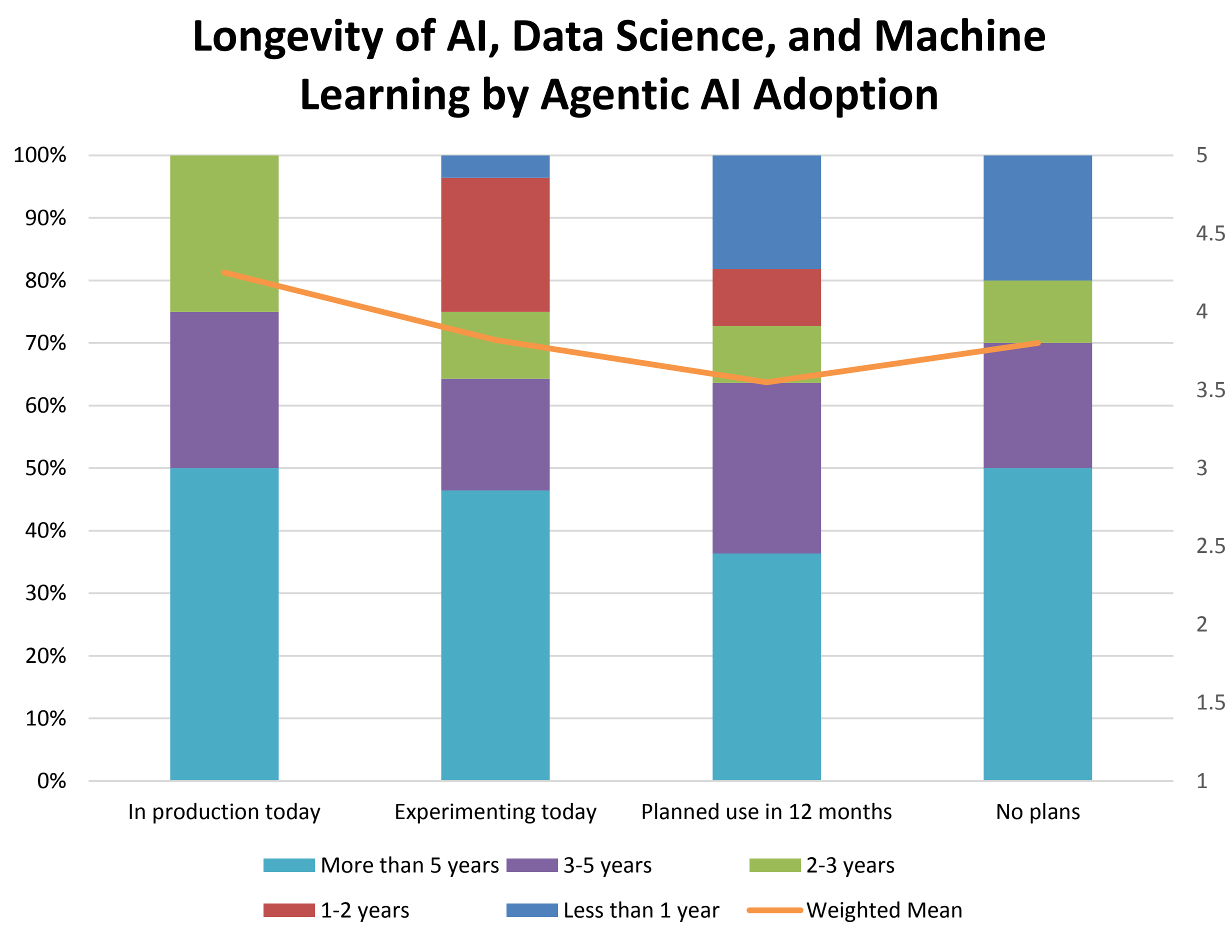


Figure 82 - Longevity of AI, data science and machine learning by agentic AI adoption

Features for AI, Data Science, and Machine Learning

Respondents express significant interest in the full range of feature requirements for AI, data science, and machine learning in 2025. All but two of 23 sampled features are at least important to two-thirds (69%) or far more (up to 79%) respondents this year (fig. 83). The most important among these mostly address traditional statistical methods: outlier detection, model explainability, range of regression models, and model management and governance, but also text analytic functions and sentiment analysis. (Also see industry support for features, figs.108-109.)

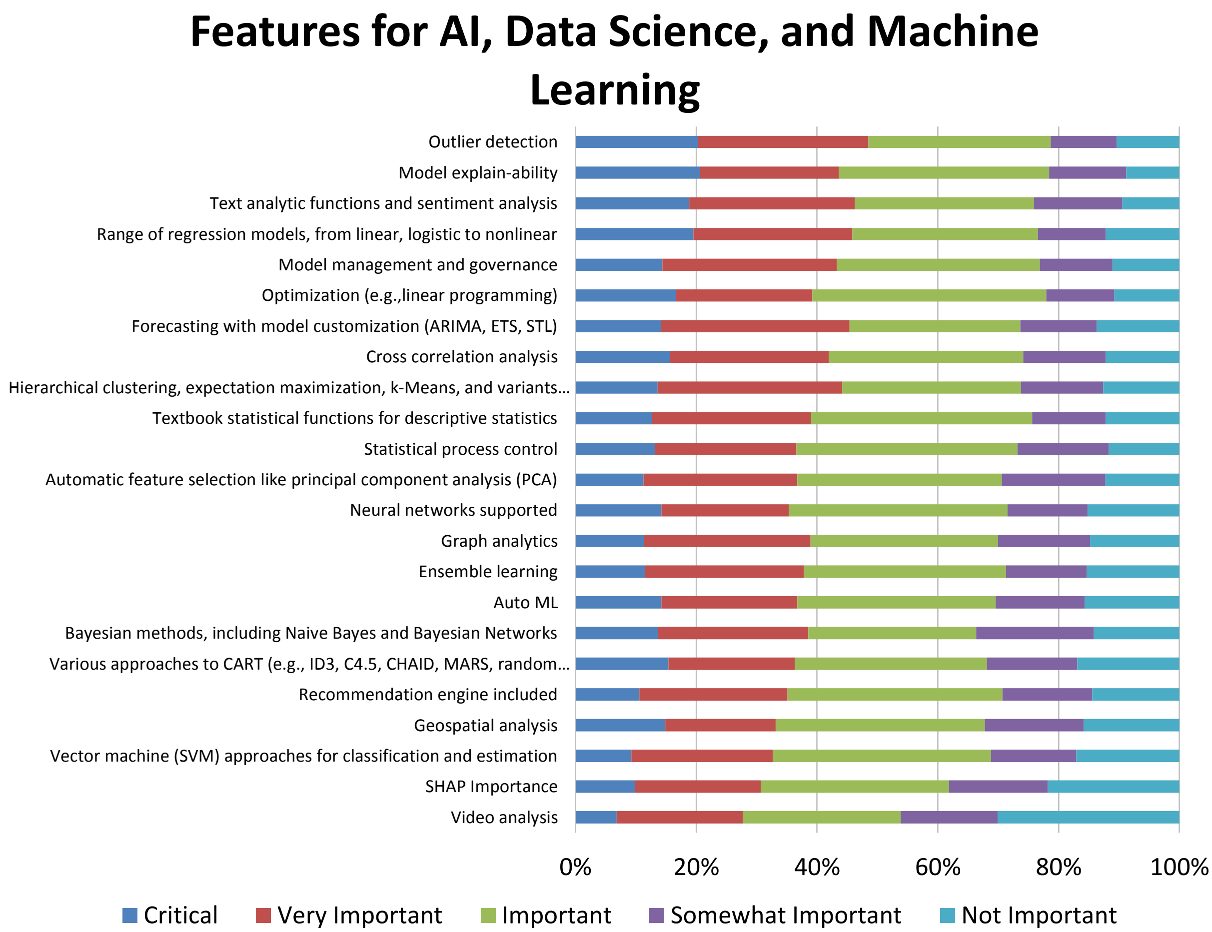


Figure 83 – Features for AI, data science, and machine learning

Features for AI, Data Science, and Machine Learning 2022-2025

Fig. 84 shows respondent interest in feature requirements for AI, data science, and machine learning across the four most recent years of our data collection. Amid ebbs and flows, we observe 2025 interest in the most popular features is flat or slightly lower year over year, while some lower-ranked features showed slight positive momentum. Thus, interest in outlier detection and model management and governance has receded slightly since 2022, while low-ranked video analysis and vector machine approaches moved onto the user radar. Overall, most features have sustained relevance in the range of the 3.0 score signifying "important" criticality.

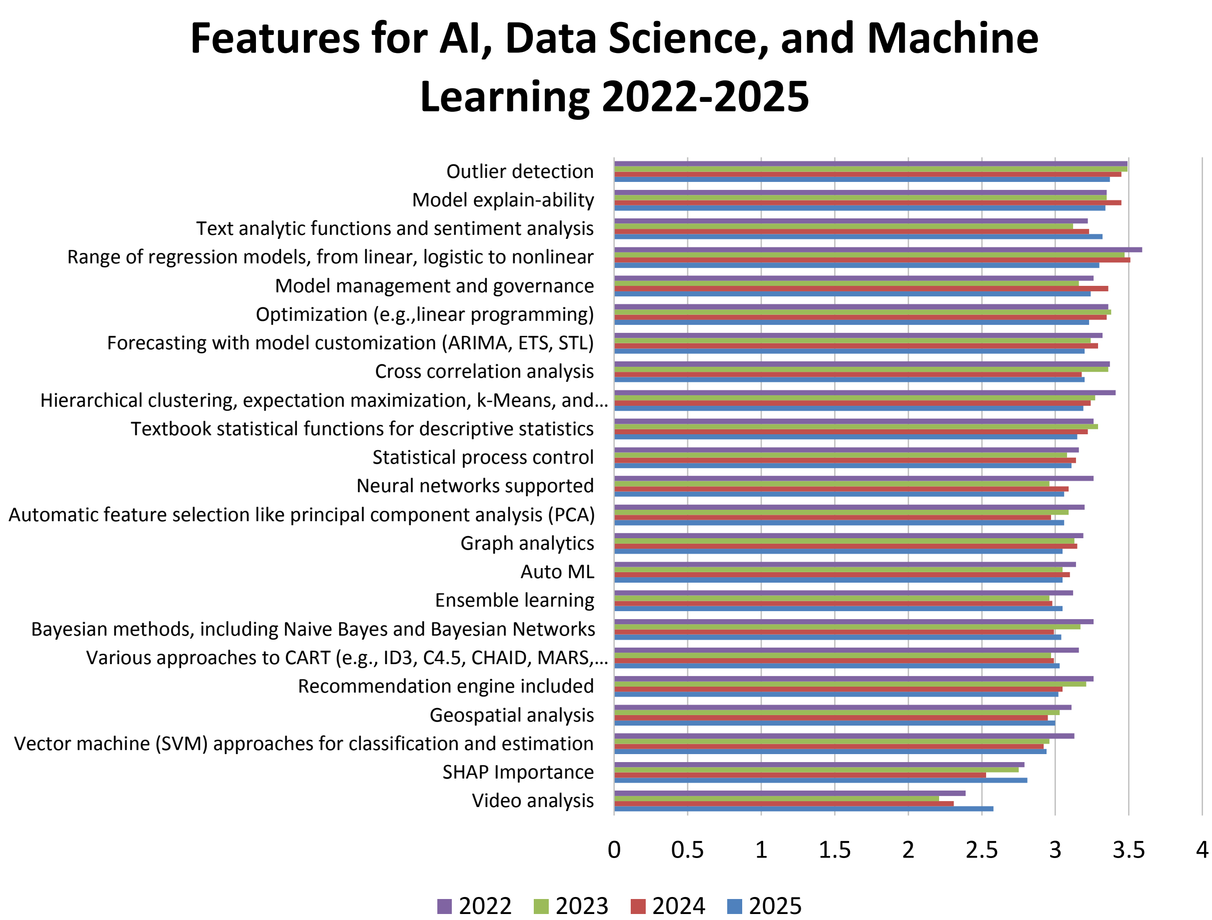


Figure 84 – Features for AI, data science, and machine learning 2022-2025

Features for AI, Data Science, and Machine Learning by Geography

In 2025, many top features for addressing AI, data science, and machine learning are broadly distributed across geographic regions, where interest is highest in Asia Pacific and Latin America (fig. 85). The top 11 features all rated at the level signifying "important" or greater across all geographies. Interest in the top feature, outlier detection, is led by Asia Pacific and well clustered in other regions solidly above the level of "important." Excluding Asia Pacific and Latin America, sentiment in EMEA is nearly always highest, and North America respondents offer the lowest score for nearly all features.

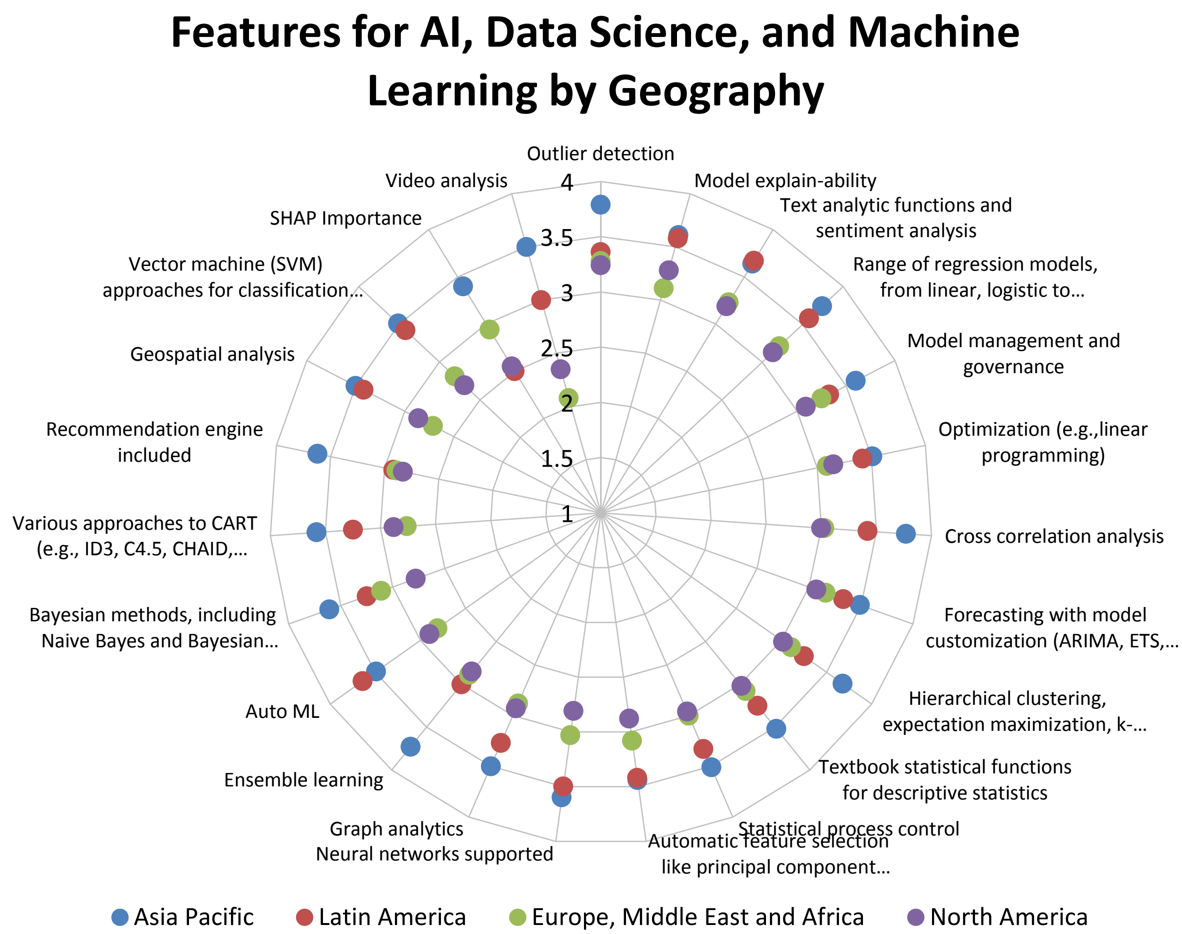


Figure 85 – Features for AI, data science, and machine learning by geography

Features for AI, Data Science, and Machine Learning by Function

2025 interest in features for AI, data science, and machine learning is led by a mix of front- and back-office roles most often found in the BICC, IT, and sales & marketing (fig. 86). In a signal of imminent or ongoing deployment, BICC respondents report the highest scores for the top five features: outlier detection, model explainability, text analytic functions and sentiment analysis, range of regression models, and model management and governance. Areas of particularly high interest to sales & marketing include optimization/linear programming, cross-correlation analysis, and lower-ranked areas including graph analytics, ensemble learning, and auto machine learning. Finance and R&D are among the least-interested functions, the latter finding possibly indicating the use of platforms or preconfigured products and services.

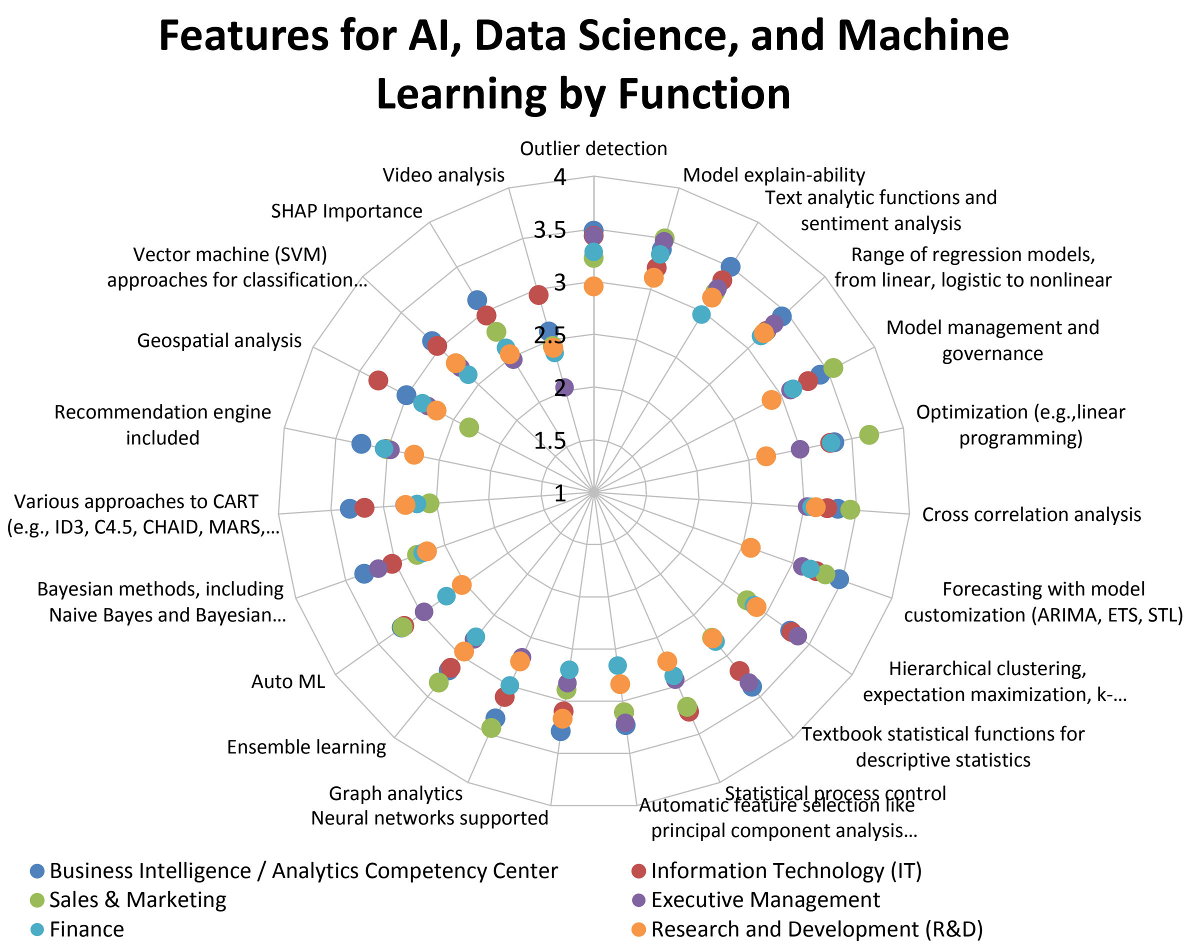


Figure 86 – Features for AI, data science, and machine learning by function

Features for AI, Data Science, and Machine Learning by Industry

Interest in features for AI, data science, and machine learning varies high and low selectively by industry in 2025 (fig. 87). This year, the most-represented industries by weighted mean include manufacturing, consumer services, and technology. For example, manufacturing respondents lead or show near-highest interest in at least 12 features, showing standout attention to areas including optimization and cross-correlation analysis. Consumer services respondents give high marks to the top two features, outlier detection and model explainability, along with neural networks, Bayesian methods, geospatial analysis, SHAP importance, and video analysis. Business services, retail and wholesale, and government are the least-interested industries.

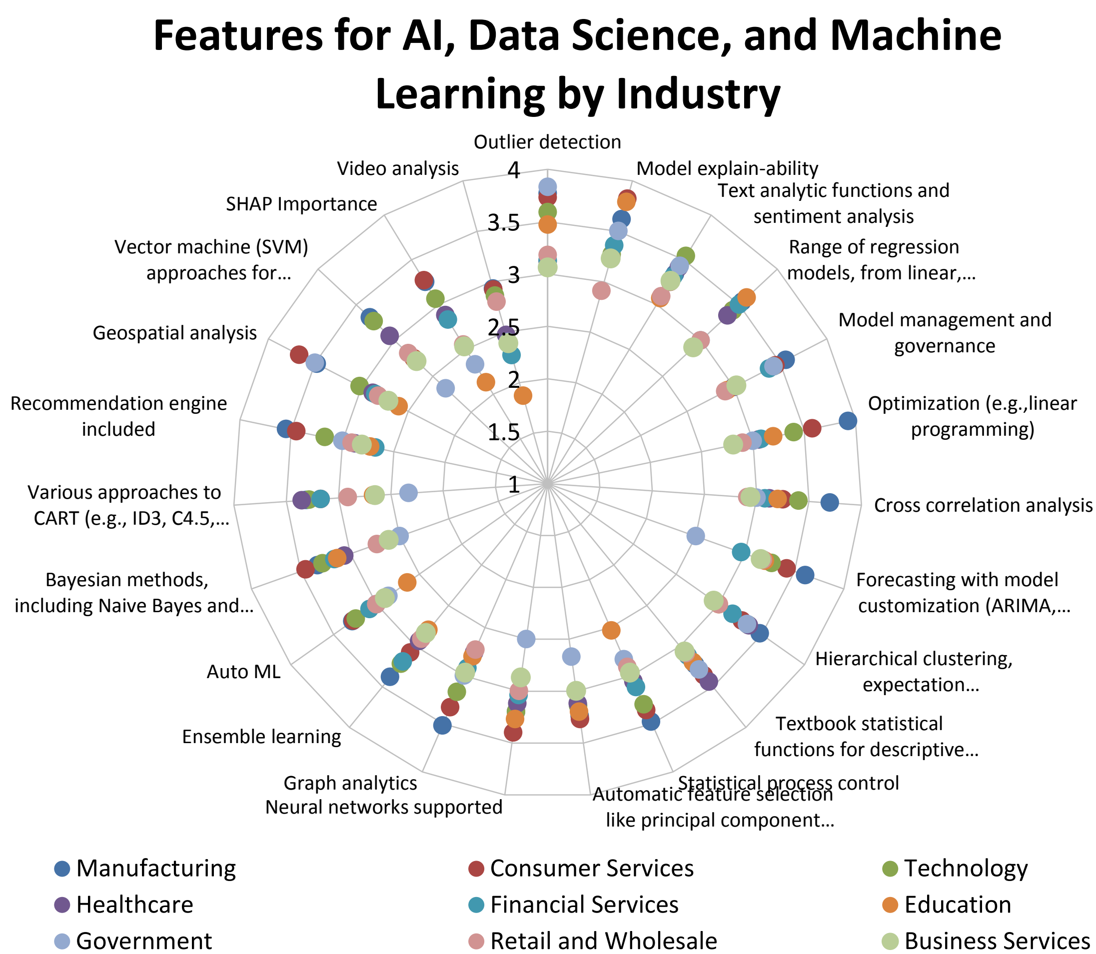


Figure 87 – Features for AI, data science, and machine learning by industry

Features for AI, Data Science, and Machine Learning by Organization Size

Interest in analytical features for AI, data science, and machine learning consistently correlates with increasing organization size in 2025 (fig. 88). Indeed, very large organizations (more than 10,000 employees) lead interest in every feature sampled this year, with some features—including model explainability, text analytic functions, hierarchical clustering, and others among lower-ranked features—receiving far outsized importance scores compared with all smaller peer organizations. Features are un-clustered and show widely varied levels of interest across organization sizes. Notably, small organizations of 1-100 employees almost always report the lowest interest in all features, contrary to other size cohorts whose interest is closer to that of very large organizations.

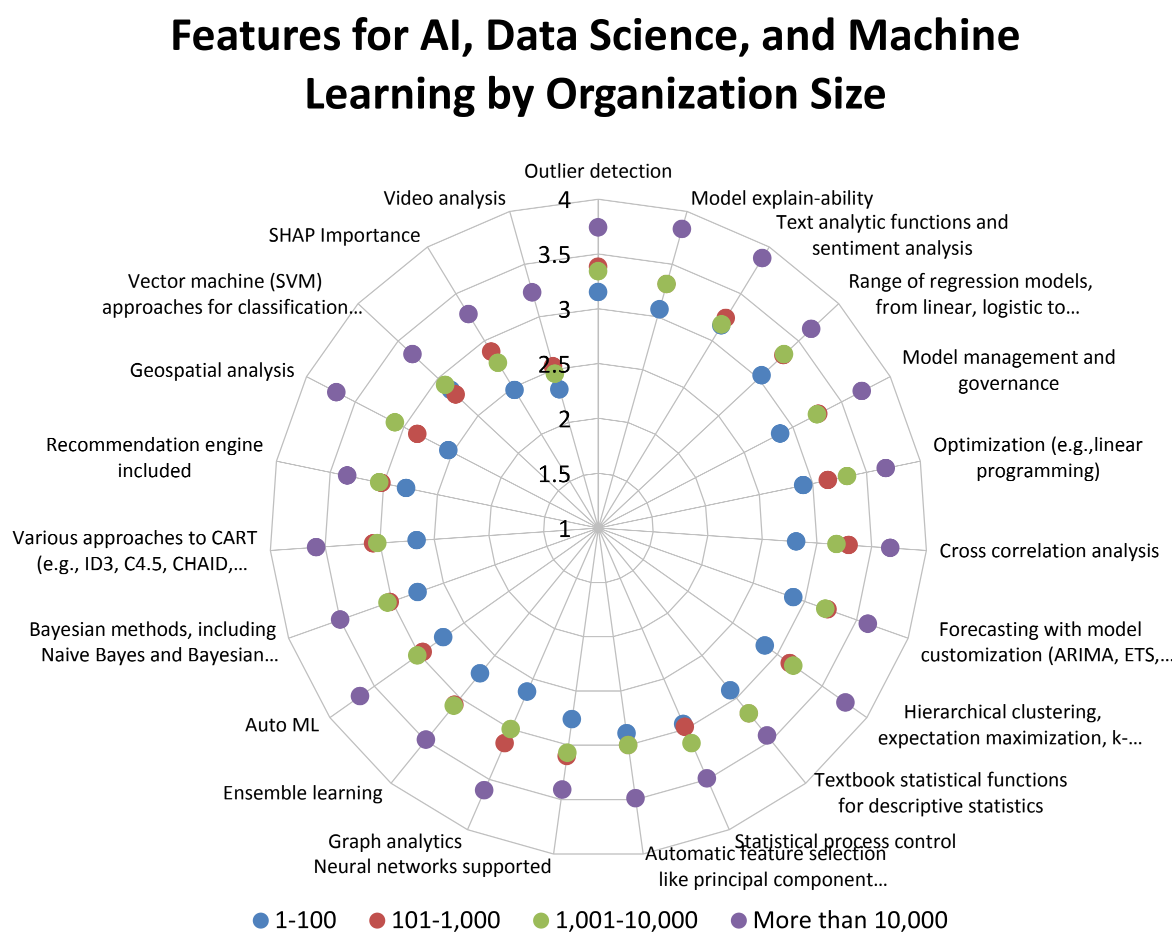


Figure 88 – Features for AI, data science, and machine learning by organization size

Usability for AI, Data Science, and Machine Learning

Our study addresses a detailed set of 16 usability benefits that support AI, data science, and machine learning activities and processes (figs. 89-93).

Usability features generally address process or activity simplification and automation. Without exception, respondents give them all significantly high importance scores. All 18 of our 2025 criteria are, at minimum, important to at least 63% and as many as 80% of respondents (fig. 89). The top three features (Python support, low code/no code, and support for easy iteration) are either critical or very important to between 49%-52% of respondents, and at least a dozen other features are critical or very important to about 40%. Half or more of the features sampled are “not important” to only 10% or fewer respondents, and all are at least somewhat important to more than 80% of all respondents. (Also see industry support for usability tools, fig.111.)

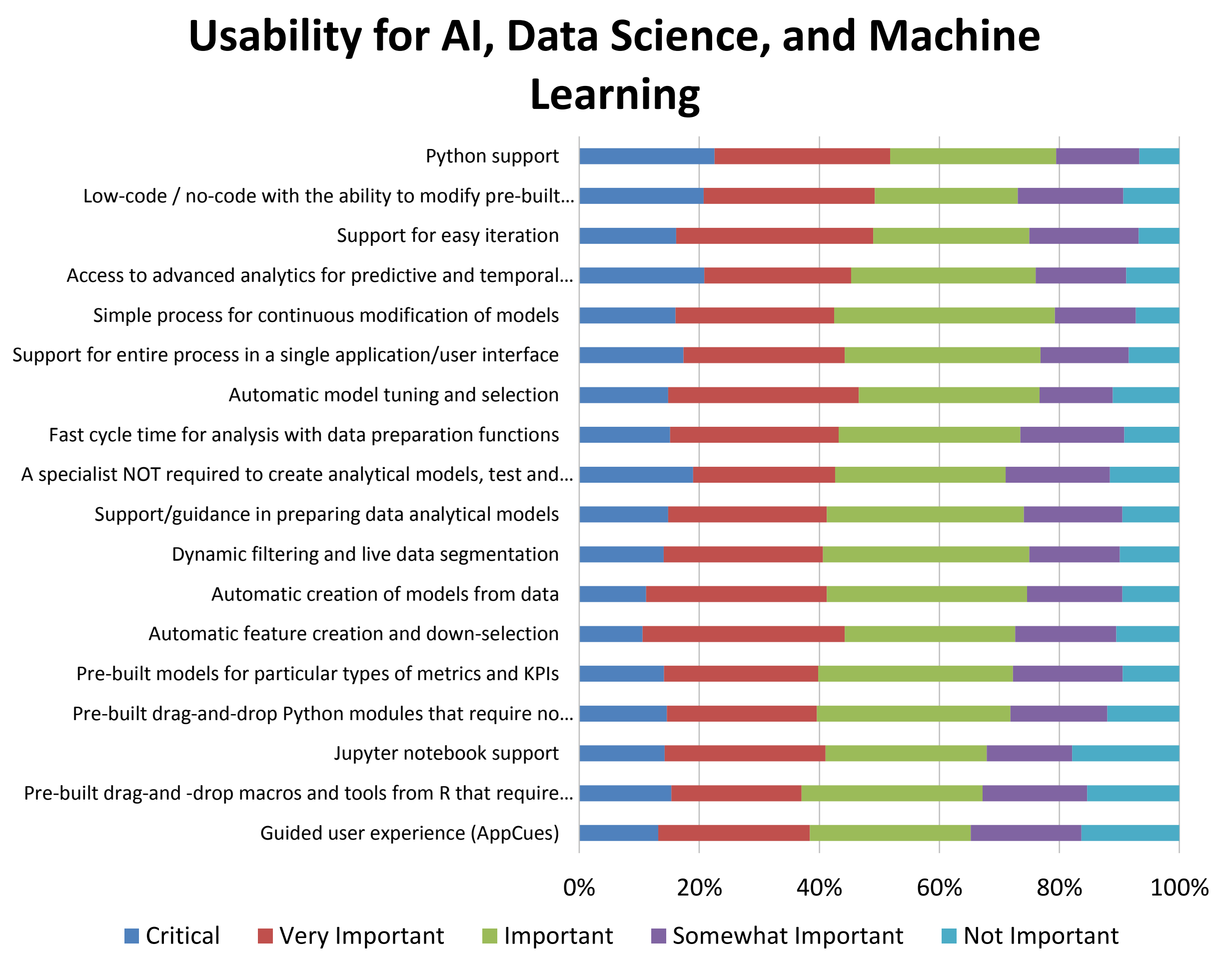


Figure 89 – Usability for AI, data science, and machine learning

Usability of AI, Data Science, and Machine Learning 2022-2025

Interest in most usability features for AI, data science, and machine learning is at or near a four-year low in 2025, but only by small degrees relative to historic measures (fig. 90). This year, only low-ranked guided user experience and Jupyter notebook support show higher year-over-year sentiment, again by very small degrees. Perhaps more important, all usability features remain at or well above the 3.0 level signifying "important" to respondents. But compared to the peak interest seen in 2018-2019 (not shown in this chart), sentiment toward nearly all usability features for AI, data science, and machine learning declines narrowly and gradually from 2022 to 2025 on a somewhat compressed scale.

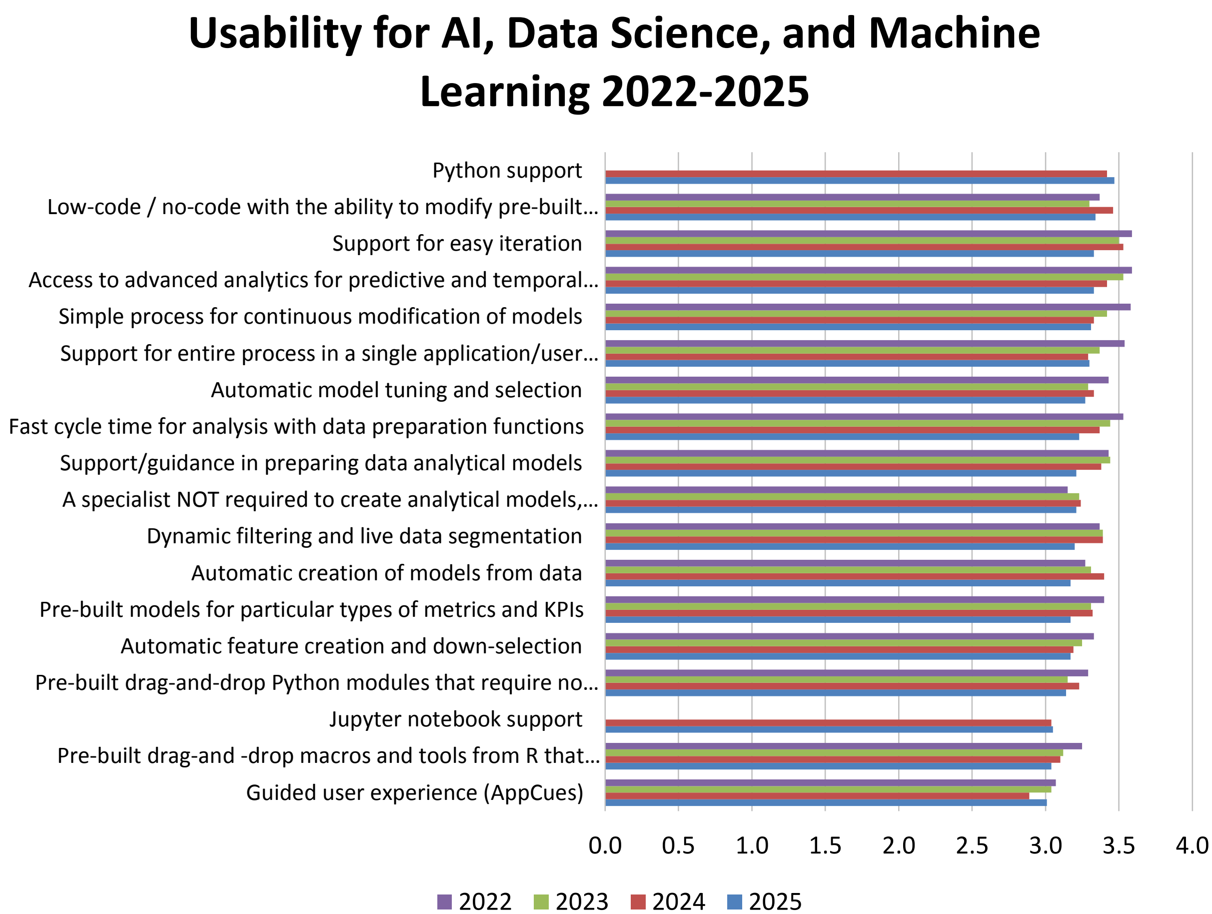


Figure 90 – Usability for AI, data science, and machine learning 2022-2025

Usability of AI, Data Science, and Machine Learning by Geography

The perceived importance of usability features for AI, data science, and machine learning in 2025 varies by geography, but is nonetheless almost always highest in Asia Pacific and usually lowest in Latin America (fig. 91). Excluding Asia Pacific, geographic interest is most often highest in North America, followed by EMEA. Even so, seven of the 18 feature scores are above the level of "important" in all geographies. The most common interest showing the tightest clustering across all geographies includes access to advanced analytics for predictive and temporal analysis, support for easy iteration, simple process for continuous modification of models, and support/guidance in preparing data analytical models.

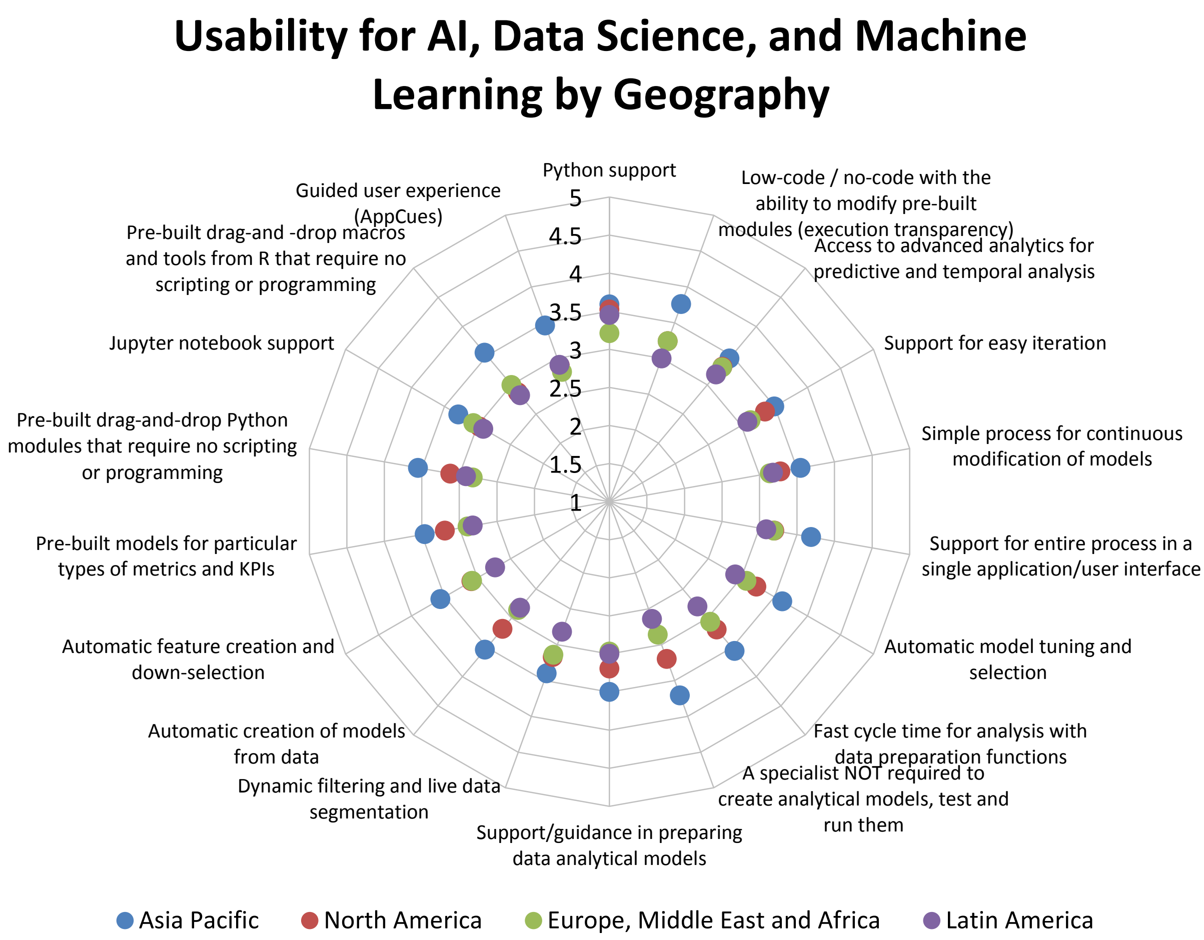


Figure 91 – Usability for AI, data science, and machine learning by geography

Usability of AI, Data Science, and Machine Learning by Function

Viewed by function, BICC and front-office sales & marketing interest combined suggest ongoing rollout momentum and uptake of AI, data science, and machine learning (fig. 92). BICC scores are highest overall, especially in areas including Python support, access to advanced analytics for predictive and temporal analysis, support for entire process in a single application/user interface, and dynamic filtering and live data segmentation. Sales & marketing posts the next-highest overall scores, with high interest in multiple discrete usability features. R&D and finance are most often least interested in usability features.

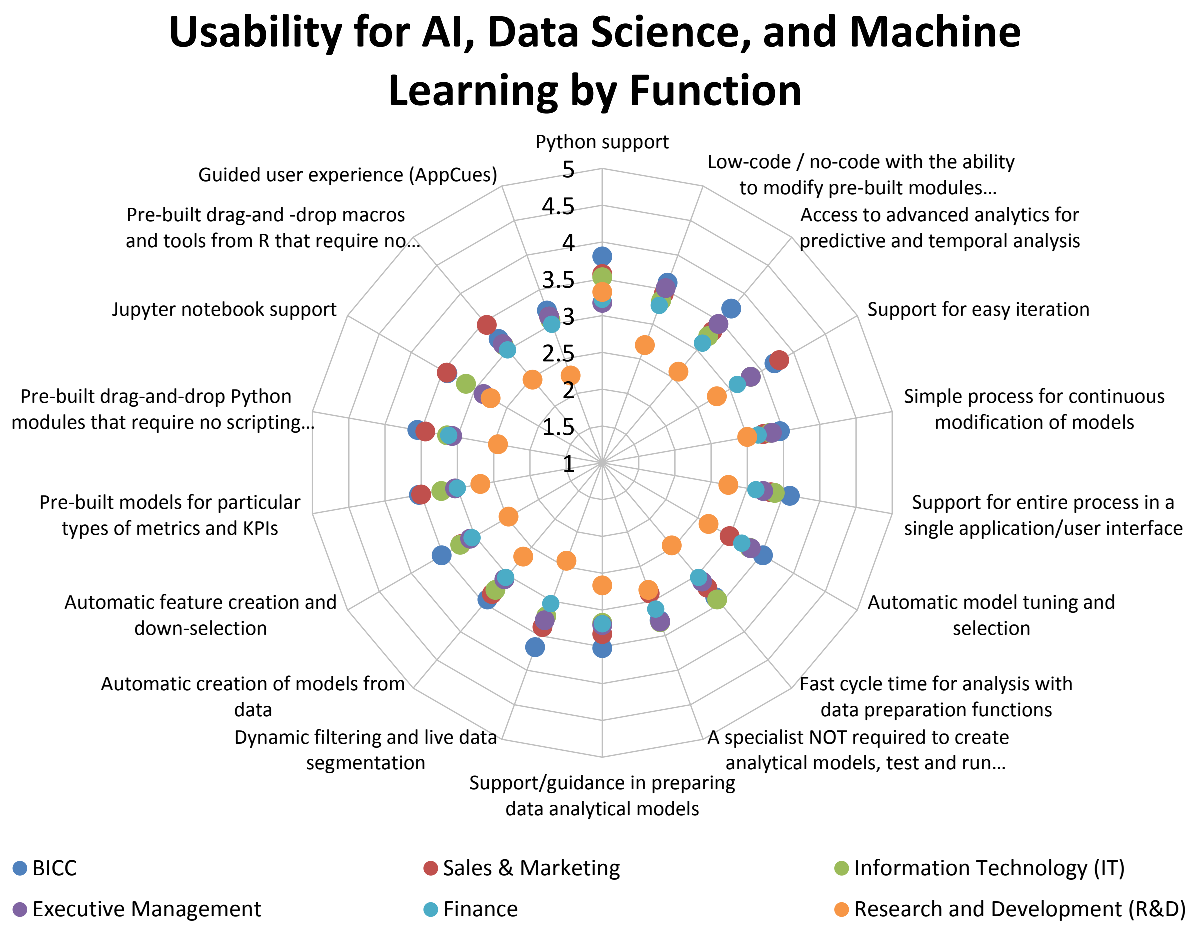


Figure 92 – Usability for AI, data science, and machine learning by function

Usability of AI, Data Science, and Machine Learning by Organization Size

Interest in usability features for AI, data science, and machine learning increases with global headcount to a visible and sometimes dramatic degree in 2025 (fig. 93). In every case, interest in features is highest at very large organizations (more than 10,000 employees), and lowest in small (1-100 employees) organizations. Between these bookends, large (1,001-10,000 employees).and midsize (101-1,000 employees) organizations mingle and overlap in interest feature by feature. Excluding small organizations, criticality toward 15 of 18 usability features is above the level signifying "important" to all organizations with 100 or more employees.

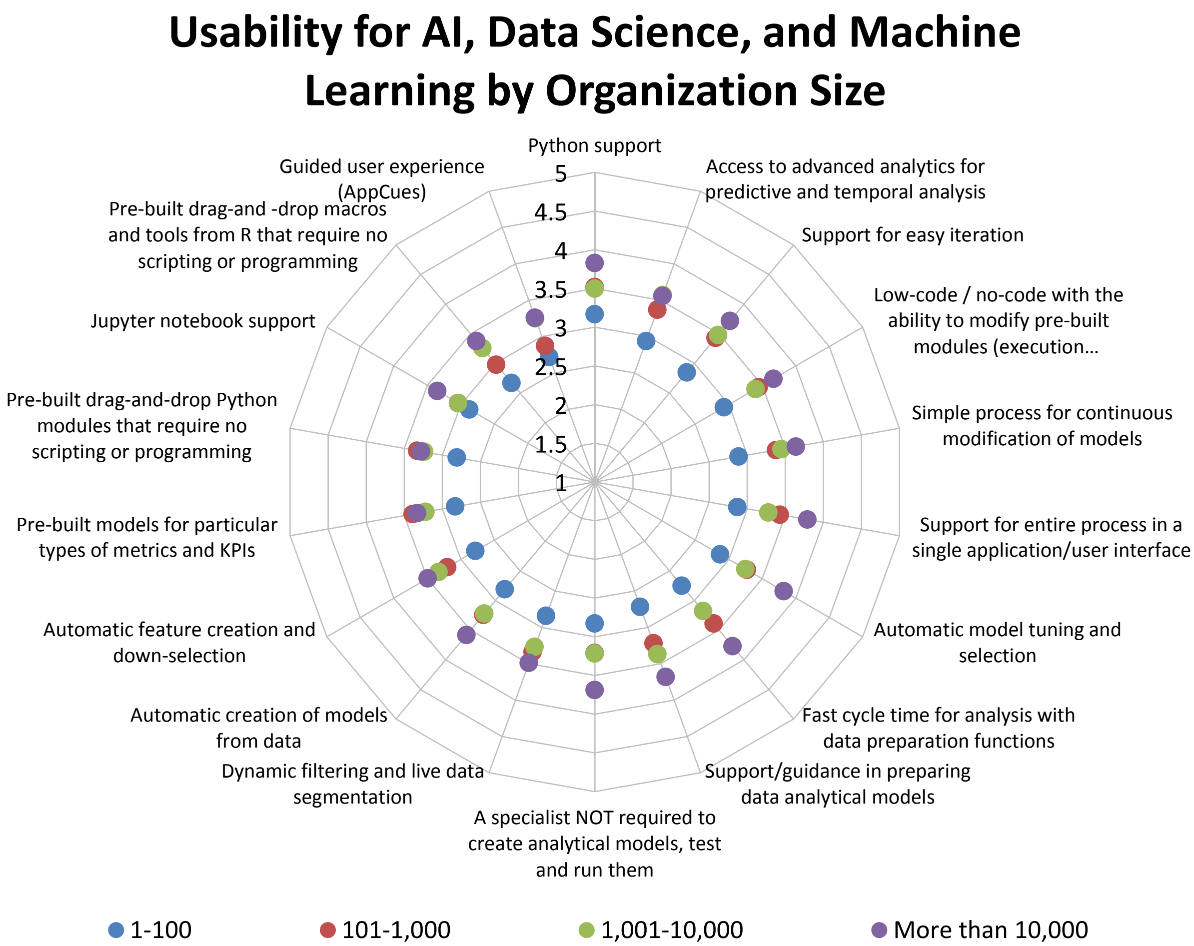


Figure 93 – Usability for AI, data science, and machine learning by organization size

Scalability of Data Science and Machine Learning

Our study addresses respondents' interest in a set of scalability technologies and architectures that support AI, data science, and machine learning (figs. 94-97). All 12 features sampled in 2025 are at least important to at least half and up to more than two-thirds of respondents (fig. 94). This year, as in the last three years, two features— in-database analytics and in-memory analytics—are most important, after which we observe rising year-over-year interest in second-tier features including horizontal scaling, code generation supported, vertical scaling, hybrid/cloud bursting, GPU acceleration, and multi-tenant cloud services. (Also see industry support for scalability, fig. 112.)

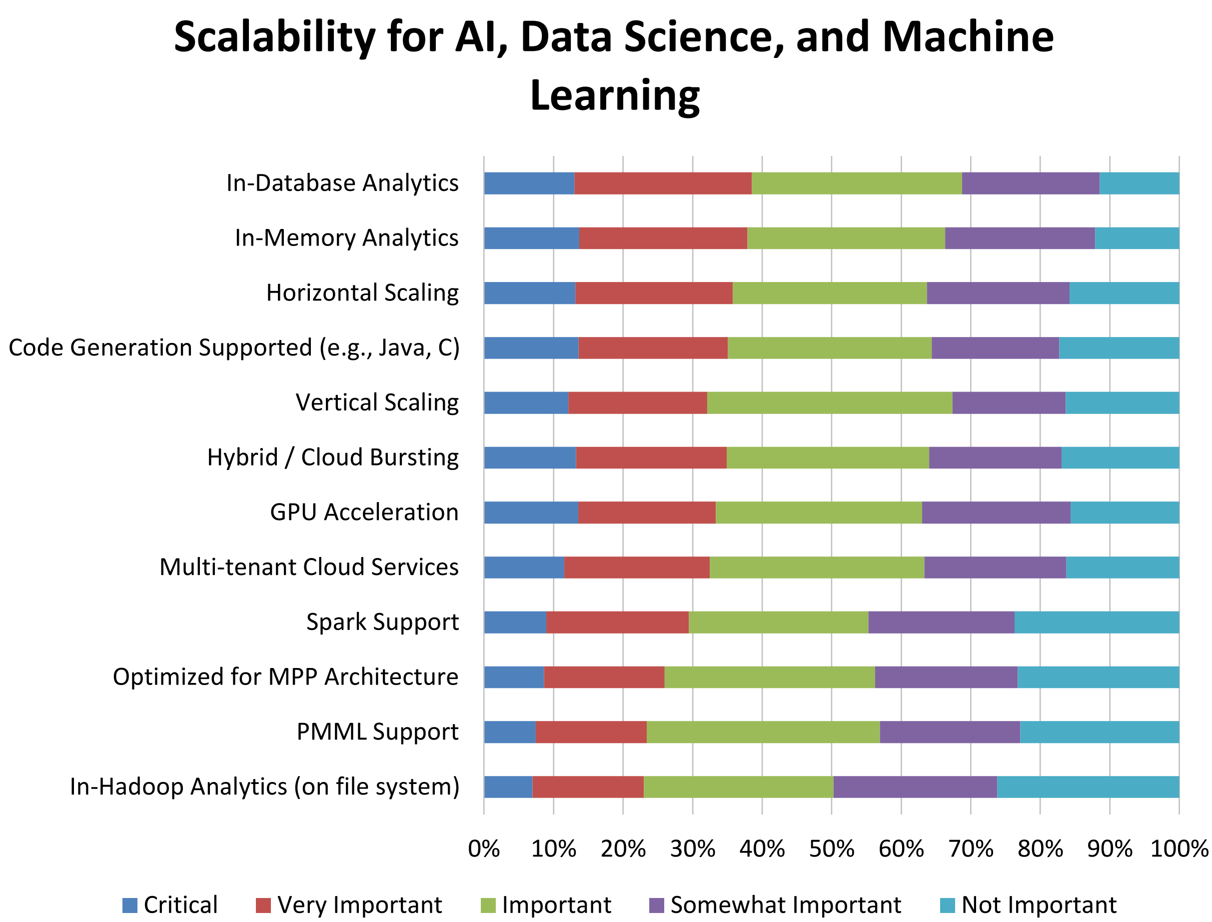


Figure 94 – Scalability for AI, data science, and machine learning

Scalability for AI, Data Science, and Machine Learning 2021-2025

Over time and in 2025, we observe mostly minor advancers and decliners in the 12 scalability capabilities for AI, data science, and machine learning (fig. 95). Interest in the top two features, in-database analytics and in-memory analytics, declined slightly year over year, which is not uncommon for the most mature category-leading features. Interest in most mid-tier features stayed flat or rose slightly. Low-ranked scalability capabilities were most likely to gap up in importance, particularly in the cases of in-Hadoop analytics and Spark support.

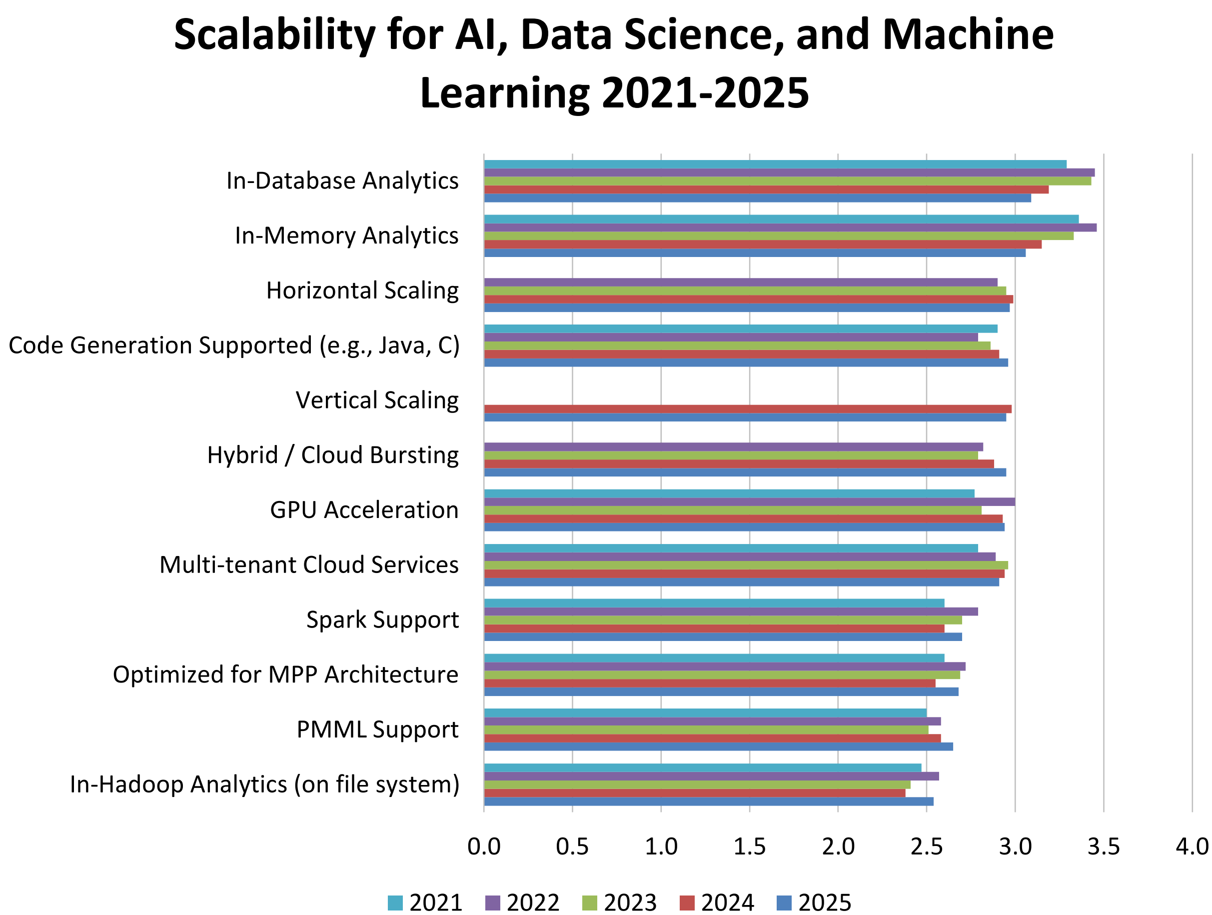


Figure 95 – Scalability for AI, data science and machine learning 2021-2025

Scalability for AI, Data Science, and Machine Learning by Function

Functional interest in scalability features for AI, data science, and machine learning is highest by weighted mean in the BICC and sales & marketing, repeating a mix that implies front-office attention and high development/deployment interest and/or incipient demand (fig. 96). In 2025, BICC activity is particularly strong in areas including in-database analytics, horizontal scaling, code generation supported, multi-tenant cloud services, Spark support, MPP architecture, and PMML support. Sales & marketing interest is highest overall for in-database analytics and in-memory analytics, and is of average or greater interest toward other scalability capabilities. Unusually, finance respondents expressed the third-highest interest by function for scalability.

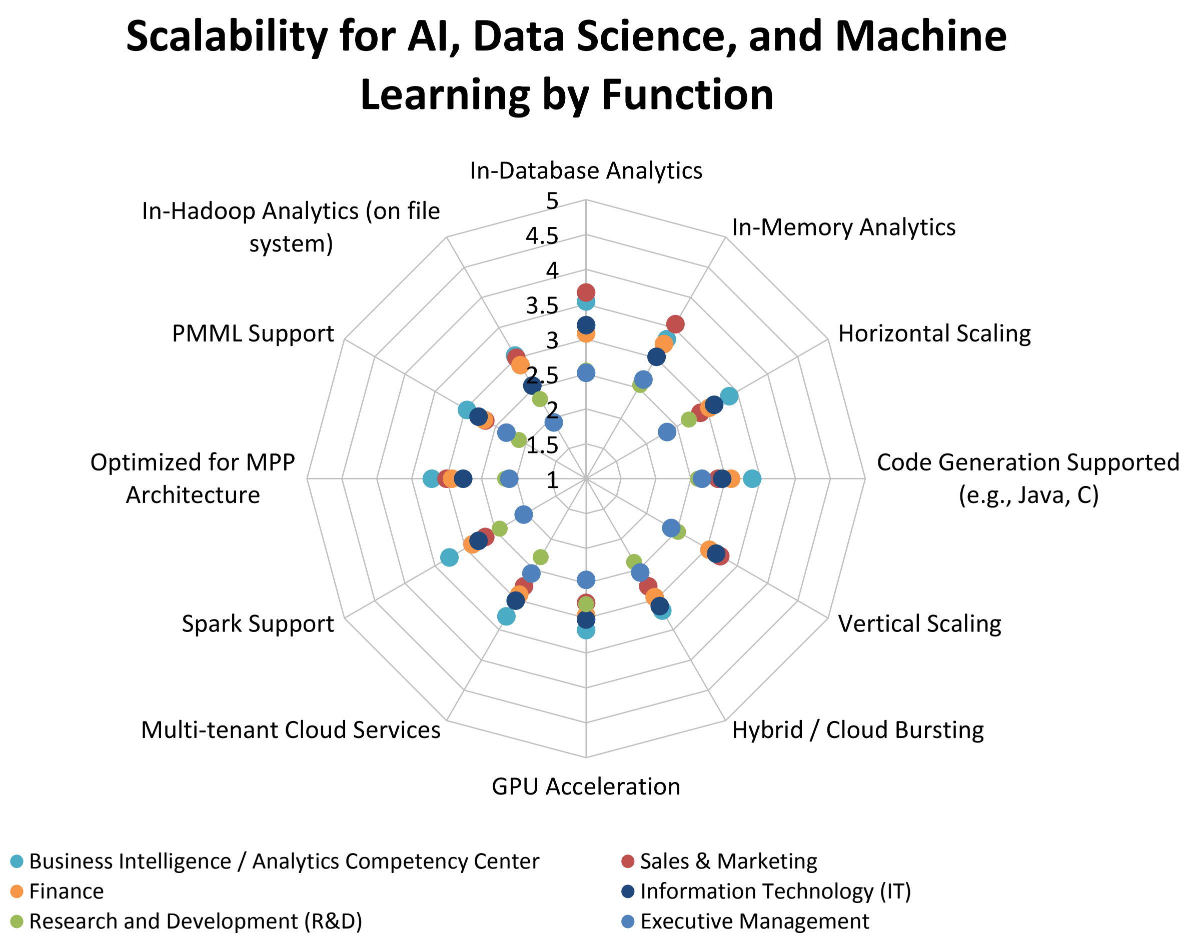


Figure 96 – Scalability for AI, data science, and machine learning by function

Scalability for AI, Data Science, and Machine Learning by Organization Size

In 2025 and in all earlier studies, interest in scalability capabilities for AI, data science, and machine learning correlates strongly to global headcount (fig. 97). In all 12 scalability categories, interest in features is highest at very large organizations (more than 10,000 employees) and lowest in small organizations (1-100 employees). Between these extremes, large (1,001-10,000 employees) or midsize (101-1,000 employees) organizations jockey and overlap in their interest, feature by feature. Excluding small organizations, criticality toward four of 12 scalability capabilities is above the level signifying "important" to all organizations with 100 or more employees.

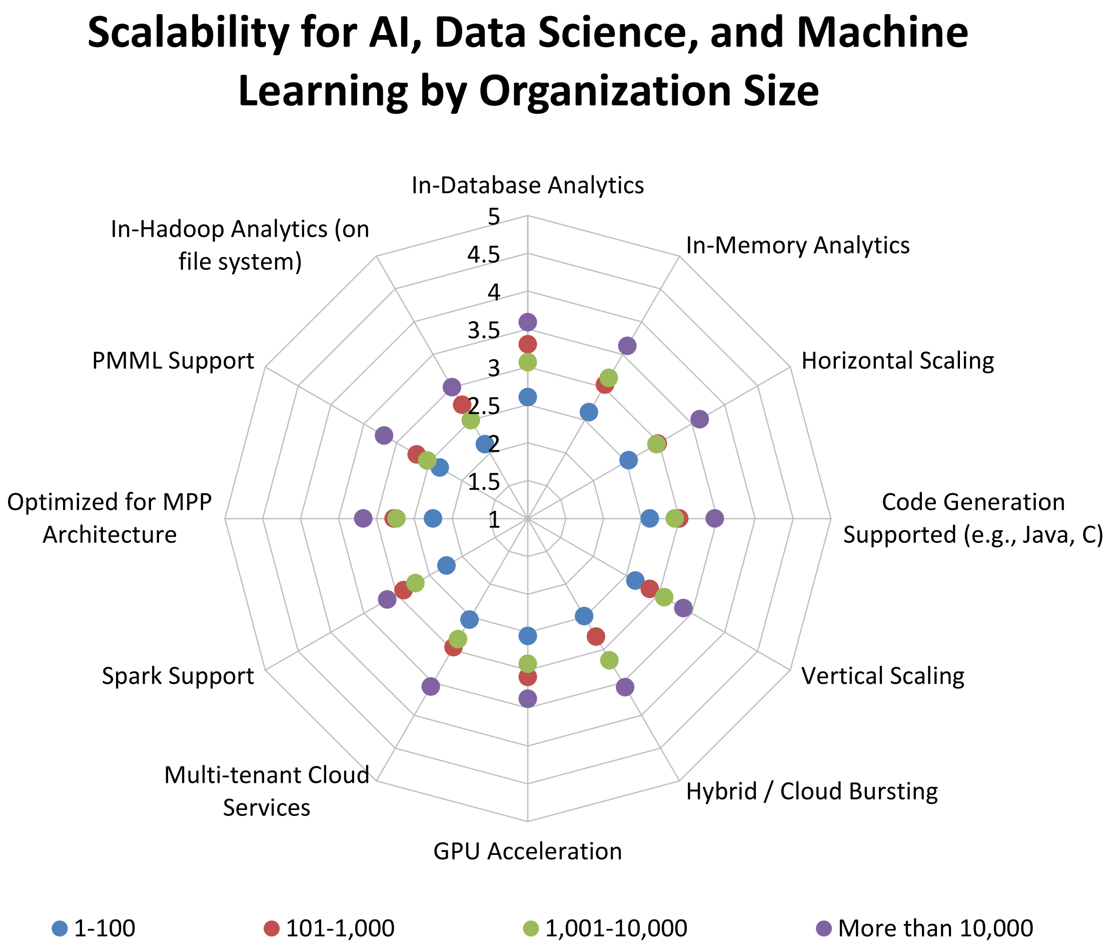


Figure 97 – Scalability for AI, data science, and machine learning by organization size

Neural Networks for AI, Data Science, and Machine Learning

We asked organizations to gauge their interest in 11 types or aspects of neural networks in the context of AI, data science, and machine learning (fig. 98). The top five in 2025 are not widely separated in importance and include recursive neural networks, artificial neural networks, transformer networks, feed-forward deep learning, and generative adversarial networks. Each of these five picks is seen as critical or very important by 36%-42% of respondents, and at least important by 66%-71% of respondents. Among the remaining networks, convolutional and recurrent neural networks are most popular. Just 10%-14% of respondents think any are not important. (Also see industry support, fig. 110.)

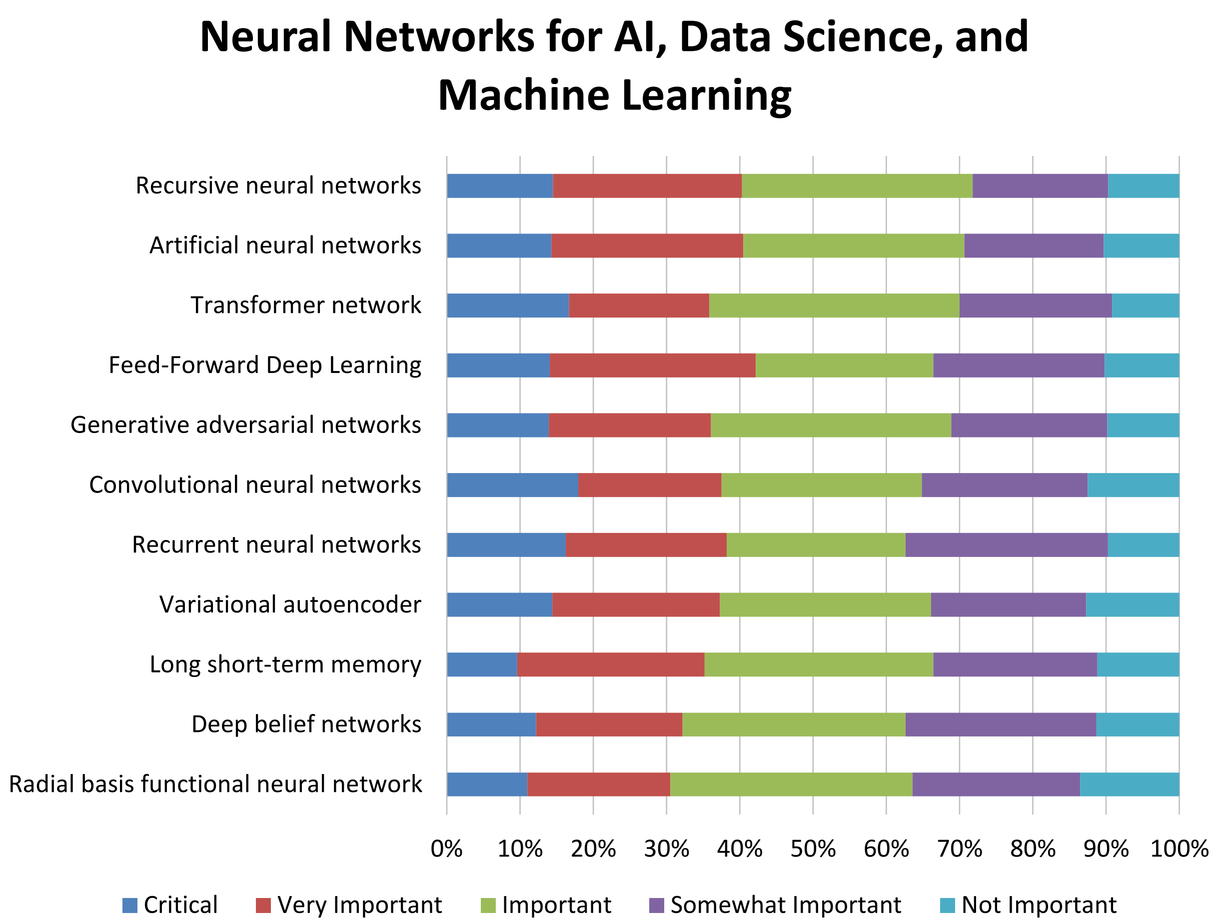


Figure 98 – Neural networks for AI, data science, and machine learning

Neural Networks for AI, Data Science, and Machine Learning 2019-2025

Attitudes toward types of neural networks in the context of AI, data science, and machine learning have varied across seven years of our study (with some networks more recently added; see fig. 99). Across the last four years, sentiment toward the six or seven network types sampled has been uneven, but has generally been sustained or increased, and 2025 sentiment is higher for all year over year.

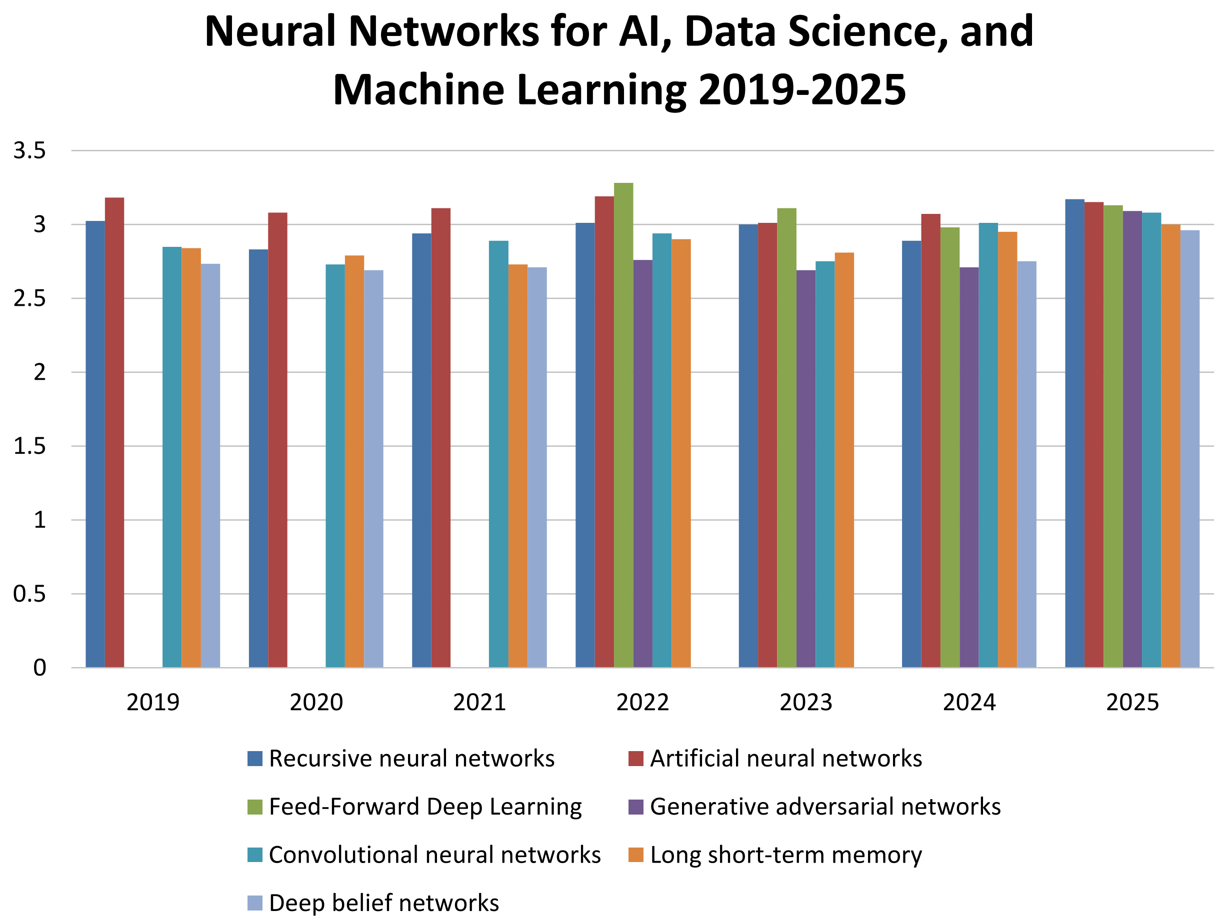


Figure 99 – Neural networks for AI, data science, and machine learning 2019-2025

Data Sources for AI, Data Science, and Machine Learning

We asked organizations to gauge their interest in 33 data sources for AI, data science, and machine learning in 2025. (Note: We have added six new data sources since 2022; see fig. 100.) Within this category, all choices have relevance to respondents; all but five are at least "important" to 40% or more; and all are at least "somewhat important" to 60% or far more respondents. Top picks Postgres and Amazon S3 are "critical" to 17%-18% of respondents and "critical" or "very important" to more than 40%. Many other technologies of interest represent diverse methods and data management practices. (Also see industry importance, fig. 133.)

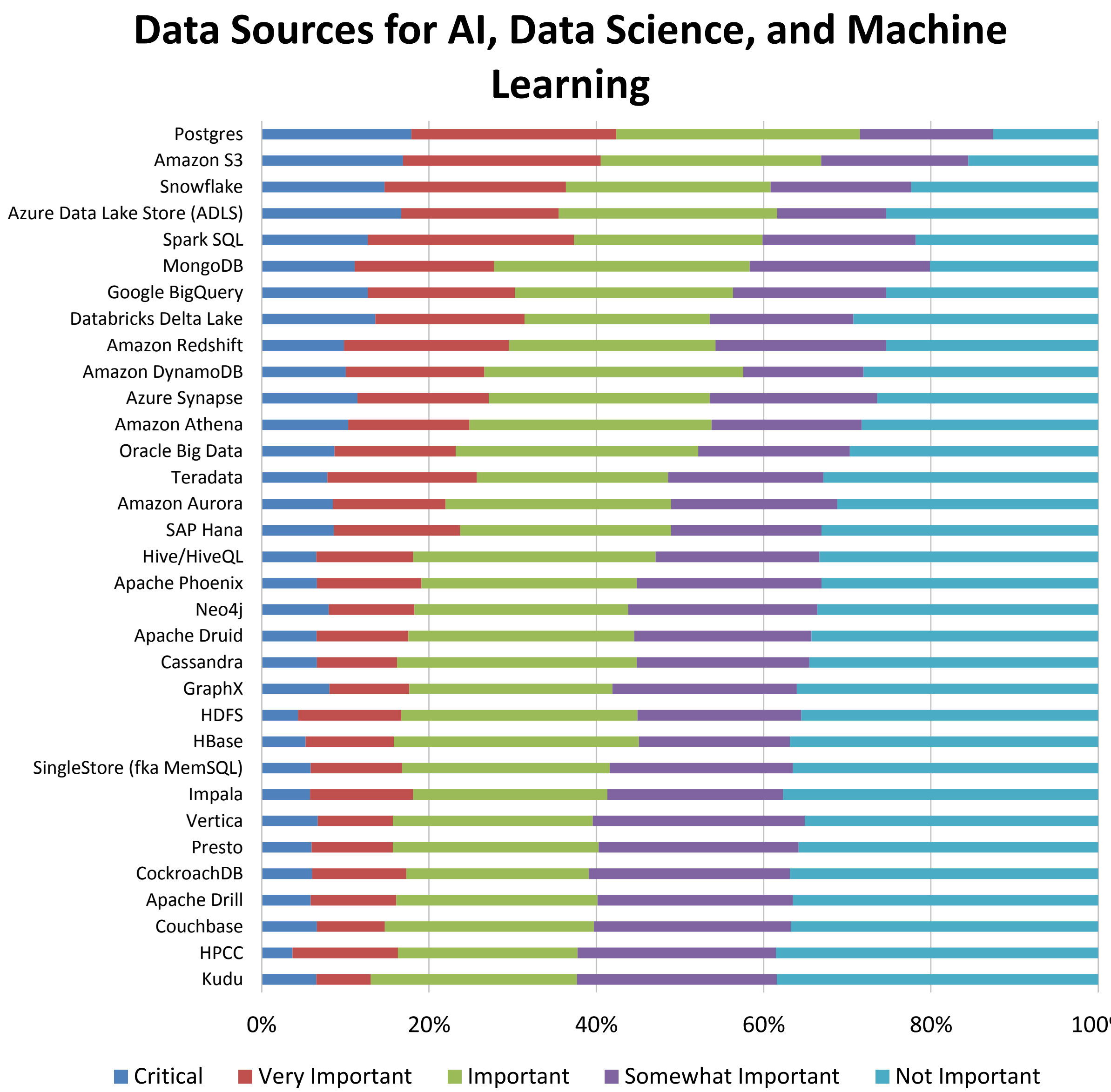


Figure 100 – Data sources for AI, data science, and machine learning

Industry and Vendor Analysis

Industry and Vendor Analysis

Industry Support of Generative AI

Industry support of generative AI took a near-quantum leap between 2024 and 2025 (fig. 101). The greatest year-over-year lift is a rise in general availability, which increased from 29% to 51%, a sure sign of a strong but still-early market. An additional 18% of industry respondents released beta testing in 2025, another strong lift for early agentic AI rollout. Offsetting gains appear in general availability and proof of concept. Countering this increase, reports of pilot programs decreased by more than half from 26% to 11%, and internal proof of concept programs, reported by 37% of industry respondents in 2024, declined to just 9% in 2025. From an industry support perspective, 2025 is the year generative AI "grew up."

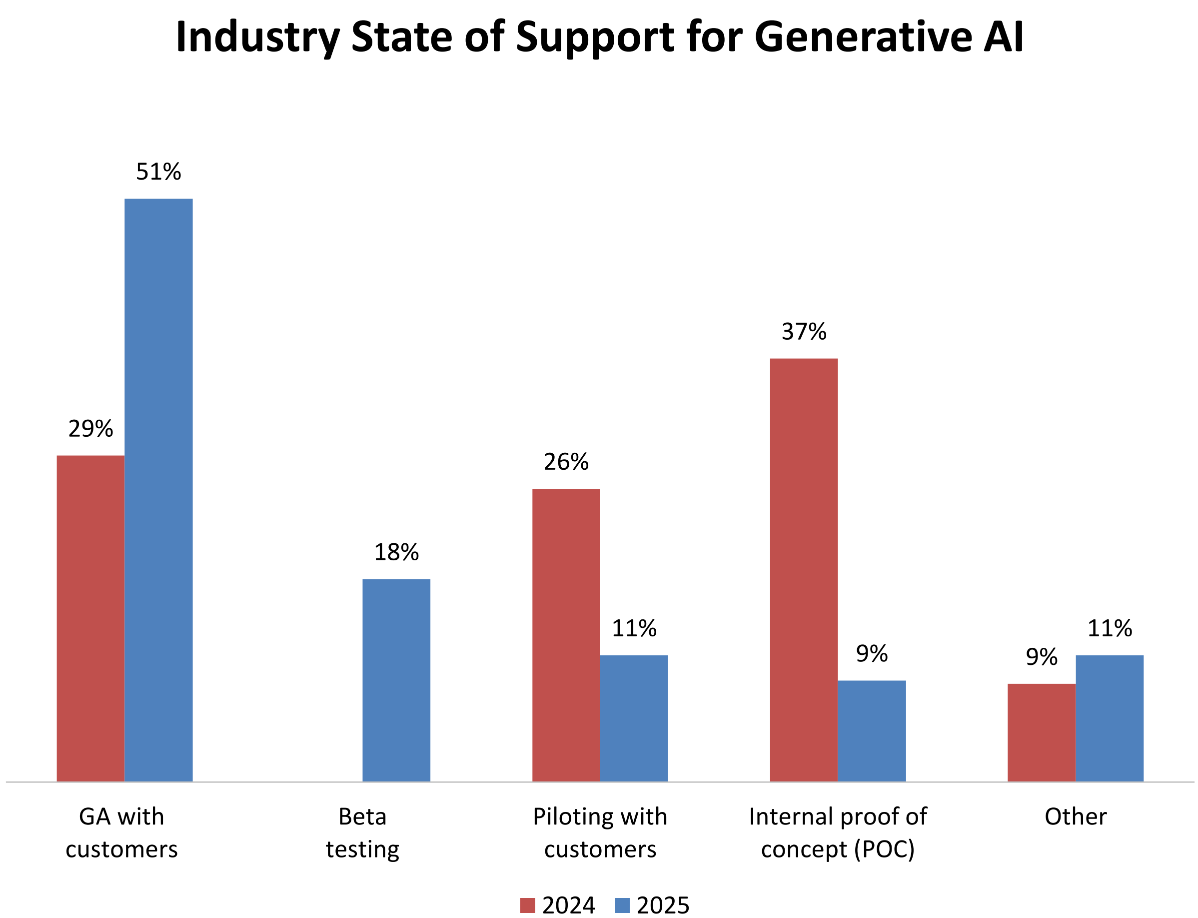


Figure 101 -- Industry state of support for generative AI

Industry Drivers for Generative AI Capabilities

We asked industry respondents about current support and future plans for specific generative AI capabilities (fig. 102). In 2025, offerings focus on development, customization, and automation. The top supported feature, APIs or SDKs for easy integration and development of generative AI models, is offered by 73% of vendors sampled; another 13% have 12-month plans. The two other leading capabilities available today include automated data-handling functionalities (60%) and transfer learning for generative AI models (53%). It is worth noting that none of the other 10 capabilities sampled is currently supported by more than 40% of industry respondents, a sign of both industry specialization and a still-immature supply side for agentic capabilities.

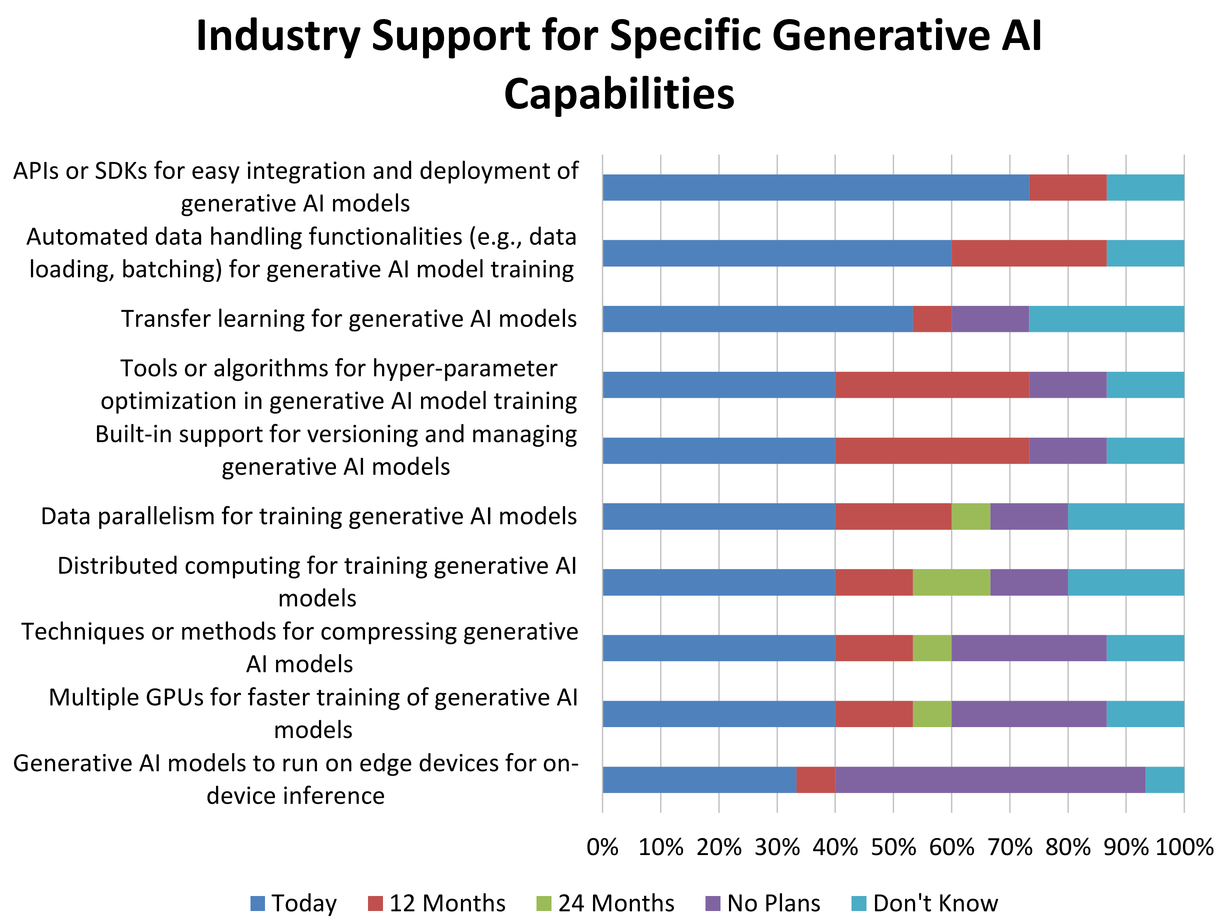


Figure 102 - Industry support for specific generative AI capabilities

Industry Impact of Generative AI

Although a significant number of vendors report either a very or somewhat positive impact from generative AI (83%), neither number represents a majority (fig. 103). Furthermore, slightly more vendors see generative AI as having a limited positive impact than do those who see it having a substantial positive impact. Overall, this data reflects the nascent state of many vendors' generative AI strategies, and the newness and relative immaturity of such capabilities compared with their other capabilities. However, for almost every vendor, the question of whether or not to "do" generative AI was never on the table for discussion. With all vendors reporting that generative AI is perceived at least as neutral (that is, no vendors report it as negatively impacting their business), the only questions regarding its availability concern timing of transitions, delivery of generative AI capabilities, and whether a vendor can successfully leverage generative AI for competitive advantage.

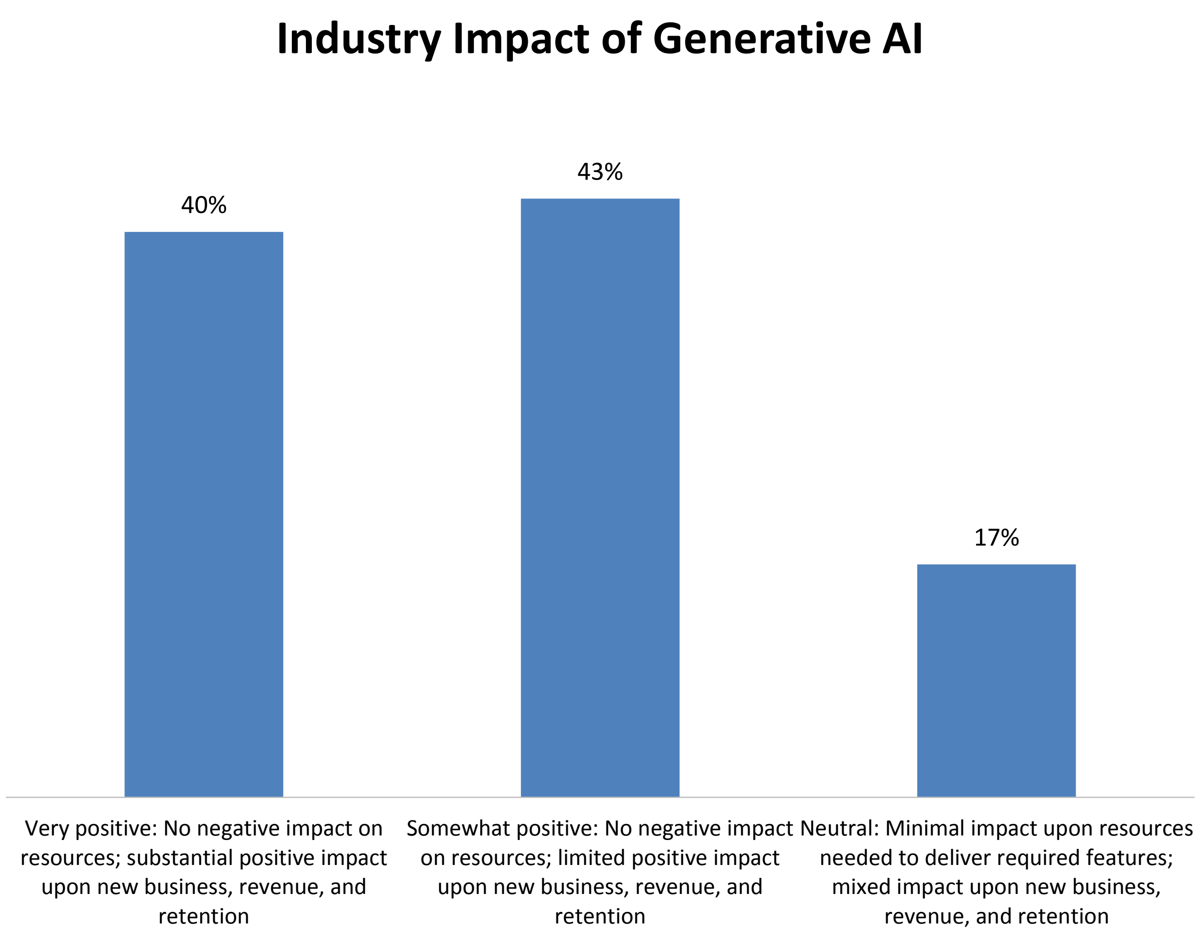


Figure 103 -- Industry impact of generative AI

Industry Support of Agentic AI

A striking 68.5% of software vendors are already supporting agentic AI (fig. 104). Another 28% plan to support it in the next 12 months, and just 2% have no plans for support. Given its recent arrival as a defined technology, the industry uptake represents an all-in affirmation of agentic AI. We can expect a range of products, platforms, and feature sets to develop and eventually better delineate its future use. We have already described user attitudes ranging from "cautiously optimistic" to "excited about the possibilities" (fig. 24), and a majority user base with a positive outlook for agentic AI. Compared with other technology waves, we feel agentic AI is arriving quickly, albeit with many loose ends, undefined strategies, and outcomes still to be determined.

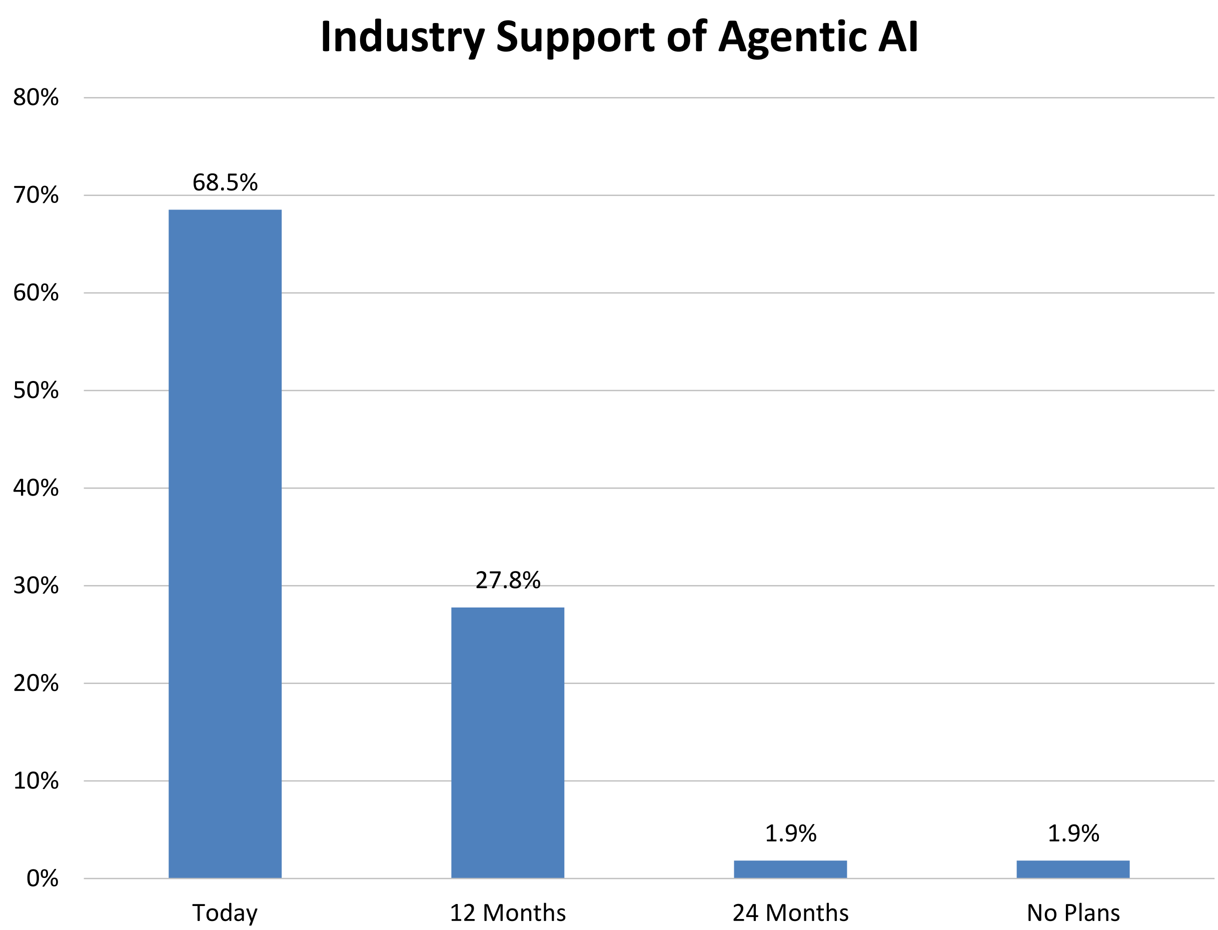


Figure 104 - Industry support of agentic AI

Industry Support of Agentic AI Capabilities

While we could have considered a broad set of capabilities, we focused on six core features that repeatedly emerged as critical to enabling enterprise-scale AI initiatives (fig. 105). These features include data integration or data virtualization, a semantic layer, workflow automation, agent chaining, integration with foundational models, and predictive and proactive systems. Each of these capabilities plays a distinct role in supporting the design and execution of intelligent, agentic workflows and decision-making systems. These findings underscore that organizations are not approaching platform modernization as a one-dimensional technical upgrade. Rather, they reflect a multifaceted strategic response—balancing near-term competitiveness with long-term resilience and operational necessity. Figure 105 lists the capabilities that vendors feel are needed to deliver an agentic AI solution.

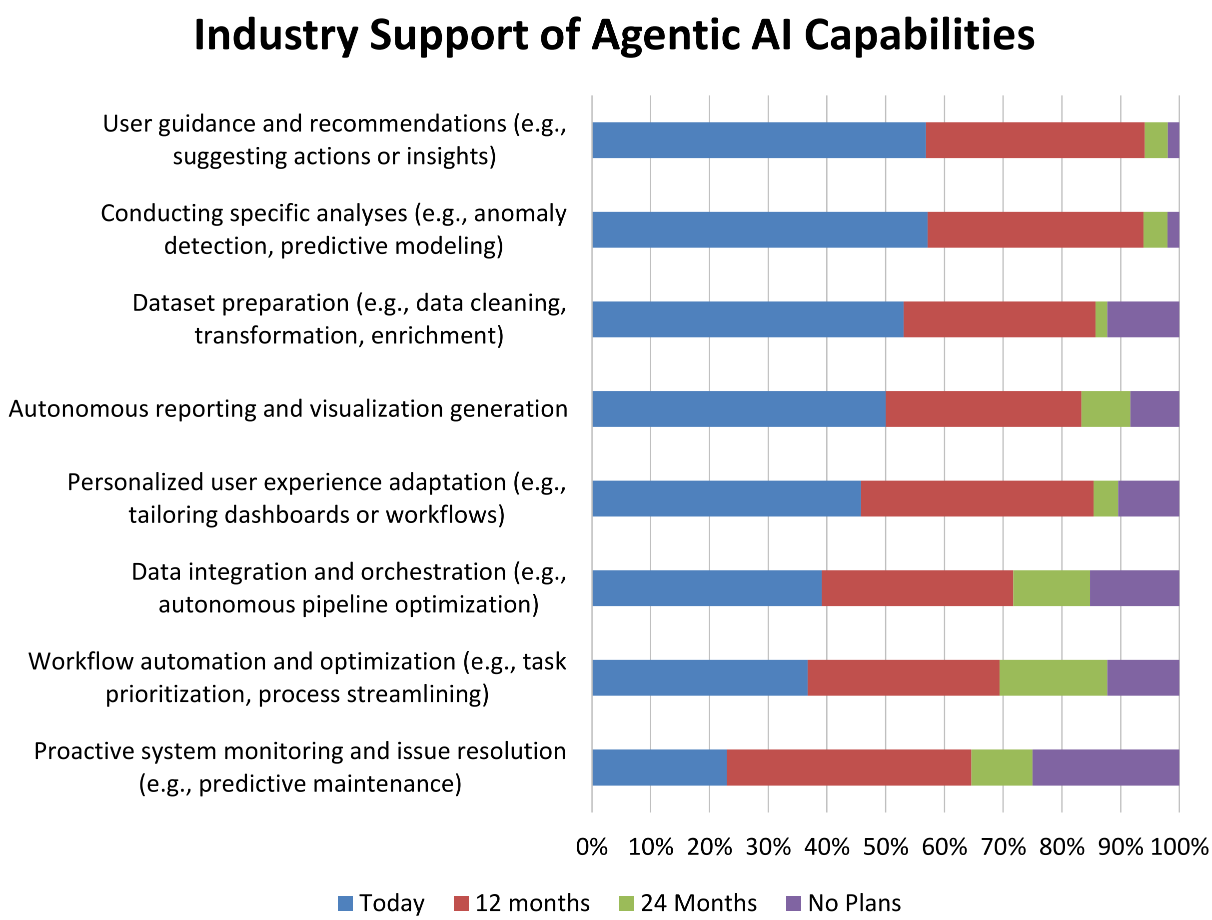


Figure 105 - Industry support of agentic AI capabilities

Industry Drivers for Agentic AI

When asked why they are prioritizing agentic AI capabilities, vendors revealed a telling hierarchy of motivations (fig. 106). Two-thirds cite market differentiation, future proofing, customer demand, and competitive necessity as their top drivers—underscoring the urgency to stay ahead of or at least at par with peers and meet rising customer expectations. At the same time, 53% pointed to operational imperatives, indicating a clear recognition that foundational changes are needed to ensure their systems and services keep pace with customer expectations.

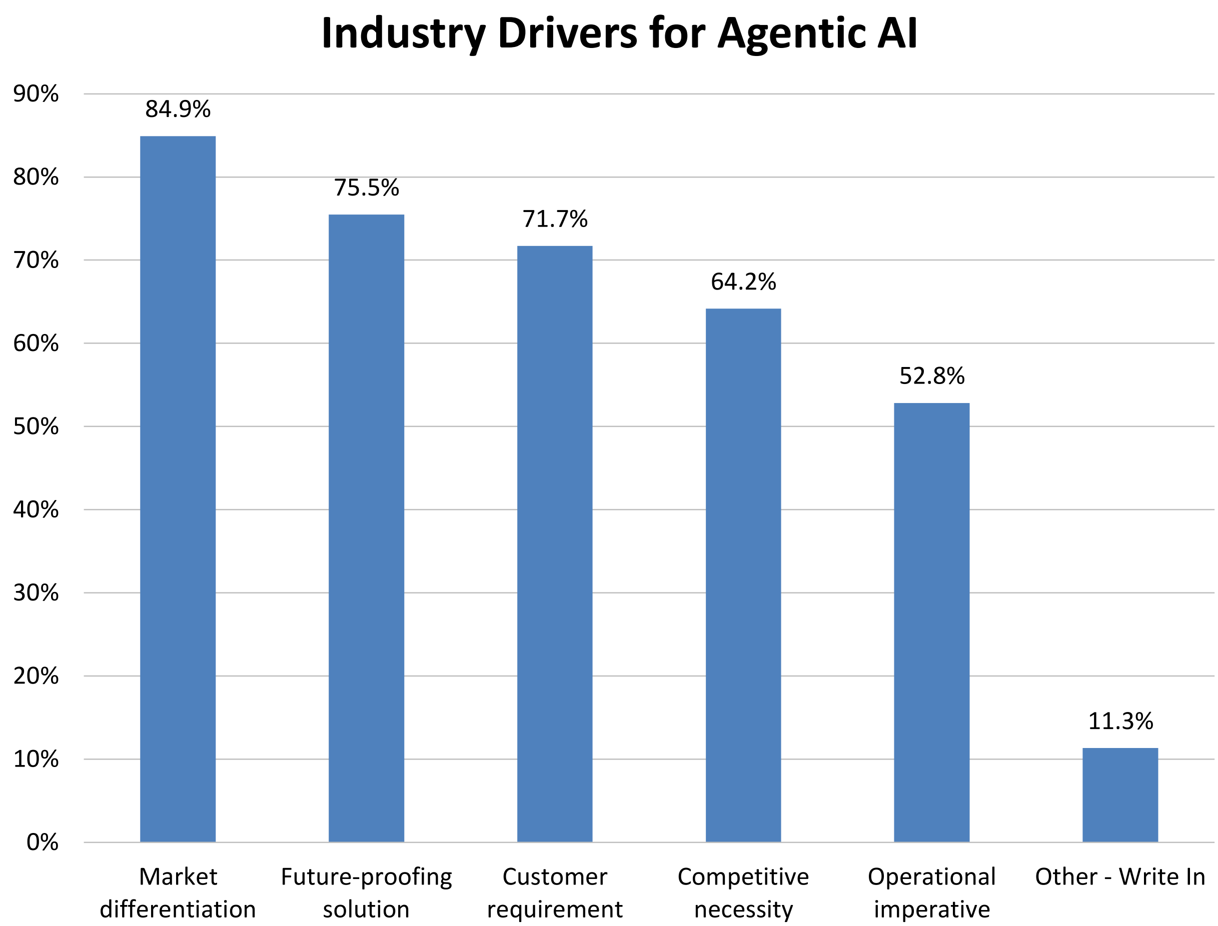


Figure 106 -- Industry drivers for agentic AI

Industry Importance of AI, Data Science and Machine Learning

Industry importance of AI, data science, and machine learning has ridden a steady 12-year rise that reached a weighted-mean score of 4.9 last year, a rarely seen consensus of effectively "critical" sentiment that slipped only slightly to 4.8 in 2025 (fig. 107). Last year's pinnacle reversed the flattening and decline during 2022 and 2023, when we speculated whether we had already seen "peak" enthusiasm for AI, data science, and machine learning. Clearly, this wasn't the case. Ongoing investments by enterprises and the industry in AI and generative AI, LLMs, and analytical solutions, and the stretched supply of processing chips is generating high demand through 2025. Though investment is unlikely to rise much further, it's also uncertain how long it will be sustained. As is often the case, industry excitement is well ahead of user sentiment (fig. 49). But we believe that mainstreaming data science and machine learning should pave a path for more widely applicable solutions from analytical vendors, hyperscalers, and chip providers.

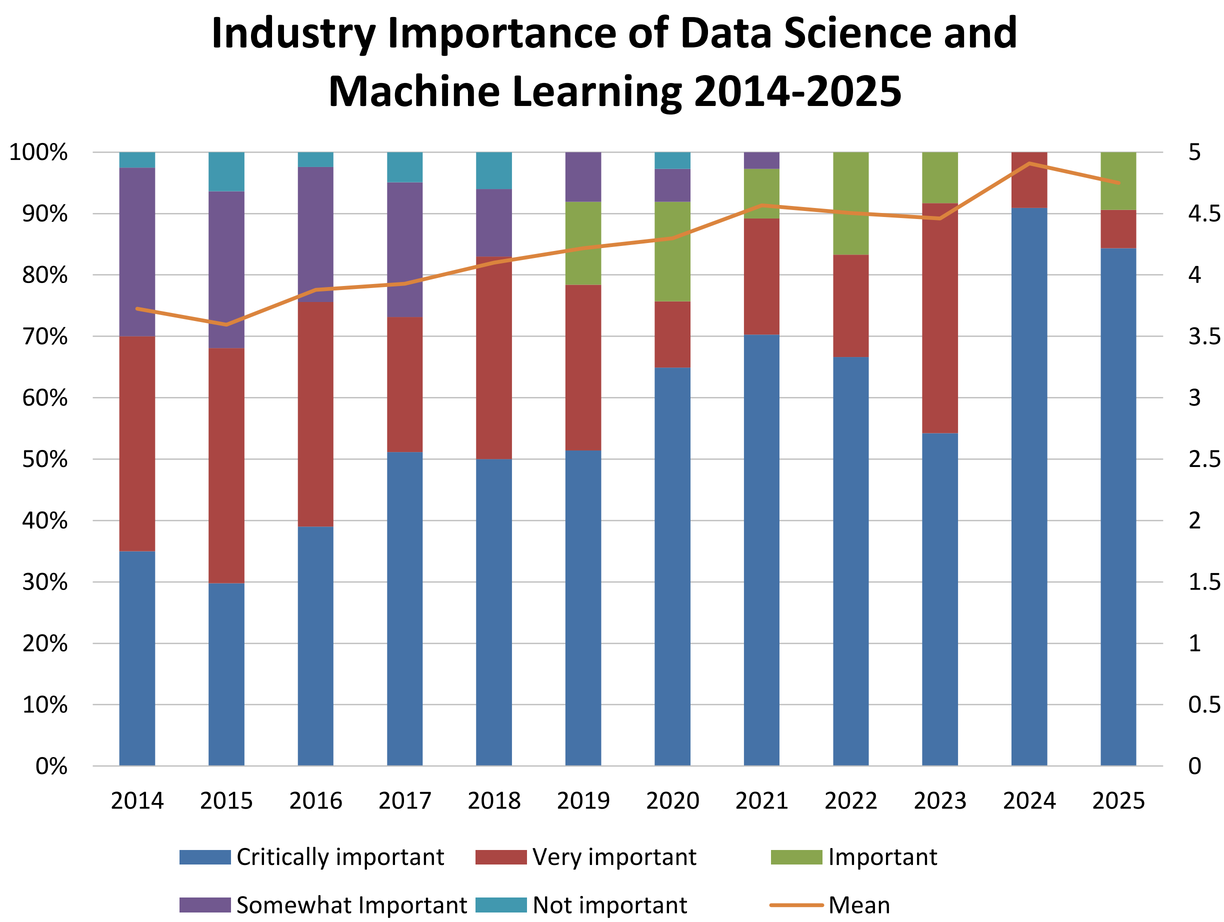


Figure 107 – Industry importance of AI, data science, and machine learning 2014-2025

Industry Support for Analytical Features and Functions

Industry respondents very strongly support multiple analytical features and functions in 2025, led by textbook statistical functions, outlier detection, cross-correlation analysis, text analytics functions, and range of regression models (fig. 108). All four features are currently supported by 77%-90% of respondents, and 90% or more expect to support them in 12 months. Support tapers only somewhat for auto machine learning, model explainability, and causal analysis, all of which are more than 70% supported. In all, 17 of 20 features and functions are currently supported by more than half of industry respondents. Vendor support and investment align well with, and far ahead of, user feature requirements this year (fig. 49).

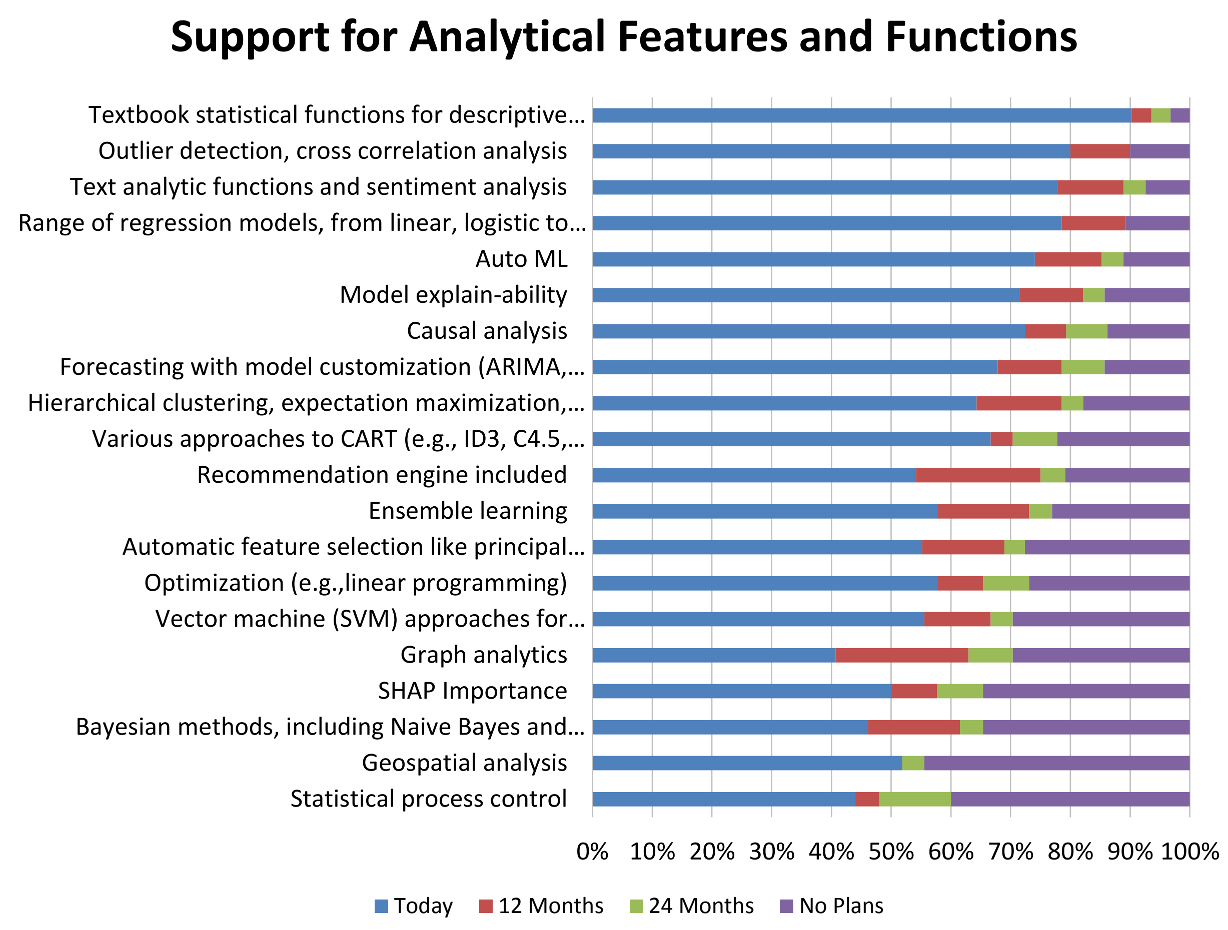


Figure 108 – Support for analytical features and functions

Viewed across 10 years of data gathering, support for many industry analytical features and functions gradually increased, jumped sharply in 2022-2024, and show more mixed results in 2025 (fig. 109). Current support for text analytic functions, causal analysis, and various approaches to CART increased year over year in 2025. Support for the top feature, textbook statistical functions, reached 90% and remained within a few points of its 2024 peak, while some other top feature support levels (range of regression models, outlier detection, auto machine learning, hierarchical clustering etc.), are lower than reported in 2024 by 10 percentage points or more. (Data for nine of 20 analytical features and functions has only been tracked since 2022-2023.) Maturing momentum in analytical features and functions nonetheless more than supports historical user demand (fig. 83).

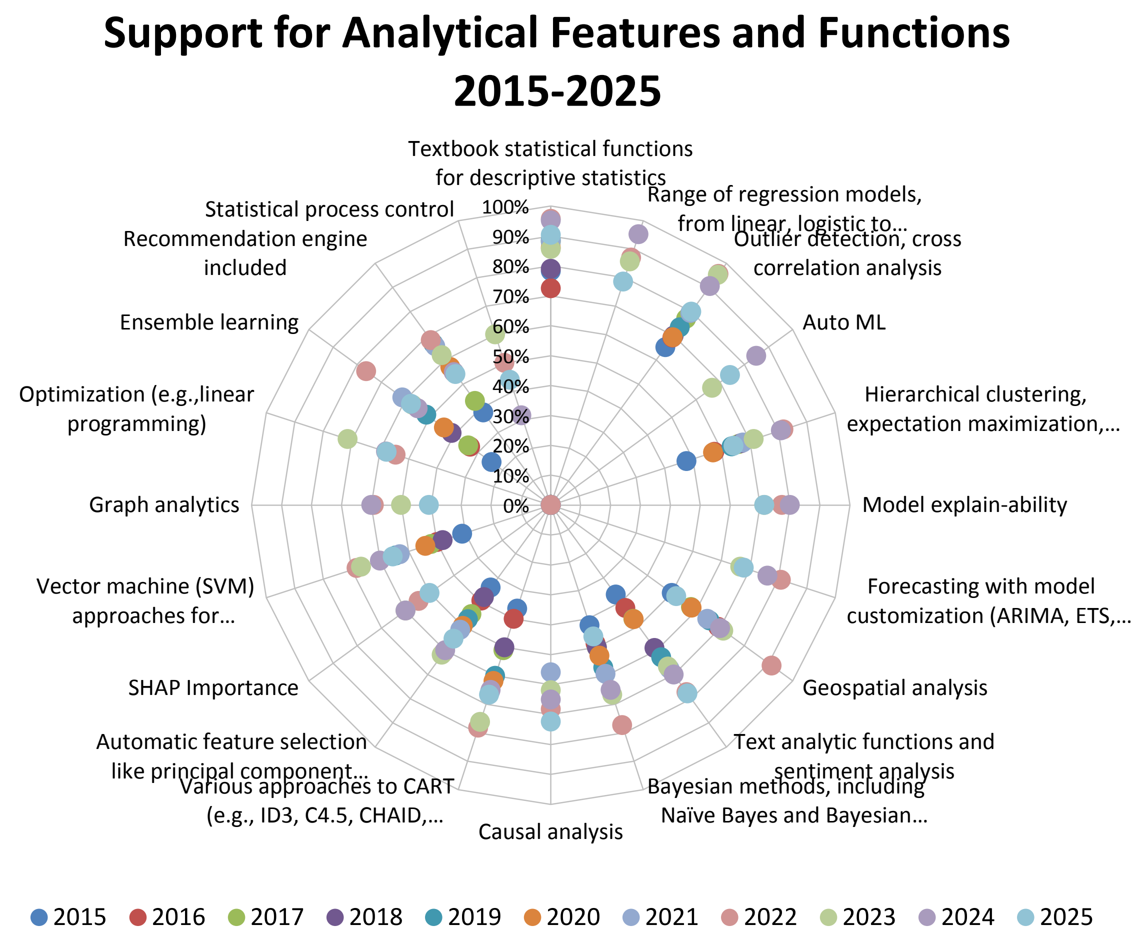


Figure 109 – Support for analytical features and functions 2015-2025

Industry Support for Neural Networks

We asked the vendor community to describe their support for 11 types of neural networks (four added since 2022). We find limited levels of support in 2025 and some rather minimal future investment planned (fig. 110). This year, current support is highest for artificial neural networks (67%, identical to 2024) and transformer network (52%, also similar to 2024). A second tier of long short-term memory, recurrent neural networks, and feed-forward deep learning is currently supported by 42%-46%. Overall current penetration is below 50% for nine of 11 neural networks and 12-month investment is not expected to lift any support level by more than about 9%. (Also see user importance, fig. 83)

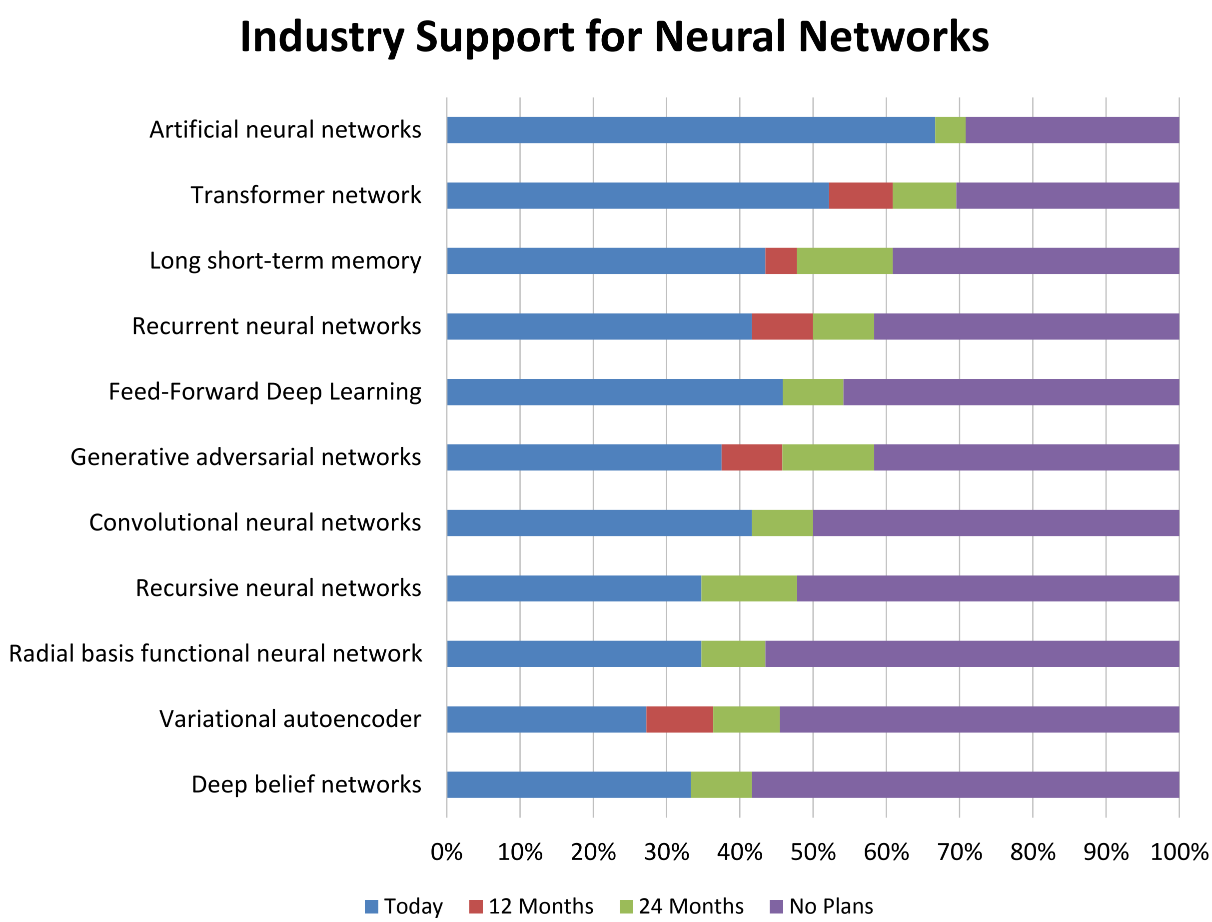


Figure 110 – Industry support for neural networks

Industry Support for Tool Usability

AI, data science, and machine learning tool usability features have very strong and mostly broad industry support in 2025 (fig. 111). Ten of 18 sampled features, led by "specialist not required" (93% current support), uphold the self-service of tool usability features. Twelve of 17 features are currently supported by at least 70% or far more of our industry sample, and vendors further expect future investments to lift support for lower-ranked features. Again, industry support well exceeds user criticality scores for every usability feature (fig. 89).

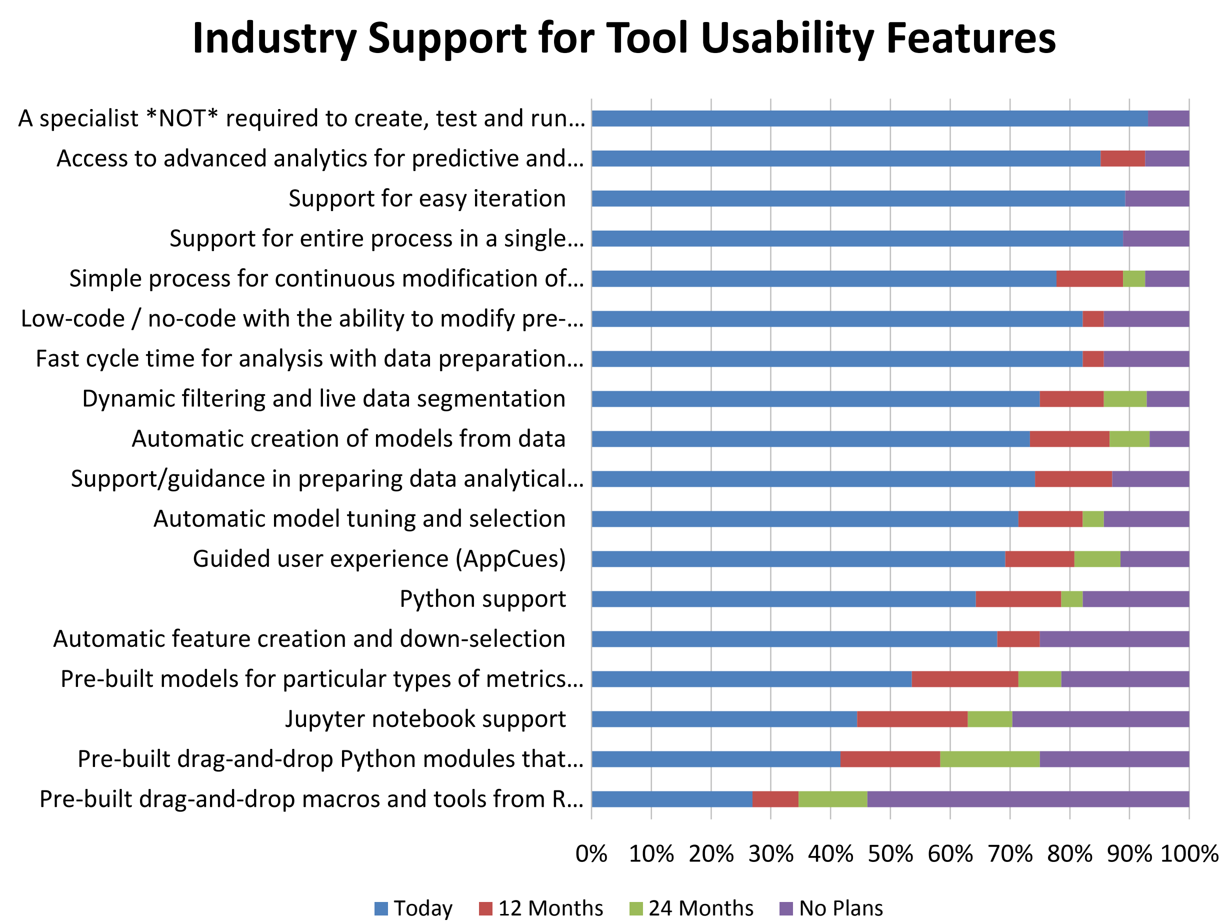


Figure 111 – Industry support for tool usability features

Industry Support for Scalability

Scalability of AI, data science, and machine learning can involve multiple different technologies and services to address high data volumes, large numbers of users, data variety, or analytic throughput. In our 2025 study, 93% of respondents currently support horizontal scaling, 81% support multi-tenant cloud services, 76% support -in-database analytics, and 73% support in-memory analytics (fig. 112). All four features (along with vertical scaling) are top-five scalability requirements for users (fig. 94). Spark support is the only other feature currently supported by half or more of industry respondents.

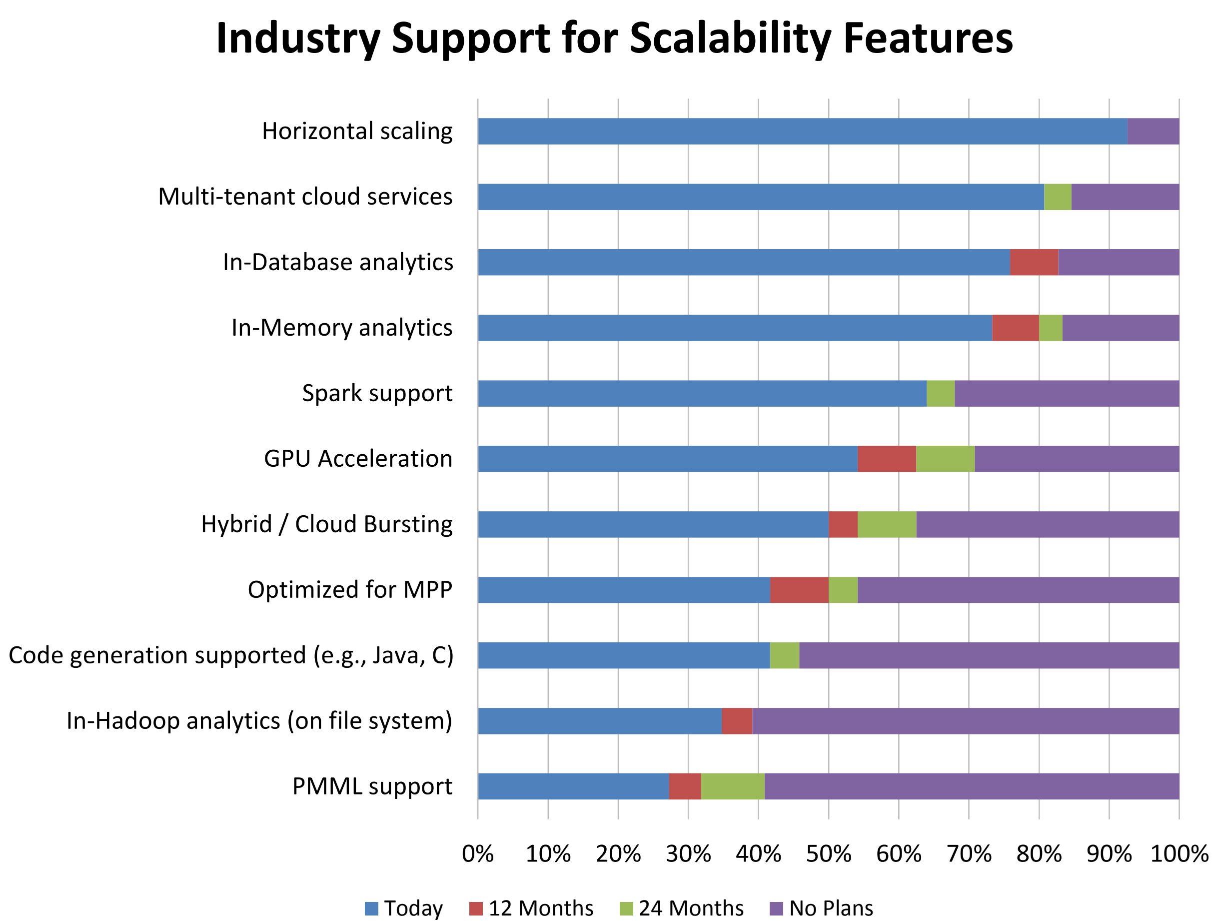


Figure 112 – Industry support for scalability features

Industry Sources of AI, Data Science, and Machine Learning Capabilities

In 2025, industry respondents most strongly support single platforms but signal openness to analytical capabilities via single or multiple products with a homogeneity of third-party/open-source features (fig. 113). This year, the largest percentage of industry respondents best describe their analytical capabilities as "single product required, some features sourced from third party/open source" (31%) and "single product required; all features proprietary." The remainder source analytical capabilities from multiple products, whether third party, open source, or proprietary.

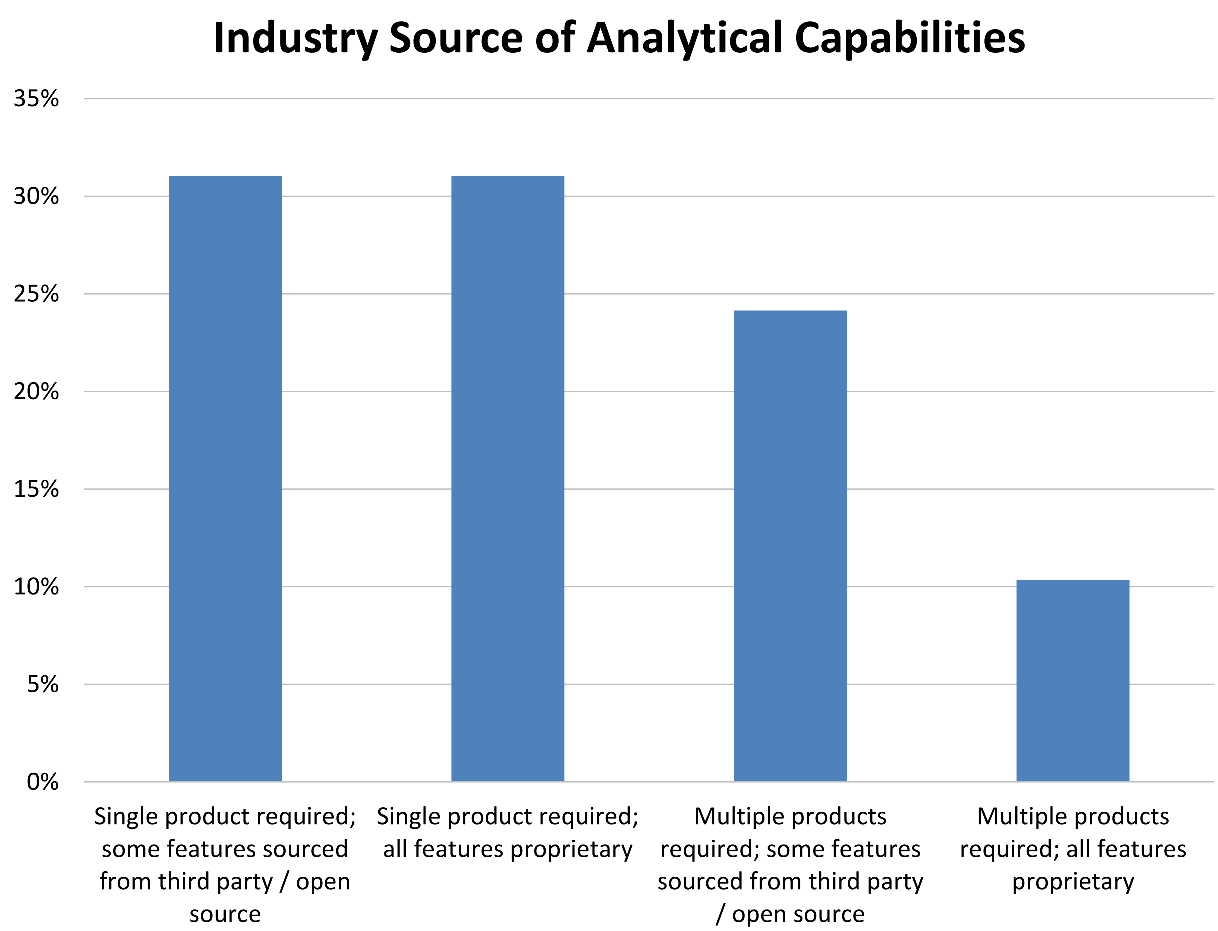


Figure 113 – Industry source of analytical capabilities

Industry Support for Analytical Data Sources

Our industry sample supports a broad range of analytical data sources in 2025, with current weighted-mean support highest for Amazon S3 (89%), Snowflake (87%), and Amazon Redshift (86%; fig. 114). Nineteen of 31 data sources are currently supported by more than half of our industry sample, with some limited future investment anticipated. (Also see user analytical data sources, fig. 43.)

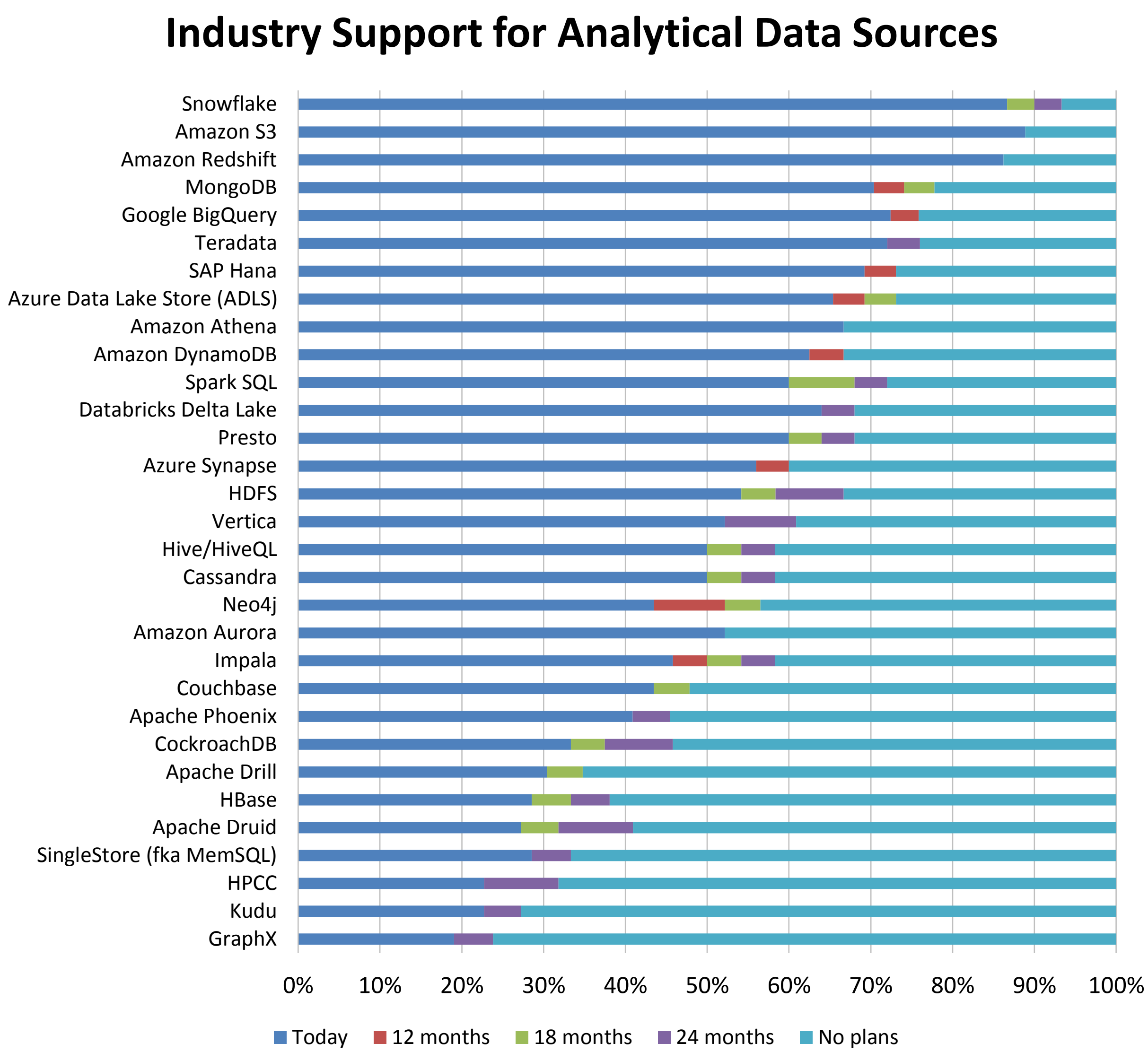


Figure 114 – Industry support for analytical data sources

AI, Data Science, and Machine Learning Vendor Ratings

In rating the vendors, we consider analytical features and functions, neural networks, data preparation, usability, ModelOps, scalability, open-source support, and access to data sources (fig. 115). It is important to scrutinize all rating categories and match vendor strengths to use cases and requirements. Ratings take into consideration vendor provided information, which is confirmed and then validated with user input.

Top-ranked vendors include Domino Data Lab (1st), Palantir (1st), Altair (2nd), Amazon AWS (3rd), Dataiku (4th), Domo (5th), Google (5th), and KNIME (5th).

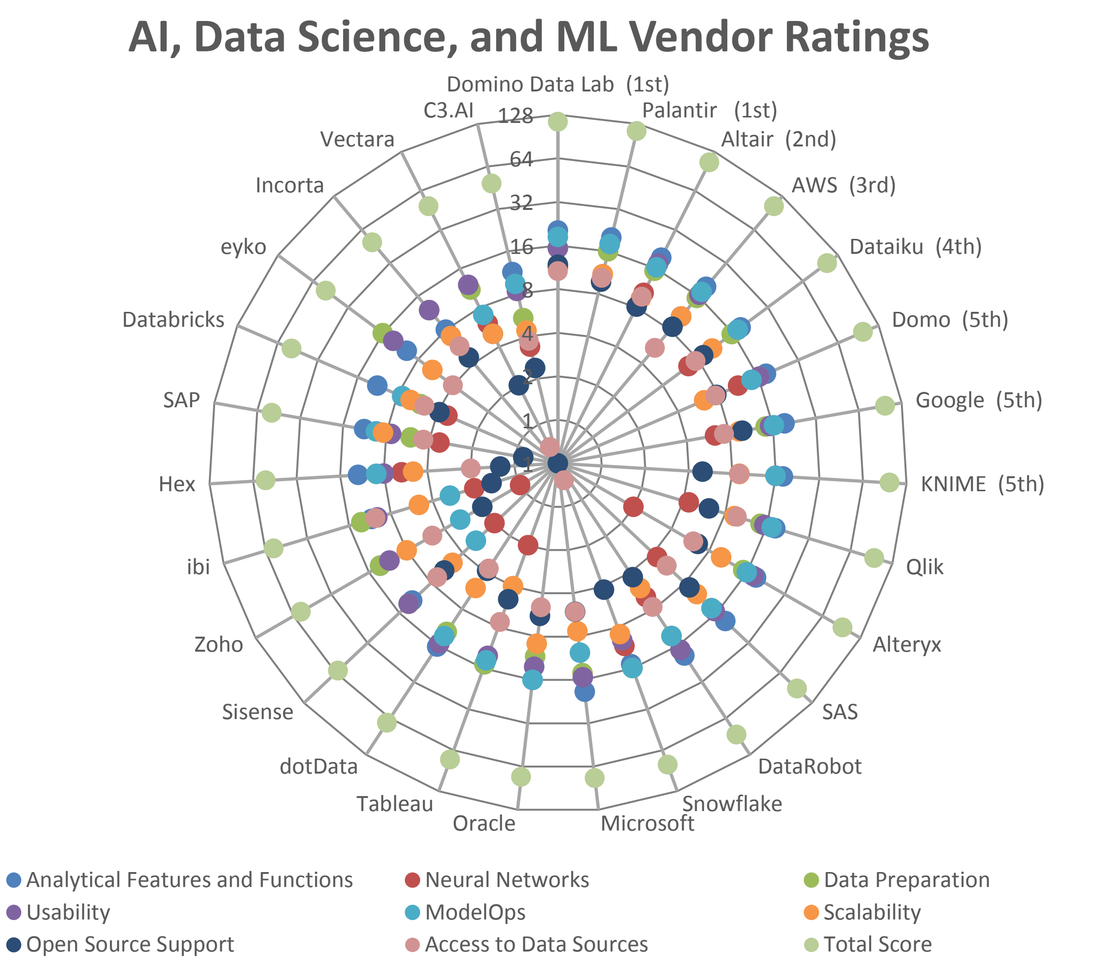


Figure 115 – AI, data science, and machine learning vendor ratings

*A logarithmic scale is used for the scoring chart to address skewness towards larger values

Other Dresner Research Reports

- Wisdom of Crowds® "Flagship" Business Intelligence Market Study
- Active Data Architecture®
- Agentic AI
- Analytical and Data Governance
- Analytical Data Infrastructure
- Analytical Platforms
- Cloud Computing and Business Intelligence
- Collective Insights®
- Data Engineering
- Embedded Business Intelligence
- Enterprise Performance Management
- ERP
- ESG Reporting
- Financial Consolidations, Close Management, and Reporting
- Generative AI
- Guided Analytics®
- Master Data Management
- ModelOps
- Sales Performance Management
- Self-Service Business Intelligence
- Semantic Layer
- Supply Chain Planning and Analysis
- Workforce Planning and Analysis

Appendix: AI, Data Science and Machine Learning Survey Instrument

Please enter your contact information below

First Name*: ____

Last Name*: ____

Title: ____

Company Name*: ____

Street Address: ____

City: ____

State: ____

Zip: ____

Country: ____

Email Address*: ____

Phone Number: ____

URL: ____

What major geography do you reside in?*

( ) North America

( ) Europe, Middle East and Africa

( ) Latin America

( ) Asia Pacific

Please identify your primary industry*

( ) Advertising

( ) Aerospace

( ) Agriculture

( ) Apparel & accessories

( ) Automotive

( ) Aviation

( ) Biotechnology

( ) Broadcasting

( ) Business services

( ) Chemical

( ) Construction

( ) Consulting

( ) Consumer products

( ) Defense

( ) Distribution & logistics

( ) Education (Higher Ed)

( ) Education (K-12)

( ) Energy

( ) Entertainment and leisure

( ) Executive search

( ) Federal government

( ) Financial services

( ) Food, beverage and tobacco

( ) Healthcare

( ) Hospitality

( ) Insurance

( ) Legal

( ) Manufacturing

( ) Mining

( ) Motion picture and video

( ) Not for profit

( ) Pharmaceuticals

( ) Publishing

( ) Real estate

( ) Retail & wholesale

( ) Sports

( ) State and local government

( ) Technology

( ) Telecommunications

( ) Transportation

( ) Utilities

( ) Other - Write In: ____

How many employees does your company employ worldwide?

( ) 1 - 100

( ) 101 - 1,000

( ) 1,001 - 2,000

( ) 2,001 - 5,000

( ) 5,001 - 10,000

( ) More than 10,000

What function do you report into?*

( ) Business Intelligence Competency Center

( ) Executive management

( ) Faculty (Education)

( ) Finance

( ) Human resources

( ) Information Technology (IT)

( ) Manufacturing

( ) Marketing

( ) Medical staff (Healthcare)

( ) Operations

( ) Research and development (R&D)

( ) Sales

( ) Strategic planning function

( ) Supply chain

( ) Other - Write In: ____

Does your organization use or intend to use data science and machine learning?

( ) Yes, we use data science and machine learning today in production

( ) Data science and machine learning is being used in very limited ways or as proof of concept

( ) We are currently evaluating data science and machine learning software

( ) No, we have no plans to use data science and machine learning at all

( ) We may use data science and machine learning in the future

What are your plans for AI, data science and machine learning in the future?

( ) Will Adopt this Year

( ) Will Adopt Next Year

( ) Will Adopt Beyond Next Year

How long have data science and machine learning been in use in your organization?

( ) Less than 1 year

( ) 1-2 years

( ) 2-3 years

( ) 3-5 years

( ) More than 5 years

Which kinds of users use (or will use) data science and machine learning within your organization?

	Constantly	**Often**	**Occasionally**	**Rarely**	**Never**
BI expert	( )	( )	( )	( )	( )

Business Analyst	( )	( )	( )	( )	( )
"Citizen" Data Scientist	( )	( )	( )	( )	( )
Executive	( )	( )	( )	( )	( )
Financial Analyst	( )	( )	( )	( )	( )
IT Staff	( )	( )	( )	( )	( )
Marketing Analyst	( )	( )	( )	( )	( )
Statistician / Data Scientist	( )	( )	( )	( )	( )
Third-Party Consultant	( )	( )	( )	( )	( )

How is data science and machine learning being used in your organization?

	Today	**12 Months**	**24 Months**	**No Plans**	**Don't know**
Churn Prevention	( )	( )	( )	( )	( )
Cognitive Robotic Process Automation	( )	( )	( )	( )	( )
Customer Lifetime	( )	( )	( )	( )	( )

Value					
Customer Segmentation	( )	( )	( )	( )	( )
Demand Forecasting	( )	( )	( )	( )	( )
Fraud Detection	( )	( )	( )	( )	( )
Next Best Action	( )	( )	( )	( )	( )
Predictive Maintenance	( )	( )	( )	( )	( )
Price Optimization	( )	( )	( )	( )	( )
Product Propensity	( )	( )	( )	( )	( )
Quality Assurance	( )	( )	( )	( )	( )
Risk Management	( )	( )	( )	( )	( )
Up and Cross-Selling	( )	( )	( )	( )	( )

Analytical Features: Which of the following features are important for AI, data science and machine learning?

	Critical	**Very Important**	**Important**	**Somewhat Important**	**Not Important**

Automatic feature selection like principal component analysis (PCA)	( )	( )	( )	( )	( )
Text analytic functions and sentiment analysis	( )	( )	( )	( )	( )
Various approaches to CART (e.g., ID3, C4.5, CHAID, MARS, random forests, gradient boosting)	( )	( )	( )	( )	( )
Vector machine (SVM) approaches for classification and estimation	( )	( )	( )	( )	( )
Neural networks supported	( )	( )	( )	( )	( )
Geospatial analysis	( )	( )	( )	( )	( )
Range of regression models, from linear, logistic to nonlinear	( )	( )	( )	( )	( )
Recommendation engine included	( )	( )	( )	( )	( )
Hierarchical clustering, expectation	( )	( )	( )	( )	( )

maximization, k-Means, and variants of self-organizing maps					
Textbook statistical functions for descriptive statistics	( )	( )	( )	( )	( )
Bayesian methods, including Naïve Bayes and Bayesian Networks	( )	( )	( )	( )	( )
Ensemble learning	( )	( )	( )	( )	( )
Video analysis	( )	( )	( )	( )	( )
Model management and governance	( )	( )	( )	( )	( )
Auto ML	( )	( )	( )	( )	( )
Model explain-ability	( )	( )	( )	( )	( )
Graph analytics	( )	( )	( )	( )	( )
Forecasting with model customization (ARIMA, ETS, STL)	( )	( )	( )	( )	( )
Optimization (e.g.,linear programming)	( )	( )	( )	( )	( )

Outlier detection	( )	( )	( )	( )	( )
Cross correlation analysis	( )	( )	( )	( )	( )
SHAP Importance	( )	( )	( )	( )	( )
Statistical process control	( )	( )	( )	( )	( )

Which types of neural networks are most important to your organization?

	Critical	**Very Important**	**Important**	**Somewhat Important**	**Not Important**	**Don't Know**
Artificial neural network	( )	( )	( )	( )	( )	( )
Convolutional neural networks	( )	( )	( )	( )	( )	( )
Long short-term memory	( )	( )	( )	( )	( )	( )
Recursive neural networks	( )	( )	( )	( )	( )	( )
Deep learning neural networks	( )	( )	( )	( )	( )	( )
Adversarial	( )	( )	( )	( )	( )	( )

neural networks						

Data Preparation: For AI, data science and machine learning, which data preparation capabilities are important?

	Critical	**Very Important**	**Important**	**Somewhat Important**	**Not Important**
Support for Cutting, Merging, and Replacing Values	( )	( )	( )	( )	( )
Complex Filtering	( )	( )	( )	( )	( )
Support for Data Type Conversions	( )	( )	( )	( )	( )
Cleansing and Enriching Source Data	( )	( )	( )	( )	( )
Detecting Duplicates or Outliers	( )	( )	( )	( )	( )
Set Operations (e.g., joins, aggregations or pivot tables)	( )	( )	( )	( )	( )
Data Flows for Multi-step Transformations	( )	( )	( )	( )	( )
Data Lineage, Profiling and Quality	( )	( )	( )	( )	( )

Differential Privacy	( )	( )	( )	( )	( )
Formula Scripting	( )	( )	( )	( )	( )
Imbalanced Classes Handling / Data Synthesis	( )	( )	( )	( )	( )
ML-Driven Data Prep Recommendations	( )	( )	( )	( )	( )
Model Data Preprocessing (imputation, encoding, feature scaling, holdout, 5-fold CV)	( )	( )	( )	( )	( )
Semi-structured Extraction and Manipulation	( )	( )	( )	( )	( )
Text Analytics and Enrichment (e.g., sentiment analytics, key word extraction)	( )	( )	( )	( )	( )
Binning	( )	( )	( )	( )	( )

Usability: Which usability features are important for AI, data science and machine learning?

	Critical	**Very Important**	**Important**	**Somewhat Important**	**Not Important**
A specialist NOT	( )	( )	( )	( )	( )

required to create analytical models, test and run them					
Support/guidance in preparing data analytical models	( )	( )	( )	( )	( )
Automatic creation of models from data	( )	( )	( )	( )	( )
Fast cycle time for analysis with data preparation functions	( )	( )	( )	( )	( )
Access to advanced analytics for predictive and temporal analysis	( )	( )	( )	( )	( )
Support for easy iteration	( )	( )	( )	( )	( )
Simple process for continuous modification of models	( )	( )	( )	( )	( )
Pre-built drag- and -drop macros and tools from R that require no scripting or programming	( )	( )	( )	( )	( )
Support for entire process in a single application/user	( )	( )	( )	( )	( )

interface					
Automatic feature creation and down-selection	( )	( )	( )	( )	( )
Automatic model tuning and selection	( )	( )	( )	( )	( )
Dynamic filtering and live data segmentation	( )	( )	( )	( )	( )
Guided user experience (AppCues)	( )	( )	( )	( )	( )
Low-code / no-code with the ability to modify pre-built modules (execution transparency)	( )	( )	( )	( )	( )
Pre-built drag-and-drop Python modules that require no scripting or programming	( )	( )	( )	( )	( )
Pre-built models for particular types of metrics and KPIs	( )	( )	( )	( )	( )

Scalability: Which scalability features are important for AI, data science and machine learning?

	Critical	Very Important	Important	Somewhat Important	Not Important
In-Memory Analytics	( )	( )	( )	( )	( )
In-Database Analytics	( )	( )	( )	( )	( )
In-Hadoop Analytics (on file system)	( )	( )	( )	( )	( )
Optimized for MPP Architecture	( )	( )	( )	( )	( )
Multi-tenant Cloud Services	( )	( )	( )	( )	( )
PMML Support	( )	( )	( )	( )	( )
Code Generation Supported (e.g., Java, C)	( )	( )	( )	( )	( )
GPU Acceleration	( )	( )	( )	( )	( )
Spark Support	( )	( )	( )	( )	( )
Hybrid / Cloud Bursting	( )	( )	( )	( )	( )
Horizontal	( )	( )	( )	( )	( )

Scaling					

What is the importance of big data/open source technologies (e.g., Apache) and architecture as a part of your data science and machine learning strategies?

( ) Critical

( ) Very Important

( ) Important

( ) Somewhat Important

( ) Not Important

Which open source/big data infrastructure features are important for AI, data science and machine learning?

	Critical	**Very Important**	**Important**	**Somewhat Important**	**Not Important**
Spark	( )	( )	( )	( )	( )
Atlas	( )	( )	( )	( )	( )
Knox Gateway	( )	( )	( )	( )	( )
Kafka	( )	( )	( )	( )	( )
Confluent KSQL	( )	( )	( )	( )	( )
Flink	( )	( )	( )	( )	( )
Amazon Kinesis	( )	( )	( )	( )	( )
Google	( )	( )	( )	( )	( )

Dataflow					
Apache Ranger	( )	( )	( )	( )	( )
Apache Sentry	( )	( )	( )	( )	( )
Spark Structured Streaming	( )	( )	( )	( )	( )
Apache Atlas	( )	( )	( )	( )	( )
Azure Data Factory	( )	( )	( )	( )	( )
Databricks	( )	( )	( )	( )	( )
Elasticsearch	( )	( )	( )	( )	( )

Which open source/big data deployment technologies are important for AI, data science and machine learning?

	Critical	**Very Important**	**Important**	**Somewhat Important**	**Not Important**
Microservices architecture	( )	( )	( )	( )	( )
Kubernetes	( )	( )	( )	( )	( )
Yarn	( )	( )	( )	( )	( )
Mesos	( )	( )	( )	( )	( )
Docker swarm	( )	( )	( )	( )	( )

Nomad	( )	( )	( )	( )	( )
Consul	( )	( )	( )	( )	( )
Etcd	( )	( )	( )	( )	( )
Zookeeper	( )	( )	( )	( )	( )
Zipkin	( )	( )	( )	( )	( )
Prometheus	( )	( )	( )	( )	( )
Grafana	( )	( )	( )	( )	( )
FluentD	( )	( )	( )	( )	( )
Akka	( )	( )	( )	( )	( )

Which open source/big data statistical and machine learning technologies are important for AI, data science and machine learning?

	Critical	**Very Important**	**Important**	**Somewhat Important**	**Not Important**
Mahout	( )	( )	( )	( )	( )
R Language	( )	( )	( )	( )	( )
Oryx	( )	( )	( )	( )	( )
Myrrix	( )	( )	( )	( )	( )
Spark MLib	( )	( )	( )	( )	( )
scikit-learn	( )	( )	( )	( )	( )
Tensorflow	( )	( )	( )	( )	( )

MLflow	( )	( )	( )	( )	( )
PyTorch	( )	( )	( )	( )	( )
Kubeflow	( )	( )	( )	( )	( )
Pandas	( )	( )	( )	( )	( )
Dask	( )	( )	( )	( )	( )
Anaconda	( )	( )	( )	( )	( )
H2O Sparkling Water	( )	( )	( )	( )	( )
Keras	( )	( )	( )	( )	( )
sparklyr	( )	( )	( )	( )	( )
SpaCy	( )	( )	( )	( )	( )

Which data sources are important for AI, data science and machine learning?

	Critical	**Very Important**	**Important**	**Somewhat Important**	**Not Important**
Google BigQuery	( )	( )	( )	( )	( )
HBase	( )	( )	( )	( )	( )
HDFS	( )	( )	( )	( )	( )
Hive/HiveQL	( )	( )	( )	( )	( )
Impala	( )	( )	( )	( )	( )
MongoDB	( )	( )	( )	( )	( )

Amazon Redshift	( )	( )	( )	( )	( )
Spark SQL	( )	( )	( )	( )	( )
Couchbase	( )	( )	( )	( )	( )
Cassandra	( )	( )	( )	( )	( )
Amazon S3	( )	( )	( )	( )	( )
Neo4j	( )	( )	( )	( )	( )
Presto	( )	( )	( )	( )	( )
Kudu	( )	( )	( )	( )	( )
Amazon DynamoDB	( )	( )	( )	( )	( )
Azure Data Lake Store (ADLS)	( )	( )	( )	( )	( )
Apache Drill	( )	( )	( )	( )	( )
SingleStore (fka MemSQL)	( )	( )	( )	( )	( )
Snowflake	( )	( )	( )	( )	( )
SAP Hana	( )	( )	( )	( )	( )
Apache Phoenix	( )	( )	( )	( )	( )
Vertica	( )	( )	( )	( )	( )
Teradata	( )	( )	( )	( )	( )
CockroachDB	( )	( )	( )	( )	( )

Apache Druid	( )	( )	( )	( )	( )
HPCC	( )	( )	( )	( )	( )
GraphX	( )	( )	( )	( )	( )
Oracle Big Data	( )	( )	( )	( )	( )
Amazon Athena	( )	( )	( )	( )	( )
Amazon Aurora	( )	( )	( )	( )	( )
Azure Synapse	( )	( )	( )	( )	( )
Databricks Delta Lake	( )	( )	( )	( )	( )

